Mathematical Prerequisites

Part I
Analysis

1 The Real Line

Remark (Elementary facts assumed)**.** Elementary calculus and linear algebra are taken as known: the product and quotient rules, integration by parts on $\mathbb{R}$, the fundamental theorem of calculus for complex-valued functions of a real variable, $(e^{it})' = ie^{it}$, the extension of a linearly independent set to a basis and the dimension of a subspace, and the fact that a right inverse of a square matrix is its inverse.

Definition P.1 (Supremum)**.** s is the **supremum** of $S \subseteq \mathbb{R}$ if s is an upper bound and no smaller number is; $\inf S = -\sup(-S)$. The **completeness axiom** states that every non-empty S bounded above has a supremum in $\mathbb{R}$.

Theorem P.2 (Monotone convergence for sequences)**.** A bounded increasing real sequence converges, to the supremum of its values.

Proof. Let $s = \sup_n a_n$, which exists by completeness since the set of values is non-empty and bounded above. For $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound, so $a_N > s - \varepsilon$ for some N. For $n \geqslant N$, monotonicity gives $a_n \geqslant a_N$, and $a_n \leqslant s$ because s is an upper bound; hence $s - \varepsilon < a_n \leqslant s$, that is $|a_n - s| < \varepsilon$, for all $n \geqslant N$. □

Theorem P.3 (Bolzano–Weierstrass)**.** Every bounded sequence in $\mathbb{R}^d$ has a convergent subsequence.

Proof. Take $d = 1$ first. Given a bounded (a_n), say $|a_n| \leqslant B$ for all n, put $b_m = \sup_{n \geqslant m} a_n$, finite by completeness. The sequence (b_m) is decreasing (a supremum over a smaller set is smaller) and bounded below by $-B$, so it converges to a limit b by Theorem P.2 applied to $(-b_m)$, and $b \leqslant b_m$ for every m. Choose indices $n_1 < n_2 < \cdots$ with $|a_{n_k} - b| < 1/k$ recursively: put $n_0 = 0$; given n_{k-1}, choose $m > n_{k-1}$ with $b_m < b + 1/k$ (possible as $b_m \downarrow b$), then $n_k \geqslant m$ with $a_{n_k} > b_m - 1/k$ (possible as b_m is a supremum, so $b_m - 1/k$ is not an upper bound of $\{a_n : n \geqslant m\}$). Then

$n_k \geqslant m > n_{k-1}$, and since $b \leqslant b_m$ and $a_{n_k} \leqslant b_m$,

$$b - 1/k \leqslant b_m - 1/k < a_{n_k} \leqslant b_m < b + 1/k,$$

that is $|a_{n_k} - b| < 1/k$. Hence $a_{n_k} \to b$.

For general d, write $a_n = (a_n^{(1)}, \ldots, a_n^{(d)})$; each coordinate sequence is bounded since $|a_n^{(j)}| \leqslant |a_n|$. Construct, for $j = 1, \ldots, d$, an infinite set of indices $N_j \subseteq \{1, 2, \ldots\}$ with $N_1 \supseteq N_2 \supseteq \cdots \supseteq N_d$ such that $(a_n^{(i)})_{n \in N_j}$, listed in increasing order of n, converges for every $i \leqslant j$. For $j = 1$, apply the case $d = 1$ to $(a_n^{(1)})_{n \geqslant 1}$ and let N_1 be the set of indices of the convergent subsequence. Given N_{j-1}, the sequence $(a_n^{(j)})_{n \in N_{j-1}}$ is bounded, so by the case $d = 1$ it has a convergent subsequence; let $N_j \subseteq N_{j-1}$ be its set of indices. Coordinates $i < j$ still converge along N_j, because a subsequence of a convergent sequence converges to the same limit, and coordinate j converges by construction. After step d, every coordinate of $(a_n)_{n \in N_d}$ converges, say $a_n^{(i)} \to c_i$; with $c = (c_1, \ldots, c_d)$, $|a_n - c| \leqslant \sum_i |a_n^{(i)} - c_i| \to 0$ along N_d, so this subsequence converges in $\mathbb{R}^d$. □

Definition P.4 (Compactness)**.** $K \subseteq \mathbb{R}^d$ is **compact** if every sequence in K has a subsequence converging to a point of K.

Theorem P.5 (Heine–Borel)**.** $K \subseteq \mathbb{R}^d$ is compact if and only if it is closed and bounded.

Proof. ($\Leftarrow$) A sequence in a bounded K is bounded, so it has a convergent subsequence by Theorem P.3, and the limit of that subsequence lies in K because K is closed (contains the limits of its convergent sequences).
($\Rightarrow$) If K were unbounded, for each n the number n would not bound K, so K would contain some x_n with $|x_n| > n$. No subsequence of such a sequence converges: if $x_{n_k} \to x$ then $|x_{n_k} - x| < 1$ for all large k, so $|x_{n_k}| < |x| + 1$, whereas $|x_{n_k}| > n_k \geqslant k \to \infty$. If K were not closed there would be $x_n \in K$ with $x_n \to x \notin K$; every subsequence also tends to x, and a sequence has at most one limit (if $y_k \to x$ and $y_k \to x'$ then $|x - x'| \leqslant |x - y_k| + |y_k - x'| \to 0$), so no subsequence converges to a point of K. Either way K would not be compact. □

Theorem P.6 (Rolle)**.** If f is continuous on $[a,b]$, differentiable on (a,b), and $f(a)=f(b)$, then $f'(\xi)=0$ for some $\xi\in(a,b)$.

Proof. $[a,b]$ is compact by Theorem P.5, so f attains a maximum and a minimum on it: taking $x_n\in[a,b]$ with $f(x_n)\to M=\sup_{[a,b]}f$, a subsequence converges to some $x\in[a,b]$ and continuity gives $f(x)=M$ (so $M<\infty$), and the same argument applied to $-f$ gives the minimum. If both are attained at endpoints then, since $f(a)=f(b)$, the maximum M and the minimum m are equal, so $m\leqslant f(x)\leqslant M=m$ for every $x\in[a,b]$: f is constant, $f'\equiv 0$ on (a,b), and any interior ξ works. Otherwise an extremum is attained at an interior ξ. Suppose it is a maximum (for a minimum apply the argument to $-f$, whose derivative is $-f'$). For $0<|h|<\min(\xi-a,b-\xi)$ the point $\xi+h$ lies in $[a,b]$, so $f(\xi+h)-f(\xi)\leqslant 0$, and the difference quotient $\{f(\xi+h)-f(\xi)\}/h$ is $\leqslant 0$ for $h>0$ and $\geqslant 0$ for $h<0$. The derivative $f'(\xi)$ is the limit of these quotients as $h\to 0$ from either side; a limit of numbers $\leqslant 0$ is $\leqslant 0$, and a limit of numbers $\geqslant 0$ is $\geqslant 0$, so $f'(\xi)=0$. □

Corollary P.7 (Mean value theorem)**.** Under the same hypotheses without $f(a)=f(b)$, there is $\xi\in(a,b)$ with $f(b)-f(a)=f'(\xi)(b-a)$.

Proof. Apply Theorem P.6 to $g(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a)$, which is continuous on $[a,b]$, differentiable on (a,b) (a difference of f and a polynomial), and has $g(a) = 0$ and $g(b) = f(b)-f(a)-\{f(b)-f(a)\} = 0$. This gives $\xi \in (a,b)$ with $0 = g'(\xi) = f'(\xi)-\{f(b)-f(a)\}/(b-a)$; multiply by $b-a$. □

2 Limits and Semi-continuity

Definition P.8 (Upper and lower limits)**.** For a real sequence (a_n), $\limsup_n a_n = \inf_m \sup_{n \geqslant m} a_n$ and $\liminf_n a_n = \sup_m \inf_{n \geqslant m} a_n$, as elements of $[-\infty, \infty]$.

Proposition P.9. $\liminf_n a_n \leqslant \limsup_n a_n$, and $a_n \to a$ if and only if $\liminf_n a_n = \limsup_n a_n = a$.

Proof. Write $\ell_m = \inf_{n \geqslant m} a_n$ and $u_m = \sup_{n \geqslant m} a_n$. Then $\ell_m \leqslant u_m$ for each m; ℓ_m increases in m and u_m decreases, because the set $\{a_n : n \geqslant m\}$ shrinks as m grows. For any m, m', with $k = \max(m, m')$, $\ell_m \leqslant \ell_k \leqslant u_k \leqslant u_{m'}$. Taking the supremum over m gives $\liminf_n a_n \leqslant u_{m'}$ for every m', and then the infimum over m' gives $\liminf_n a_n \leqslant \limsup_n a_n$.

If both equal a, let $\varepsilon > 0$. Since $\sup_m \ell_m = a$, the number $a - \varepsilon$ is not an upper bound of the ℓ_m, so $\ell_{m_1} > a - \varepsilon$ for some m_1; since $\inf_m u_m = a$, $u_{m_2} < a + \varepsilon$ for some m_2. With $m = \max(m_1, m_2)$ and the monotonicity of ℓ and u, $a - \varepsilon < \ell_m \leqslant a_n \leqslant u_m < a + \varepsilon$ for $n \geqslant m$, so $|a_n - a| < \varepsilon$ for $n \geqslant m$: $a_n \to a$.

Conversely, let $a_n \to a$ and $\varepsilon > 0$, and take m with $a - \varepsilon < a_n < a + \varepsilon$ for $n \geqslant m$. Then $u_m \leqslant a + \varepsilon$ and $\ell_m \geqslant a - \varepsilon$, so $\limsup_n a_n = \inf_k u_k \leqslant u_m \leqslant a + \varepsilon$ and $\liminf_n a_n = \sup_k \ell_k \geqslant \ell_m \geqslant a - \varepsilon$. With the first part, $a - \varepsilon \leqslant \liminf_n a_n \leqslant \limsup_n a_n \leqslant a + \varepsilon$, and $\varepsilon > 0$ is arbitrary, so both equal a. □

Definition P.10 (Semi-continuity)**.** $f : \mathbb{R}^d \to \mathbb{R}$ is **lower semi-continuous** at x if $\liminf_{y \to x} f(y) \geqslant f(x)$, and **upper semi-continuous** if $\limsup_{y \to x} f(y) \leqslant f(x)$. Here $\liminf_{y \to x} f(y) = \sup_{\delta > 0} \inf_{|y - x| < \delta} f(y)$ and $\limsup_{y \to x} f(y) = \inf_{\delta > 0} \sup_{|y - x| < \delta} f(y)$, the point $y = x$ included.

Lemma P.11 (Semi-continuous envelopes)**.** For bounded $f : \mathbb{R}^d \to \mathbb{R}$ put $f_*(x) = \liminf_{y \to x} f(y)$ and $f^*(x) = \limsup_{y \to x} f(y)$. Then f_* is lower semi-continuous, f^* is upper

semi-continuous, $f_* \leqslant f \leqslant f^*$, and the three agree exactly at the continuity points of f.

Proof. Since f is bounded, say $|f| \leqslant B$, every infimum and supremum over a ball lies in $[-B, B]$, so f_* and f^* are real-valued.

Lower semi-continuity of f_*. Fix x and $\delta > 0$. Since $f_*(x) = \sup_{r>0} \inf_{|y-x|<r} f(y)$, the number $f_*(x) - \delta$ is not an upper bound of these infima, so there is $r > 0$ with $\inf_V f > f_*(x) - \delta$ for the open ball $V = \{y : |y - x| < r\}$. For $z \in V$ put $r' = r - |z - x| > 0$; the ball $\{y : |y - z| < r'\}$ lies in V by the triangle inequality, so the infimum of f over it is at least $\inf_V f$, and $f_*(z) \geqslant \inf_{|y-z|<r'} f(y) \geqslant \inf_V f > f_*(x) - \delta$. Hence $\inf_{|z-x|<r} f_*(z) \geqslant f_*(x) - \delta$, so $\liminf_{z\to x} f_*(z) \geqslant f_*(x) - \delta$ for every $\delta > 0$, which is lower semi-continuity of f_* at x.

Upper semi-continuity of f^*. Infima and suprema satisfy $\inf(-g) = -\sup g$, so $(-f)_* = -f^*$. The function $-f$ is bounded, so $(-f)_*$ is lower semi-continuous by the first step, and therefore $f^* = -(-f)_*$ is upper semi-continuous.

The inequalities. The point $y = x$ belongs to every ball about x, so $\inf_{|y-x|<r} f(y) \leqslant f(x) \leqslant \sup_{|y-x|<r} f(y)$ for every $r > 0$; taking the supremum, respectively infimum, over r gives $f_*(x) \leqslant f(x) \leqslant f^*(x)$.

Agreement. If f is continuous at x and $\varepsilon > 0$, there is $r > 0$ with $f(x) - \varepsilon < f(y) < f(x) + \varepsilon$ for $|y - x| < r$, so $f_*(x) \geqslant f(x) - \varepsilon$ and $f^*(x) \leqslant f(x) + \varepsilon$; as ε is arbitrary, $f_*(x) \geqslant f(x) \geqslant f^*(x)$, and with the inequalities all three are equal. Conversely, if $f_*(x) = f(x) = f^*(x)$ and $\varepsilon > 0$, the argument of the first step gives $r_1 > 0$ with $\inf_{|y-x|<r_1} f > f(x) - \varepsilon$ and, applied to $-f$, $r_2 > 0$ with $\sup_{|y-x|<r_2} f < f(x) + \varepsilon$; for $|y - x| < \min(r_1, r_2)$, $|f(y) - f(x)| < \varepsilon$, so f is continuous at x. □

Lemma P.12 (Approximation from below)**.** A lower semi-continuous f bounded below is the pointwise increasing limit of the bounded continuous functions $f_k(x) = \inf_y \{f(y) \wedge k + k|x - y|\}$.

Proof. For each y put $g_y(x) = f(y) \wedge k + k|x - y|$, so $f_k = \inf_y g_y$. Every g_y is at least $f(y) \wedge k \geqslant \inf f \wedge k$, so $f_k \geqslant \inf f \wedge k > -\infty$; and $f_k(x) \leqslant g_x(x) = f(x) \wedge k \leqslant k$. So f_k is bounded, and $f_k \leqslant f \wedge k \leqslant f$. By the triangle inequality, $g_y(x) \leqslant g_y(x') + k|x - x'|$ for all x, x', y; taking the infimum over y on both sides gives $f_k(x) \leqslant f_k(x') + k|x - x'|$, and exchanging x and x' gives $|f_k(x) - f_k(x')| \leqslant k|x - x'|$. Hence f_k is continuous. Since $f(y) \wedge k \leqslant f(y) \wedge (k+1)$ and $k|x - y| \leqslant (k+1)|x - y|$, every g_y increases with k, and so does the infimum: $f_k \leqslant f_{k+1}$.

For the limit, fix x and $c < f(x)$. Lower semi-continuity gives $\delta > 0$ with $\inf_{|y-x|<\delta} f(y) > c$ (as in the proof of Lemma P.11), so $f(y) > c$ whenever $|y - x| < \delta$. Take k with $k > c$ and $k\delta > c - \inf f$, and bound $g_y(x)$ below in two cases. If $|y - x| < \delta$ then $f(y) > c$ and $k > c$, so $f(y) \wedge k > c$ and $g_y(x) \geqslant f(y) \wedge k > c$. If $|y - x| \geqslant \delta$ then $g_y(x) \geqslant (\inf f \wedge k) + k\delta$; when $k \geqslant \inf f$ this is $\inf f + k\delta > c$ by the choice of k, and when $k < \inf f$ it is $k + k\delta > k > c$. Hence $g_y(x) > c$ for every y, and $f_k(x) \geqslant c$. The sequence $(f_k(x))_k$ is increasing and bounded above by $f(x)$, so it converges by Theorem P.2, and its limit lies in $[c, f(x)]$. As $c < f(x)$ was arbitrary, the limit is $f(x)$. □

3 Continuity, Compactness, Differentiation

Theorem P.13 (Uniform continuity on compacts)**.** A continuous f on a compact $K \subseteq \mathbb{R}^d$ (Definition P.4) is bounded, uniformly continuous, and attains its supremum.

Proof. Attainment. Let $M = \sup_K f$, finite or not, and take $x_n \in K$ with $f(x_n) \to M$: if $M < \infty$, $M - 1/n$ is not an upper bound, so some x_n has $M - 1/n < f(x_n) \leqslant M$; if $M = \infty$, n is not an upper bound, so some x_n has $f(x_n) > n$. By compactness a subsequence x_{n_k} converges

to some $x \in K$, and continuity at x gives $f(x_{n_k}) \to f(x)$; also $f(x_{n_k}) \to M$, so $M = f(x)$. In particular $M < \infty$, so f is bounded above and attains its supremum; applying this to the continuous function $-f$ bounds f below.

Uniform continuity. If it failed there would be $\varepsilon > 0$ such that for every n some points $x_n, y_n \in K$ have $|x_n - y_n| < 1/n$ but $|f(x_n) - f(y_n)| \geqslant \varepsilon$. By compactness a subsequence x_{n_k} converges to some $x \in K$, and $|y_{n_k} - x| \leqslant |y_{n_k} - x_{n_k}| + |x_{n_k} - x| < 1/n_k + |x_{n_k} - x| \to 0$, so $y_{n_k} \to x$ too. Continuity at x makes both $f(x_{n_k})$ and $f(y_{n_k})$ tend to $f(x)$, so $|f(x_{n_k}) - f(y_{n_k})| \to 0$, contradicting $|f(x_{n_k}) - f(y_{n_k})| \geqslant \varepsilon$ for every k. □

Theorem P.14 (Taylor with Lagrange remainder)**.** If $f : \mathbb{R} \to \mathbb{R}$ is $m+1$ times differentiable on an interval containing 0 and t, then for some ξ between 0 and t,

$$f(t) = \sum_{k=0}^{m} \frac{f^{(k)}(0)}{k!} t^k + \frac{f^{(m+1)}(\xi)}{(m+1)!} t^{m+1}. \tag{P.1}$$

Proof. For $t = 0$ both sides of (P.1) equal $f(0)$, and $\xi = 0$ works. Fix $t \neq 0$ and put

$$R = \frac{1}{t^{m+1}}\Big\{ f(t) - \sum_{k\leqslant m} \frac{f^{(k)}(0)}{k!}t^k \Big\}, \qquad g(s) = f(t) - \sum_{k\leqslant m} \frac{f^{(k)}(s)}{k!}(t-s)^k - R\,(t-s)^{m+1},$$

so that $g(0) = 0$. At $s = t$ every term with $k \geqslant 1$ and the last term vanish, leaving $g(t) = f(t) - f(t) = 0$. Each $f^{(k)}$ with $k \leqslant m$ is differentiable on the interval, hence continuous, so g is continuous on the closed interval between 0 and t and differentiable inside it. By Rolle's theorem (Theorem P.6) $g'(\xi) = 0$ for some ξ strictly between 0 and t. By the product rule, for $k \geqslant 1$,

$$\frac{\mathrm{d}}{\mathrm{d}s}\Big\{ \frac{f^{(k)}(s)}{k!}(t-s)^k \Big\} = \frac{f^{(k+1)}(s)}{k!}(t-s)^k - \frac{f^{(k)}(s)}{(k-1)!}(t-s)^{k-1},$$

and for $k = 0$ the derivative is $f'(s)$. Summing over $k = 0, \dots, m$, the negative term of index k cancels the positive term of index $k-1$, and only the positive term of index m survives:

$$\frac{\mathrm{d}}{\mathrm{d}s}\sum_{k\leqslant m} \frac{f^{(k)}(s)}{k!}(t-s)^k = \frac{f^{(m+1)}(s)}{m!}(t-s)^m.$$

So $g'(\xi) = 0$ reads $-\frac{f^{(m+1)}(\xi)}{m!}(t-\xi)^m + R(m+1)(t-\xi)^m = 0$. As ξ lies strictly between 0 and t, $t - \xi \neq 0$; dividing by $(m+1)(t-\xi)^m$ gives $R = f^{(m+1)}(\xi)/(m+1)!$. Multiplying the definition of R by t^{m+1} and substituting this value is (P.1). □

Lemma P.15 (Taylor bound for the complex exponential)**.** For every $t \in \mathbb{R}$ and integer $m \geqslant 0$,

$$\left| e^{it} - \sum_{k=0}^{m} \frac{(it)^k}{k!} \right| \leqslant \min\left\{ \frac{2|t|^m}{m!},\ \frac{|t|^{m+1}}{(m+1)!} \right\}. \tag{P.2}$$

Proof. For $m \geqslant -1$ let $h_m(t) = e^{it} - \sum_{k=0}^{m} (it)^k/k!$ (the empty sum for $m = -1$, so $h_{-1}(t) = e^{it}$ and $|h_{-1}| \equiv 1$). For $m \geqslant 0$, $h_m(0) = 1 - 1 = 0$, and differentiating term by term (a finite sum), using $(e^{it})' = ie^{it}$,

$$h_m'(t) = ie^{it} - \sum_{k=1}^{m} \frac{ik(it)^{k-1}}{k!} = i\left\{ e^{it} - \sum_{j=0}^{m-1} \frac{(it)^j}{j!} \right\} = ih_{m-1}(t).$$

By the fundamental theorem of calculus, $h_m(t) = i\int_0^t h_{m-1}(s)\,\mathrm{d}s$ for $m \geqslant 0$. Consequently, if

$|h_{m-1}(s)| \leqslant c\,|s|^p$ for all s between 0 and t, then

$$|h_m(t)| \leqslant \left|\int_0^t |h_{m-1}(s)|\,\mathrm{d}s\right| \leqslant c\int_0^{|t|} u^p\,\mathrm{d}u = \frac{c\,|t|^{p+1}}{p+1}. \qquad (*)$$

Second bound. Claim: $|h_m(t)| \leqslant |t|^{m+1}/(m+1)!$ for all t and all $m \geqslant -1$. Base case $m=-1$: $|h_{-1}(t)| = 1 = |t|^0/0!$. Step: if the claim holds for $m-1$, then $(*)$ with $c = 1/m!$ and $p = m$ gives $|h_m(t)| \leqslant |t|^{m+1}/\{m!(m+1)\} = |t|^{m+1}/(m+1)!$.
First bound. Claim: $|h_m(t)| \leqslant 2|t|^m/m!$ for all t and all $m \geqslant 0$. Base case $m=0$: $|h_0(t)| = |e^{it}-1| \leqslant |e^{it}|+1 = 2$. Step: if the claim holds for $m-1 \geqslant 0$, then $(*)$ with $c = 2/(m-1)!$ and $p = m-1$ gives $|h_m(t)| \leqslant 2|t|^m/\{(m-1)!\,m\} = 2|t|^m/m!$.
Both claims hold for every $m \geqslant 0$, so $|h_m(t)|$ is at most their minimum, which is (P.2). □

Lemma P.16 (Product comparison)**.** For complex $z_1, \ldots, z_m$ and $z'_1, \ldots, z'_m$ of modulus at most 1, $\left|\prod_k z_k - \prod_k z'_k\right| \leqslant \sum_k |z_k - z'_k|$.

Proof. Induction on m. For $m = 1$ both sides equal $|z_1 - z_1'|$. For $m = 2$, inserting $z_1' z_2$,

$$|z_1 z_2 - z_1' z_2'| = |(z_1 - z_1')z_2 + z_1'(z_2 - z_2')| \leqslant |z_2||z_1 - z_1'| + |z_1'||z_2 - z_2'| \leqslant |z_1 - z_1'| + |z_2 - z_2'|,$$

using $|z_2| \leqslant 1$ and $|z_1'| \leqslant 1$. Suppose the claim holds for $m - 1$ factors, $m \geqslant 3$, and put $w = \prod_{k<m} z_k$ and $w' = \prod_{k<m} z_k'$. Then $|w| \leqslant 1$ and $|w'| \leqslant 1$, as products of numbers of modulus at most 1, so the case $m = 2$ applied to w, z_m and w', z_m', followed by the hypothesis for $m - 1$, gives

$$\Big|\prod_{k\leqslant m} z_k - \prod_{k\leqslant m} z_k'\Big| = |w z_m - w' z_m'| \leqslant |w - w'| + |z_m - z_m'| \leqslant \sum_{k<m} |z_k - z_k'| + |z_m - z_m'|.$$

□

Lemma P.17 (Sum of reciprocal squares)**.** $\sum_{k=1}^{\infty} k^{-2} = \pi^2/6$ and $\sum_{k=1}^{\infty} (-1)^{k+1} k^{-2} = \pi^2/12$.

Proof. For $n \geqslant 0$ let $I_n = \int_0^{\pi/2} \cos^{2n} x \, \mathrm{d}x$ and $J_n = \int_0^{\pi/2} x^2 \cos^{2n} x \, \mathrm{d}x$, both positive. For $n \geqslant 1$,

integrating by parts twice, the boundary terms vanishing at 0 and $\pi/2$,

$$\begin{aligned} I_n &= 2n \int_0^{\pi/2} x \sin x \cos^{2n-1} x \, \mathrm{d}x \\ &= -n \int_0^{\pi/2} x^2 \{\cos^{2n} x - (2n-1) \sin^2 x \cos^{2n-2} x\} \, \mathrm{d}x \\ &= -n \int_0^{\pi/2} x^2 \{2n \cos^{2n} x - (2n-1) \cos^{2n-2} x\} \, \mathrm{d}x \\ &= n(2n-1) J_{n-1} - 2n^2 J_n. \end{aligned}$$

The first line takes $u = \cos^{2n} x$, $\mathrm{d}v = \mathrm{d}x$, $v = x$, with $u' = -2n \sin x \cos^{2n-1} x$; the second takes $u = \sin x \cos^{2n-1} x$, $\mathrm{d}v = x \, \mathrm{d}x$, $v = x^2/2$, with $u' = \cos^{2n} x - (2n-1) \sin^2 x \cos^{2n-2} x$; the boundary terms contain the factor x or x^2, which vanishes at 0, and the factor $\cos^{2n-1} x$ or $\cos^{2n} x$, which vanishes at $\pi/2$ as $n \geqslant 1$. The third line uses $\sin^2 x = 1 - \cos^2 x$. Integrating $I_n = \int_0^{\pi/2} \cos x \cos^{2n-1} x \, \mathrm{d}x$ by parts with $\mathrm{d}v = \cos x \, \mathrm{d}x$, $v = \sin x$, the boundary term $\sin x \cos^{2n-1} x$ vanishing at both ends, gives $I_n = (2n-1) \int_0^{\pi/2} \sin^2 x \cos^{2n-2} x \, \mathrm{d}x = (2n-1)(I_{n-1} - I_n)$, so

$2nI_n = (2n-1)I_{n-1}$, that is $I_n = \frac{2n-1}{2n} I_{n-1}$. Dividing the first identity by $n^2 I_n$ gives $1/n^2 = (2n-1)J_{n-1}/(nI_n) - 2J_n/I_n$, and substituting $(2n-1)/(nI_n) = 2/I_{n-1}$ from the second,

$$\frac{1}{n^2} = \frac{2J_{n-1}}{I_{n-1}} - \frac{2J_n}{I_n}, \qquad \sum_{k=1}^{n} \frac{1}{k^2} = \frac{2J_0}{I_0} - \frac{2J_n}{I_n} = \frac{\pi^2}{6} - \frac{2J_n}{I_n},$$

the first sum telescoping, as $J_0 = \int_0^{\pi/2} x^2 \,\mathrm{d}x = \pi^3/24$ and $I_0 = \pi/2$. The function $h(x) = \sin x - 2x/\pi$ vanishes at 0 and $\pi/2$, and $h'(x) = \cos x - 2/\pi$ is strictly decreasing on $[0, \pi/2]$ by the mean value theorem (Corollary P.7), as $\cos' = -\sin < 0$ on $(0, \pi/2]$. Since $h'(0) = 1 - 2/\pi > 0$, the set $\{x : h'(x) > 0\}$ is non-empty; let c be its supremum. As h' is strictly decreasing, $h' > 0$ on $[0, c)$ and $h' \leqslant 0$ on $(c, \pi/2]$, and by the mean value theorem h is nondecreasing on $[0, c]$ and nonincreasing on $[c, \pi/2]$. Hence $h(x) \geqslant h(0) = 0$ for $x \leqslant c$ and $h(x) \geqslant h(\pi/2) = 0$ for $x \geqslant c$, that is $\sin x \geqslant 2x/\pi \geqslant 0$ on $[0, \pi/2]$, so $x^2 \leqslant (\pi^2/4)\sin^2 x$ there. Since $\cos^{2n} x - \cos^{2n+2} x = \sin^2 x \cos^{2n} x$ and $I_{n+1} = \frac{2n+1}{2n+2} I_n$,

$$0 < J_n \leqslant \frac{\pi^2}{4} \int_0^{\pi/2} \sin^2 x \cos^{2n} x \,\mathrm{d}x = \frac{\pi^2}{4}(I_n - I_{n+1}) = \frac{\pi^2}{4} \cdot \frac{I_n}{2n+2},$$

so $0 < 2J_n/I_n \leqslant \pi^2/\{4(n+1)\} \to 0$, and the partial sums tend to $\pi^2/6$, which is the first sum. For the second, split the partial sum with $2N$ terms into odd and even k:

$$\sum_{k=1}^{2N} \frac{(-1)^{k+1}}{k^2} = \sum_{k=1}^{2N} \frac{1}{k^2} - 2\sum_{j=1}^{N} \frac{1}{(2j)^2} = \sum_{k=1}^{2N} \frac{1}{k^2} - \frac{1}{2}\sum_{j=1}^{N} \frac{1}{j^2} \longrightarrow \frac{\pi^2}{6} - \frac{\pi^2}{12} = \frac{\pi^2}{12},$$

and the partial sum with $2N+1$ terms differs from this by $(2N+1)^{-2} \to 0$. □

4 Convexity and Separation

Definition P.18 (Convex set, convex function)**.** $C \subseteq \mathbb{R}^d$ is **convex** if $\lambda x + (1-\lambda)y \in C$ whenever $x, y \in C$ and $\lambda \in [0,1]$. A function $f : C \to \mathbb{R}$ on a convex C is **convex** if $f(\lambda x + (1-\lambda)y) \leqslant \lambda f(x) + (1-\lambda)f(y)$.

Lemma P.19 (Convexity from the derivative)**.** Let g be continuous on an interval I and differentiable at its interior points. If g' is nondecreasing there, g is convex on I; if g' is strictly increasing, g is **strictly convex**: $g(\lambda x + (1-\lambda)y) < \lambda g(x) + (1-\lambda)g(y)$ for $x \neq y$ in I and $0 < \lambda < 1$. A

second derivative $g'' \geqslant 0$ (> 0) in the interior suffices. With the inequalities reversed, g is concave (strictly concave), that is, $-g$ is convex (strictly convex). A function f on a convex $C \subseteq \mathbb{R}^d$ is strictly convex if $f(\lambda x + (1-\lambda)y) < \lambda f(x) + (1-\lambda)f(y)$ for $x \neq y$ in C and $0 < \lambda < 1$; it is convex, strictly convex, concave or strictly concave iff every restriction $t \mapsto f(x+tv)$ to a line is.

Proof. Step 1: one variable. Let $x \neq y$ in I and $0 < \lambda < 1$; exchanging the roles of (x, λ) and $(y, 1-\lambda)$ if necessary, assume $x < y$. Put $z = \lambda x + (1-\lambda)y$, so $x < z < y$. By the mean value theorem (Corollary P.7) on $[x, z]$ and on $[z, y]$ there are $\xi_1 \in (x, z)$ and $\xi_2 \in (z, y)$ with $g(z) - g(x) = g'(\xi_1)(z - x)$ and $g(y) - g(z) = g'(\xi_2)(y - z)$. Since $\xi_1 < \xi_2$ are interior points and g' is nondecreasing, $g'(\xi_1) \leqslant g'(\xi_2)$, strictly in the strict case. As $z - x = (1-\lambda)(y-x)$ and $y - z = \lambda(y-x)$,

$$\lambda\{g(z) - g(x)\} = \lambda(1-\lambda)(y-x)\, g'(\xi_1) \leqslant \lambda(1-\lambda)(y-x)\, g'(\xi_2) = (1-\lambda)\{g(y) - g(z)\},$$

strictly in the strict case because $\lambda(1-\lambda)(y-x) > 0$. Collecting the terms in $g(z)$, whose coefficients add to $\lambda + (1-\lambda) = 1$, gives $g(z) \leqslant \lambda g(x) + (1-\lambda)g(y)$, strictly in the strict case. Convexity also holds trivially when $x = y$ or $\lambda \in \{0, 1\}$. If $g'' \geqslant 0$ (> 0) in the interior, then for

interior $u < v$ the mean value theorem applied to g' gives $g'(v) - g'(u) = g''(\eta)(v-u) \geqslant 0$ (> 0) for some $\eta \in (u, v)$, so g' is nondecreasing (strictly increasing). The concave statements are the convex ones for $-g$, whose derivatives are $-g'$ and $-g''$.

Step 2: restriction to lines. For $x \in C$ and $v \in \mathbb{R}^d$ the set $T = \{t : x + tv \in C\}$ is an interval, as C is convex, and $\varphi(t) = f(x + tv)$ is defined on T. If f is convex and $s, t \in T$, then $\varphi(\lambda s + (1-\lambda)t) = f\big(\lambda(x+sv) + (1-\lambda)(x+tv)\big) \leqslant \lambda\varphi(s) + (1-\lambda)\varphi(t)$, so φ is convex; for $v \neq \mathbf{0}$ and $s \neq t$ the points $x + sv \neq x + tv$, so strictness passes over too. Conversely, given $x \neq y$ in C, take $v = y - x$: then $0, 1 \in T$, $\varphi(0) = f(x)$, $\varphi(1) = f(y)$ and $f(\lambda x + (1-\lambda)y) = \varphi\big(\lambda \cdot 0 + (1-\lambda)\cdot 1\big) \leqslant \lambda\varphi(0) + (1-\lambda)\varphi(1) = \lambda f(x) + (1-\lambda) f(y)$, strictly if φ is strictly convex. The concave cases follow by applying this to $-f$. □

Theorem P.20 (Nearest point in a closed convex set)**.** Let $C \subseteq \mathbb{R}^d$ be non-empty, closed and convex, and $z \in \mathbb{R}^d$. Then there is a unique $p \in C$ minimising $|z - x|$ over $x \in C$, and it is characterised by

$$\langle z - p,\ x - p \rangle \leqslant 0 \qquad \text{for every } x \in C. \tag{P.3}$$

Proof. Existence. Let $\delta = \inf_{x\in C} |z-x|$, which exists as C is non-empty and the distances are bounded below by 0, and take $x_n \in C$ with $|z-x_n| \to \delta$ (for each n, $\delta + 1/n$ is not a lower bound). The parallelogram law $|a-b|^2 = 2|a|^2 + 2|b|^2 - |a+b|^2$, with $a = z - x_m$ and $b = z - x_n$, gives

$$|x_n - x_m|^2 = 2|z-x_n|^2 + 2|z-x_m|^2 - 4\left|z - \tfrac{x_n+x_m}{2}\right|^2 \leqslant 2|z-x_n|^2 + 2|z-x_m|^2 - 4\delta^2,$$

using $\frac{1}{2}(x_n+x_m) \in C$ by convexity, so that its distance to z is at least δ. Given $\varepsilon > 0$, take N with $|z-x_n|^2 < \delta^2 + \varepsilon$ for $n \geqslant N$; then the right side is less than 4ε for $m, n \geqslant N$, so (x_n) is Cauchy. It converges to some p: the sequence is bounded, as $|x_n| \leqslant |z| + |z - x_n|$ and $|z-x_n| \to \delta$, so by Theorem P.3 a subsequence x_{n_k} converges to some p, and then $|x_n - p| \leqslant |x_n - x_{n_k}| + |x_{n_k} - p|$, where the first term is small once n and n_k exceed the Cauchy index and the second once k is large. The limit p lies in C because C is closed, and $|z-p| \leqslant |z-x_n| + |x_n - p| \to \delta$, so $|z-p| = \delta$.
Characterisation. Let $p \in C$ be any minimiser. For $x \in C$ and $\lambda \in (0,1]$, the point $p + \lambda(x-p) = (1-\lambda)p + \lambda x$ lies in C by convexity, so

$$\delta^2 \leqslant |z - p - \lambda(x-p)|^2 = \delta^2 - 2\lambda\langle z-p, x-p\rangle + \lambda^2|x-p|^2.$$

Subtracting δ^2 and dividing by $2\lambda > 0$ gives $\langle z-p, x-p\rangle \leqslant \lambda|x-p|^2/2$, and letting $\lambda \to 0^+$ gives (P.3). Conversely if $p \in C$ satisfies (P.3) then for any $x \in C$, expanding $|(z-p)-(x-p)|^2$, $|z-x|^2 = |z-p|^2 - 2\langle z-p, x-p\rangle + |x-p|^2 \geqslant |z-p|^2$, so p is minimising.

Uniqueness. If p and p' both minimise, both satisfy (P.3) by the characterisation. Taking $x = p'$ in the inequality for p and $x = p$ in the inequality for p' gives $\langle z-p, p'-p\rangle \leqslant 0$ and $\langle z-p', p-p'\rangle \leqslant 0$. Adding, $\langle (z-p)-(z-p'), p'-p\rangle = |p'-p|^2 \leqslant 0$, so $p = p'$. □

Theorem P.21 (Separating hyperplane)**.** Let $C \subseteq \mathbb{R}^d$ be non-empty, closed and convex, and $z \notin C$. Then there exist $\mathbf{w} \neq \mathbf{0}$ and $\alpha \in \mathbb{R}$ with $\langle \mathbf{w}, z\rangle > \alpha \geqslant \langle \mathbf{w}, x\rangle$ for every $x \in C$.

Proof. Let p be the nearest point of Theorem P.20 and put $\mathbf{w} = z - p \neq \mathbf{0}$. By (P.3), $\langle \mathbf{w}, x\rangle \leqslant \langle \mathbf{w}, p\rangle$ for every $x \in C$; take $\alpha = \langle \mathbf{w}, p\rangle$. Then $\langle \mathbf{w}, z\rangle - \alpha = \langle \mathbf{w}, z-p\rangle = |\mathbf{w}|^2 > 0$. □

Proposition P.22 (First-order condition)**.** A differentiable f on an open convex C is convex if and only if $f(y) \geqslant f(x) + \langle \nabla f(x), y-x\rangle$ for all $x, y \in C$. A convex differentiable f attains a global minimum at x if and only if $\nabla f(x) = \mathbf{0}$.

Proof. If f is convex then for $\lambda \in (0,1]$, $f(x+\lambda(y-x)) \leqslant f(x) + \lambda\{f(y)-f(x)\}$, so

$$\frac{f(x+\lambda(y-x))-f(x)}{\lambda} \leqslant f(y)-f(x);$$

letting $\lambda \to 0^+$, the left side tends to $\langle \nabla f(x), y-x\rangle$, because differentiability at x means $f(x+h) = f(x) + \langle \nabla f(x), h\rangle + r(h)$ with $r(h)/|h| \to 0$, and with $h = \lambda(y-x)$, $y \neq x$, the remainder divided by λ is $|y-x|\, r(h)/|h| \to 0$ (for $y = x$ there is nothing to prove). A non-strict inequality survives the limit, which gives the gradient inequality. Conversely, given the gradient inequality, let $x, y \in C$, $\lambda \in [0,1]$ and $u = \lambda x + (1-\lambda)y \in C$. The gradient inequality at u gives

$$f(x) \geqslant f(u) + \langle \nabla f(u), x-u\rangle, \qquad f(y) \geqslant f(u) + \langle \nabla f(u), y-u\rangle.$$

Multiplying the first by λ, the second by $1-\lambda$ and adding, $\lambda f(x) + (1-\lambda) f(y) \geqslant f(u) + \langle \nabla f(u), \lambda x + (1-\lambda)y - u\rangle = f(u)$, since $\lambda x + (1-\lambda)y - u = \mathbf{0}$.
The minimum statement follows: $\nabla f(x) = \mathbf{0}$ makes the gradient inequality read $f(y) \geqslant f(x)$ for all $y \in C$. Conversely, let x be a minimiser. Since C is open there is $r > 0$ with $x + t\mathbf{e}_j \in C$ for $|t| < r$, where $\mathbf{e}_j$ is the jth standard basis vector, and $\varphi_j(t) = f(x + t\mathbf{e}_j)$ has a minimum at the

interior point $t = 0$ of $(-r, r)$, with $\varphi_j'(0) = \partial_j f(x)$. The difference quotients $\{\varphi_j(t) - \varphi_j(0)\}/t$ are $\geqslant 0$ for $0 < t < r$ and $\leqslant 0$ for $-r < t < 0$, so their limit $\partial_j f(x)$ is both $\geqslant 0$ and $\leqslant 0$, as in the proof of Theorem P.6. Hence every partial derivative vanishes: $\nabla f(x) = \mathbf{0}$. □

Lemma P.23 (Coercive minimisation)**.** Let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous and **coercive**, meaning $f(x) \to \infty$ as $|x| \to \infty$. Then f attains its infimum. If f is in addition strictly convex (Lemma P.19), the minimiser is unique.

Proof. Fix x_0 and put $S = \{x : f(x) \leqslant f(x_0)\}$, which contains x_0. It is closed: if $x_n \in S$ and $x_n \to x$, continuity gives $f(x) = \lim_n f(x_n) \leqslant f(x_0)$, so $x \in S$. Coercivity gives R with $f(x) > f(x_0)$ whenever $|x| > R$, so $S \subseteq \{|x| \leqslant R\}$ is bounded, hence compact by Theorem P.5. By Theorem P.13 applied to $-f$, f attains a minimum on S at some $x^\star$, and for $x \notin S$ one has $f(x) > f(x_0) \geqslant f(x^\star)$, so $x^\star$ minimises f on $\mathbb{R}^d$.
If $x^\star \neq y^\star$ both minimise, with common value m, strict convexity with $\lambda = \tfrac12$ gives $f\big(\tfrac12(x^\star + y^\star)\big) < \tfrac12 f(x^\star) + \tfrac12 f(y^\star) = m$, contradicting minimality. □

Lemma P.24 (Directional characterisation of a convex minimum)**.** Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex. For each x and v the difference quotient $t \mapsto \{f(x+tv) - f(x)\}/t$ is non-decreasing on $t > 0$, so

$$f'(x; v) = \lim_{t \downarrow 0} \frac{f(x+tv) - f(x)}{t} \tag{P.4}$$

exists in $[-\infty, \infty)$. Moreover x minimises f over $\mathbb{R}^d$ if and only if $f'(x; v) \geqslant 0$ for every v.

Proof. Write $q(t) = \{f(x + tv) - f(x)\}/t$ for $t > 0$. For $0 < s < t$ write $x + sv = (1 - s/t)x + (s/t)(x + tv)$, a convex combination since $0 < s/t < 1$; convexity gives $f(x + sv) \leqslant (1 - s/t)f(x) + (s/t)f(x + tv)$, that is $f(x+sv) - f(x) \leqslant (s/t)\{f(x+tv) - f(x)\}$, and dividing by $s > 0$ gives $q(s) \leqslant q(t)$. Put $\ell = \inf_{t>0} q(t) \in [-\infty, \infty)$. For any $c > \ell$ there is $t_0 > 0$ with $q(t_0) < c$, and then $\ell \leqslant q(t) \leqslant q(t_0) < c$ for $0 < t < t_0$ by monotonicity; so $q(t) \to \ell$ as $t \downarrow 0$ (when $\ell = -\infty$ this says $q(t) \to -\infty$). Thus the limit (P.4) exists and equals ℓ, and $f'(x; v) = \ell \leqslant q(t)$ for every $t > 0$; multiplying by t,

$$f(x + tv) \geqslant f(x) + t\, f'(x; v) \qquad (t > 0). \tag{P.5}$$

If $f'(x;v) \geqslant 0$ for every v, then (P.5) with $t = 1$ and $v = y - x$ gives $f(y) \geqslant f(x) + f'(x; y-x) \geqslant f(x)$ for every y. Conversely, if x is a minimiser, then $f(x+tv) - f(x) \geqslant 0$, so every difference quotient in (P.4) is non-negative, hence so is their infimum $f'(x;v)$. □

Lemma P.25 (Interior of the closure of a convex set)**.** If $E \subseteq \mathbb{R}^d$ is convex and $\overline{E}$ contains an open ball about x, then $x \in E$. Here $\overline{E}$, the closure of E, is the set of limits of convergent sequences of points of E.

Proof. Translating, take $x = \mathbf{0}$ and an open ball B of radius $r > 0$ about $\mathbf{0}$ with $B \subseteq \overline{E}$. First, E has an interior point. E lies in no hyperplane $H = \{y : \langle \mathbf{a}, y\rangle = c\}$ with $\mathbf{a} \neq \mathbf{0}$: H is closed, so $E \subseteq H$ would give $B \subseteq \overline{E} \subseteq H$, yet $\pm\frac{r}{2}\mathbf{a}/|\mathbf{a}| \in B$ give different values of $\langle \mathbf{a}, \cdot\rangle$ (namely $\pm\frac{r}{2}|\mathbf{a}|$). ($E$ is non-empty, since $\overline{E} \supseteq B$ is.)
Pick $w_0 \in E$ and then, for $j = 1, \ldots, d$ in turn, $w_j \in E$ with $w_j - w_0 \notin U_{j-1} = \operatorname{span}\{w_1 - w_0, \ldots, w_{j-1} - w_0\}$ ($U_0 = \{\mathbf{0}\}$). Such a w_j exists: otherwise $E \subseteq w_0 + U_{j-1}$, where U_{j-1} has dimension at most $j - 1 < d$. Take a basis $u_1, \ldots, u_k$ of U_{j-1}, $k < d$; the linear map $T\mathbf{a} = (\langle \mathbf{a}, u_1\rangle, \ldots, \langle \mathbf{a}, u_k\rangle)$ from $\mathbb{R}^d$ to $\mathbb{R}^k$ is not injective, since an injective linear map would carry the d

standard basis vectors to d linearly independent vectors of $\mathbb{R}^k$, impossible as $k < d$. So some $\mathbf{a} \neq \mathbf{0}$ has $\langle \mathbf{a}, u_i \rangle = 0$ for all i, hence $\langle \mathbf{a}, u \rangle = 0$ for $u \in U_{j-1}$, and $E \subseteq w_0 + U_{j-1} \subseteq \{y : \langle \mathbf{a}, y \rangle = \langle \mathbf{a}, w_0 \rangle\}$, a hyperplane, contradicting the previous paragraph. The vectors $w_1 - w_0, \ldots, w_d - w_0$ are linearly independent: in a vanishing combination $\sum_i c_i(w_i - w_0) = \mathbf{0}$ with some $c_i \neq 0$, the largest such index j would give $w_j - w_0 = -\sum_{i<j}(c_i/c_j)(w_i - w_0) \in U_{j-1}$. Being d independent vectors in $\mathbb{R}^d$ they form a basis, so the matrix V with columns $w_j - w_0$ satisfies $V\mathbf{c}_i = \mathbf{e}_i$ for some $\mathbf{c}_i$, $i = 1, \ldots, d$; the matrix with columns $\mathbf{c}_i$ is a right inverse of V, hence its inverse V^{-1}. The map $\mu(y) = V^{-1}(y - w_0)$ is continuous in y, each coordinate being linear. Every y with $\mu(y) \geqslant \mathbf{0}$ and $\sum_j \mu_j(y) \leqslant 1$ satisfies $y = w_0 + V\mu(y) = (1 - \sum_j \mu_j)w_0 + \sum_j \mu_j w_j$, a convex combination of points of E, so $y \in E$ (a convex set contains the convex combinations of finitely many of its points, by induction on their number: $\sum_{j=0}^{k} \theta_j w_j = (1 - \theta_k) \sum_{j<k} \frac{\theta_j}{1-\theta_k} w_j + \theta_k w_k$ for $\theta_k < 1$). At $z = w_0 + V\mathbf{1}/(d+1)$ the coordinates of $\mu(z)$ equal $1/(d+1) > 0$ and sum to $d/(d+1) < 1$. Each of the $d+1$ functions $\mu_1, \ldots, \mu_d$ and $\sum_j \mu_j$ is continuous, so each strict inequality holds on some open ball about z; on the smallest of these balls, of radius $\rho > 0$ say, all hold, and that ball therefore lies in E.

Now take $t > 0$ with $t|z| < r$, so $y = -tz \in B \subseteq \overline{E}$, and, y being a limit of points of E, pick $y' \in E$ with $|y' - y| < t\rho$. Then $z' = z - (y' - y)/t$ satisfies $|z' - z| < \rho$, so $z' \in E$, and

$$\tfrac{1}{1+t}\, y' + \tfrac{t}{1+t}\, z' = \tfrac{y'+tz-(y'-y)}{1+t} = \tfrac{y+tz}{1+t} = \mathbf{0},$$

a convex combination of the points $y', z' \in E$; by convexity $\mathbf{0} \in E$. □

Theorem P.26 (Separation of two convex sets)**.** Let $C, D \subseteq \mathbb{R}^d$ be non-empty disjoint convex sets. Then there are $\mathbf{w} \neq \mathbf{0}$ and α with $\langle \mathbf{w}, x\rangle \leqslant \alpha \leqslant \langle \mathbf{w}, y\rangle$ for all $x \in C$, $y \in D$.

Proof. $E = C - D = \{x - y : x \in C,\ y \in D\}$ is non-empty and convex: for $x, x' \in C$, $y, y' \in D$ and $\theta \in [0,1]$, $\theta(x-y) + (1-\theta)(x'-y') = \{\theta x + (1-\theta)x'\} - \{\theta y + (1-\theta)y'\} \in C - D$. Also $\mathbf{0} \notin E$ because C and D are disjoint. Let $\overline{E}$ be the set of limits of convergent sequences in E. It contains E (constant sequences), and it is convex: if $a_n \to a$ and $b_n \to b$ with $a_n, b_n \in E$, then $\theta a_n + (1-\theta)b_n \in E$ tends to $\theta a + (1-\theta)b$. It is closed: if $c_k \in \overline{E}$ and $c_k \to c$, pick $e_k \in E$ with $|e_k - c_k| < 1/k$; then $e_k \to c$, so $c \in \overline{E}$.

Case $\mathbf{0} \notin \overline{E}$. Theorem P.21 applied to the non-empty closed convex $\overline{E}$ and the point $\mathbf{0}$ gives

$\mathbf{w} \neq \mathbf{0}$ and α' with $\langle \mathbf{w}, z\rangle \leqslant \alpha' < \langle \mathbf{w}, \mathbf{0}\rangle = 0$ for all $z \in \overline{E}$.

Case $\mathbf{0} \in \overline{E} \setminus E$. No open ball about $\mathbf{0}$ lies inside $\overline{E}$: otherwise Lemma P.25 would place $\mathbf{0}$ in E. So for each k the ball of radius $1/k$ about $\mathbf{0}$ contains some $z_k \notin \overline{E}$, and $z_k \to \mathbf{0}$. For each k, Theorem P.21 applied to $\overline{E}$ and z_k gives $\mathbf{w}_k \neq \mathbf{0}$ with $\langle \mathbf{w}_k, z\rangle < \langle \mathbf{w}_k, z_k\rangle$ for all $z \in \overline{E}$; dividing by $|\mathbf{w}_k|$, take $|\mathbf{w}_k| = 1$. The $\mathbf{w}_k$ are bounded, so by Theorem P.3 a subsequence $\mathbf{w}_{k_j}$ converges to some $\mathbf{w}$, and $|\mathbf{w}| = \lim_j |\mathbf{w}_{k_j}| = 1$. For fixed $z \in \overline{E}$, $\langle \mathbf{w}_{k_j}, z\rangle < \langle \mathbf{w}_{k_j}, z_{k_j}\rangle \leqslant d\,|z_{k_j}| \to 0$, since each coordinate of the unit vector $\mathbf{w}_{k_j}$ has modulus at most 1 and each coordinate of z_{k_j} has modulus at most $|z_{k_j}|$; meanwhile $\langle \mathbf{w}_{k_j}, z\rangle \to \langle \mathbf{w}, z\rangle$; so $\langle \mathbf{w}, z\rangle \leqslant 0$ for all $z \in \overline{E}$, in particular for all $z \in E$. Either way $\mathbf{w} \neq \mathbf{0}$ and $\langle \mathbf{w}, x - y\rangle \leqslant 0$ for $x \in C$, $y \in D$, that is $\langle \mathbf{w}, x\rangle \leqslant \langle \mathbf{w}, y\rangle$. Fixing y, the set $\{\langle \mathbf{w}, x\rangle : x \in C\}$ is bounded above, so its supremum s exists and $s \leqslant \langle \mathbf{w}, y\rangle$ for every $y \in D$; hence $s \leqslant \inf_{y\in D}\langle \mathbf{w}, y\rangle$, and $\alpha = s$ works. □

5 Convex Programming and Duality

Consider

$$p^\star = \inf\ f_0(x) \quad \text{subject to} \quad f_i(x) \leqslant 0,\ i = 1, \dots, m, \qquad x \in \mathcal{D} \subseteq \mathbb{R}^d, \tag{P.6}$$

with $\mathcal{D}$ convex and $f_0, \dots, f_m$ convex on $\mathcal{D}$.

Definition P.27 (Lagrangian and dual function)**.** For $\boldsymbol{\lambda} \in \mathbb{R}^m$,

$$L(x, \boldsymbol{\lambda}) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x), \qquad g(\boldsymbol{\lambda}) = \inf_{x \in \mathcal{D}} L(x, \boldsymbol{\lambda}), \tag{P.7}$$

and the **dual problem** is $d^\star = \sup_{\boldsymbol{\lambda} \geqslant \mathbf{0}} g(\boldsymbol{\lambda})$.

Proposition P.28 (Concavity of the dual)**.** g is concave on $\mathbb{R}^m$, whatever the f_i.

Proof. For each fixed x, $\boldsymbol{\lambda} \mapsto L(x, \boldsymbol{\lambda})$ is affine. An infimum of affine functions is concave: if $g(\boldsymbol{\lambda}) = \inf_x L(x, \boldsymbol{\lambda})$ then for $\theta \in [0, 1]$,

$$L(x, \theta\boldsymbol{\lambda} + (1-\theta)\boldsymbol{\mu}) = \theta L(x, \boldsymbol{\lambda}) + (1-\theta) L(x, \boldsymbol{\mu}) \geqslant \theta g(\boldsymbol{\lambda}) + (1-\theta) g(\boldsymbol{\mu}),$$

and taking the infimum over x on the left preserves the bound. □

Theorem P.29 (Weak duality)**.** $d^\star \leqslant p^\star$.

Proof. Let x be feasible and $\boldsymbol{\lambda} \geqslant \mathbf{0}$. Then $\lambda_i f_i(x) \leqslant 0$ for every i, so $L(x, \boldsymbol{\lambda}) \leqslant f_0(x)$ and hence $g(\boldsymbol{\lambda}) \leqslant L(x, \boldsymbol{\lambda}) \leqslant f_0(x)$. Taking the infimum over feasible x gives $g(\boldsymbol{\lambda}) \leqslant p^\star$, and then the supremum over $\boldsymbol{\lambda} \geqslant \mathbf{0}$. □

Theorem P.30 (Slater implies strong duality)**.** Suppose $p^\star$ is finite and **Slater's condition** holds: there is $\tilde{x} \in \mathcal{D}$ with $f_i(\tilde{x}) < 0$ for every i. Then $d^\star = p^\star$, and the dual optimum is attained.

Proof. Put

$$\mathcal{A} = \big\{(\mathbf{u}, t) \in \mathbb{R}^m \times \mathbb{R} : \ \exists x \in \mathcal{D} \text{ with } f_i(x) \leqslant u_i \ \forall i, \ f_0(x) \leqslant t\big\}.$$

$\mathcal{A}$ is convex: given $(\mathbf{u}, t)$ witnessed by x and $(\mathbf{u}', t')$ by x', and $\theta \in [0, 1]$, the point $x_\theta = \theta x + (1-\theta)x'$ lies in $\mathcal{D}$ by convexity of $\mathcal{D}$, and convexity of each f_i gives $f_i(x_\theta) \leqslant \theta f_i(x) + (1-\theta) f_i(x') \leqslant \theta u_i + (1-\theta) u_i'$, and likewise $f_0(x_\theta) \leqslant \theta t + (1-\theta) t'$; so x_θ witnesses $\theta(\mathbf{u}, t) + (1-\theta)(\mathbf{u}', t') \in \mathcal{A}$. Also $\mathcal{B} = \{(\mathbf{0}, s) : s < p^\star\}$ is convex (a segment of a line), both sets are non-empty ($\mathcal{A}$ contains $(f_1(\tilde{x}), \dots, f_m(\tilde{x}), f_0(\tilde{x}))$, and $\mathcal{B}$ is non-empty as $p^\star$ is finite), and $\mathcal{A} \cap \mathcal{B} = \emptyset$, since a point $(\mathbf{0}, s)$ of the intersection would be witnessed by an $x \in \mathcal{D}$ with $f_i(x) \leqslant 0$ for all i, a feasible point, with $f_0(x) \leqslant s < p^\star$, contradicting the definition of $p^\star$.

By Theorem P.26 applied with $C = \mathcal{B}$ and $D = \mathcal{A}$ there are $(\boldsymbol{\lambda}, \mu) \neq \mathbf{0}$ and α with

$$\langle \boldsymbol{\lambda}, \mathbf{u} \rangle + \mu t \geqslant \alpha \ \text{ on } \mathcal{A}, \qquad \mu s \leqslant \alpha \ \text{ for all } s < p^\star. \tag{P.8}$$

Increasing any u_i or t keeps a point in $\mathcal{A}$ (the same witness x serves). If some $\lambda_i < 0$, then starting from a point $(\mathbf{u}, t) \in \mathcal{A}$ and replacing u_i by $u_i + r$, the left side of (P.8) becomes $\langle \boldsymbol{\lambda}, \mathbf{u} \rangle + \mu t + \lambda_i r \to -\infty$ as $r \to \infty$, contradicting the bound $\geqslant \alpha$; so $\boldsymbol{\lambda} \geqslant \mathbf{0}$, and in the same way, increasing t, $\mu \geqslant 0$. Letting $s \uparrow p^\star$ in $\mu s \leqslant \alpha$ gives $\mu p^\star \leqslant \alpha$.

Case $\mu > 0$. For arbitrary $x \in \mathcal{D}$ the point $(\mathbf{u}, t) = (f_1(x), \dots, f_m(x), f_0(x))$ lies in $\mathcal{A}$, witnessed by x itself. Dividing the first inequality of (P.8) at this point by μ, and using $\alpha/\mu \geqslant p^\star$,

$$f_0(x) + \sum_i \frac{\lambda_i}{\mu} f_i(x) \geqslant \frac{\alpha}{\mu} \geqslant p^\star.$$

The left side is $L(x, \boldsymbol{\lambda}/\mu)$, so taking the infimum over $x \in \mathcal{D}$ gives $g(\boldsymbol{\lambda}/\mu) \geqslant p^\star$. As $\boldsymbol{\lambda}/\mu \geqslant \mathbf{0}$, $d^\star \geqslant g(\boldsymbol{\lambda}/\mu) \geqslant p^\star$, and Theorem P.29 gives $d^\star \leqslant p^\star$; so $d^\star = p^\star = g(\boldsymbol{\lambda}/\mu)$, and the dual optimum is attained at $\boldsymbol{\lambda}/\mu$.

Case $\mu = 0$. Then $\langle \boldsymbol{\lambda}, \mathbf{u} \rangle \geqslant \alpha \geqslant \mu p^\star = 0$ on $\mathcal{A}$, so $\sum_i \lambda_i f_i(x) \geqslant 0$ for every $x \in \mathcal{D}$. Taking $x = \tilde{x}$: every term $\lambda_i f_i(\tilde{x})$ is $\leqslant 0$, as $\lambda_i \geqslant 0$ and $f_i(\tilde{x}) < 0$, and the terms add to a number $\geqslant 0$, so every term is 0; since $f_i(\tilde{x}) \neq 0$, every $\lambda_i = 0$. Then $(\boldsymbol{\lambda}, \mu) = \mathbf{0}$, a contradiction. So this case cannot occur. □

Theorem P.31 (Karush–Kuhn–Tucker)**.** Assume $\mathcal{D}$ open, $f_0, \ldots, f_m$ differentiable and strong duality, with $x^\star$ primal and $\boldsymbol{\lambda}^\star$ dual optimal. Then

$$f_i(x^\star) \leqslant 0, \qquad \lambda_i^\star \geqslant 0, \tag{P.9}$$

$$\lambda_i^\star f_i(x^\star) = 0 \quad (i = 1, \ldots, m), \qquad \nabla f_0(x^\star) + \sum_i \lambda_i^\star \nabla f_i(x^\star) = \mathbf{0}. \tag{P.10}$$

Conversely, if the problem is convex and $(x^\star, \boldsymbol{\lambda}^\star)$ satisfies (P.9)–(P.10), then they are primal and dual optimal with zero gap.

Proof. Necessity. Primal optimality includes feasibility, $f_i(x^\star) \leqslant 0$, and dual optimality in-

cludes $\boldsymbol{\lambda}^\star \geqslant \mathbf{0}$: that is (P.9). Strong duality, $f_0(x^\star) = p^\star = d^\star = g(\boldsymbol{\lambda}^\star)$, gives

$$f_0(x^\star) = g(\boldsymbol{\lambda}^\star) = \inf_{x\in\mathcal{D}} \Big\{ f_0(x) + \sum_i \lambda_i^\star f_i(x) \Big\} \leqslant f_0(x^\star) + \sum_i \lambda_i^\star f_i(x^\star) \leqslant f_0(x^\star),$$

the last step because each $\lambda_i^\star f_i(x^\star) \leqslant 0$. So both inequalities are equalities. Equality on the right forces $\sum_i \lambda_i^\star f_i(x^\star) = 0$, a sum of non-positive terms, hence each term is zero: that is **complementary slackness**. Equality on the left says $x^\star$ minimises the differentiable function $x \mapsto L(x, \boldsymbol{\lambda}^\star)$ over $\mathcal{D}$, whose gradient is $\nabla f_0 + \sum_i \lambda_i^\star \nabla f_i$. Since $\mathcal{D}$ is open, the argument at the end of the proof of Proposition P.22 (a minimum at an interior point of each coordinate line forces the partial derivative to vanish, which uses no convexity) shows this gradient vanishes at $x^\star$: that is the stationarity condition in (P.10).

Sufficiency. Complementary slackness gives $L(x^\star, \boldsymbol{\lambda}^\star) = f_0(x^\star) + \sum_i \lambda_i^\star f_i(x^\star) = f_0(x^\star)$. Under convexity, $L(\cdot, \boldsymbol{\lambda}^\star) = f_0 + \sum_i \lambda_i^\star f_i$ is convex, as a sum of convex functions with non-negative weights $\lambda_i^\star$, and differentiable on the open convex $\mathcal{D}$ with vanishing gradient at $x^\star$ by (P.10); so by Proposition P.22 $x^\star$ minimises it, i.e. $g(\boldsymbol{\lambda}^\star) = L(x^\star, \boldsymbol{\lambda}^\star) = f_0(x^\star)$. Since $x^\star$ is feasible and $\boldsymbol{\lambda}^\star \geqslant \mathbf{0}$,

the definitions and Theorem P.29 give

$$f_0(x^\star) \geqslant p^\star \geqslant d^\star \geqslant g(\boldsymbol{\lambda}^\star) = f_0(x^\star),$$

so all four are equal: $x^\star$ attains $p^\star$, $\boldsymbol{\lambda}^\star$ attains $d^\star$, and the gap $p^\star - d^\star$ is zero. □

Remark. In the support vector classifier of Unit 14, $x = (\mathbf{w}, b)$, $f_0(x) = \frac{1}{2}|\mathbf{w}|^2$, and each observation $(\mathbf{x}_i, y_i)$, $y_i \in \{-1, 1\}$, contributes the margin constraint $f_i(x) = 1 - y_i(\mathbf{w}'\mathbf{x}_i + b) \leqslant 0$; the stationarity condition in (P.10), read in the $\mathbf{w}$ coordinates, gives $\mathbf{w}^\star = \sum_i \lambda_i^\star y_i \mathbf{x}_i$. Here Theorem P.31 applies when the two classes are linearly separable: the problem is convex with $0 \leqslant p^\star < \infty$ (separability makes it feasible), and rescaling a separating $(\mathbf{w}, b)$ makes every $f_i < 0$, which is Slater's condition, so Theorem P.30 supplies the strong duality and the dual optimum that Theorem P.31 assumes at a primal optimum $x^\star$. Complementary slackness produces the support vectors: $\lambda_i^\star > 0$ forces $f_i(x^\star) = 0$, so only the observations whose margin constraint is active carry a nonzero multiplier, and $\mathbf{w}^\star$ depends on those alone.

Part II
Linear Algebra

6 Rank, Column Space, Rank–Nullity

For $\mathbf{A} \in \mathbb{R}^{n\times p}$ write $\mathcal{C}(\mathbf{A}) = \{\mathbf{A}\mathbf{v} : \mathbf{v} \in \mathbb{R}^p\}$ for the column space and $\mathcal{N}(\mathbf{A}) = \{\mathbf{v} : \mathbf{A}\mathbf{v} = \mathbf{0}\}$ for the null space, and $\operatorname{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$.

Theorem P.32 (Rank–nullity)**.** $\operatorname{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = p$.

Proof. Take a basis $\mathbf{v}_1, \dots, \mathbf{v}_q$ of $\mathcal{N}(\mathbf{A})$ and extend it to a basis $\mathbf{v}_1, \dots, \mathbf{v}_p$ of $\mathbb{R}^p$. The vectors $\mathbf{A}\mathbf{v}_{q+1}, \dots, \mathbf{A}\mathbf{v}_p$ span $\mathcal{C}(\mathbf{A})$: any $\mathbf{v} \in \mathbb{R}^p$ is $\sum_{j\leqslant p} c_j \mathbf{v}_j$, and since $\mathbf{A}\mathbf{v}_j = \mathbf{0}$ for $j \leqslant q$, $\mathbf{A}\mathbf{v} = \sum_{j>q} c_j \mathbf{A}\mathbf{v}_j$. They are independent: if $\sum_{j>q} c_j \mathbf{A}\mathbf{v}_j = \mathbf{0}$ then $\mathbf{A}\big(\sum_{j>q} c_j \mathbf{v}_j\big) = \mathbf{0}$, so $\sum_{j>q} c_j \mathbf{v}_j \in \mathcal{N}(\mathbf{A})$ is a combination $\sum_{j\leqslant q} d_j \mathbf{v}_j$; then $\sum_{j\leqslant q} d_j \mathbf{v}_j - \sum_{j>q} c_j \mathbf{v}_j = \mathbf{0}$, and independence of the full

basis forces every $c_j = 0$. So $\mathbf{A}\mathbf{v}_{q+1}, \ldots, \mathbf{A}\mathbf{v}_p$ is a basis of $\mathcal{C}(\mathbf{A})$, and $\operatorname{rank}(\mathbf{A}) = p - q$. □

Proposition P.33 (Gram matrices)**.** $\mathcal{N}(\mathbf{A}'\mathbf{A}) = \mathcal{N}(\mathbf{A})$ and $\operatorname{rank}(\mathbf{A}'\mathbf{A}) = \operatorname{rank}(\mathbf{A})$. In particular $\mathbf{A}'\mathbf{A}$ is invertible if and only if $\mathbf{A}$ has full column rank.

Proof. If $\mathbf{A}\mathbf{v} = \mathbf{0}$ then $\mathbf{A}'\mathbf{A}\mathbf{v} = \mathbf{0}$. Conversely $\mathbf{A}'\mathbf{A}\mathbf{v} = \mathbf{0}$ gives $\mathbf{v}'\mathbf{A}'\mathbf{A}\mathbf{v} = |\mathbf{A}\mathbf{v}|^2 = 0$, so $\mathbf{A}\mathbf{v} = \mathbf{0}$. The null spaces coincide, and both matrices have p columns, so Theorem P.32 gives $\operatorname{rank}(\mathbf{A}'\mathbf{A}) = p - \dim \mathcal{N}(\mathbf{A}'\mathbf{A}) = p - \dim \mathcal{N}(\mathbf{A}) = \operatorname{rank}(\mathbf{A})$. $\mathbf{A}'\mathbf{A}$ is $p \times p$. If its rank is p, its column space is a p-dimensional subspace of $\mathbb{R}^p$, hence all of $\mathbb{R}^p$; so for each standard basis vector $\mathbf{e}_i$ there is $\mathbf{c}_i$ with $\mathbf{A}'\mathbf{A}\mathbf{c}_i = \mathbf{e}_i$, and the matrix with columns $\mathbf{c}_i$ is a right inverse, hence the inverse. Conversely, if $\mathbf{A}'\mathbf{A}$ is invertible then $\mathbf{A}'\mathbf{A}\mathbf{v} = \mathbf{0}$ forces $\mathbf{v} = \mathbf{0}$, so $\dim \mathcal{N}(\mathbf{A}'\mathbf{A}) = 0$ and the rank is p. Full column rank of $\mathbf{A}$ means $\operatorname{rank}(\mathbf{A}) = p$, which by the first part is $\operatorname{rank}(\mathbf{A}'\mathbf{A}) = p$. □

7 The Trace

Definition P.34 (Trace)**.** $\operatorname{tr}(\mathbf{A}) = \sum_i a_{ii}$ for square $\mathbf{A}$.

Proposition P.35 (Properties of the trace)**.**

(i) tr is linear: $\operatorname{tr}(\alpha\mathbf{A} + \beta\mathbf{B}) = \alpha \operatorname{tr}(\mathbf{A}) + \beta \operatorname{tr}(\mathbf{B})$.

(ii) $\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA})$ whenever both products are defined, even when $\mathbf{AB}$ and $\mathbf{BA}$ have different sizes.

(iii) $\operatorname{tr}(\mathbf{PAP}^{-1}) = \operatorname{tr}(\mathbf{A})$ for invertible $\mathbf{P}$: the trace is a similarity invariant.

(iv) $\operatorname{tr}(\mathbf{A}'\mathbf{A}) = \sum_{i,j} a_{ij}^2$, so $\operatorname{tr}(\mathbf{A}'\mathbf{A}) = 0$ forces $\mathbf{A} = \mathbf{0}$.

Proof. (i) is immediate. For (ii), with $\mathbf{A}$ of size $m \times n$ and $\mathbf{B}$ of size $n \times m$,

$$\operatorname{tr}(\mathbf{AB}) = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} b_{ji} = \sum_{j=1}^{n} \sum_{i=1}^{m} b_{ji} a_{ij} = \operatorname{tr}(\mathbf{BA}),$$

the two double sums differing only in the order of summation, which is finite. (iii) follows by

applying (ii) with the pair $\mathbf{PA}$ and $\mathbf{P}^{-1}$: $\operatorname{tr}\big((\mathbf{PA})\mathbf{P}^{-1}\big) = \operatorname{tr}\big(\mathbf{P}^{-1}(\mathbf{PA})\big) = \operatorname{tr}(\mathbf{A})$. For (iv), the (i,i) entry of $\mathbf{A}'\mathbf{A}$ is $\sum_j a_{ji}^2$. □

Remark. Property (iii) makes $\operatorname{tr}(\mathbf{A})$ well defined on a linear map rather than on a matrix.

8 Inner Products, Orthogonality, Projection

Theorem P.36 (Cauchy–Schwarz)**.** $|\langle \mathbf{u}, \mathbf{v}\rangle| \leqslant |\mathbf{u}|\,|\mathbf{v}|$, with equality iff $\mathbf{u}$ and $\mathbf{v}$ are proportional.

Proof. If $\mathbf{v} = \mathbf{0}$ both sides are 0 and $\mathbf{v} = 0\cdot\mathbf{u}$, so $\mathbf{u}$ and $\mathbf{v}$ are proportional. For $\mathbf{v} \neq \mathbf{0}$ and $t \in \mathbb{R}$, $0 \leqslant |\mathbf{u} - t\mathbf{v}|^2 = |\mathbf{u}|^2 - 2t\langle \mathbf{u}, \mathbf{v}\rangle + t^2|\mathbf{v}|^2$. The right side is a quadratic in t with non-negative values; at its minimising point $t_0 = \langle \mathbf{u}, \mathbf{v}\rangle / |\mathbf{v}|^2$ it equals $|\mathbf{u}|^2 - \langle \mathbf{u}, \mathbf{v}\rangle^2/|\mathbf{v}|^2$, so

$$0 \leqslant |\mathbf{u} - t_0\mathbf{v}|^2 = |\mathbf{u}|^2 - \frac{\langle \mathbf{u}, \mathbf{v}\rangle^2}{|\mathbf{v}|^2},$$

that is $\langle \mathbf{u}, \mathbf{v}\rangle^2 \leqslant |\mathbf{u}|^2|\mathbf{v}|^2$, which is the inequality after taking square roots. Equality holds exactly when $|\mathbf{u} - t_0\mathbf{v}| = 0$, that is $\mathbf{u} = t_0\mathbf{v}$, so equality forces proportionality. Conversely, if $\mathbf{u} = c\mathbf{v}$ both sides equal $|c|\,|\mathbf{v}|^2$, and if $\mathbf{v} = c\mathbf{u}$ both sides equal $|c|\,|\mathbf{u}|^2$. □

Theorem P.37 (Orthogonal projection)**.** Let $S \subseteq \mathbb{R}^n$ be a subspace and $\mathbf{y} \in \mathbb{R}^n$. There is a unique $\hat{\mathbf{y}} \in S$ minimising $|\mathbf{y} - \mathbf{u}|$ over $\mathbf{u} \in S$, characterised by $\mathbf{y} - \hat{\mathbf{y}} \perp S$. If $S = \mathcal{C}(\mathbf{X})$ with $\mathbf{X}$ of full column rank, then $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ with $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

Proof. A subspace is convex, and it is closed: this is trivial for $S = \{\mathbf{0}\}$; otherwise let the columns of $\mathbf{B}$ form a basis of S, so that $\mathbf{B}'\mathbf{B}$ is invertible by Proposition P.33, and if $\mathbf{B}\mathbf{c}_m \to \mathbf{y}$ then $\mathbf{c}_m = (\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'\mathbf{B}\mathbf{c}_m \to \mathbf{c} := (\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'\mathbf{y}$ (each entry of $(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'\mathbf{z}$ is a fixed linear combination of the entries of $\mathbf{z}$, hence continuous in $\mathbf{z}$). Then $\mathbf{B}\mathbf{c}_m \to \mathbf{B}\mathbf{c}$ in the same way, and since a sequence has one limit, $\mathbf{y} = \mathbf{B}\mathbf{c} \in S$. A subspace is non-empty, so Theorem P.20 gives a unique minimiser $\hat{\mathbf{y}}$, characterised by $\langle \mathbf{y} - \hat{\mathbf{y}}, \mathbf{u} - \hat{\mathbf{y}} \rangle \leqslant 0$ for all $\mathbf{u} \in S$. For a subspace this is equivalent to $\mathbf{y} - \hat{\mathbf{y}} \perp S$. Indeed, for $\mathbf{s} \in S$ both $\mathbf{u} = \hat{\mathbf{y}} + \mathbf{s}$ and $\mathbf{u} = \hat{\mathbf{y}} - \mathbf{s}$ lie in S, and the variational inequality (P.3) at these two points reads $\langle \mathbf{y} - \hat{\mathbf{y}}, \mathbf{s} \rangle \leqslant 0$ and $-\langle \mathbf{y} - \hat{\mathbf{y}}, \mathbf{s} \rangle \leqslant 0$, forcing $\langle \mathbf{y} - \hat{\mathbf{y}}, \mathbf{s} \rangle = 0$. Conversely, if $\mathbf{y} - \hat{\mathbf{y}} \perp S$ with $\hat{\mathbf{y}} \in S$, then $\mathbf{u} - \hat{\mathbf{y}} \in S$ for $\mathbf{u} \in S$, so $\langle \mathbf{y} - \hat{\mathbf{y}}, \mathbf{u} - \hat{\mathbf{y}} \rangle = 0 \leqslant 0$, and Theorem P.20 identifies $\hat{\mathbf{y}}$ as the minimiser.

For the formula, $\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{C}(\mathbf{X})$ means $\langle \mathbf{y} - \hat{\mathbf{y}}, \mathbf{X}\mathbf{e}_j \rangle = 0$ for every column, that is $\mathbf{X}'(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{0}$. As $\hat{\mathbf{y}} \in \mathcal{C}(\mathbf{X})$, write $\hat{\mathbf{y}} = \mathbf{X}\mathbf{c}$; then $\mathbf{X}'\mathbf{X}\mathbf{c} = \mathbf{X}'\mathbf{y}$. Since $\mathbf{X}'\mathbf{X}$ is invertible by Proposition P.33,

$\mathbf{c} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ and $\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y}$. □

Theorem P.38 (Pythagoras)**.** If $\mathbf{u} \perp \mathbf{v}$ then $|\mathbf{u}+\mathbf{v}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2$.

Proof. $|\mathbf{u}+\mathbf{v}|^2 = \langle \mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{v}\rangle = |\mathbf{u}|^2 + 2\langle \mathbf{u}, \mathbf{v}\rangle + |\mathbf{v}|^2$, and $\langle \mathbf{u}, \mathbf{v}\rangle = 0$. □

9 The Spectral Theorem

Theorem P.39 (Spectral theorem)**.** A symmetric $\mathbf{A} \in \mathbb{R}^{n\times n}$ has an orthonormal basis of eigenvectors: there are an orthogonal $\mathbf{P}$ and a diagonal $\mathbf{\Lambda}$ with $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$, the diagonal entries of $\mathbf{\Lambda}$ being the eigenvalues of $\mathbf{A}$, all real.

Proof. Write $q(\mathbf{v}) = \mathbf{v}'\mathbf{A}\mathbf{v}$, a polynomial in the coordinates of $\mathbf{v}$, hence continuous. For a subspace $W \neq \{\mathbf{0}\}$ of $\mathbb{R}^n$ let $\Sigma_W = \{\mathbf{v} \in W : |\mathbf{v}| = 1\}$ be its unit sphere, which is non-empty ($\mathbf{w}/|\mathbf{w}|$ for any $\mathbf{w} \in W \setminus \{\mathbf{0}\}$).

Step 1: the maximum of q on Σ_W is attained. Σ_W is bounded, and it is closed: if $\mathbf{v}_m \in \Sigma_W$ and $\mathbf{v}_m \to \mathbf{v}$, then $\mathbf{v} \in W$ because W is closed (as shown in the proof of Theorem P.37), and

$|\mathbf{v}| = \lim_m |\mathbf{v}_m| = 1$. So Σ_W is compact by Theorem P.5, and by Theorem P.13 the continuous q attains its maximum on Σ_W.

Step 2: a maximiser on an invariant subspace is an eigenvector. Let $W \neq \{\mathbf{0}\}$ be $\mathbf{A}$-invariant ($\mathbf{A}W \subseteq W$), let $\mathbf{v}_1 \in \Sigma_W$ maximise q on Σ_W, and put $\lambda_1 = q(\mathbf{v}_1) = \mathbf{v}_1'\mathbf{A}\mathbf{v}_1$, a real number. Fix $\mathbf{u} \in W$ with $\mathbf{u} \perp \mathbf{v}_1$. By Theorem P.38, $|\mathbf{v}_1 + t\mathbf{u}|^2 = 1 + t^2|\mathbf{u}|^2 > 0$ for every $t \in \mathbb{R}$, so $\mathbf{w}(t) = (\mathbf{v}_1 + t\mathbf{u})/|\mathbf{v}_1 + t\mathbf{u}|$ lies in Σ_W and $q(\mathbf{w}(t)) \leqslant \lambda_1 = q(\mathbf{w}(0))$. Expanding with $\mathbf{u}'\mathbf{A}\mathbf{v}_1 = \mathbf{v}_1'\mathbf{A}\mathbf{u}$ (symmetry of $\mathbf{A}$),

$$\varphi(t) := q(\mathbf{w}(t)) = \frac{N(t)}{D(t)}, \qquad N(t) = \lambda_1 + 2t\,\mathbf{u}'\mathbf{A}\mathbf{v}_1 + t^2\,\mathbf{u}'\mathbf{A}\mathbf{u}, \qquad D(t) = 1 + t^2|\mathbf{u}|^2,$$

a quotient of polynomials with $D > 0$, hence differentiable, with a maximum at the interior point $t = 0$. As in the proof of Theorem P.6, $\varphi'(0) = 0$. By the quotient rule, with $N(0) = \lambda_1$, $N'(0) = 2\mathbf{u}'\mathbf{A}\mathbf{v}_1$, $D(0) = 1$, $D'(0) = 0$,

$$0 = \varphi'(0) = \frac{N'(0)D(0) - N(0)D'(0)}{D(0)^2} = 2\mathbf{u}'\mathbf{A}\mathbf{v}_1.$$

So $\mathbf{u}'\mathbf{A}\mathbf{v}_1 = 0$ for every $\mathbf{u} \in W$ with $\mathbf{u} \perp \mathbf{v}_1$. Apply this to $\mathbf{u} = \mathbf{A}\mathbf{v}_1 - \lambda_1\mathbf{v}_1$: it lies in W, since $\mathbf{A}\mathbf{v}_1 \in W$ by invariance and $\mathbf{v}_1 \in W$, and $\mathbf{v}_1'\mathbf{u} = \mathbf{v}_1'\mathbf{A}\mathbf{v}_1 - \lambda_1|\mathbf{v}_1|^2 = \lambda_1 - \lambda_1 = 0$. Hence

$$0 = \mathbf{u}'\mathbf{A}\mathbf{v}_1 = \mathbf{u}'(\mathbf{u} + \lambda_1\mathbf{v}_1) = |\mathbf{u}|^2 + \lambda_1\mathbf{u}'\mathbf{v}_1 = |\mathbf{u}|^2,$$

so $\mathbf{u} = \mathbf{0}$, that is $\mathbf{A}\mathbf{v}_1 = \lambda_1\mathbf{v}_1$.

Step 3: the orthogonal complement of an eigenvector stays invariant. With W, $\mathbf{v}_1$, λ_1 as in Step 2, the subspace $W \cap \mathbf{v}_1^{\perp} = \{\mathbf{u} \in W : \mathbf{u} \perp \mathbf{v}_1\}$ is $\mathbf{A}$-invariant: for $\mathbf{u}$ in it, $\mathbf{A}\mathbf{u} \in W$ by invariance of W, and by symmetry $\langle \mathbf{A}\mathbf{u}, \mathbf{v}_1\rangle = \mathbf{u}'\mathbf{A}'\mathbf{v}_1 = \mathbf{u}'\mathbf{A}\mathbf{v}_1 = \lambda_1\langle \mathbf{u}, \mathbf{v}_1\rangle = 0$.

Step 4: the recursion. Put $W_0 = \mathbb{R}^n$, which is $\mathbf{A}$-invariant. For $k = 1, 2, \ldots, n$ in turn, suppose orthonormal $\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}$ have been chosen (none when $k = 1$) with $\mathbf{A}\mathbf{v}_j = \lambda_j\mathbf{v}_j$, and that

$$W_{k-1} = \{\mathbf{u} \in \mathbb{R}^n : \mathbf{u} \perp \mathbf{v}_1, \ldots, \mathbf{u} \perp \mathbf{v}_{k-1}\}$$

is $\mathbf{A}$-invariant (for $k = 1$ this is $W_0 = \mathbb{R}^n$). For $k \geqslant 2$ let $\mathbf{V}_{k-1}$ be the $n \times (k-1)$ matrix with columns $\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}$; then $W_{k-1} = \mathcal{N}(\mathbf{V}_{k-1}')$, a subspace. Orthonormality means $\mathbf{V}_{k-1}'\mathbf{V}_{k-1} = \mathbf{I}_{k-1}$, so every $\mathbf{c} \in \mathbb{R}^{k-1}$ equals $\mathbf{V}_{k-1}'(\mathbf{V}_{k-1}\mathbf{c})$ and $\mathcal{C}(\mathbf{V}_{k-1}') = \mathbb{R}^{k-1}$; by Theorem P.32, $\dim W_{k-1} =$

$n-(k-1)\geqslant 1$. So $W_{k-1}\neq\{\mathbf{0}\}$ for every $k\leqslant n$. By Step 1 choose $\mathbf{v}_k\in\Sigma_{W_{k-1}}$ maximising q there, and put $\lambda_k=q(\mathbf{v}_k)$. By Step 2, $\mathbf{A}\mathbf{v}_k=\lambda_k\mathbf{v}_k$ with λ_k real. Since $\mathbf{v}_k\in W_{k-1}$, it is orthogonal to $\mathbf{v}_1,\dots,\mathbf{v}_{k-1}$, and $|\mathbf{v}_k|=1$, so $\mathbf{v}_1,\dots,\mathbf{v}_k$ are orthonormal. Finally $W_k=W_{k-1}\cap\mathbf{v}_k^{\perp}$ is $\mathbf{A}$-invariant by Step 3. This is the hypothesis for $k+1$, and the recursion continues until $k=n$. Since $W_k\subseteq W_{k-1}$, the maximum over the smaller sphere is no larger, so $\lambda_1\geqslant\lambda_2\geqslant\cdots\geqslant\lambda_n$.

Step 5: the diagonalisation. Let $\mathbf{P}$ have columns $\mathbf{v}_1,\dots,\mathbf{v}_n$ and $\mathbf{\Lambda}=\operatorname{diag}(\lambda_1,\dots,\lambda_n)$. Orthonormality gives $\mathbf{P}'\mathbf{P}=\mathbf{I}$, so $\mathbf{P}$ is a right inverse of the square matrix $\mathbf{P}'$, hence its inverse, and $\mathbf{P}\mathbf{P}'=\mathbf{I}$ as well: $\mathbf{P}$ is orthogonal. Column by column, $\mathbf{A}\mathbf{P}=(\mathbf{A}\mathbf{v}_1,\dots,\mathbf{A}\mathbf{v}_n)=(\lambda_1\mathbf{v}_1,\dots,\lambda_n\mathbf{v}_n)=\mathbf{P}\mathbf{\Lambda}$, and multiplying on the right by $\mathbf{P}'$ gives $\mathbf{A}=\mathbf{P}\mathbf{\Lambda}\mathbf{P}'$. Each λ_k is an eigenvalue, with eigenvector $\mathbf{v}_k$. Conversely, if $\mathbf{A}\mathbf{x}=\mu\mathbf{x}$ for some $\mathbf{x}\neq\mathbf{0}$ (real or complex, with μ real or complex), then $\mathbf{w}=\mathbf{P}'\mathbf{x}\neq\mathbf{0}$ (as $\mathbf{x}=\mathbf{P}\mathbf{w}$) and $\mathbf{P}\mathbf{\Lambda}\mathbf{w}=\mathbf{A}\mathbf{x}=\mu\mathbf{x}=\mu\mathbf{P}\mathbf{w}$; multiplying by $\mathbf{P}'$ gives $\lambda_j w_j=\mu w_j$ for every j, and a j with $w_j\neq 0$ gives $\mu=\lambda_j$. So the eigenvalues of $\mathbf{A}$ are exactly $\lambda_1,\dots,\lambda_n$, all real. □

Corollary P.40 (Rayleigh)**.** For symmetric $\mathbf{A}$ with eigenvalues $\lambda_1\geqslant\cdots\geqslant\lambda_n$, $\max_{|\mathbf{v}|=1}\mathbf{v}'\mathbf{A}\mathbf{v}=\lambda_1$ and $\min_{|\mathbf{v}|=1}\mathbf{v}'\mathbf{A}\mathbf{v}=\lambda_n$.

Proof. By Theorem P.39, $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ with $\mathbf{P}$ orthogonal; permuting the columns of $\mathbf{P}$ and the diagonal entries of $\mathbf{\Lambda}$ in the same way (which preserves $\mathbf{P}\mathbf{\Lambda}\mathbf{P}' = \sum_j \lambda_j \mathbf{p}_j \mathbf{p}_j'$, $\mathbf{p}_j$ the jth column), take the diagonal of $\mathbf{\Lambda}$ in the order $\lambda_1 \geqslant \cdots \geqslant \lambda_n$. For $|\mathbf{v}| = 1$ put $\mathbf{w} = \mathbf{P}'\mathbf{v}$; then $\mathbf{v} = \mathbf{P}\mathbf{w}$ and $|\mathbf{w}|^2 = \mathbf{v}'\mathbf{P}\mathbf{P}'\mathbf{v} = |\mathbf{v}|^2 = 1$, and

$$\mathbf{v}'\mathbf{A}\mathbf{v} = \mathbf{w}'\mathbf{P}'\mathbf{P}\mathbf{\Lambda}\mathbf{P}'\mathbf{P}\mathbf{w} = \mathbf{w}'\mathbf{\Lambda}\mathbf{w} = \sum_j \lambda_j w_j^2,$$

a convex combination of the eigenvalues, the weights $w_j^2 \geqslant 0$ adding to 1. Hence $\lambda_n = \sum_j \lambda_n w_j^2 \leqslant \mathbf{v}'\mathbf{A}\mathbf{v} \leqslant \sum_j \lambda_1 w_j^2 = \lambda_1$. The bounds are attained at the first and last columns of $\mathbf{P}$, unit eigenvectors with $\mathbf{p}_1'\mathbf{A}\mathbf{p}_1 = \lambda_1 |\mathbf{p}_1|^2 = \lambda_1$ and likewise $\mathbf{p}_n'\mathbf{A}\mathbf{p}_n = \lambda_n$. □

10 Positive Definiteness and Square Roots

Definition P.41. Symmetric $\mathbf{A}$ is **positive semi-definite** if $\mathbf{v}'\mathbf{A}\mathbf{v} \geqslant 0$ for all $\mathbf{v}$, and **positive definite** if $\mathbf{v}'\mathbf{A}\mathbf{v} > 0$ for $\mathbf{v} \neq \mathbf{0}$.

Proposition P.42. A symmetric $\mathbf{A}$ is positive semi-definite iff all its eigenvalues are non-negative, and positive definite iff all are strictly positive; in the latter case $\mathbf{A}$ is invertible.

Proof. By Corollary P.40 the extreme values of $\mathbf{v}'\mathbf{A}\mathbf{v}$ on the unit sphere are λ_n and λ_1. For $\mathbf{v} \neq \mathbf{0}$, $\mathbf{v}'\mathbf{A}\mathbf{v} = |\mathbf{v}|^2\,(\mathbf{v}/|\mathbf{v}|)'\mathbf{A}(\mathbf{v}/|\mathbf{v}|) \geqslant |\mathbf{v}|^2\lambda_n$; so $\lambda_n \geqslant 0$ makes the form non-negative and $\lambda_n > 0$ makes it positive. Conversely, a unit eigenvector $\mathbf{u}$ for λ_n has $\mathbf{u}'\mathbf{A}\mathbf{u} = \lambda_n$, so a non-negative (positive) form forces $\lambda_n \geqslant 0$ ($\lambda_n > 0$). As λ_n is the smallest eigenvalue, this is the condition on all eigenvalues. If every eigenvalue is nonzero, $\mathbf{\Lambda}^{-1} = \operatorname{diag}(1/\lambda_j)$ exists, and with $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ and $\mathbf{P}'\mathbf{P} = \mathbf{I}$, $\mathbf{A}\,\mathbf{P}\mathbf{\Lambda}^{-1}\mathbf{P}' = \mathbf{P}\mathbf{\Lambda}\mathbf{\Lambda}^{-1}\mathbf{P}' = \mathbf{P}\mathbf{P}' = \mathbf{I}$; so $\mathbf{A}^{-1} = \mathbf{P}\mathbf{\Lambda}^{-1}\mathbf{P}'$. □

Lemma P.43 (Inverses of convergent positive definite matrices)**.** Let $\mathbf{A}_n$ be symmetric matrices whose entries converge to those of a positive definite $\mathbf{Q}$, and let $\lambda > 0$ be the smallest eigenvalue of $\mathbf{Q}$. For all large n, $\mathbf{A}_n$ is positive definite with every eigenvalue at least $\lambda/2$, every entry of $\mathbf{A}_n^{-1}$ is at most $2/\lambda$ in absolute value, and $\mathbf{A}_n^{-1} \to \mathbf{Q}^{-1}$ entrywise.

Proof. For $|\mathbf{v}| = 1$ every coordinate has $|v_r| \leqslant 1$, so $|v_r v_s| \leqslant 1$ and $|\mathbf{v}'(\mathbf{A}_n - \mathbf{Q})\mathbf{v}| = \left|\sum_{r,s} v_r(\mathbf{A}_n - \mathbf{Q})_{rs}v_s\right| \leqslant \sum_{r,s}|(\mathbf{A}_n - \mathbf{Q})_{rs}| =: \delta_n$, and $\delta_n \to 0$ as a finite sum of terms tending to 0. By

Corollary P.40, $\mathbf{v}'\mathbf{Q}\mathbf{v} \geqslant \lambda$, hence $\mathbf{v}'\mathbf{A}_n\mathbf{v} = \mathbf{v}'\mathbf{Q}\mathbf{v} + \mathbf{v}'(\mathbf{A}_n - \mathbf{Q})\mathbf{v} \geqslant \lambda - \delta_n \geqslant \lambda/2$ for every unit $\mathbf{v}$ once $\delta_n \leqslant \lambda/2$, that is for all large n. By the same corollary applied to $\mathbf{A}_n$, its smallest eigenvalue is the minimum of $\mathbf{v}'\mathbf{A}_n\mathbf{v}$ over unit $\mathbf{v}$, so every eigenvalue of $\mathbf{A}_n$ is at least $\lambda/2$, and Proposition P.42 makes $\mathbf{A}_n$ positive definite and invertible. With $\mathbf{A}_n = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ (Theorem P.39; $\mathbf{P}$, $\mathbf{\Lambda}$ depend on n), $\mathbf{A}_n^{-1} = \mathbf{P}\mathbf{\Lambda}^{-1}\mathbf{P}'$ by the proof of Proposition P.42. For unit vectors $\mathbf{u}, \mathbf{v}$ put $\mathbf{w} = \mathbf{P}'\mathbf{v}$, so $|\mathbf{w}| = 1$; then $|\mathbf{A}_n^{-1}\mathbf{v}|^2 = \mathbf{w}'\mathbf{\Lambda}^{-1}\mathbf{P}'\mathbf{P}\mathbf{\Lambda}^{-1}\mathbf{w} = \sum_j \lambda_j^{-2} w_j^2 \leqslant (2/\lambda)^2 \sum_j w_j^2 = (2/\lambda)^2$, the $\lambda_j \geqslant \lambda/2$ being the eigenvalues of $\mathbf{A}_n$, and $|\mathbf{u}'\mathbf{A}_n^{-1}\mathbf{v}| \leqslant |\mathbf{u}|\,|\mathbf{A}_n^{-1}\mathbf{v}| \leqslant 2/\lambda$ by Theorem P.36. Taking $\mathbf{u} = \mathbf{e}_r$ and $\mathbf{v} = \mathbf{e}_s$ gives $|(\mathbf{A}_n^{-1})_{rs}| \leqslant 2/\lambda$. Finally $\mathbf{A}_n^{-1} - \mathbf{Q}^{-1} = \mathbf{A}_n^{-1}(\mathbf{Q} - \mathbf{A}_n)\mathbf{Q}^{-1}$ (multiply out the right side), whose (r, s) entry $\sum_{a,b}(\mathbf{A}_n^{-1})_{ra}(\mathbf{Q} - \mathbf{A}_n)_{ab}(\mathbf{Q}^{-1})_{bs}$ is a finite sum of products of an entry bounded by $2/\lambda$, an entry tending to 0, and a fixed entry; each product tends to 0, hence so does the sum. □

Theorem P.44 (Symmetric square root)**.** A positive semi-definite $\mathbf{A}$ has a unique positive semi-definite $\mathbf{A}^{1/2}$ with $(\mathbf{A}^{1/2})^2 = \mathbf{A}$. If $\mathbf{A}$ is positive definite so is $\mathbf{A}^{1/2}$, and $\mathbf{A}^{-1/2} := (\mathbf{A}^{1/2})^{-1}$ satisfies $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{I}$.

Proof. Write $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ and set $\mathbf{A}^{1/2} = \mathbf{P}\mathbf{\Lambda}^{1/2}\mathbf{P}'$ with $\mathbf{\Lambda}^{1/2}$ the diagonal matrix of non-negative square roots, legitimate by Proposition P.42. It is symmetric, since $(\mathbf{P}\mathbf{\Lambda}^{1/2}\mathbf{P}')' = \mathbf{P}\mathbf{\Lambda}^{1/2}\mathbf{P}'$; it squares to $\mathbf{A}$, since $\mathbf{P}\mathbf{\Lambda}^{1/2}\mathbf{P}'\mathbf{P}\mathbf{\Lambda}^{1/2}\mathbf{P}' = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ by $\mathbf{P}'\mathbf{P} = \mathbf{I}$; and it is positive semi-definite, since with $\mathbf{w} = \mathbf{P}'\mathbf{v}$, $\mathbf{v}'\mathbf{A}^{1/2}\mathbf{v} = \sum_j \lambda_j^{1/2} w_j^2 \geqslant 0$.

Uniqueness. Let $\mathbf{B}$ be positive semi-definite with $\mathbf{B}^2 = \mathbf{A}$. By Theorem P.39, $\mathbf{B} = \mathbf{Q}\mathbf{M}\mathbf{Q}'$ with $\mathbf{Q}$ orthogonal, columns $\mathbf{q}_i$, and $\mathbf{M} = \operatorname{diag}(\mu_1, \ldots, \mu_n)$, where $\mu_i \geqslant 0$ by Proposition P.42; then $\mathbf{A} = \mathbf{B}^2 = \mathbf{Q}\mathbf{M}^2\mathbf{Q}'$, so $\mathbf{A}\mathbf{q}_i = \mu_i^2\mathbf{q}_i$. Let $\alpha \geqslant 0$ and $\mathbf{x} \in \mathbb{R}^n$ with $\mathbf{A}\mathbf{x} = \alpha\mathbf{x}$. Writing $\mathbf{x} = \sum_i d_i\mathbf{q}_i$ (the $\mathbf{q}_i$ form a basis), $\mathbf{0} = \mathbf{A}\mathbf{x} - \alpha\mathbf{x} = \sum_i d_i(\mu_i^2 - \alpha)\mathbf{q}_i$, so by independence $d_i = 0$ unless $\mu_i^2 = \alpha$, that is unless $\mu_i = \sqrt{\alpha}$ (both being non-negative). Hence $\mathbf{B}\mathbf{x} = \sum_i d_i\mu_i\mathbf{q}_i = \sqrt{\alpha}\sum_i d_i\mathbf{q}_i = \sqrt{\alpha}\,\mathbf{x}$. The same argument with $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ and $\mathbf{A}^{1/2} = \mathbf{P}\mathbf{\Lambda}^{1/2}\mathbf{P}'$ in place of $\mathbf{Q}\mathbf{M}^2\mathbf{Q}'$ and $\mathbf{Q}\mathbf{M}\mathbf{Q}'$ gives $\mathbf{A}^{1/2}\mathbf{x} = \sqrt{\alpha}\,\mathbf{x}$. Apply both to the jth column $\mathbf{p}_j$ of $\mathbf{P}$, an eigenvector of $\mathbf{A}$ with eigenvalue $\alpha = \lambda_j \geqslant 0$: $\mathbf{B}\mathbf{p}_j = \sqrt{\lambda_j}\,\mathbf{p}_j = \mathbf{A}^{1/2}\mathbf{p}_j$. So $\mathbf{B}\mathbf{P} = \mathbf{A}^{1/2}\mathbf{P}$, and multiplying on the right by $\mathbf{P}'$ gives $\mathbf{B} = \mathbf{A}^{1/2}$.

The positive definite case. If $\mathbf{A}$ is positive definite, every $\lambda_j > 0$ by Proposition P.42, so the eigenvalues $\lambda_j^{1/2}$ of $\mathbf{A}^{1/2}$ are positive and $\mathbf{A}^{1/2}$ is positive definite by the same proposition, hence

invertible, with $\mathbf{A}^{-1/2} = \mathbf{P}\mathbf{\Lambda}^{-1/2}\mathbf{P}'$ (as in the proof of Proposition P.42). Then $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{P}\mathbf{\Lambda}^{-1/2}\mathbf{P}'\mathbf{P}\mathbf{\Lambda}\mathbf{P}'\mathbf{P}\mathbf{\Lambda}^{-1/2}\mathbf{P}' = \mathbf{P}\mathbf{\Lambda}^{-1/2}\mathbf{\Lambda}\mathbf{\Lambda}^{-1/2}\mathbf{P}' = \mathbf{P}\mathbf{P}' = \mathbf{I}$. □

11 Idempotent Matrices

Definition P.45. $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$.

Lemma P.46 (Trace equals rank)**.** An idempotent $\mathbf{A} \in \mathbb{R}^{n\times n}$ has $\operatorname{tr}(\mathbf{A}) = \operatorname{rank}(\mathbf{A})$, and its eigenvalues lie in $\{0, 1\}$.

Proof. If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$ then $\lambda\mathbf{v} = \mathbf{A}^2\mathbf{v} = \lambda^2\mathbf{v}$, so $\lambda \in \{0, 1\}$. Next, $\mathbb{R}^n = \mathcal{N}(\mathbf{A}) \oplus \mathcal{C}(\mathbf{A})$: any $\mathbf{v}$ splits as $(\mathbf{v} - \mathbf{A}\mathbf{v}) + \mathbf{A}\mathbf{v}$ with $\mathbf{A}(\mathbf{v} - \mathbf{A}\mathbf{v}) = \mathbf{0}$, and if $\mathbf{v} = \mathbf{A}\mathbf{w}$ lies in both then $\mathbf{v} = \mathbf{A}\mathbf{w} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{v} = \mathbf{0}$. Every $\mathbf{u} = \mathbf{A}\mathbf{w} \in \mathcal{C}(\mathbf{A})$ has $\mathbf{A}\mathbf{u} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{w} = \mathbf{u}$. Let $\mathbf{n}_1, \ldots, \mathbf{n}_{n-r}$ be a basis of $\mathcal{N}(\mathbf{A})$ and $\mathbf{c}_1, \ldots, \mathbf{c}_r$ one of $\mathcal{C}(\mathbf{A})$, where $r = \operatorname{rank}(\mathbf{A})$ and $\dim \mathcal{N}(\mathbf{A}) = n - r$ by Theorem P.32. Together they are linearly independent: if $\sum_i a_i\mathbf{n}_i + \sum_j b_j\mathbf{c}_j = \mathbf{0}$, the vector $\sum_i a_i\mathbf{n}_i = -\sum_j b_j\mathbf{c}_j$ lies in both subspaces, hence is $\mathbf{0}$, and independence within each basis forces all a_i and b_j to vanish. Being n independent vectors in $\mathbb{R}^n$ they form a basis, so the matrix $\mathbf{P}$ with these columns

is invertible (as in the proof of Proposition P.33). As $\mathbf{A}\mathbf{n}_i = \mathbf{0}$ and $\mathbf{A}\mathbf{c}_j = \mathbf{c}_j$, $\mathbf{AP} = \mathbf{P}\boldsymbol{\Lambda}$ with $\boldsymbol{\Lambda}$ diagonal carrying $n - r$ zeros and r ones, so $\mathbf{A} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}^{-1}$. The trace is invariant under similarity (Proposition P.35(iii)), so $\operatorname{tr}(\mathbf{A}) = \operatorname{tr}(\boldsymbol{\Lambda}) = r$. □

Proposition P.47 (Projections are the symmetric idempotents)**.** $\mathbf{H}$ is the matrix of the orthogonal projection onto a subspace S if and only if $\mathbf{H}$ is symmetric, idempotent, and $\mathcal{C}(\mathbf{H}) = S$.

Proof. If $S = \{\mathbf{0}\}$ the projection is $\mathbf{H} = \mathbf{0}$ and both sides hold trivially. Otherwise the projection is $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ by Theorem P.37, with $\mathbf{X}$ a matrix whose columns form a basis of S (so $\mathbf{X}$ has full column rank). The matrix $(\mathbf{X}'\mathbf{X})^{-1}$ is symmetric, being the inverse of a symmetric matrix: $\{(\mathbf{X}'\mathbf{X})^{-1}\}'\mathbf{X}'\mathbf{X} = \{\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\}' = \mathbf{I}$. Hence $\mathbf{H}' = \mathbf{X}\{(\mathbf{X}'\mathbf{X})^{-1}\}'\mathbf{X}' = \mathbf{H}$, and $\mathbf{H}^2 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{H}$. Every $\mathbf{H}\mathbf{y} = \mathbf{X}\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}\}$ lies in $\mathcal{C}(\mathbf{X})$, and $\mathbf{H}\mathbf{X} = \mathbf{X}$ shows each column of $\mathbf{X}$ lies in $\mathcal{C}(\mathbf{H})$; so $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X}) = S$. Conversely let $\mathbf{H}$ be symmetric idempotent with column space S. For $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{H}\mathbf{y} \in S$ and $\mathbf{y} - \mathbf{H}\mathbf{y} \perp S$, because for $\mathbf{u} = \mathbf{H}\mathbf{w} \in S$, $\langle \mathbf{y} - \mathbf{H}\mathbf{y}, \mathbf{H}\mathbf{w} \rangle = \mathbf{w}'\mathbf{H}'(\mathbf{I} - \mathbf{H})\mathbf{y} = \mathbf{w}'(\mathbf{H} - \mathbf{H}^2)\mathbf{y} = 0$. By the characterisation in Theorem P.37, $\mathbf{H}\mathbf{y}$ is the projection. □

12 Block Matrices

Theorem P.48 (Schur complement)**.** Let $\mathbf{M} = \left(\begin{smallmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{C} & \mathbf{D}\end{smallmatrix}\right)$ with $\mathbf{A}$ invertible, and put $\mathbf{S} = \mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$. If $\mathbf{S}$ is invertible then $\mathbf{M}$ is, with

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1} \\ -\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} & \mathbf{S}^{-1} \end{pmatrix}, \tag{P.11}$$

so the lower-right block of $\mathbf{M}^{-1}$ is $\mathbf{S}^{-1}$.

Proof. Verify $\mathbf{M}\mathbf{M}^{-1} = \mathbf{I}$ blockwise. The $(1,1)$ block is $\mathbf{A}(\mathbf{A}^{-1}+\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1}) - \mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} = \mathbf{I} + \mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} - \mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} = \mathbf{I}$; the $(1,2)$ block is $-\mathbf{A}\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1} + \mathbf{B}\mathbf{S}^{-1} = \mathbf{0}$; the $(2,1)$ block is $\mathbf{C}\mathbf{A}^{-1} + \mathbf{C}\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} - \mathbf{D}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} = \mathbf{C}\mathbf{A}^{-1} - \mathbf{S}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} = \mathbf{0}$; and the $(2,2)$ block is $-\mathbf{C}\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1} + \mathbf{D}\mathbf{S}^{-1} = (\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B})\mathbf{S}^{-1} = \mathbf{S}\mathbf{S}^{-1} = \mathbf{I}$. So the matrix in (P.11) is a right inverse of the square matrix $\mathbf{M}$, hence its inverse. □

Corollary P.49. For $\mathbf{W} = [\mathbf{1}\ \mathbf{Z}]$ with n rows and of full column rank, the lower-right $p \times p$ block of $(\mathbf{W}'\mathbf{W})^{-1}$ is $(\mathbf{Z}'\mathbf{M}_0\mathbf{Z})^{-1}$, where $\mathbf{M}_0 = \mathbf{I}_n - n^{-1}\mathbf{1}\mathbf{1}'$.

Proof. $\mathbf{W}'\mathbf{W} = \left(\begin{smallmatrix} n & \mathbf{1}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{1} & \mathbf{Z}'\mathbf{Z} \end{smallmatrix}\right)$, whose $(1,1)$ entry n is invertible and whose Schur complement of that entry is $\mathbf{Z}'\mathbf{Z} - \mathbf{Z}'\mathbf{1}\, n^{-1}\mathbf{1}'\mathbf{Z} = \mathbf{Z}'(\mathbf{I}_n - n^{-1}\mathbf{1}\mathbf{1}')\mathbf{Z} = \mathbf{Z}'\mathbf{M}_0\mathbf{Z}$. This square matrix is invertible. Indeed, $\mathbf{M}_0$ is symmetric, and as $\mathbf{1}'\mathbf{1} = n$, $\mathbf{M}_0^2 = \mathbf{I}_n - 2n^{-1}\mathbf{1}\mathbf{1}' + n^{-2}\mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = \mathbf{M}_0$. If $\mathbf{Z}'\mathbf{M}_0\mathbf{Z}\mathbf{c} = \mathbf{0}$ then $|\mathbf{M}_0\mathbf{Z}\mathbf{c}|^2 = \mathbf{c}'\mathbf{Z}'\mathbf{M}_0'\mathbf{M}_0\mathbf{Z}\mathbf{c} = \mathbf{c}'\mathbf{Z}'\mathbf{M}_0\mathbf{Z}\mathbf{c} = 0$, so $\mathbf{0} = \mathbf{M}_0\mathbf{Z}\mathbf{c} = \mathbf{Z}\mathbf{c} - n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{Z}\mathbf{c})$, that is $\mathbf{Z}\mathbf{c} = a\mathbf{1}$ with $a = n^{-1}\mathbf{1}'\mathbf{Z}\mathbf{c}$. Then $\mathbf{W}(-a, \mathbf{c}')' = -a\mathbf{1} + \mathbf{Z}\mathbf{c} = \mathbf{0}$, and full column rank of $\mathbf{W}$ ($\mathcal{N}(\mathbf{W}) = \{\mathbf{0}\}$ by Theorem P.32) forces $(-a, \mathbf{c}')' = \mathbf{0}$, in particular $\mathbf{c} = \mathbf{0}$. So $\mathcal{N}(\mathbf{Z}'\mathbf{M}_0\mathbf{Z}) = \{\mathbf{0}\}$, the $p \times p$ matrix $\mathbf{Z}'\mathbf{M}_0\mathbf{Z}$ has rank p, and it is invertible as in the proof of Proposition P.33. Theorem P.48, with $\mathbf{A} = n$, $\mathbf{B} = \mathbf{1}'\mathbf{Z}$, $\mathbf{C} = \mathbf{Z}'\mathbf{1}$, $\mathbf{D} = \mathbf{Z}'\mathbf{Z}$, gives the lower-right block $(\mathbf{Z}'\mathbf{M}_0\mathbf{Z})^{-1}$. □

Lemma P.50 (Sherman–Morrison)**.** Let $\mathbf{A}$ be invertible and $\mathbf{u}, \mathbf{v}$ vectors with $1 - \mathbf{v}'\mathbf{A}^{-1}\mathbf{u} \neq 0$. Then

$$(\mathbf{A} - \mathbf{u}\mathbf{v}')^{-1} = \mathbf{A}^{-1} + \frac{\mathbf{A}^{-1}\mathbf{u}\,\mathbf{v}'\mathbf{A}^{-1}}{1 - \mathbf{v}'\mathbf{A}^{-1}\mathbf{u}}. \tag{P.12}$$

Proof. Write $c = 1 - \mathbf{v}'\mathbf{A}^{-1}\mathbf{u} \neq 0$, let $\mathbf{B}$ denote the right side of (P.12), and multiply $\mathbf{B}$ on the right by $\mathbf{A} - \mathbf{uv}'$:

$$\mathbf{B}(\mathbf{A} - \mathbf{uv}') = \mathbf{I} - \mathbf{A}^{-1}\mathbf{uv}' + \frac{\mathbf{A}^{-1}\mathbf{uv}' - \mathbf{A}^{-1}\mathbf{u}(\mathbf{v}'\mathbf{A}^{-1}\mathbf{u})\mathbf{v}'}{c} = \mathbf{I} - \mathbf{A}^{-1}\mathbf{uv}' + \frac{\mathbf{A}^{-1}\mathbf{uv}'\,c}{c} = \mathbf{I},$$

using that $\mathbf{v}'\mathbf{A}^{-1}\mathbf{u}$ is a scalar, so that $\mathbf{A}^{-1}\mathbf{u}(\mathbf{v}'\mathbf{A}^{-1}\mathbf{u})\mathbf{v}' = (\mathbf{v}'\mathbf{A}^{-1}\mathbf{u})\mathbf{A}^{-1}\mathbf{uv}'$ and the numerator is $(1 - \mathbf{v}'\mathbf{A}^{-1}\mathbf{u})\mathbf{A}^{-1}\mathbf{uv}'$. So $\mathbf{A} - \mathbf{uv}'$ is a right inverse of the square matrix $\mathbf{B}$, hence $\mathbf{B}$ is invertible with $\mathbf{B}^{-1} = \mathbf{A} - \mathbf{uv}'$, and therefore $\mathbf{A} - \mathbf{uv}'$ is invertible with inverse $\mathbf{B}$. □

13 The Singular Value Decomposition

Theorem P.51 (SVD)**.** Every $\mathbf{A} \in \mathbb{R}^{n\times p}$ factors as $\mathbf{A} = \mathbf{UDV}'$ with $\mathbf{U} \in \mathbb{R}^{n\times r}$ and $\mathbf{V} \in \mathbb{R}^{p\times r}$ having orthonormal columns, $r = \operatorname{rank}(\mathbf{A})$, and $\mathbf{D} = \operatorname{diag}(d_1, \ldots, d_r)$ with $d_1 \geqslant \cdots \geqslant d_r > 0$. The d_j^2 are the nonzero eigenvalues of $\mathbf{A}'\mathbf{A}$.

Proof. $\mathbf{A}'\mathbf{A}$ is symmetric, and positive semi-definite since $\mathbf{x}'\mathbf{A}'\mathbf{Ax} = |\mathbf{Ax}|^2 \geqslant 0$, so by Theorem P.39 and Proposition P.42 it has an orthonormal eigenbasis $\mathbf{v}_1, \ldots, \mathbf{v}_p$ with eigenvalues

$\lambda_1 \geqslant \cdots \geqslant \lambda_p \geqslant 0$ (ordered as in the proof of Corollary P.40). Write $\mathbf{A}'\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ with $\mathbf{P}$ the orthogonal matrix of columns $\mathbf{v}_j$. Since $\mathbf{P}$ and $\mathbf{P}'$ are invertible, $\mathcal{C}(\mathbf{A}'\mathbf{A}) = \mathcal{C}(\mathbf{P}\mathbf{\Lambda}\mathbf{P}') = \mathbf{P}\mathcal{C}(\mathbf{\Lambda})$ has the same dimension as $\mathcal{C}(\mathbf{\Lambda})$, which is spanned by the $\mathbf{e}_j$ with $\lambda_j \neq 0$; so the number of positive λ_j is $\operatorname{rank}(\mathbf{A}'\mathbf{A})$, which equals $r = \operatorname{rank}(\mathbf{A})$ by Proposition P.33. By the ordering, $\lambda_1, \ldots, \lambda_r > 0$ and $\lambda_{r+1} = \cdots = \lambda_p = 0$. Also $|\mathbf{A}\mathbf{v}_j|^2 = \mathbf{v}_j'\mathbf{A}'\mathbf{A}\mathbf{v}_j = \lambda_j|\mathbf{v}_j|^2 = \lambda_j$, so $\mathbf{A}\mathbf{v}_j = \mathbf{0}$ for $j > r$. Put $d_j = \sqrt{\lambda_j}$, so $d_1 \geqslant \cdots \geqslant d_r > 0$, and $\mathbf{u}_j = \mathbf{A}\mathbf{v}_j/d_j$ for $j \leqslant r$; these are orthonormal, since $\mathbf{u}_j'\mathbf{u}_k = \mathbf{v}_j'\mathbf{A}'\mathbf{A}\mathbf{v}_k/(d_jd_k) = \lambda_k\mathbf{v}_j'\mathbf{v}_k/(d_jd_k) = \lambda_k\delta_{jk}/(d_jd_k)$, which is 1 for $j = k$ and 0 otherwise. Let $\mathbf{U}$ and $\mathbf{V}$ have columns $\mathbf{u}_1, \ldots, \mathbf{u}_r$ and $\mathbf{v}_1, \ldots, \mathbf{v}_r$, and $\mathbf{D} = \operatorname{diag}(d_1, \ldots, d_r)$.

For any $\mathbf{x}$, $\mathbf{x} = \mathbf{P}\mathbf{P}'\mathbf{x} = \sum_{j\leqslant p}\langle\mathbf{v}_j, \mathbf{x}\rangle\mathbf{v}_j$. The terms with $j > r$ have $\mathbf{A}\mathbf{v}_j = \mathbf{0}$, and $\mathbf{A}\mathbf{v}_j = d_j\mathbf{u}_j$ for $j \leqslant r$, so

$$\mathbf{A}\mathbf{x} = \sum_{j\leqslant r}\langle\mathbf{v}_j, \mathbf{x}\rangle d_j\mathbf{u}_j = \mathbf{U}\mathbf{D}\mathbf{V}'\mathbf{x},$$

because $\mathbf{V}'\mathbf{x}$ has entries $\langle\mathbf{v}_j, \mathbf{x}\rangle$, $\mathbf{D}$ multiplies the jth by d_j, and $\mathbf{U}$ forms the combination of the $\mathbf{u}_j$ with these coefficients. As this holds for every $\mathbf{x}$, $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}'$. The nonzero eigenvalues of $\mathbf{A}'\mathbf{A}$ are $\lambda_1, \ldots, \lambda_r$ (Theorem P.39), that is $d_1^2, \ldots, d_r^2$. □

14 Norms and Conditioning

Definition P.52 (Operator norm, condition number)**.** $\|\mathbf{A}\| = \sup_{|\mathbf{x}|=1} |\mathbf{A}\mathbf{x}|$. For invertible $\mathbf{A}$ the **condition number** is $\kappa(\mathbf{A}) = \|\mathbf{A}\|\,\|\mathbf{A}^{-1}\|$.

Proposition P.53. For $\mathbf{A} \neq \mathbf{0}$, $\|\mathbf{A}\| = d_1$, the largest singular value of $\mathbf{A}$ (and $\|\mathbf{0}\| = 0$). For symmetric positive definite $\mathbf{A}$ with eigenvalues $\lambda_1 \geqslant \cdots \geqslant \lambda_n > 0$, $\|\mathbf{A}\| = \lambda_1$, $\|\mathbf{A}^{-1}\| = \lambda_n^{-1}$ and $\kappa(\mathbf{A}) = \lambda_1/\lambda_n$.

Proof. For $\mathbf{A} = \mathbf{0}$, $|\mathbf{A}\mathbf{x}| = 0$ for every $\mathbf{x}$. For $\mathbf{A} \neq \mathbf{0}$, $r = \operatorname{rank}(\mathbf{A}) \geqslant 1$. $|\mathbf{A}\mathbf{x}|^2 = \mathbf{x}'\mathbf{A}'\mathbf{A}\mathbf{x}$, and by Corollary P.40 applied to the symmetric $\mathbf{A}'\mathbf{A}$ its maximum over $|\mathbf{x}| = 1$ is the largest eigenvalue of $\mathbf{A}'\mathbf{A}$, which is d_1^2 by Theorem P.51 (the largest eigenvalue is positive, since $\mathbf{A}'\mathbf{A} \neq \mathbf{0}$ by Proposition P.33, so it is the largest nonzero one). As $s \mapsto \sqrt{s}$ is increasing, $\sup_{|\mathbf{x}|=1} |\mathbf{A}\mathbf{x}| = \sqrt{d_1^2} = d_1$. For symmetric positive definite $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$, $\mathbf{A}'\mathbf{A} = \mathbf{A}^2 = \mathbf{P}\mathbf{\Lambda}^2\mathbf{P}'$ has eigenvalues λ_i^2, the largest being λ_1^2, so $\|\mathbf{A}\| = \lambda_1$; and $\mathbf{A}^{-1} = \mathbf{P}\mathbf{\Lambda}^{-1}\mathbf{P}'$ (Proposition P.42) is symmetric positive definite with eigenvalues λ_i^{-1}, the largest being λ_n^{-1}, so by the same argument $\|\mathbf{A}^{-1}\| = \lambda_n^{-1}$.

Multiplying, $\kappa(\mathbf{A}) = \lambda_1/\lambda_n$. □

Theorem P.54 (Conditioning of a linear system)**.** Let $\mathbf{A}$ be invertible, $\mathbf{Ax} = \mathbf{b} \neq \mathbf{0}$, and $\mathbf{A}(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}$. Then

$$\frac{|\delta\mathbf{x}|}{|\mathbf{x}|} \leqslant \kappa(\mathbf{A}) \frac{|\delta\mathbf{b}|}{|\mathbf{b}|}. \tag{P.13}$$

Proof. For any matrix $\mathbf{B}$ and vector $\mathbf{y} \neq \mathbf{0}$, $\mathbf{y}/|\mathbf{y}|$ is a unit vector, so $|\mathbf{By}| = |\mathbf{y}|\,|\mathbf{B}(\mathbf{y}/|\mathbf{y}|)| \leqslant \|\mathbf{B}\|\,|\mathbf{y}|$; this also holds for $\mathbf{y} = \mathbf{0}$. Subtracting the two systems gives $\mathbf{A}\,\delta\mathbf{x} = \delta\mathbf{b}$, so $|\delta\mathbf{x}| = |\mathbf{A}^{-1}\delta\mathbf{b}| \leqslant \|\mathbf{A}^{-1}\|\,|\delta\mathbf{b}|$. Also $|\mathbf{b}| = |\mathbf{Ax}| \leqslant \|\mathbf{A}\|\,|\mathbf{x}|$, where $\mathbf{x} \neq \mathbf{0}$ because $\mathbf{b} \neq \mathbf{0}$; dividing by $|\mathbf{x}|\,|\mathbf{b}| > 0$, $1/|\mathbf{x}| \leqslant \|\mathbf{A}\|/|\mathbf{b}|$. Multiplying the two bounds, both sides of each being non-negative, gives $|\delta\mathbf{x}|/|\mathbf{x}| \leqslant \|\mathbf{A}^{-1}\|\,\|\mathbf{A}\|\,|\delta\mathbf{b}|/|\mathbf{b}|$, which is (P.13). □

Corollary P.55 (Why the normal equations square the difficulty)**.** For $\mathbf{X}$ of full column rank, $\kappa(\mathbf{X}'\mathbf{X}) = \kappa(\mathbf{X})^2$, where $\kappa(\mathbf{X})$ is the ratio of the largest to the smallest singular value of $\mathbf{X}$.

Proof. Let $\mathbf{X}$ have p columns. By Theorem P.51, $\mathbf{X} = \mathbf{UDV}'$ with $r = p$ by full column rank, so $\mathbf{V}$ is $p \times p$ with orthonormal columns, that is orthogonal, and $\mathbf{X}'\mathbf{X} = \mathbf{VDU}'\mathbf{UDV}' = \mathbf{VD}^2\mathbf{V}'$ as $\mathbf{U}'\mathbf{U} = \mathbf{I}$. So $\mathbf{X}'\mathbf{X}$ is symmetric positive definite (invertible by Proposition P.33) with eigenvalues $d_1^2 \geqslant \cdots \geqslant d_p^2 > 0$, and by Proposition P.53 its condition number is $d_1^2/d_p^2 = (d_1/d_p)^2 = \kappa(\mathbf{X})^2$. □

Remark. Corollary P.55 is why least squares is computed from a QR factorisation of $\mathbf{X}$ rather than by forming $\mathbf{X}'\mathbf{X}$: the latter squares the condition number, so a design that is merely ill-conditioned becomes numerically singular. After centring and scaling the predictors, so that $\mathbf{X}'\mathbf{X}$ becomes a multiple of their correlation matrix $\mathbf{R}$, a large $\kappa(\mathbf{R})$ and a large variance inflation factor $\mathrm{VIF}_j = [\mathbf{R}^{-1}]_{jj}$ (Unit 4) detect the same near-degeneracy, the first through the spectrum and the second one predictor at a time: by Corollary P.40, $\max_j \mathrm{VIF}_j \leqslant 1/\lambda_{\min}(\mathbf{R}) \leqslant \sum_j \mathrm{VIF}_j$, and the eigenvalues of $\mathbf{R} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}'$ are positive with sum $\operatorname{tr}\boldsymbol{\Lambda} = \operatorname{tr}\mathbf{R} = p$ (Theorem P.39, Proposition P.35(ii)), so $1 \leqslant \lambda_{\max}(\mathbf{R}) \leqslant p$ and, by Proposition P.53, $1/\lambda_{\min}(\mathbf{R}) \leqslant \kappa(\mathbf{R}) \leqslant p/\lambda_{\min}(\mathbf{R})$. Without the scaling they can disagree: two orthogonal predictors on scales 1 and 10^4 give $\kappa(\mathbf{X}'\mathbf{X}) = 10^8$, yet both variance inflation factors equal 1.

15 Matrix Differentiation

Definition P.56 (Differential and derivative)**.** For $f:\mathbb{R}^p\to\mathbb{R}$, $Df(\boldsymbol{\beta}):=\partial f/\partial\boldsymbol{\beta}'$ is the row vector of partial derivatives, and f is **differentiable** at $\boldsymbol{\beta}$ if $f(\boldsymbol{\beta}+\mathbf{h})-f(\boldsymbol{\beta})=Df(\boldsymbol{\beta})\mathbf{h}+o(|\mathbf{h}|)$, meaning that the remainder divided by $|\mathbf{h}|$ tends to 0 as $\mathbf{h}\to\mathbf{0}$; for vector-valued f the same is required of each coordinate, and $Df(\boldsymbol{\beta})$ is the matrix whose rows are the coordinates' derivatives. The **differential** is the linear form $\mathrm{d}f=\sum_j(\partial f/\partial b_j)\,\mathrm{d}b_j$.

Proposition P.57 (First identification theorem)**.** If $\mathrm{d}f(\boldsymbol{\beta})=\mathbf{c}(\boldsymbol{\beta})'\mathrm{d}\boldsymbol{\beta}$ for a vector $\mathbf{c}(\boldsymbol{\beta})$, then $Df(\boldsymbol{\beta})=\mathbf{c}(\boldsymbol{\beta})'$.

Proof. Both sides are the coefficient vector of the same linear form in $\mathrm{d}\boldsymbol{\beta}$: $\sum_j(\partial f/\partial b_j)\,\mathrm{d}b_j=\sum_j c_j(\boldsymbol{\beta})\,\mathrm{d}b_j$ for every $\mathrm{d}\boldsymbol{\beta}$. A linear form determines its coefficients: applying both sides to the jth standard basis vector $\mathrm{d}\boldsymbol{\beta}=\mathbf{e}_j$ gives $\partial f/\partial b_j=c_j(\boldsymbol{\beta})$ for each j, that is $Df(\boldsymbol{\beta})=\mathbf{c}(\boldsymbol{\beta})'$. □

Lemma P.58 (The two rules used in the course)**.** For constant $\mathbf{a} \in \mathbb{R}^p$ and constant $\mathbf{A} \in \mathbb{R}^{p\times p}$,

(i) $D(\mathbf{a}'\boldsymbol{\beta}) = D(\boldsymbol{\beta}'\mathbf{a}) = \mathbf{a}'$;

(ii) $D(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = \boldsymbol{\beta}'(\mathbf{A}+\mathbf{A}')$, which is $2\boldsymbol{\beta}'\mathbf{A}$ when $\mathbf{A}$ is symmetric.

Proof. **(i)** $\mathbf{a}'\boldsymbol{\beta} = \sum_j a_j b_j$, so $\partial(\mathbf{a}'\boldsymbol{\beta})/\partial b_j = a_j$ and $\mathrm{d}(\mathbf{a}'\boldsymbol{\beta}) = \sum_j a_j\,\mathrm{d}b_j = \mathbf{a}'\mathrm{d}\boldsymbol{\beta}$ (the coefficients $\mathbf{a}$ being constant); Proposition P.57 gives $D(\mathbf{a}'\boldsymbol{\beta}) = \mathbf{a}'$, and $\boldsymbol{\beta}'\mathbf{a} = \mathbf{a}'\boldsymbol{\beta}$. The function is differentiable in the sense of Definition P.56, as $\mathbf{a}'(\boldsymbol{\beta}+\mathbf{h}) - \mathbf{a}'\boldsymbol{\beta} = \mathbf{a}'\mathbf{h}$ exactly, with zero remainder.
(ii) Expanding, $(\boldsymbol{\beta}+\mathbf{h})'\mathbf{A}(\boldsymbol{\beta}+\mathbf{h}) - \boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{A}\mathbf{h} + \mathbf{h}'\mathbf{A}\boldsymbol{\beta} + \mathbf{h}'\mathbf{A}\mathbf{h}$, and the scalar $\mathbf{h}'\mathbf{A}\boldsymbol{\beta}$ equals its transpose $\boldsymbol{\beta}'\mathbf{A}'\mathbf{h}$, so the difference is $\boldsymbol{\beta}'(\mathbf{A}+\mathbf{A}')\mathbf{h} + \mathbf{h}'\mathbf{A}\mathbf{h}$. By Theorem P.36 and the bound $|\mathbf{A}\mathbf{h}| \leqslant \|\mathbf{A}\|\,|\mathbf{h}|$ (Definition P.52, as in the proof of Theorem P.54), $|\mathbf{h}'\mathbf{A}\mathbf{h}| \leqslant |\mathbf{h}|\,|\mathbf{A}\mathbf{h}| \leqslant \|\mathbf{A}\|\,|\mathbf{h}|^2$, so the remainder divided by $|\mathbf{h}|$ is at most $\|\mathbf{A}\|\,|\mathbf{h}| \to 0$: it is $o(|\mathbf{h}|)$. Taking $\mathbf{h} = t\mathbf{e}_j$, with $\mathbf{e}_j$ the jth standard basis vector, and dividing by $t \neq 0$, the difference quotient is $\boldsymbol{\beta}'(\mathbf{A}+\mathbf{A}')\mathbf{e}_j + t\,\mathbf{e}_j'\mathbf{A}\mathbf{e}_j$, which tends to $\boldsymbol{\beta}'(\mathbf{A}+\mathbf{A}')\mathbf{e}_j$ as $t \to 0$; so $\partial(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta})/\partial b_j$ is the jth entry of $\boldsymbol{\beta}'(\mathbf{A}+\mathbf{A}')$. When $\mathbf{A}' = \mathbf{A}$ this row vector is $2\boldsymbol{\beta}'\mathbf{A}$. □

Proposition P.59 (Chain rule for differentials)**.** Let $\mathbf{u} = g(\mathbf{x})$ be differentiable at $\mathbf{x}$ and f differentiable at $\mathbf{u}$. Then the differential of the composition is obtained by substitution: $\mathrm{d}f = (Df)(\mathbf{u})\,\mathrm{d}\mathbf{u}$ with $\mathrm{d}\mathbf{u} = (Dg)(\mathbf{x})\,\mathrm{d}\mathbf{x}$, hence $D(f\circ g)(\mathbf{x}) = (Df)(g(\mathbf{x}))\,(Dg)(\mathbf{x})$.

Proof. It suffices to treat scalar f, a vector-valued f being handled one coordinate at a time. Write $\mathbf{F} = (Df)(\mathbf{u})$ (a row vector) and $\mathbf{G} = (Dg)(\mathbf{x})$ (a matrix). Differentiability of f at $\mathbf{u}$ gives $f(\mathbf{u}+\mathbf{h}) - f(\mathbf{u}) = \mathbf{F}\mathbf{h} + r_f(\mathbf{h})$ with $r_f(\mathbf{h})/|\mathbf{h}| \to 0$ as $\mathbf{h}\to\mathbf{0}$, and of g at $\mathbf{x}$ gives $g(\mathbf{x}+\mathbf{k}) - g(\mathbf{x}) = \mathbf{G}\mathbf{k} + \mathbf{r}_g(\mathbf{k})$ with each coordinate of $\mathbf{r}_g(\mathbf{k})/|\mathbf{k}|$ tending to 0, hence $|\mathbf{r}_g(\mathbf{k})|/|\mathbf{k}| \to 0$. Put $\rho(\mathbf{h}) = |r_f(\mathbf{h})|/|\mathbf{h}|$ for $\mathbf{h}\neq\mathbf{0}$ and $\rho(\mathbf{0}) = 0$ (note $r_f(\mathbf{0}) = 0$), so $\rho(\mathbf{h})\to 0$ as $\mathbf{h}\to\mathbf{0}$ and $|r_f(\mathbf{h})| = \rho(\mathbf{h})|\mathbf{h}|$ for every $\mathbf{h}$. Substitute $\mathbf{h} = \mathbf{h}(\mathbf{k}) = g(\mathbf{x}+\mathbf{k}) - g(\mathbf{x}) = \mathbf{G}\mathbf{k} + \mathbf{r}_g(\mathbf{k})$:

$$f(g(\mathbf{x}+\mathbf{k})) - f(g(\mathbf{x})) = \mathbf{F}\mathbf{G}\mathbf{k} + \mathbf{F}\mathbf{r}_g(\mathbf{k}) + r_f(\mathbf{h}(\mathbf{k})).$$

The second term is at most $|\mathbf{F}|\,|\mathbf{r}_g(\mathbf{k})|$ by Theorem P.36, which divided by $|\mathbf{k}|$ tends to 0. For the third, take $\mathbf{k}$ so small that $|\mathbf{r}_g(\mathbf{k})| \leqslant |\mathbf{k}|$; then $|\mathbf{h}(\mathbf{k})| \leqslant \|\mathbf{G}\|\,|\mathbf{k}| + |\mathbf{k}| = (\|\mathbf{G}\| + 1)|\mathbf{k}|$

(Definition P.52), so $\mathbf{h}(\mathbf{k}) \to \mathbf{0}$ as $\mathbf{k} \to \mathbf{0}$ and

$$\frac{|r_f(\mathbf{h}(\mathbf{k}))|}{|\mathbf{k}|} = \rho(\mathbf{h}(\mathbf{k}))\frac{|\mathbf{h}(\mathbf{k})|}{|\mathbf{k}|} \leqslant (\|\mathbf{G}\| + 1)\,\rho(\mathbf{h}(\mathbf{k})) \to 0.$$

Hence $f(g(\mathbf{x}+\mathbf{k})) - f(g(\mathbf{x})) = \mathbf{FGk} + o(|\mathbf{k}|)$, so $f \circ g$ is differentiable at $\mathbf{x}$. Taking $\mathbf{k} = t\mathbf{e}_j$, with $\mathbf{e}_j$ the jth standard basis vector, dividing by $t \neq 0$ and letting $t \to 0$ shows that $\partial(f \circ g)/\partial x_j$ at $\mathbf{x}$ is the jth entry of $\mathbf{FG} = (Df)(\mathbf{u})(Dg)(\mathbf{x})$, and the substitution rule for the differentials restates this identity. □

Example P.1 (The form used for likelihoods)**.** For a scalar h and the linear index $\eta = \mathbf{x}'\boldsymbol{\beta}$,

$$\frac{\partial}{\partial \boldsymbol{\beta}'}\, h(\mathbf{x}'\boldsymbol{\beta}) = h'(\mathbf{x}'\boldsymbol{\beta})\,\mathbf{x}',$$

since $\mathrm{d}\eta = \mathbf{x}'\mathrm{d}\boldsymbol{\beta}$ by Lemma P.58(i) and $\mathrm{d}h = h'(\eta)\,\mathrm{d}\eta$. Summing over observations gives the score vector of a generalised linear model, and differentiating once more with Proposition P.59 gives the information matrix.

Part III
Measure-Theoretic Probability

16 Measure and Measurable Functions

Definition P.60 (Extended real line)**.** $[-\infty,\infty]=\mathbb{R}\cup\{-\infty,\infty\}$ is ordered by $-\infty<x<\infty$ for $x\in\mathbb{R}$. Every $S\subseteq[-\infty,\infty]$ has a supremum: $\sup S=\infty$ if $\infty\in S$ or $S\cap\mathbb{R}$ is unbounded above, $\sup S=-\infty$ if $S\subseteq\{-\infty\}$, and otherwise $\sup S=\sup(S\cap\mathbb{R})$, which exists by the completeness axiom (Definition P.1); $\inf S=-\sup(-S)$. On $[0,\infty]$, $a+\infty=\infty+a=\infty$, $a\cdot\infty=\infty\cdot a=\infty$ for $a>0$, and $0\cdot\infty=\infty\cdot 0=0$. A sequence $a_n\in[-\infty,\infty]$ **tends to** ∞ if for each $M\in\mathbb{R}$ eventually $a_n>M$; $\limsup_n a_n$ and $\liminf_n a_n$ are defined by the formulas of Definition P.8. For $a_n\in[0,\infty]$, $\sum_n a_n=\sup_N\sum_{n\leqslant N}a_n$.

Remark. An increasing sequence in $[-\infty, \infty]$ converges to its supremum: by Theorem P.2 when the supremum is finite and the terms are eventually finite, and directly from the definition otherwise. In particular a series of terms in $[0, \infty]$ is the limit of its partial sums, so $\sum_n (a_n + b_n) = \sum_n a_n + \sum_n b_n$ and $\sum_n ca_n = c\sum_n a_n$ for $a_n, b_n, c \in [0, \infty]$, and a finite sum of such series may be summed term by term in either order.

Definition P.61 (σ-field)**.** A collection $\mathcal{F}$ of subsets of a set Ω is a **σ-field** if $\Omega \in \mathcal{F}$; $A \in \mathcal{F}$ implies $A^c = \Omega \setminus A \in \mathcal{F}$; and $A_1, A_2, \ldots \in \mathcal{F}$ implies $\bigcup_n A_n \in \mathcal{F}$. The pair $(\Omega, \mathcal{F})$ is a **measurable space** and the members of $\mathcal{F}$ are **measurable sets**.

Remark. A σ-field contains $\emptyset = \Omega^c$; it is closed under countable intersections, since $\bigcap_n A_n = (\bigcup_n A_n^c)^c$; under finite unions and intersections, by padding a finite list with copies of $\emptyset$ or of Ω; and under differences, $A \setminus B = A \cap B^c$.

Definition P.62 (Generated and Borel σ-fields)**.** For a collection $\mathcal{C}$ of subsets of Ω, $\sigma(\mathcal{C})$ is the intersection of all σ-fields on Ω containing $\mathcal{C}$. The **Borel σ-field** $\mathcal{B}(\mathbb{R}^d)$ is $\sigma(\mathcal{O}_d)$ with $\mathcal{O}_d$ the open subsets of $\mathbb{R}^d$; its members are **Borel sets**. A set $B \subseteq [-\infty, \infty]$ is Borel if $B \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})$.

Remark. The collection of all subsets of Ω is a σ-field containing $\mathcal{C}$, so the intersection is over a non-empty family. Each of the three defining properties holds in every member of the family, hence in the intersection: $\sigma(\mathcal{C})$ is a σ-field, and it is contained in every σ-field containing $\mathcal{C}$. If $\mathcal{C} \subseteq \mathcal{D}$ then $\sigma(\mathcal{C}) \subseteq \sigma(\mathcal{D})$, and if $\mathcal{C} \subseteq \sigma(\mathcal{D})$ then $\sigma(\mathcal{C}) \subseteq \sigma(\mathcal{D})$. Closed sets are Borel, being complements of open sets. The Borel sets of $[-\infty, \infty]$ form a σ-field, since $B^c \cap \mathbb{R} = \mathbb{R} \setminus (B \cap \mathbb{R})$ and $(\bigcup_n B_n) \cap \mathbb{R} = \bigcup_n (B_n \cap \mathbb{R})$.

Lemma P.63 (Generators of the Borel σ-field)**.** (i) Every open $U \subseteq \mathbb{R}^d$ is the union of the countably many open boxes $\prod_{i \leqslant d}(a_i, b_i)$ with rational $a_i < b_i$ that are contained in U.

(ii) $\mathcal{B}(\mathbb{R}^d) = \sigma(\mathcal{R}_d)$, where $\mathcal{R}_d$ is the class of half-open boxes $\prod_{i \leqslant d}(a_i, b_i]$ with $a_i < b_i$ real.

(iii) $\mathcal{B}(\mathbb{R}) = \sigma\big(\{(-\infty, a] : a \in \mathbb{R}\}\big)$.

Proof. (i) The boxes with rational corners form a countable family, as $\mathbb{Q}^{2d}$ is countable. Each box in the union lies in U, so the union is contained in U. Conversely let $x \in U$. As U is open there is $\delta > 0$ with $|y - x| < \delta$ implying $y \in U$. Choose rationals $a_i < x_i < b_i$ with $b_i - a_i < \delta/\sqrt{d}$. If y lies in the box $\prod_i (a_i, b_i)$, then x_i and y_i both lie in (a_i, b_i), so $|y_i - x_i| < \delta/\sqrt{d}$ for each i and

$|y-x|<\delta$; thus the box lies in U, and it contains x.
(ii) An open box is a countable union of half-open boxes, $\prod_i (a_i,b_i) = \bigcup_{n>n_0} \prod_i (a_i, b_i - 1/n]$ with n_0 so large that $a_i < b_i - 1/n_0$ for every i. By (i) every open set is therefore a countable union of members of $\mathcal{R}_d$, so $\mathcal{O}_d \subseteq \sigma(\mathcal{R}_d)$ and $\mathcal{B}(\mathbb{R}^d) \subseteq \sigma(\mathcal{R}_d)$. Conversely a half-open box is the countable intersection of the open boxes $\prod_i (a_i, b_i + 1/n)$, hence Borel, so $\sigma(\mathcal{R}_d) \subseteq \mathcal{B}(\mathbb{R}^d)$.
(iii) Let $\mathcal{G} = \sigma(\{(-\infty, a]\})$. The sets $(-\infty, a]$ are closed, hence Borel, so $\mathcal{G} \subseteq \mathcal{B}(\mathbb{R})$. Conversely $(a,b] = (-\infty, b] \setminus (-\infty, a]$ lies in $\mathcal{G}$, so $\mathcal{R}_1 \subseteq \mathcal{G}$ and $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{R}_1) \subseteq \mathcal{G}$ by (ii). □

Definition P.64 (Measure)**.** A **measure** on a measurable space $(\Omega, \mathcal{F})$ is a function $\mu : \mathcal{F} \to [0,\infty]$ with $\mu(\emptyset) = 0$ that is **countably additive**: $\mu(\bigcup_n A_n) = \sum_n \mu(A_n)$ for pairwise disjoint $A_1, A_2, \ldots \in \mathcal{F}$. The triple $(\Omega, \mathcal{F}, \mu)$ is a **measure space**. It is a **probability space**, written $(\Omega, \mathcal{F}, \mathsf{P})$, if $\mathsf{P}(\Omega) = 1$; μ is **finite** if $\mu(\Omega) < \infty$, and **σ-finite** if $\Omega = \bigcup_n \Omega_n$ for some $\Omega_n \in \mathcal{F}$ with $\mu(\Omega_n) < \infty$. A set $N \in \mathcal{F}$ with $\mu(N) = 0$ is a **null set**.

Proposition P.65 (Properties of measures)**.** In a measure space $(\Omega, \mathcal{F}, \mu)$ let $A, B, A_1, A_2, \ldots \in \mathcal{F}$.

(i) **Finite additivity.** If $A_1, \dots, A_k$ are pairwise disjoint then $\mu(\bigcup_{i\leqslant k} A_i) = \sum_{i\leqslant k} \mu(A_i)$.
(ii) **Monotonicity.** If $A \subseteq B$ then $\mu(A) \leqslant \mu(B)$, and $\mu(B \setminus A) = \mu(B) - \mu(A)$ when $\mu(A) < \infty$.
(iii) **Countable subadditivity.** $\mu(\bigcup_n A_n) \leqslant \sum_n \mu(A_n)$.
(iv) **Continuity from below.** If $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \bigcup_n A_n$, then $\mu(A_n) \uparrow \mu(A)$.
(v) **Continuity from above.** If $A_1 \supseteq A_2 \supseteq \cdots$, $A = \bigcap_n A_n$ and $\mu(A_1) < \infty$, then $\mu(A_n) \downarrow \mu(A)$.

Proof. (i) Put $A_i = \emptyset$ for $i > k$. The sequence $A_1, A_2, \dots$ is pairwise disjoint with union $\bigcup_{i\leqslant k} A_i$, and countable additivity gives $\mu(\bigcup_{i\leqslant k} A_i) = \sum_{i\leqslant k} \mu(A_i) + \sum_{i>k} 0$.
(ii) B is the disjoint union of A and $B \setminus A$, so by (i) $\mu(B) = \mu(A) + \mu(B \setminus A) \geqslant \mu(A)$. If $\mu(A) < \infty$, subtracting $\mu(A)$ from both sides gives the second statement.
(iii) Put $C_1 = A_1$ and $C_n = A_n \setminus \bigcup_{i<n} A_i$ for $n \geqslant 2$. The C_n lie in $\mathcal{F}$, are pairwise disjoint, satisfy $C_n \subseteq A_n$, and have the same union as the A_n: a point of $\bigcup_n A_n$ lies in C_n for the least n with that point in A_n. By countable additivity and (ii), $\mu(\bigcup_n A_n) = \sum_n \mu(C_n) \leqslant \sum_n \mu(A_n)$.
(iv) Put $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geqslant 2$. These are pairwise disjoint, $\bigcup_{i\leqslant n} B_i = A_n$, and

$\bigcup_i B_i = A$. By countable additivity and (i),

$$\mu(A) = \sum_i \mu(B_i) = \lim_n \sum_{i \leqslant n} \mu(B_i) = \lim_n \mu(A_n),$$

and the $\mu(A_n)$ increase by (ii).
(v) The sets $A_1 \setminus A_n$ increase with union $A_1 \setminus A$, so by (iv) $\mu(A_1 \setminus A_n) \uparrow \mu(A_1 \setminus A)$. Every subset of A_1 in $\mathcal{F}$ has finite measure, so by (ii) this reads $\mu(A_1) - \mu(A_n) \to \mu(A_1) - \mu(A)$; subtracting from the finite number $\mu(A_1)$ gives $\mu(A_n) \to \mu(A)$, and the $\mu(A_n)$ decrease by (ii). □

Remark. The finiteness in (v) cannot be dropped: for the counting measure on the positive integers, $\mu(A) =$ the number of elements of A, the sets $A_n = \{n, n+1, \ldots\}$ decrease to $\emptyset$ while $\mu(A_n) = \infty$ for every n.

Definition P.66 (Measurable function)**.** Let $(\Omega, \mathcal{F})$ and $(E, \mathcal{E})$ be measurable spaces. A map $f : \Omega \to E$ is **$\mathcal{F}/\mathcal{E}$-measurable** if $f^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{E}$. When E is $\mathbb{R}^d$ or $[-\infty, \infty]$ with its Borel sets, f is simply called **measurable**, and **Borel** when moreover $\Omega = \mathbb{R}^k$ and

$\mathcal{F} = \mathcal{B}(\mathbb{R}^k)$. A **non-negative measurable function** is a measurable $f : \Omega \to [-\infty, \infty]$ with values in $[0, \infty]$. A complex $f = u + iv$ is measurable if u and v are.

Lemma P.67 (Criteria for measurability)**.** Let $(\Omega, \mathcal{F})$ be a measurable space.

(i) If $\mathcal{E} = \sigma(\mathcal{C})$, then $f : \Omega \to E$ is $\mathcal{F}/\mathcal{E}$-measurable if and only if $f^{-1}(C) \in \mathcal{F}$ for every $C \in \mathcal{C}$.

(ii) $f : \Omega \to [-\infty, \infty]$ is measurable if and only if $\{f \leqslant a\} \in \mathcal{F}$ for every $a \in \mathbb{R}$, and then $\{f < a\}$, $\{f \geqslant a\}$, $\{f > a\}$ and $\{f = a\}$ lie in $\mathcal{F}$ for every $a \in [-\infty, \infty]$. A real-valued f is measurable as a map to $\mathbb{R}$ if and only if it is measurable as a map to $[-\infty, \infty]$.

(iii) A continuous $g : \mathbb{R}^k \to \mathbb{R}^m$ is Borel, and if f is $\mathcal{F}/\mathcal{E}$-measurable and g is $\mathcal{E}/\mathcal{E}'$-measurable then $g \circ f$ is $\mathcal{F}/\mathcal{E}'$-measurable.

(iv) $f = (f_1, \ldots, f_d) : \Omega \to \mathbb{R}^d$ is measurable if and only if every f_i is.

Proof. (i) Necessity is clear as $\mathcal{C} \subseteq \mathcal{E}$. For sufficiency let $\mathcal{D} = \{B \subseteq E : f^{-1}(B) \in \mathcal{F}\}$. Since $f^{-1}(E) = \Omega$, $f^{-1}(E \setminus B) = \Omega \setminus f^{-1}(B)$ and $f^{-1}(\bigcup_n B_n) = \bigcup_n f^{-1}(B_n)$, $\mathcal{D}$ is a σ-field. It contains $\mathcal{C}$ by hypothesis, hence contains $\sigma(\mathcal{C}) = \mathcal{E}$.

(ii) Let $\mathcal{G} = \sigma(\{[-\infty, a] : a \in \mathbb{R}\})$, a σ-field on $[-\infty, \infty]$; by (i) it suffices to show that $\mathcal{G}$ is the Borel σ-field of $[-\infty, \infty]$. Each $[-\infty, a]$ is Borel, its trace on $\mathbb{R}$ being the closed set $(-\infty, a]$, so $\mathcal{G}$ is contained in the Borel sets. Conversely $\{-\infty\} = \bigcap_n [-\infty, -n]$ and $\{\infty\} = [-\infty, \infty] \setminus \bigcup_n [-\infty, n]$ lie in $\mathcal{G}$, so $\mathbb{R} \in \mathcal{G}$ and the subsets of $\mathbb{R}$ belonging to $\mathcal{G}$ form a σ-field on $\mathbb{R}$ (the complement in $\mathbb{R}$ of such a set B is $\mathbb{R} \setminus B \in \mathcal{G}$). This σ-field contains $(-\infty, a] = [-\infty, a] \setminus \{-\infty\}$, hence contains $\mathcal{B}(\mathbb{R})$ by Lemma P.63(iii). A Borel $B \subseteq [-\infty, \infty]$ is the union of $B \cap \mathbb{R}$ and a subset of $\{-\infty, \infty\}$, so $B \in \mathcal{G}$. For the other sets: $\{f > a\} = \Omega \setminus \{f \leqslant a\}$, $\{f < a\} = \bigcup_n \{f \leqslant a - 1/n\}$, $\{f \geqslant a\} = \Omega \setminus \{f < a\}$ and $\{f = a\} = \{f \leqslant a\} \cap \{f \geqslant a\}$ for real a, and $\{f = \pm\infty\} = f^{-1}(\{\pm\infty\})$; the cases $a = \pm\infty$ follow, as $\{f < \infty\} = \Omega \setminus \{f = \infty\}$, $\{f \geqslant \infty\} = \{f = \infty\}$, $\{f > \infty\} = \emptyset$, and symmetrically at $-\infty$. Finally, for real-valued f and Borel $B \subseteq [-\infty, \infty]$, $f^{-1}(B) = f^{-1}(B \cap \mathbb{R})$, and every Borel subset of $\mathbb{R}$ is a Borel subset of $[-\infty, \infty]$.

(iii) Let $V \subseteq \mathbb{R}^m$ be open and $x \in g^{-1}(V)$. There is $\varepsilon > 0$ with the ball of radius ε about $g(x)$ inside V, and by continuity $\delta > 0$ with $|g(y) - g(x)| < \varepsilon$ whenever $|y - x| < \delta$; so the ball of radius δ about x lies in $g^{-1}(V)$, and $g^{-1}(V)$ is open, hence Borel. By (i) with $\mathcal{C} = \mathcal{O}_m$, g is Borel. For the composition, $(g \circ f)^{-1}(B) = f^{-1}(g^{-1}(B))$ and $g^{-1}(B) \in \mathcal{E}$.

(iv) If f is measurable then $f_i = p_i \circ f$ with $p_i(x) = x_i$ continuous, so f_i is measurable by (iii).

Conversely, $f^{-1}(\prod_i (a_i, b_i]) = \bigcap_i f_i^{-1}((a_i, b_i]) \in \mathcal{F}$, and the half-open boxes generate $\mathcal{B}(\mathbb{R}^d)$ by Lemma P.63(ii); apply (i). □

Proposition P.68 (Operations preserving measurability)**.** Let $(\Omega, \mathcal{F})$ be a measurable space.

(i) If $f, g : \Omega \to \mathbb{R}$ are measurable and $c \in \mathbb{R}$, then $f + g$, cf, fg, $f \vee g$, $f \wedge g$, $f^+ = f \vee 0$, $f^- = (-f) \vee 0$ and $|f|$ are measurable.

(ii) If $f_1, f_2, \ldots : \Omega \to [-\infty, \infty]$ are measurable, so are $\sup_n f_n$, $\inf_n f_n$, $\limsup_n f_n$ and $\liminf_n f_n$. In particular, if $f_n \to f$ pointwise with f real-valued, then f is measurable.

(iii) If f, g are non-negative measurable functions, so are $f + g$ and fg.

(iv) The indicator $\mathbb{1}_A$ is measurable if and only if $A \in \mathcal{F}$.

Proof. (i) The map $\omega \mapsto (f(\omega), g(\omega))$ is measurable into $\mathbb{R}^2$ by Lemma P.67(iv). The functions $(x, y) \mapsto x + y$, cx, xy, $\max(x, y)$, $\min(x, y)$ are continuous on $\mathbb{R}^2$, so the compositions are measurable by Lemma P.67(iii). Then $f^+ = f \vee 0$ and $f^- = (-f) \vee 0$ are measurable, and $|f| = f^+ + f^-$.

(ii) For real a,

$$\{\sup_n f_n \leqslant a\} = \bigcap_n \{f_n \leqslant a\}, \qquad \{\inf_n f_n \leqslant a\} = \bigcap_k \bigcup_n \{f_n < a + 1/k\},$$

so both are measurable by Lemma P.67(ii). Then $\limsup_n f_n = \inf_m \sup_{n \geqslant m} f_n$ and $\liminf_n f_n = \sup_m \inf_{n \geqslant m} f_n$ are measurable. If $f_n \to f$ pointwise with f real, then $f = \limsup_n f_n$ by Proposition P.9.

(iii) Let $a \in \mathbb{R}$. If $a < 0$ then $\{f + g > a\} = \Omega$. For $a \geqslant 0$ the claim is

$$\{f + g > a\} = \bigcup_{r \in \mathbb{Q}} \big(\{f > r\} \cap \{g > a - r\}\big),$$

a countable union of measurable sets. If $f(\omega) > r$ and $g(\omega) > a - r$ then $f(\omega) + g(\omega) > a$, also when a value is infinite. Conversely suppose $f(\omega) + g(\omega) > a$. If $f(\omega) = \infty$, take rational $r > a$, so that $a - r < 0 \leqslant g(\omega)$. If $f(\omega) < \infty = g(\omega)$, take rational $r < f(\omega)$. If both are finite, then $a - g(\omega) < f(\omega)$ and any rational r strictly between them works. For the product, $f \wedge n$ and $g \wedge n$ are real and measurable, as $\{f \wedge n \leqslant a\}$ is $\{f \leqslant a\}$ for $a < n$ and Ω for $a \geqslant n$; so

$(f \wedge n)(g \wedge n)$ is measurable by (i). It increases in n to fg: if $f(\omega) = 0$ or $g(\omega) = 0$ every term and $fg(\omega)$ vanish; if both are finite and positive, the terms equal $fg(\omega)$ once n exceeds both; if $f(\omega) = \infty$ and $g(\omega) > 0$, the terms are at least $n\,(g(\omega) \wedge 1) \to \infty$, and symmetrically. By (ii), $fg = \sup_n (f \wedge n)(g \wedge n)$ is measurable.
(iv) $\mathbb{1}_A^{-1}(B)$ is one of $\emptyset$, A, A^c, Ω, and equals A for $B = \{1\}$. □

Definition P.69 (Simple function)**.** A **simple function** is a measurable $s : \Omega \to [0, \infty)$ taking finitely many values. Its **canonical form** is $s = \sum_{a \in s(\Omega)} a\, \mathbb{1}_{\{s=a\}}$, the sets $\{s = a\} \in \mathcal{F}$ partitioning Ω.

Remark. If $A_1, \ldots, A_k \in \mathcal{F}$ and $a_1, \ldots, a_k \in [0, \infty)$, then $\sum_i a_i \mathbb{1}_{A_i}$ is simple: it is measurable by Proposition P.68(i),(iv), and takes at most 2^k values.

Theorem P.70 (Approximation by simple functions)**.** Let f be a non-negative measurable function. Put $\varphi_n(t) = n$ for $t \geqslant n$ and, for $0 \leqslant t < n$, $\varphi_n(t) = (k-1)2^{-n}$ where k is the integer with $(k-1)2^{-n} \leqslant t < k2^{-n}$; and put $s_n = \varphi_n \circ f$. Then each s_n is simple, $0 \leqslant s_1 \leqslant s_2 \leqslant \cdots \leqslant f$, and $s_n \to f$ pointwise.

Proof. Simple. Explicitly

$$s_n = \sum_{k=1}^{n2^n} (k-1)2^{-n}\ \mathbb{1}\{(k-1)2^{-n} \leqslant f < k2^{-n}\} + n\ \mathbb{1}\{f \geqslant n\},$$

and the sets on the right lie in $\mathcal{F}$ by Lemma P.67(ii), being intersections of sets of the form $\{f \geqslant c\}$ and $\{f < c\}$. So s_n is simple.

Below f. For $t \in [0,\infty]$, $\varphi_n(t) \leqslant t$: if $t \geqslant n$ then $\varphi_n(t) = n \leqslant t$, and otherwise $\varphi_n(t)$ is the largest multiple of 2^{-n} not exceeding t.

Increasing. Fix $t \in [0,\infty]$. If $t \geqslant n+1$ then $\varphi_n(t) = n < n+1 = \varphi_{n+1}(t)$. If $n \leqslant t < n+1$ then $\varphi_n(t) = n$, and n is a multiple of $2^{-(n+1)}$ not exceeding t, so $\varphi_{n+1}(t) \geqslant n$. If $t < n$ then $\varphi_n(t)$ is a multiple of 2^{-n}, hence of $2^{-(n+1)}$, not exceeding t, and $\varphi_{n+1}(t)$ is the largest such multiple, so $\varphi_{n+1}(t) \geqslant \varphi_n(t)$.

Convergence. If $t = \infty$ then $\varphi_n(t) = n \to \infty$. If $t < \infty$ then for $n > t$, $0 \leqslant t - \varphi_n(t) < 2^{-n}$, so $\varphi_n(t) \to t$. Applying this to $t = f(\omega)$ gives $s_n(\omega) \to f(\omega)$. □

Definition P.71 (π-system, λ-system)**.** A collection $\mathcal{P}$ of subsets of Ω is a **π-system** if $A, B \in \mathcal{P}$ implies $A \cap B \in \mathcal{P}$. A collection $\mathcal{L}$ is a **λ-system** if $\Omega \in \mathcal{L}$; $A, B \in \mathcal{L}$ and $A \subseteq B$ imply $B \setminus A \in \mathcal{L}$; and $A_1 \subseteq A_2 \subseteq \cdots$ in $\mathcal{L}$ imply $\bigcup_n A_n \in \mathcal{L}$.

Theorem P.72 (Dynkin's π–λ theorem)**.** If $\mathcal{P}$ is a π-system, $\mathcal{L}$ is a λ-system and $\mathcal{P} \subseteq \mathcal{L}$, then $\sigma(\mathcal{P}) \subseteq \mathcal{L}$.

Proof. Step 1: a π-system that is a λ-system is a σ-field. Let $\mathcal{D}$ be both. Then $\Omega \in \mathcal{D}$, and $A^c = \Omega \setminus A \in \mathcal{D}$ for $A \in \mathcal{D}$. For $A, B \in \mathcal{D}$, $A \cup B = (A^c \cap B^c)^c \in \mathcal{D}$, so $\mathcal{D}$ is closed under finite unions. For $A_1, A_2, \ldots \in \mathcal{D}$ the sets $B_n = \bigcup_{i \leqslant n} A_i$ lie in $\mathcal{D}$ and increase to $\bigcup_i A_i$, which therefore lies in $\mathcal{D}$.

Step 2: the smallest λ-system. Let $\lambda(\mathcal{P})$ be the intersection of all λ-systems containing $\mathcal{P}$ (the collection of all subsets of Ω is one). Each defining property of a λ-system holds in every member of the family, hence in the intersection, so $\lambda(\mathcal{P})$ is a λ-system, and $\lambda(\mathcal{P}) \subseteq \mathcal{L}$. By Step 1 it suffices to show that $\lambda(\mathcal{P})$ is a π-system: it is then a σ-field containing $\mathcal{P}$, so $\sigma(\mathcal{P}) \subseteq \lambda(\mathcal{P}) \subseteq \mathcal{L}$.

Step 3: an auxiliary λ-system. For $A \in \lambda(\mathcal{P})$ let $\mathcal{L}_A = \{B \subseteq \Omega : A \cap B \in \lambda(\mathcal{P})\}$. This

is a λ-system: $A \cap \Omega = A \in \lambda(\mathcal{P})$; if $B \subseteq C$ lie in $\mathcal{L}_A$ then $A \cap (C \setminus B) = (A \cap C) \setminus (A \cap B)$, a difference of two members of $\lambda(\mathcal{P})$ with the first containing the second, hence in $\lambda(\mathcal{P})$; and if $B_n \uparrow B$ with $B_n \in \mathcal{L}_A$ then $A \cap B_n \uparrow A \cap B$, so $A \cap B \in \lambda(\mathcal{P})$.

Step 4: closure under intersection. If $A \in \mathcal{P}$ then $\mathcal{P} \subseteq \mathcal{L}_A$, as $\mathcal{P}$ is a π-system; $\mathcal{L}_A$ being a λ-system, $\lambda(\mathcal{P}) \subseteq \mathcal{L}_A$. Thus $A \cap B \in \lambda(\mathcal{P})$ whenever $A \in \mathcal{P}$ and $B \in \lambda(\mathcal{P})$. Now fix $B \in \lambda(\mathcal{P})$. What was just shown says $\mathcal{P} \subseteq \mathcal{L}_B$, so $\lambda(\mathcal{P}) \subseteq \mathcal{L}_B$, that is, $B \cap C \in \lambda(\mathcal{P})$ for every $C \in \lambda(\mathcal{P})$. □

Corollary P.73 (Uniqueness of measures)**.** Let $\mathcal{P}$ be a π-system on Ω and μ, ν measures on $\sigma(\mathcal{P})$ with $\mu(A) = \nu(A)$ for $A \in \mathcal{P}$. If there are $\Omega_1 \subseteq \Omega_2 \subseteq \cdots$ in $\mathcal{P}$ with $\bigcup_n \Omega_n = \Omega$ and $\mu(\Omega_n) < \infty$, then $\mu = \nu$ on $\sigma(\mathcal{P})$. In particular two finite measures with $\mu(\Omega) = \nu(\Omega)$, such as two probability measures, that agree on $\mathcal{P}$ agree on $\sigma(\mathcal{P})$.

Proof. Fix n and let $\mathcal{L}_n = \{A \in \sigma(\mathcal{P}) : \mu(A \cap \Omega_n) = \nu(A \cap \Omega_n)\}$. It is a λ-system. First, $\Omega \in \mathcal{L}_n$ as $\Omega_n \in \mathcal{P}$. Next, if $A \subseteq B$ lie in $\mathcal{L}_n$, then $(B \setminus A) \cap \Omega_n = (B \cap \Omega_n) \setminus (A \cap \Omega_n)$, and all these sets have finite measure, at most $\mu(\Omega_n) = \nu(\Omega_n)$; by Proposition P.65(ii) both μ and ν give

it the measure $\mu(B\cap\Omega_n)-\mu(A\cap\Omega_n)=\nu(B\cap\Omega_n)-\nu(A\cap\Omega_n)$. Finally if $A_k\uparrow A$ with $A_k\in\mathcal{L}_n$ then $A_k\cap\Omega_n\uparrow A\cap\Omega_n$, and continuity from below (Proposition P.65(iv)) passes the equality to the limit. Also $\mathcal{P}\subseteq\mathcal{L}_n$, since $A\cap\Omega_n\in\mathcal{P}$ for $A\in\mathcal{P}$. Theorem P.72 gives $\sigma(\mathcal{P})\subseteq\mathcal{L}_n$. For $A\in\sigma(\mathcal{P})$, $A\cap\Omega_n\uparrow A$, so by continuity from below $\mu(A)=\lim_n\mu(A\cap\Omega_n)=\lim_n\nu(A\cap\Omega_n)=\nu(A)$.
For the last statement, $\mathcal{P}\cup\{\Omega\}$ is a π-system generating the same σ-field, on which μ and ν agree; take $\Omega_n=\Omega$. □

17 The Integral and Its Convergence Theorems

Throughout, $(\Omega,\mathcal{F},\mu)$ is a measure space.

Definition P.74 (Integral of a simple function)**.** For a simple s with canonical form $\sum_a a\,\mathbb{1}_{\{s=a\}}$, $\int s\,\mathrm{d}\mu=\sum_a a\,\mu\{s=a\}$, a finite sum in $[0,\infty]$.

Lemma P.75 (Integral of simple functions)**.** Let s,t be simple.

(i) If $s=\sum_{i\leqslant k}a_i\,\mathbb{1}_{A_i}$ with $a_i\in[0,\infty)$ and $A_i\in\mathcal{F}$, not necessarily disjoint, then $\int s\,\mathrm{d}\mu=\sum_{i\leqslant k}a_i\mu(A_i)$.

(ii) $\int(as+bt)\,\mathrm{d}\mu=a\int s\,\mathrm{d}\mu+b\int t\,\mathrm{d}\mu$ for $a,b\in[0,\infty)$.

(iii) If $s \leqslant t$ then $\int s\,\mathrm{d}\mu \leqslant \int t\,\mathrm{d}\mu$.
(iv) $\nu(A) = \int s\,\mathbb{1}_A\,\mathrm{d}\mu$ defines a measure ν on $\mathcal{F}$.

Proof. (i) **Disjoint case.** Suppose first that $A_1, \dots, A_k$ partition Ω. A point of A_i then has $s = a_i$, so for each value a of s the set $\{s = a\}$ is the disjoint union of the A_i with $a_i = a$, and $\mu\{s = a\} = \sum_{i: a_i = a} \mu(A_i)$ by finite additivity; an A_i whose a_i is not a value of s is empty and contributes $a_i \mu(\emptyset) = 0$. Hence

$$\sum_a a\,\mu\{s = a\} = \sum_a \sum_{i:\, a_i = a} a_i \mu(A_i) = \sum_i a_i \mu(A_i).$$

If the A_i are disjoint but do not cover Ω, adjoin $A_0 = \Omega \setminus \bigcup_i A_i$ with $a_0 = 0$, which adds $0 \cdot \mu(A_0) = 0$.

General case. For $\epsilon = (\epsilon_1, \dots, \epsilon_k) \in \{0,1\}^k$ let $B_\epsilon = \bigcap_i A_i^{(\epsilon_i)}$, where $A^{(1)} = A$ and $A^{(0)} = A^c$. The 2^k sets B_ϵ lie in $\mathcal{F}$ and partition Ω, each point lying in exactly the B_ϵ with $\epsilon_i = \mathbb{1}_{A_i}(\omega)$; and A_i is the disjoint union of the B_ϵ with $\epsilon_i = 1$. On B_ϵ, $s = c_\epsilon := \sum_i a_i \epsilon_i$. By the disjoint case and

finite additivity,

$$\int s\,\mathrm{d}\mu = \sum_{\epsilon} c_\epsilon \mu(B_\epsilon) = \sum_i a_i \sum_{\epsilon:\,\epsilon_i=1} \mu(B_\epsilon) = \sum_i a_i \mu(A_i),$$

the rearrangement of a finite sum of terms in $[0,\infty]$ being permitted.
(ii) Let $s = \sum_i a_i \mathbb{1}_{A_i}$ and $t = \sum_j b_j \mathbb{1}_{B_j}$ be the canonical forms. Then $as + bt = \sum_i (aa_i)\mathbb{1}_{A_i} + \sum_j (bb_j)\mathbb{1}_{B_j}$, and (i) gives $\int (as+bt)\,\mathrm{d}\mu = a\sum_i a_i\mu(A_i) + b\sum_j b_j\mu(B_j)$.
(iii) $t - s$ is measurable with finitely many values in $[0,\infty)$, so it is simple, and by (ii) $\int t\,\mathrm{d}\mu = \int s\,\mathrm{d}\mu + \int (t-s)\,\mathrm{d}\mu \geqslant \int s\,\mathrm{d}\mu$.
(iv) With $s = \sum_i a_i \mathbb{1}_{A_i}$, $s\,\mathbb{1}_A = \sum_i a_i \mathbb{1}_{A_i \cap A}$, so by (i) $\nu(A) = \sum_i a_i \mu(A_i \cap A)$. Then $\nu(\emptyset) = \sum_i a_i \mu(\emptyset) = 0$. If $C_1, C_2, \ldots$ are pairwise disjoint with union C, so are the $A_i \cap C_n$, with union $A_i \cap C$, and

$$\nu(C) = \sum_i a_i \sum_n \mu(A_i \cap C_n) = \sum_n \sum_i a_i \mu(A_i \cap C_n) = \sum_n \nu(C_n),$$

a finite sum of series with terms in $[0,\infty]$ being summed term by term. □

Definition P.76 (Integral of a non-negative function)**.** For a non-negative measurable f,

$$\int f \,\mathrm{d}\mu = \sup\Big\{ \int s \,\mathrm{d}\mu : \ s \text{ simple}, \ s \leqslant f \Big\} \in [0,\infty],$$

also written $\int f(\omega)\,\mu(\mathrm{d}\omega)$; and $\int_A f \,\mathrm{d}\mu = \int f \,\mathbb{1}_A \,\mathrm{d}\mu$ for $A \in \mathcal{F}$.

Lemma P.77 (Monotonicity and scaling)**.** Let f, g be non-negative measurable functions.

(i) For simple f, Definitions P.74 and P.76 agree.
(ii) If $f \leqslant g$ then $\int f \,\mathrm{d}\mu \leqslant \int g \,\mathrm{d}\mu$.
(iii) $\int cf \,\mathrm{d}\mu = c \int f \,\mathrm{d}\mu$ for $c \in [0,\infty)$.
(iv) $\int c\,\mathbb{1}_A \,\mathrm{d}\mu = c\,\mu(A)$ for $c \in [0,\infty)$ and $A \in \mathcal{F}$.

Proof. (i) If f is simple, every simple $s \leqslant f$ has $\int s \leqslant \int f$ by Lemma P.75(iii), and $s = f$ is admissible, so the supremum is the integral of Definition P.74. (ii) Every simple $s \leqslant f$ is also $\leqslant g$, so the supremum defining $\int g$ is over a larger set. (iii) For $c = 0$ both sides vanish. For $c > 0$, $s \leqslant f$ if and only if $cs \leqslant cf$, and $\int cs = c \int s$ by Lemma P.75(ii), so the two suprema differ by

the factor c. (iv) is (i) with Lemma P.75(i). □

Theorem P.78 (Monotone convergence theorem)**.** Let $f_1 \leqslant f_2 \leqslant \cdots$ be non-negative measurable functions and $f = \lim_n f_n = \sup_n f_n$. Then f is measurable and $\int f_n \,\mathrm{d}\mu \uparrow \int f \,\mathrm{d}\mu$.

Proof. Step 1: the easy inequality. f is measurable by Proposition P.68(ii). Since $f_n \leqslant f_{n+1} \leqslant f$, Lemma P.77(ii) shows that $\int f_n$ increases and is at most $\int f$; so $L = \lim_n \int f_n$ exists in $[0, \infty]$ and $L \leqslant \int f$.

Step 2: a set on which f_n is nearly as large as s. By Definition P.76 it remains to show $\int s \leqslant L$ for every simple $s \leqslant f$. Fix such s and $c \in (0, 1)$, and let $E_n = \{f_n \geqslant cs\}$, which lies in $\mathcal{F}$ as the finite union over the values a of s of $\{s = a\} \cap \{f_n \geqslant ca\}$. The E_n increase, since the f_n do. Their union is Ω: if $s(\omega) = 0$ then $\omega \in E_1$, and if $s(\omega) > 0$ then $cs(\omega) < s(\omega) \leqslant f(\omega) = \lim_n f_n(\omega)$, so $f_n(\omega) > cs(\omega)$ for all large n.

Step 3: the reverse inequality. The simple function $cs\,\mathbb{1}_{E_n}$ satisfies $cs\,\mathbb{1}_{E_n} \leqslant f_n$, so by

Definition P.76 and Lemma P.75(ii)

$$\int f_n \,\mathrm{d}\mu \geqslant \int cs\,\mathbb{1}_{E_n}\,\mathrm{d}\mu = c\,\nu(E_n), \qquad \nu(A) = \int s\,\mathbb{1}_A\,\mathrm{d}\mu.$$

By Lemma P.75(iv) ν is a measure, so continuity from below (Proposition P.65(iv)) gives $\nu(E_n) \uparrow \nu(\Omega) = \int s$. Letting $n \to \infty$, $L \geqslant c\int s$; letting $c \uparrow 1$, $L \geqslant \int s$. □

Theorem P.79 (Additivity of the integral)**.** For non-negative measurable f, g and $a, b \in [0, \infty)$, $\int (af + bg)\,\mathrm{d}\mu = a\int f\,\mathrm{d}\mu + b\int g\,\mathrm{d}\mu$.

Proof. By Theorem P.70 there are simple $s_n \uparrow f$ and $t_n \uparrow g$. Then $as_n + bt_n$ is simple and increases. At every point $as_n \to af$ (both sides vanish if $a = 0$, and $as_n \to \infty$ if $a > 0$ and $f = \infty$), and likewise $bt_n \to bg$, so $as_n + bt_n \uparrow af + bg$. By Theorem P.78 applied three times and Lemma P.75(ii),

$$\int (af + bg)\,\mathrm{d}\mu = \lim_n \int (as_n + bt_n)\,\mathrm{d}\mu = \lim_n \Big(a\int s_n\,\mathrm{d}\mu + b\int t_n\,\mathrm{d}\mu\Big) = a\int f\,\mathrm{d}\mu + b\int g\,\mathrm{d}\mu,$$

the limit of a sum of increasing sequences in $[0,\infty]$ being the sum of the limits. □

Corollary P.80 (Integral of a series)**.** (i) If $f_1, f_2, \ldots$ are non-negative and measurable, then $\int \sum_n f_n \,\mathrm{d}\mu = \sum_n \int f_n \,\mathrm{d}\mu$.
(ii) For non-negative measurable f, $\nu(A) = \int_A f \,\mathrm{d}\mu$ defines a measure on $\mathcal{F}$.

Proof. (i) The partial sums $g_N = \sum_{n\leqslant N} f_n$ are measurable by Proposition P.68(iii), increase in N, and converge to $\sum_n f_n$ (Definition P.60). By induction on N, $\int g_N = \sum_{n\leqslant N} \int f_n$: for $N = 1$ both sides are $\int f_1$, and if it holds for $N-1$ then $g_N = g_{N-1} + f_N$ and Theorem P.79 gives $\int g_N = \int g_{N-1} + \int f_N = \sum_{n\leqslant N} \int f_n$. Theorem P.78 lets $N \to \infty$ on the left, and the right side tends to $\sum_n \int f_n$ by Definition P.60.
(ii) $\nu(\emptyset) = \int 0 = 0$. If $A_1, A_2, \ldots$ are pairwise disjoint with union A, then $f\,\mathbb{1}_A = \sum_n f\,\mathbb{1}_{A_n}$ pointwise, as each point of A lies in exactly one A_n; apply (i). □

Definition P.81 (Almost everywhere)**.** A property of points of Ω holds **μ-almost everywhere** (a.e.), or **almost surely** (a.s.) when μ is a probability, if the set where it fails is contained in a null set.

Proposition P.82 (Null sets and the integral)**.** Let f, g be non-negative measurable functions.

(i) $a\,\mu\{f \geqslant a\} \leqslant \int f\,\mathrm{d}\mu$ for $a \in (0, \infty)$.
(ii) If N is a null set then $\int_N f\,\mathrm{d}\mu = 0$.
(iii) If $f = g$ a.e. then $\int f\,\mathrm{d}\mu = \int g\,\mathrm{d}\mu$.
(iv) $\int f\,\mathrm{d}\mu = 0$ if and only if $f = 0$ a.e.
(v) If $\int f\,\mathrm{d}\mu < \infty$ then $f < \infty$ a.e.
(vi) Countable unions of null sets are null; so if each of countably many properties holds a.e., they hold simultaneously a.e.

Proof. (i) $a\,\mathbb{1}_{\{f \geqslant a\}} \leqslant f$ pointwise; apply Lemma P.77(ii),(iv).
(ii) A simple $s \leqslant f\,\mathbb{1}_N$ vanishes off N, so for each value $a > 0$ of s, $\{s = a\} \subseteq N$ and $\mu\{s = a\} = 0$ by Proposition P.65(ii); hence $\int s = 0$, and the supremum is 0.
(iii) Let N be a null set with $f = g$ off N. By Theorem P.79 and (ii), $\int f = \int f\,\mathbb{1}_{N^c} + \int f\,\mathbb{1}_N = \int f\,\mathbb{1}_{N^c}$, and likewise $\int g = \int g\,\mathbb{1}_{N^c}$; and $f\,\mathbb{1}_{N^c} = g\,\mathbb{1}_{N^c}$.
(iv) If $f = 0$ a.e., (iii) with $g = 0$ gives $\int f = 0$. Conversely, if $\int f = 0$, (i) gives $\mu\{f \geqslant 1/k\} \leqslant k \int f = 0$ for each k, and $\{f > 0\} = \bigcup_k \{f \geqslant 1/k\}$ is null by countable subadditivity

(Proposition P.65(iii)).
(v) $n\,\mathbb{1}_{\{f=\infty\}} \leqslant f$, so $n\,\mu\{f=\infty\} \leqslant \int f < \infty$ for every n, which forces $\mu\{f=\infty\} = 0$.
(vi) If $N_1, N_2, \ldots$ are null, countable subadditivity (Proposition P.65(iii)) gives $\mu(\bigcup_k N_k) \leqslant \sum_k 0 = 0$. If the kth property fails only inside a null set N_k, then the set where some property fails lies in $\bigcup_k N_k$, which is null. □

Theorem P.83 (Fatou's lemma)**.** For non-negative measurable $f_1, f_2, \ldots$, $\int \liminf_n f_n \,\mathrm{d}\mu \leqslant \liminf_n \int f_n \,\mathrm{d}\mu$.

Proof. Let $g_m = \inf_{n \geqslant m} f_n$, measurable by Proposition P.68(ii). The g_m are non-negative and increase in m (an infimum over fewer terms), with $\sup_m g_m = \liminf_n f_n$ by Definition P.8. Since $g_m \leqslant f_n$ for every $n \geqslant m$, Lemma P.77(ii) gives $\int g_m \leqslant \inf_{n \geqslant m} \int f_n$. By Theorem P.78,

$$\int \liminf_n f_n \,\mathrm{d}\mu = \lim_m \int g_m \,\mathrm{d}\mu \leqslant \lim_m \inf_{n \geqslant m} \int f_n \,\mathrm{d}\mu = \liminf_n \int f_n \,\mathrm{d}\mu. \qquad \square$$

Definition P.84 (Integrable function)**.** A measurable $f : \Omega \to \mathbb{R}$ is **integrable** if $\int |f| \, d\mu < \infty$, and then $\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu$, both terms being finite as $f^\pm \leqslant |f|$. More generally the integral of a measurable real f **exists**, with this value in $[-\infty, \infty]$, when at least one of $\int f^\pm$ is finite. A measurable complex $f = u + iv$ is integrable if $\int |f| \, d\mu < \infty$, and then $\int f \, d\mu = \int u \, d\mu + i \int v \, d\mu$, the real functions u, v being integrable as $|u|, |v| \leqslant |f|$.

Theorem P.85 (Properties of the integral)**.** Let f, g be integrable, real or complex.

(i) For constants a, b (real, or complex in the complex case), $af + bg$ is integrable and $\int (af + bg) \, d\mu = a \int f \, d\mu + b \int g \, d\mu$.

(ii) If f, g are real and $f \leqslant g$ then $\int f \, d\mu \leqslant \int g \, d\mu$.

(iii) $\left| \int f \, d\mu \right| \leqslant \int |f| \, d\mu$.

(iv) If h is measurable and $h = f$ a.e., then h is integrable and $\int h \, d\mu = \int f \, d\mu$.

Proof. (i) **Real case.** $|af + bg| \leqslant |a||f| + |b||g|$, so $af + bg$ is integrable by Lemma P.77 and Theorem P.79. For $a \geqslant 0$, $(af)^\pm = af^\pm$, so $\int af = a \int f^+ - a \int f^- = a \int f$ by Lemma P.77(iii); for $a < 0$, $(af)^+ = |a| f^-$ and $(af)^- = |a| f^+$, so $\int af = |a| \int f^- - |a| \int f^+ = a \int f$. For the

sum, let $h = f + g$. From $h^+ - h^- = f^+ - f^- + g^+ - g^-$,

$$h^+ + f^- + g^- = h^- + f^+ + g^+,$$

an identity between non-negative functions. Theorem P.79 gives $\int h^+ + \int f^- + \int g^- = \int h^- + \int f^+ + \int g^+$, all six terms finite; rearranging, $\int h = \int f + \int g$.

Complex case. With $f = u + iv$, $g = w + iz$, $f + g = (u + w) + i(v + z)$, and the real case gives additivity. For $a = \alpha + i\beta$, $af = (\alpha u - \beta v) + i(\beta u + \alpha v)$, and by the real case $\int af = \alpha \int u - \beta \int v + i(\beta \int u + \alpha \int v) = (\alpha + i\beta)(\int u + i \int v)$.

(ii) $g - f \geqslant 0$ is integrable with $(g - f)^- = 0$, so by (i) $\int g - \int f = \int (g - f) = \int (g - f)^+ \geqslant 0$.

(iii) Real case: $\int f = \int f^+ - \int f^-$ and $-\int f$ are both at most $\int f^+ + \int f^- = \int |f|$, by Theorem P.79. Complex case: let $z = \int f$; if $z = 0$ there is nothing to prove. Otherwise put $\theta = \bar{z}/|z|$, so $|\theta| = 1$ and $\theta z = |z|$. By (i), $|z| = \int \theta f$, a real number, so it equals the real part $\int \mathrm{Re}(\theta f)$, and by (ii) and the real case $\int \mathrm{Re}(\theta f) \leqslant \int |\theta f| = \int |f|$.

(iv) $|h| = |f|$ a.e., so $\int |h| = \int |f| < \infty$ by Proposition P.82(iii). For real f, $h^\pm = f^\pm$ a.e., and Proposition P.82(iii) gives $\int h^\pm = \int f^\pm$; the complex case follows from the real and imaginary parts. □

Theorem P.86 (Dominated convergence theorem)**.** Let $f, f_1, f_2, \dots$ be measurable, real or complex, with $f_n \to f$ a.e., and let g be a non-negative integrable function with $|f_n| \leqslant g$ a.e. for every n. Then f and the f_n are integrable, $\int |f_n - f|\,\mathrm{d}\mu \to 0$, and $\int f_n\,\mathrm{d}\mu \to \int f\,\mathrm{d}\mu$.

Proof. Step 1: reduction to everywhere. By Proposition P.82(vi) there is a null set N off which $f_n \to f$ and $|f_n| \leqslant g$ for every n. Replacing f, f_n, g by $f\,\mathbb{1}_{N^c}$, $f_n\,\mathbb{1}_{N^c}$, $g\,\mathbb{1}_{N^c}$ changes none of the integrals in the statement, by Theorem P.85(iv) and Proposition P.82(iii), and the new functions satisfy the hypotheses at every point.

Step 2: integrability. $|f_n| \leqslant g$ gives $\int |f_n| \leqslant \int g < \infty$. Letting $n \to \infty$ in $|f_n| \leqslant g$ gives $|f| \leqslant g$, so f is integrable.

Step 3: Fatou applied to $2g - |f_n - f|$. Since $|f_n - f| \leqslant |f_n| + |f| \leqslant 2g$, the functions $h_n = 2g - |f_n - f|$ are non-negative and measurable, and $h_n \to 2g$ pointwise. Theorem P.83 and Theorem P.85(i) give

$$\int 2g\,\mathrm{d}\mu \leqslant \liminf_n \int h_n\,\mathrm{d}\mu = \int 2g\,\mathrm{d}\mu - \limsup_n \int |f_n - f|\,\mathrm{d}\mu,$$

using $\liminf_n (c - x_n) = c - \limsup_n x_n$, which holds because $\inf_{n \geqslant m}(c - x_n) = c - \sup_{n \geqslant m} x_n$. As $\int 2g$ is finite it may be cancelled, leaving $\limsup_n \int |f_n - f| \leqslant 0$. The terms are non-negative, so $\int |f_n - f| \to 0$.

Step 4. By Theorem P.85(i),(iii), $|\int f_n - \int f| = |\int (f_n - f)| \leqslant \int |f_n - f| \to 0$. □

Theorem P.87 (Differentiation under the integral sign)**.** Let $(\mathcal{X}, \mu)$ be a measure space, $I \subseteq \mathbb{R}$ an open interval, and $f : I \times \mathcal{X} \to \mathbb{R}$ such that

(i) $x \mapsto f(t, x)$ is μ-integrable for each $t \in I$;

(ii) $\partial f / \partial t$ exists for every (t, x);

(iii) $|\partial f / \partial t(t, x)| \leqslant g(x)$ for all $t \in I$, with $\int g \, \mathrm{d}\mu < \infty$.

Then $F(t) = \int f(t, x) \, \mathrm{d}\mu(x)$ is differentiable on I with $F'(t) = \int \frac{\partial f}{\partial t}(t, x) \, \mathrm{d}\mu(x)$.

Proof. Fix $t \in I$ and any sequence $h_n \to 0$ with $t + h_n \in I$, $h_n \neq 0$. Put $q_n(x) = \{f(t + h_n, x) - f(t, x)\}/h_n$, measurable by (i) and Proposition P.68(i). By (ii) and the mean value theorem (Corollary P.7) applied in t, for each x there is ξ between t and $t + h_n$ with $q_n(x) = \partial f / \partial t(\xi, x)$, so $|q_n(x)| \leqslant g(x)$ by (iii). Also $q_n(x) \to \partial f / \partial t(t, x)$ pointwise by (ii), so $x \mapsto \partial f / \partial t(t, x)$ is

measurable by Proposition P.68(ii). The dominated convergence theorem (Theorem P.86) gives $\int q_n \,\mathrm{d}\mu \to \int \partial f/\partial t \,\mathrm{d}\mu$, and by Theorem P.85(i) the left side is $\{F(t+h_n) - F(t)\}/h_n$, F being defined by (i). As the sequence (h_n) was arbitrary, the difference quotient of F tends to $L = \int \partial f/\partial t(t,x)\,\mathrm{d}\mu(x)$ as $h \to 0$: otherwise there would be $\varepsilon > 0$ and, for each n, some h_n with $0 < |h_n| < 1/n$, $t + h_n \in I$ and $|\{F(t+h_n) - F(t)\}/h_n - L| \geqslant \varepsilon$, a sequence along which the quotients do not tend to L. So $F'(t) = L$. □

Definition P.88 (Random vectors, laws and expectation)**.** On a probability space $(\Omega, \mathcal{F}, \mathsf{P})$ a **random vector** is a measurable $X : \Omega \to \mathbb{R}^d$, a **random variable** when $d = 1$. Its **law** is $\mu_X(B) = \mathsf{P}\{X \in B\}$ for $B \in \mathcal{B}(\mathbb{R}^d)$. The **expectation** of a random variable Y is $\mathsf{E}[Y] = \int Y \,\mathrm{d}\,\mathsf{P}$ whenever the integral exists, and likewise $\mathsf{E}[Y] = \int Y \,\mathrm{d}\,\mathsf{P}$ for measurable $Y : \Omega \to [0,\infty]$ and for integrable complex Y (Definition P.84). For an integrable random variable Y, $\operatorname{var}(Y) = \mathsf{E}[(Y - \mathsf{E}\,Y)^2] \in [0,\infty]$ and $\operatorname{sd}(Y) = \operatorname{var}(Y)^{1/2}$; for a random vector $\mathbf{z} = (z_1, \ldots, z_d)'$ with integrable coordinates, $\mathsf{E}\,\mathbf{z} = (\mathsf{E}\,z_1, \ldots, \mathsf{E}\,z_d)'$, $\operatorname{cov}(z_i, z_j) = \mathsf{E}[(z_i - \mathsf{E}\,z_i)(z_j - \mathsf{E}\,z_j)]$ whenever this expectation exists, and $\operatorname{var}(\mathbf{z})$ is the $d \times d$ matrix with entries $\operatorname{cov}(z_i, z_j)$.

Remark. μ_X is a probability measure on $\mathcal{B}(\mathbb{R}^d)$: $\mu_X(\mathbb{R}^d) = \mathsf{P}(\Omega) = 1$, and preimages of pairwise disjoint sets are pairwise disjoint with $X^{-1}(\bigcup_n B_n) = \bigcup_n X^{-1}(B_n)$.

Theorem P.89 (Change of variables)**.** Let $T : \Omega \to E$ be $\mathcal{F}/\mathcal{E}$-measurable and $\mu T^{-1}(B) = \mu(T^{-1}(B))$ for $B \in \mathcal{E}$. For $\mathcal{E}$-measurable $g : E \to [0, \infty]$,

$$\int g \circ T \, \mathrm{d}\mu = \int g \, \mathrm{d}(\mu T^{-1}),$$

and the same holds for real or complex g whenever either side is finite with $|g|$ in place of g. In particular $\mathsf{E}[g(X)] = \int g \, \mathrm{d}\mu_X$ for a random vector X and a Borel $g \geqslant 0$, or a Borel g with $\mathsf{E}\,|g(X)| < \infty$.

Proof. μT^{-1} is a measure on $\mathcal{E}$, by the argument of the preceding remark. For $g = \mathbb{1}_B$, $g \circ T = \mathbb{1}_{T^{-1}(B)}$, and both sides equal $\mu(T^{-1}(B))$ by Lemma P.77(iv). For simple $g = \sum_i a_i \mathbb{1}_{B_i}$, $g \circ T = \sum_i a_i \mathbb{1}_{T^{-1}(B_i)}$ is simple on Ω, and both sides equal $\sum_i a_i \mu(T^{-1}(B_i))$ by Lemma P.75(i). For measurable $g \geqslant 0$, take simple $s_n \uparrow g$ (Theorem P.70); then $s_n \circ T$ is simple, $s_n \circ T \uparrow g \circ T$,

and Theorem P.78 on both sides passes the identity to the limit. For real g with $\int |g| \circ T \, d\mu = \int |g| \, d(\mu T^{-1}) < \infty$, apply this to g^+ and g^-, noting $(g \circ T)^{\pm} = g^{\pm} \circ T$, and subtract; for complex g apply the real case to the real and imaginary parts. □

Lemma P.90 (Cauchy–Schwarz and Minkowski for integrals)**.** Let u, v be real measurable with $\int u^2 \, d\mu$ and $\int v^2 \, d\mu$ finite, and put $\|u\| = (\int u^2 \, d\mu)^{1/2}$. Then uv is integrable, $|\int uv \, d\mu| \leqslant \|u\| \, \|v\|$, $\int (u+v)^2 \, d\mu < \infty$, and $\|u+v\| \leqslant \|u\| + \|v\|$.

Proof. From $0 \leqslant (|u| - |v|)^2$, $|uv| \leqslant \frac{1}{2}(u^2 + v^2)$, so uv is integrable by Theorem P.79; and $(u+v)^2 \leqslant 2u^2 + 2v^2$, so $(u+v)^2$ is integrable. For real t, Theorem P.85(i) and (ii) give

$$0 \leqslant \int (u - tv)^2 \, d\mu = \|u\|^2 - 2t \int uv \, d\mu + t^2 \|v\|^2.$$

If $\|v\| = 0$ the right side is affine in t and non-negative for every t, so its slope $-2 \int uv$ vanishes and the inequality holds. If $\|v\| > 0$ the quadratic in t cannot have two distinct real roots, as it would be negative between them, so its discriminant $4(\int uv)^2 - 4\|u\|^2\|v\|^2$ is at most 0. In both cases $(\int uv)^2 \leqslant \|u\|^2\|v\|^2$, and taking square roots gives $|\int uv \, d\mu| \leqslant \|u\| \, \|v\|$. Then, by

Theorem P.85(i),

$$\|u+v\|^2 = \|u\|^2 + 2\int uv\,\mathrm{d}\mu + \|v\|^2 \leqslant \|u\|^2 + 2\|u\|\,\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2. \qquad \square$$

Theorem P.91 (Completeness of L^2)**.** Let $f_1, f_2, \ldots$ be real measurable functions with $\int f_n^2\,\mathrm{d}\mu < \infty$, and suppose that for every $\varepsilon > 0$ there is N with $\int (f_n - f_m)^2\,\mathrm{d}\mu \leqslant \varepsilon^2$ for all $m, n \geqslant N$. Then there is a real measurable f with $\int f^2\,\mathrm{d}\mu < \infty$ and $\int (f_n - f)^2\,\mathrm{d}\mu \to 0$, and a subsequence of (f_n) converges to f a.e.

Proof. Step 1: a fast subsequence. Choose $n_1 < n_2 < \cdots$ such that $\int (f_n - f_m)^2 \leqslant 4^{-k}$ for all $m, n \geqslant n_k$: with $N(\varepsilon)$ the index of the hypothesis, put $n_1 = N(1/2)$ and $n_k = \max\{N(2^{-k}), n_{k-1}+1\}$ for $k \geqslant 2$. As $n_{k+1} > n_k$, taking $m = n_k$ and $n = n_{k+1}$ gives $\int (f_{n_{k+1}} - f_{n_k})^2 \leqslant 4^{-k}$. With the notation of Lemma P.90, $\|f_{n_{k+1}} - f_{n_k}\| \leqslant 2^{-k}$.

Step 2: the increments are summable a.e. Let $G_K = \sum_{k \leqslant K} |f_{n_{k+1}} - f_{n_k}|$ and $G = \lim_K G_K \in [0, \infty]$ (an increasing limit). Claim: $\int G_K^2 < \infty$ and $\|G_K\| \leqslant \sum_{k \leqslant K} 2^{-k}$. For $K = 1$, $\|G_1\| = \|f_{n_2} - f_{n_1}\| \leqslant 2^{-1}$. If the claim holds for $K-1$, then $G_K = G_{K-1} + |f_{n_{K+1}} - f_{n_K}|$, and

Lemma P.90 gives $\int G_K^2 < \infty$ and $\|G_K\| \leqslant \|G_{K-1}\| + 2^{-K} \leqslant \sum_{k\leqslant K} 2^{-k}$. Hence $\|G_K\| \leqslant 1$, that is, $\int G_K^2 \leqslant 1$. As $G_K^2 \uparrow G^2$, Theorem P.78 gives $\int G^2 \leqslant 1$, and Proposition P.82(v) gives a null set $N \in \mathcal{F}$ (namely $\{G = \infty\}$) off which $G < \infty$.

Step 3: the subsequence converges off N. Fix $\omega \notin N$ and put $a_k = f_{n_{k+1}}(\omega) - f_{n_k}(\omega)$. The partial sums of a_k^+ and of a_k^- increase and are bounded by $G(\omega) < \infty$, so both converge by Theorem P.2, and hence so does $f_{n_{K+1}}(\omega) = f_{n_1}(\omega) + \sum_{k\leqslant K} a_k^+ - \sum_{k\leqslant K} a_k^-$. Define $f = \lim_k f_{n_k}$ on N^c and $f = 0$ on N. Then f is the pointwise limit of the measurable real functions $f_{n_k}\mathbb{1}_{N^c}$, hence measurable by Proposition P.68(ii), and $f_{n_k} \to f$ a.e.

Step 4: L^2 convergence. Let $\varepsilon > 0$ and N_ε be as in the hypothesis, and fix $n \geqslant N_\varepsilon$. On N^c, $(f_{n_k} - f_n)^2 \to (f - f_n)^2$ as $k \to \infty$. By Proposition P.82(iii) and Theorem P.83,

$$\int (f - f_n)^2 \,\mathrm{d}\mu = \int \liminf_k (f_{n_k} - f_n)^2 \,\mathbb{1}_{N^c} \,\mathrm{d}\mu \leqslant \liminf_k \int (f_{n_k} - f_n)^2 \,\mathrm{d}\mu \leqslant \varepsilon^2,$$

using Lemma P.77(ii) to drop $\mathbb{1}_{N^c}$, since $n_k \geqslant k \geqslant N_\varepsilon$ for large k. Thus $f - f_n$ is square integrable, so is $f = f_n + (f - f_n)$ by Lemma P.90, and $\int (f_n - f)^2 \leqslant \varepsilon^2$ for every $n \geqslant N_\varepsilon$. □

18 Product Measures

Throughout, $(\Omega_1, \mathcal{F}_1, \mu_1)$ and $(\Omega_2, \mathcal{F}_2, \mu_2)$ are measure spaces.

Definition P.92 (Product σ-field)**.** A **measurable rectangle** is a set $A \times B$ with $A \in \mathcal{F}_1$, $B \in \mathcal{F}_2$. The **product σ-field** $\mathcal{F}_1 \otimes \mathcal{F}_2$ is the σ-field on $\Omega_1 \times \Omega_2$ generated by the measurable rectangles. For $E \subseteq \Omega_1 \times \Omega_2$ the **sections** are $E_x = \{y : (x, y) \in E\}$ for $x \in \Omega_1$ and $E^y = \{x : (x, y) \in E\}$ for $y \in \Omega_2$.

Remark. The measurable rectangles form a π-system: $(A \times B) \cap (A' \times B') = (A \cap A') \times (B \cap B')$.

Lemma P.93 (Borel sets of a product)**.** $\mathcal{B}(\mathbb{R}^{a+b}) = \mathcal{B}(\mathbb{R}^a) \otimes \mathcal{B}(\mathbb{R}^b)$, identifying $\mathbb{R}^{a+b}$ with $\mathbb{R}^a \times \mathbb{R}^b$.

Proof. ($\subseteq$) An open box in $\mathbb{R}^{a+b}$ is the product of an open box in $\mathbb{R}^a$ and one in $\mathbb{R}^b$, both open, hence a measurable rectangle. By Lemma P.63(i) every open subset of $\mathbb{R}^{a+b}$ is a countable union of open boxes and so lies in $\mathcal{B}(\mathbb{R}^a) \otimes \mathcal{B}(\mathbb{R}^b)$; hence so does $\mathcal{B}(\mathbb{R}^{a+b})$, the σ-field the open sets generate. ($\supseteq$) For $A \in \mathcal{B}(\mathbb{R}^a)$, $A \times \mathbb{R}^b = p^{-1}(A)$ with $p(x, y) = x$ continuous, hence Borel by Lemma P.67(iii); so $A \times \mathbb{R}^b \in \mathcal{B}(\mathbb{R}^{a+b})$, and likewise $\mathbb{R}^a \times B$ for $B \in \mathcal{B}(\mathbb{R}^b)$. Their intersection $A \times B$ is therefore

Borel, and the σ-field the rectangles generate is contained in $\mathcal{B}(\mathbb{R}^{a+b})$. □

Lemma P.94 (Measurability of sections)**.** If $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$ then $E_x \in \mathcal{F}_2$ for every x and $E^y \in \mathcal{F}_1$ for every y. If $f : \Omega_1 \times \Omega_2 \to [-\infty, \infty]$ is $\mathcal{F}_1 \otimes \mathcal{F}_2$-measurable, then $y \mapsto f(x, y)$ is $\mathcal{F}_2$-measurable for every x, and $x \mapsto f(x, y)$ is $\mathcal{F}_1$-measurable for every y.

Proof. Let $\mathcal{G} = \{E \subseteq \Omega_1 \times \Omega_2 : E_x \in \mathcal{F}_2 \text{ for every } x\}$. A rectangle lies in $\mathcal{G}$, as $(A \times B)_x$ is B for $x \in A$ and $\emptyset$ otherwise. $\mathcal{G}$ is a σ-field, since $(\Omega_1 \times \Omega_2)_x = \Omega_2$, $(E^c)_x = (E_x)^c$ and $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$. Hence $\mathcal{F}_1 \otimes \mathcal{F}_2 \subseteq \mathcal{G}$. For f, $\{y : f(x, y) \leqslant a\} = (\{f \leqslant a\})_x \in \mathcal{F}_2$; apply Lemma P.67(ii). The statements for E^y and $x \mapsto f(x, y)$ follow by exchanging the roles of the coordinates: the class of E with $E^y \in \mathcal{F}_1$ for every y contains the rectangles, as $(A \times B)^y$ is A for $y \in B$ and $\emptyset$ otherwise, and is a σ-field by the same three identities for E^y; and $\{x : f(x, y) \leqslant a\} = (\{f \leqslant a\})^y$. □

Lemma P.95 (Measurability of section measures)**.** If μ_2 is σ-finite and $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$, then $x \mapsto \mu_2(E_x)$ is a non-negative $\mathcal{F}_1$-measurable function. If μ_1 is σ-finite, $y \mapsto \mu_1(E^y)$ is $\mathcal{F}_2$-measurable.

Proof. Step 1: μ_2 finite. Let $\mathcal{L}$ be the class of $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$ for which $x \mapsto \mu_2(E_x)$ is measurable; the function is defined by Lemma P.94. A rectangle lies in $\mathcal{L}$, as $\mu_2((A \times B)_x) = \mu_2(B)\,\mathbb{1}_A(x)$. $\mathcal{L}$ is a λ-system: the whole space gives the constant $\mu_2(\Omega_2)$; if $E \subseteq F$ lie in $\mathcal{L}$ then $(F \setminus E)_x = F_x \setminus E_x$ with $E_x \subseteq F_x$, so $\mu_2((F \setminus E)_x) = \mu_2(F_x) - \mu_2(E_x)$ by Proposition P.65(ii), a difference of real measurable functions; and if $E_n \uparrow E$ with $E_n \in \mathcal{L}$ then $(E_n)_x \uparrow E_x$, so $\mu_2((E_n)_x) \uparrow \mu_2(E_x)$ by continuity from below, a supremum of measurable functions. The rectangles form a π-system generating $\mathcal{F}_1 \otimes \mathcal{F}_2$, so Theorem P.72 gives $\mathcal{L} = \mathcal{F}_1 \otimes \mathcal{F}_2$.
Step 2: μ_2 σ-finite. Let $\Omega_2 = \bigcup_k C_k$ with $\mu_2(C_k) < \infty$ and put $D_n = \bigcup_{k \leqslant n} C_k$, so $D_n \uparrow \Omega_2$ and $\mu_2(D_n) < \infty$ by subadditivity. The finite measure $B \mapsto \mu_2(B \cap D_n)$ satisfies the hypothesis of Step 1, so $x \mapsto \mu_2(E_x \cap D_n)$ is measurable. As $n \to \infty$ it increases to $\mu_2(E_x)$ by continuity from below, and a supremum of measurable functions is measurable. The second statement is Steps 1 and 2 with the roles of the coordinates exchanged, using $(A \times B)^y$ (A for $y \in B$, $\emptyset$ otherwise) and the sections E^y of Lemma P.94. □

Theorem P.96 (Product measure)**.** Let μ_1, μ_2 be σ-finite. There is a unique measure $\mu_1 \otimes \mu_2$ on $\mathcal{F}_1 \otimes \mathcal{F}_2$ with $(\mu_1 \otimes \mu_2)(A \times B) = \mu_1(A)\mu_2(B)$ for all $A \in \mathcal{F}_1$, $B \in \mathcal{F}_2$. It is σ-finite, a

probability if μ_1, μ_2 are, and for every $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$

$$(\mu_1 \otimes \mu_2)(E) = \int \mu_2(E_x)\, \mu_1(\mathrm{d}x) = \int \mu_1(E^y)\, \mu_2(\mathrm{d}y). \tag{P.14}$$

Proof. Step 1: two measures. By Lemma P.95 the integrands in (P.14) are non-negative measurable functions, so $\pi_1(E) = \int \mu_2(E_x)\, \mu_1(\mathrm{d}x)$ and $\pi_2(E) = \int \mu_1(E^y)\, \mu_2(\mathrm{d}y)$ are defined. $\pi_1(\emptyset) = 0$. If $E_1, E_2, \ldots$ are pairwise disjoint with union E, then for each x the sections $(E_n)_x$ are pairwise disjoint with union E_x, so $\mu_2(E_x) = \sum_n \mu_2((E_n)_x)$, and Corollary P.80(i) gives $\pi_1(E) = \sum_n \pi_1(E_n)$. So π_1 is a measure, and likewise π_2.

Step 2: values on rectangles. $\pi_1(A \times B) = \int \mu_2(B)\, \mathbb{1}_A \, \mathrm{d}\mu_1$. If $\mu_2(B) < \infty$ this is $\mu_2(B)\mu_1(A)$ by Lemma P.77(iv). If $\mu_2(B) = \infty$, the functions $n\, \mathbb{1}_A$ increase to $\mu_2(B)\, \mathbb{1}_A$, and Theorem P.78 gives $\lim_n n\mu_1(A)$, which is 0 if $\mu_1(A) = 0$ and ∞ otherwise, that is $\mu_1(A)\mu_2(B)$ under the conventions of Definition P.60. In the same way $\pi_2(A \times B) = \mu_1(A)\mu_2(B)$.

Step 3: uniqueness and the equality $\pi_1 = \pi_2$. As in Lemma P.95, Step 2, there are $A_n \uparrow \Omega_1$ and $B_n \uparrow \Omega_2$ with $\mu_1(A_n), \mu_2(B_n) < \infty$. The rectangles $A_n \times B_n$ increase, have finite

measure $\mu_1(A_n)\mu_2(B_n)$ under any measure taking the stated values on rectangles, and their union is $\Omega_1 \times \Omega_2$, since a point (x, y) lies in $A_n \times B_n$ as soon as $x \in A_n$ and $y \in B_n$. By Corollary P.73, applied to the π-system of rectangles, any two measures on $\mathcal{F}_1 \otimes \mathcal{F}_2$ taking the values $\mu_1(A)\mu_2(B)$ on rectangles coincide. This gives uniqueness and $\pi_1 = \pi_2$, which is (P.14). The sets $A_n \times B_n$ show σ-finiteness, and $(\mu_1 \otimes \mu_2)(\Omega_1 \times \Omega_2) = \mu_1(\Omega_1)\mu_2(\Omega_2)$. □

Theorem P.97 (Fubini–Tonelli theorem)**.** Let μ_1, μ_2 be σ-finite and $\pi = \mu_1 \otimes \mu_2$.

(i) **Tonelli.** If $f : \Omega_1 \times \Omega_2 \to [0, \infty]$ is measurable, then $x \mapsto \int f(x, y)\, \mu_2(\mathrm{d}y)$ is $\mathcal{F}_1$-measurable, $y \mapsto \int f(x, y)\, \mu_1(\mathrm{d}x)$ is $\mathcal{F}_2$-measurable, and

$$\int f \,\mathrm{d}\pi = \int \Big[\int f(x, y)\, \mu_2(\mathrm{d}y)\Big] \mu_1(\mathrm{d}x) = \int \Big[\int f(x, y)\, \mu_1(\mathrm{d}x)\Big] \mu_2(\mathrm{d}y).$$

(ii) **Fubini.** If f is measurable, real or complex, with $\int |f| \,\mathrm{d}\pi < \infty$, then $y \mapsto f(x, y)$ is μ_2-integrable for x outside a μ_1-null set N_1; $I_1(x) = \int f(x, y)\, \mu_2(\mathrm{d}y)$ for $x \notin N_1$ and $I_1(x) = 0$ for $x \in N_1$ defines a μ_1-integrable function; and $\int f \,\mathrm{d}\pi = \int I_1 \,\mathrm{d}\mu_1$. The same holds with the roles of the coordinates exchanged, so both iterated integrals equal $\int f \,\mathrm{d}\pi$.

Proof. (i) **Step 1: indicators.** For $f = \mathbb{1}_E$ with $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$, $f(x,\cdot) = \mathbb{1}_{E_x}$, so $\int f(x,y)\,\mu_2(\mathrm{d}y) = \mu_2(E_x)$, measurable by Lemma P.95, and its μ_1-integral is $\pi(E) = \int f\,\mathrm{d}\pi$ by Theorem P.96. The other order is the same.

Step 2: simple functions. For simple $f = \sum_i a_i\,\mathbb{1}_{E_i}$, the inner integral is $\sum_i a_i \mu_2((E_i)_x)$ by Theorem P.79, measurable by Proposition P.68(iii); integrating in x and using Theorem P.79 and Step 1 gives $\sum_i a_i \pi(E_i) = \int f\,\mathrm{d}\pi$, by Lemma P.75(i).

Step 3: monotone limits. For general $f \geqslant 0$ take simple $s_n \uparrow f$ (Theorem P.70). For each x, $s_n(x,\cdot) \uparrow f(x,\cdot)$, these functions being measurable by Lemma P.94, so by Theorem P.78

$$\int s_n(x,y)\,\mu_2(\mathrm{d}y) \uparrow \int f(x,y)\,\mu_2(\mathrm{d}y).$$

The inner integral of f is thus an increasing limit of measurable functions (Step 2), hence measurable by Proposition P.68(ii). Applying Theorem P.78 in x, and also on $\Omega_1 \times \Omega_2$ to $s_n \uparrow f$,

$$\int \Big[\int f(x,y)\,\mu_2(\mathrm{d}y)\Big]\mu_1(\mathrm{d}x) = \lim_n \int \Big[\int s_n(x,y)\,\mu_2(\mathrm{d}y)\Big]\mu_1(\mathrm{d}x) = \lim_n \int s_n\,\mathrm{d}\pi = \int f\,\mathrm{d}\pi.$$

The other order is the same.

(ii) **Step 1: the exceptional set.** By (i) applied to $|f|$, $J(x) = \int |f(x,y)|\,\mu_2(\mathrm{d}y)$ is measurable with $\int J\,\mathrm{d}\mu_1 = \int |f|\,\mathrm{d}\pi < \infty$. By Proposition P.82(v), $N_1 = \{J = \infty\}$ is a μ_1-null set in $\mathcal{F}_1$, and for $x \notin N_1$ the function $f(x,\cdot)$, measurable by Lemma P.94 applied to the real and imaginary parts, is μ_2-integrable.

Step 2: real f. Apply (i) to f^+ and f^-: $h^\pm(x) = \int f^\pm(x,y)\,\mu_2(\mathrm{d}y)$ are measurable, $h^\pm \leqslant J$, and $\int h^\pm\,\mathrm{d}\mu_1 = \int f^\pm\,\mathrm{d}\pi < \infty$. Off N_1 both are finite and $I_1 = h^+ - h^-$; so $I_1 = h^+\,\mathbb{1}_{N_1^c} - h^-\,\mathbb{1}_{N_1^c}$ is a difference of real measurable functions, it is integrable as $|I_1| \leqslant h^+ + h^-$, and by Theorem P.85(i) and Proposition P.82(ii),(iii)

$$\int I_1\,\mathrm{d}\mu_1 = \int h^+\,\mathrm{d}\mu_1 - \int h^-\,\mathrm{d}\mu_1 = \int f^+\,\mathrm{d}\pi - \int f^-\,\mathrm{d}\pi = \int f\,\mathrm{d}\pi.$$

Step 3: complex f. The real and imaginary parts u, v of f satisfy $|u|, |v| \leqslant |f|$, and the exceptional set of Step 1 for each of them is contained in N_1. Applying Step 2 to u and v, with N_1 as the exceptional set (enlarging a null exceptional set changes no integral, by Proposition P.82(iii)), and Theorem P.85(i) gives the claim. Exchanging the coordinates gives the last statement. □

Corollary P.98 (Finite products)**.** Let $\mu_1, \dots, \mu_k$ be σ-finite measures on $\mathcal{B}(\mathbb{R}^{d_1}), \dots, \mathcal{B}(\mathbb{R}^{d_k})$ and $d = d_1 + \cdots + d_k$. There is a unique measure $\mu_1 \otimes \cdots \otimes \mu_k$ on $\mathcal{B}(\mathbb{R}^d)$ with

$$(\mu_1 \otimes \cdots \otimes \mu_k)(B_1 \times \cdots \times B_k) = \prod_{i \leqslant k} \mu_i(B_i) \qquad (B_i \in \mathcal{B}(\mathbb{R}^{d_i})).$$

It is σ-finite, a probability if every μ_i is, and for $k \geqslant 2$ it is $(\mu_1 \otimes \cdots \otimes \mu_{k-1}) \otimes \mu_k$.

Proof. Existence, by induction on k, the case $k = 1$ being trivial. If $\nu = \mu_1 \otimes \cdots \otimes \mu_{k-1}$ exists on $\mathcal{B}(\mathbb{R}^{d-d_k})$ and is σ-finite, then Lemma P.93 identifies $\mathcal{B}(\mathbb{R}^{d-d_k}) \otimes \mathcal{B}(\mathbb{R}^{d_k})$ with $\mathcal{B}(\mathbb{R}^d)$, and Theorem P.96 gives the σ-finite measure $\nu \otimes \mu_k$ on it, with $(\nu \otimes \mu_k)(B_1 \times \cdots \times B_k) = \nu(B_1 \times \cdots \times B_{k-1})\mu_k(B_k) = \prod_i \mu_i(B_i)$; it is a probability if ν and μ_k are.
Uniqueness. The sets $B_1 \times \cdots \times B_k$ form a π-system, as intersections are taken coordinatewise. It generates $\mathcal{B}(\mathbb{R}^d)$: it contains the half-open boxes of $\mathbb{R}^d$, which generate $\mathcal{B}(\mathbb{R}^d)$ by Lemma P.63(ii); and each $B_1 \times \cdots \times B_k$ is the intersection of the sets $p_i^{-1}(B_i)$, with p_i the continuous projection onto the i-th block of coordinates, which are Borel by Lemma P.67(iii). Taking $C_n^{(i)} \uparrow \mathbb{R}^{d_i}$ with $\mu_i(C_n^{(i)}) < \infty$, as in Lemma P.95, the sets $C_n^{(1)} \times \cdots \times C_n^{(k)}$ increase to $\mathbb{R}^d$ and have finite measure,

so Corollary P.73 applies. □

Remark (Assumed facts)**.** Besides the elementary calculus and linear algebra of Remark 1, one fact of calculus in several variables is taken as known: a function whose partial derivatives up to order two exist and are continuous on an open subset of $\mathbb{R}^p$ (a **twice continuously differentiable** function) is differentiable there in the sense of Definition P.56, so is its gradient, and its matrix of second partial derivatives is symmetric. The following measure-theoretic facts are used without proof, except (a) and (b), which have been proved:

(a) square-integrable random variables form a complete space: Theorem P.91;

(b) two probability measures that agree on a π-system agree on the σ-field it generates: Theorem P.72 and Corollary P.73;

(c) product probability measures exist for countably many factors (for finitely many σ-finite factors this is Corollary P.98); Lebesgue measure λ on $\mathcal{B}(\mathbb{R})$, the measure with $\lambda((a,b]) = b-a$ for $a < b$, exists (it is unique by Corollary P.73 applied to the π-system of these intervals and $\emptyset$, which generates $\mathcal{B}(\mathbb{R})$ by Lemma P.63(ii), with $\Omega_n = (-n, n]$), and Lebesgue measure on $\mathbb{R}^d$ is its d-fold product, λ being σ-finite; $\int g(x)\,\mathrm{d}x$ denotes $\int g\,\mathrm{d}\lambda$, and for continuous g

on $[a,b]$, $\int_{[a,b]} g\,\mathrm{d}\lambda$ is the Riemann integral of calculus; a **density** with respect to a measure μ on $\mathcal{B}(\mathbb{R}^d)$ is a Borel $f \geqslant 0$ with $\int f\,\mathrm{d}\mu = 1$, and a random vector X in $\mathbb{R}^d$ **has density** f if $\mathsf{P}\{X \in B\} = \int_B f\,\mathrm{d}\mu$ for every $B \in \mathcal{B}(\mathbb{R}^d)$ (this is a definition, not an assumption; $B \mapsto \int_B f\,\mathrm{d}\mu$ is a measure by Corollary P.80(ii)); with no measure named, μ is Lebesgue measure;

(d) a **standard normal** variable is one with density $\phi(x) = (2\pi)^{-1/2}e^{-x^2/2}$, and $\int \phi = 1$, $\int x^2\phi(x)\,\mathrm{d}x = 1$;

(e) if Z has density f on $\mathbb{R}$ and $a \neq 0$, then $aZ + b$ has density $|a|^{-1}f\big((x-b)/a\big)$;

(f) for a Borel map g, the set of points where g is discontinuous is a Borel set;

(g) if $F(\omega, t)$ is real, measurable in ω for each t and continuous in t for each ω, with t ranging over a compact $K \subseteq \mathbb{R}^k$, then a point of K at which $F(\omega,\cdot)$ attains its maximum, and a zero of $F(\omega,\cdot)$ on the event that one exists, can be chosen as measurable functions of ω (measurable selection).

Lemma P.99 (Integration against a density)**.** Let X be a random vector in $\mathbb{R}^d$ with density f with respect to a measure μ on $\mathcal{B}(\mathbb{R}^d)$. For Borel $g : \mathbb{R}^d \to [0, \infty]$,

$$\mathsf{E}[g(X)] = \int gf \, \mathrm{d}\mu.$$

For real or complex Borel g, $\mathsf{E}\,|g(X)| = \int |g| f \, \mathrm{d}\mu$, and when this is finite, $\mathsf{E}[g(X)] = \int gf \, \mathrm{d}\mu$.

Proof. By Theorem P.89, $\mathsf{E}[g(X)] = \int g \, \mathrm{d}\mu_X$ for Borel $g \geqslant 0$, so for such g it suffices to show $\int g \, \mathrm{d}\mu_X = \int gf \, \mathrm{d}\mu$. The function gf is non-negative and measurable by Proposition P.68(iii).
Indicators. For $B \in \mathcal{B}(\mathbb{R}^d)$, Lemma P.77(iv) gives $\int \mathbb{1}_B \, \mathrm{d}\mu_X = \mu_X(B) = \mathsf{P}\{X \in B\}$, and $\int \mathbb{1}_B f \, \mathrm{d}\mu = \int_B f \, \mathrm{d}\mu$ by Definition P.76. The two are equal by the definition of a density in Remark 18(c).
Simple functions. Let $s = \sum_{i \leqslant k} a_i \mathbb{1}_{B_i}$ with $a_i \in [0, \infty)$ and $B_i \in \mathcal{B}(\mathbb{R}^d)$. By Lemma P.75(i)

and the indicator case,

$$\int s \, \mathrm{d}\mu_X = \sum_{i \leqslant k} a_i \mu_X(B_i) = \sum_{i \leqslant k} a_i \int \mathbb{1}_{B_i} f \, \mathrm{d}\mu.$$

By Lemma P.77(iii), $a_i \int \mathbb{1}_{B_i} f \, \mathrm{d}\mu = \int a_i \mathbb{1}_{B_i} f \, \mathrm{d}\mu$, and by Corollary P.80(i), applied to the functions $a_i \mathbb{1}_{B_i} f$ for $i \leqslant k$ and 0 for $i > k$, the sum of these integrals is $\int \sum_{i \leqslant k} a_i \mathbb{1}_{B_i} f \, \mathrm{d}\mu = \int sf \, \mathrm{d}\mu$.

Non-negative functions. For Borel $g : \mathbb{R}^d \to [0, \infty]$, let s_n be the simple functions of Theorem P.70, so that $0 \leqslant s_1 \leqslant s_2 \leqslant \cdots$ and $s_n \to g$ pointwise. Then each $s_n f$ is non-negative and measurable, and $s_n f \uparrow gf$ pointwise. Where $f(x) > 0$, this is $s_n(x) \uparrow g(x)$ multiplied by $f(x)$, with $s_n(x)f(x) \to \infty$ when $g(x) = \infty$. Where $f(x) = 0$, every $s_n(x)f(x)$ and $g(x)f(x)$ are 0, by the convention $0 \cdot \infty = 0$ of Definition P.60. Theorem P.78, applied on both sides, and the simple case give

$$\int g \, \mathrm{d}\mu_X = \lim_n \int s_n \, \mathrm{d}\mu_X = \lim_n \int s_n f \, \mathrm{d}\mu = \int gf \, \mathrm{d}\mu.$$

Real and complex functions. For Borel real or complex g, $|g|$ is Borel and non-negative, so the case just proved gives $\mathsf{E}\,|g(X)| = \int |g| f \, \mathrm{d}\mu$. Suppose this is finite. For real g, the non-negative

case applied to g^+ and g^- gives $\mathsf{E}[g^{\pm}(X)] = \int g^{\pm} f \, \mathrm{d}\mu$, and all four quantities are finite since $g^{\pm} \leqslant |g|$. Since $f \geqslant 0$, $(gf)^{\pm} = g^{\pm} f$, and $(g \circ X)^{\pm} = g^{\pm} \circ X$. Subtracting therefore gives $\mathsf{E}[g(X)] = \int gf \, \mathrm{d}\mu$, by Definition P.84. For complex $g = u + iv$, apply the real case to u and v, which satisfy $|u|, |v| \leqslant |g|$. □

19 Expectation and Inequalities

Theorem P.100 (Markov, Chebyshev)**.** For a random variable $X \geqslant 0$ and $a > 0$, $\mathsf{P}\{X \geqslant a\} \leqslant \mathsf{E}[X]/a$. Consequently $\mathsf{P}\{|Y - \mathsf{E}\,Y| \geqslant a\} \leqslant \operatorname{var}(Y)/a^2$.

Proof. $a\,\mathbb{1}\{X \geqslant a\} \leqslant X$ pointwise (at points with $X \geqslant a$ the left side is $a \leqslant X$, elsewhere it is $0 \leqslant X$). Taking expectations, Lemma P.77(ii),(iv) give $a\,\mathsf{P}\{X \geqslant a\} \leqslant \mathsf{E}[X]$; divide by $a > 0$. For the second, $\{|Y - \mathsf{E}\,Y| \geqslant a\} = \{(Y - \mathsf{E}\,Y)^2 \geqslant a^2\}$, so apply the first to $X = (Y - \mathsf{E}\,Y)^2$, whose expectation is $\operatorname{var}(Y)$, at the level a^2. □

Theorem P.101 (Jensen)**.** If ϕ is convex on an open interval containing the range of an integrable X, then $\phi(\mathsf{E}\,X) \leqslant \mathsf{E}[\phi(X)]$.

Proof. Let I be the open interval and $m = \mathsf{E}\,X$, which lies in I: if, say, $I = (\alpha, \beta)$ with $\beta < \infty$ and $m \geqslant \beta$, then $\beta - X > 0$ everywhere, so $\mathsf{E}[\beta - X] \geqslant 0$, while $\mathsf{E}[\beta - X] = \beta - m \leqslant 0$; hence $\mathsf{E}[\beta - X] = 0$, and Proposition P.82(iv) forces $\beta - X = 0$ almost surely, contradicting $\beta - X > 0$ as $\mathsf{P}(\Omega) = 1$. The same argument applied to $X - \alpha$ gives $m > \alpha$ when $\alpha > -\infty$.

Step 1: slopes. For $x < m < y$ in I write $m = \lambda x + (1-\lambda)y$ with $\lambda = (y-m)/(y-x) \in (0,1)$. Convexity gives $\phi(m) \leqslant \lambda\phi(x) + (1-\lambda)\phi(y)$, that is $\lambda\{\phi(m) - \phi(x)\} \leqslant (1-\lambda)\{\phi(y) - \phi(m)\}$, and since $\lambda/(1-\lambda) = (y-m)/(m-x)$,

$$\frac{\phi(m) - \phi(x)}{m - x} \leqslant \frac{\phi(y) - \phi(m)}{y - m} \qquad (x < m < y). \tag{P.15}$$

The argument of Lemma P.24, which uses convexity only on the segment involved, shows that the right slope $\{\phi(y) - \phi(m)\}/(y-m)$ is non-decreasing in $y > m$, and, applied to $t \mapsto \phi(m-t)$, that the left slope $\{\phi(m) - \phi(x)\}/(m-x)$ is non-decreasing in $x < m$.

Step 2: one-sided derivatives and a supporting line. As $y \downarrow m$ the right slopes decrease and are bounded below by any fixed left slope, by (P.15); so (as in the proof of Lemma P.24) they converge to their infimum $\phi'_+(m)$, which is finite. As $x \uparrow m$ the left slopes increase and are bounded

above by any right slope, so they converge to their supremum $\phi'_-(m)$, also finite. Passing to the limits in (P.15), $\phi'_-(m) \leqslant \phi'_+(m)$. For $c \in [\phi'_-(m), \phi'_+(m)]$, $\{\phi(y)-\phi(m)\}/(y-m) \geqslant \phi'_+(m) \geqslant c$ for $y > m$ and $\{\phi(m) - \phi(x)\}/(m-x) \leqslant \phi'_-(m) \leqslant c$ for $x < m$; multiplying by $y - m > 0$, respectively $m - x > 0$, both rearrange to

$$\phi(x) \geqslant \phi(m) + c(x - m) \qquad \text{for all } x \in I. \tag{P.16}$$

In the differentiable case this is Proposition P.22, with $c = \phi'(m)$.

Step 3: measurability of $\phi(X)$. The same construction at any point $z \in I$ shows ϕ is continuous at z: for $z < y < y_0$ in I, Step 1 at z gives $\phi'_+(z)(y-z) \leqslant \phi(y)-\phi(z) \leqslant \frac{\phi(y_0)-\phi(z)}{y_0-z}(y-z)$, so $\phi(y) \to \phi(z)$ as $y \downarrow z$, and symmetrically as $y \uparrow z$. Hence for real a the set $U_a = \{x \in I : \phi(x) > a\}$ is open (each of its points has a neighbourhood in I on which $\phi > a$), so $\{\phi(X) > a\} = X^{-1}(U_a) \in \mathcal{F}$, and $\phi(X)$ is measurable by Lemma P.67(ii).

Step 4: expectations. By (P.16) at $x = X$, $\phi(X) \geqslant \phi(m) + c(X - m)$, whose right side is integrable. So $\phi(X)^- \leqslant |\phi(m)| + |c|\,|X - m|$ is integrable and $\mathsf{E}[\phi(X)]$ exists in $(-\infty, \infty]$. If it is ∞ there is nothing to prove; otherwise $\phi(X)$ is integrable, and Theorem P.85(i),(ii) give $\mathsf{E}[\phi(X)] \geqslant \phi(m) + c(\mathsf{E}\,X - m) = \phi(m)$: taking expectations kills the linear term. □

Proposition P.102 (Variance under a linear map)**.** If $\mathbf{z}$ has covariance $\operatorname{var}(\mathbf{z})$, and $\mathbf{A}$, $\mathbf{b}$ are constant, then $\operatorname{var}(\mathbf{Az}+\mathbf{b}) = \mathbf{A}\operatorname{var}(\mathbf{z})\mathbf{A}'$.

Proof. With $\boldsymbol{\mu} = \mathsf{E}[\mathbf{z}]$, linearity of the expectation applied to each coordinate (Theorem P.85(i)) gives $\mathsf{E}[\mathbf{Az}+\mathbf{b}] = \mathbf{A}\boldsymbol{\mu}+\mathbf{b}$, so $\mathbf{Az}+\mathbf{b}-\mathsf{E}[\mathbf{Az}+\mathbf{b}] = \mathbf{A}(\mathbf{z}-\boldsymbol{\mu})$. Each entry of $\mathbf{A}(\mathbf{z}-\boldsymbol{\mu})(\mathbf{z}-\boldsymbol{\mu})'\mathbf{A}'$ is the fixed linear combination $\sum_{k,l} a_{ik}a_{jl}(z_k-\mu_k)(z_l-\mu_l)$ of integrable products, so linearity again gives

$$\operatorname{var}(\mathbf{Az}+\mathbf{b}) = \mathsf{E}\big[\mathbf{A}(\mathbf{z}-\boldsymbol{\mu})(\mathbf{z}-\boldsymbol{\mu})'\mathbf{A}'\big] = \mathbf{A}\,\mathsf{E}[(\mathbf{z}-\boldsymbol{\mu})(\mathbf{z}-\boldsymbol{\mu})']\,\mathbf{A}'. \qquad \square$$

Lemma P.103 (Expectation of a quadratic form)**.** If $\mathsf{E}[\mathbf{z}] = \boldsymbol{\mu}$ and $\operatorname{var}(\mathbf{z}) = \boldsymbol{\Sigma}$, and $\mathbf{A}$ is a constant square matrix, then $\mathsf{E}[\mathbf{z}'\mathbf{Az}] = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$.

Proof. $\mathbf{z}'\mathbf{Az} = \sum_{i,j} a_{ij}z_iz_j$, each z_iz_j being integrable as the second moments are finite (Lemma P.90), so by linearity (Theorem P.85(i)) $\mathsf{E}[\mathbf{z}'\mathbf{Az}] = \sum_{i,j} a_{ij}\,\mathsf{E}[z_iz_j]$; and $\Sigma_{ij} = \operatorname{cov}(z_i,z_j) = \mathsf{E}[(z_i-\mu_i)(z_j-\mu_j)] = \mathsf{E}[z_iz_j] - \mu_i\mu_j$, so $\mathsf{E}[z_iz_j] = \Sigma_{ij}+\mu_i\mu_j$. The first part is $\sum_{i,j} a_{ij}\Sigma_{ij} = \sum_i\sum_j a_{ij}\Sigma_{ji} = \sum_i(\mathbf{A}\boldsymbol{\Sigma})_{ii} = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma})$ by symmetry of $\boldsymbol{\Sigma}$, and the second is $\sum_{i,j}\mu_i a_{ij}\mu_j = \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$. $\square$

20 Independence and Borel–Cantelli

Definition P.104 (Independence)**.** Random vectors $X_1, \dots, X_k$ on one probability space are **independent** if $\mathsf{P}\{X_1 \in B_1, \dots, X_k \in B_k\} = \prod_{i \leqslant k} \mathsf{P}\{X_i \in B_i\}$ for all Borel sets $B_1, \dots, B_k$; equivalently, by Corollary P.98, their joint law is the product measure $\mu_{X_1} \otimes \cdots \otimes \mu_{X_k}$. An infinite family is independent if every finite subfamily is, and **independent of** N means that N and each finite subfamily, taken together, are independent.

Lemma P.105 (Disjoint functions of independent variables)**.** Let $\varepsilon_1, \dots, \varepsilon_N$ be independent, let U, V be disjoint subsets of $\{1, \dots, N\}$ and let f, g be measurable functions of $(\varepsilon_u)_{u \in U}$ and $(\varepsilon_v)_{v \in V}$ respectively. Then f and g are independent, and if both are integrable then so is fg, with $\mathsf{E}[fg] = \mathsf{E}[f]\,\mathsf{E}[g]$.

Proof. Write $\varepsilon_U = (\varepsilon_u)_{u \in U}$, $\varepsilon_V = (\varepsilon_v)_{v \in V}$, and $f = \tilde{f}(\varepsilon_U)$, $g = \tilde{g}(\varepsilon_V)$ with $\tilde{f}, \tilde{g}$ Borel; let μ_U, μ_V be the laws of ε_U, ε_V.

Step 1: the joint law of the two blocks is $\mu_U \otimes \mu_V$. The subfamilies $(\varepsilon_u)_{u \in U}$, $(\varepsilon_v)_{v \in V}$

and $(\varepsilon_u)_{u\in U\cup V}$ are independent (Definition P.104, taking $B_i = \mathbb{R}^{d_i}$ for the indices left out), so by Corollary P.98 μ_U, μ_V and the law $\mu_{U\cup V}$ of $(\varepsilon_U, \varepsilon_V)$ are the products of the laws μ_{ε_i} over U, V and $U \cup V$. For product sets $B_U \times B_V$ with $B_U = \prod_{u\in U} B_u$, $B_V = \prod_{v\in V} B_v$,

$$\mu_{U\cup V}(B_U \times B_V) = \prod_{i\in U\cup V} \mu_{\varepsilon_i}(B_i) = \mu_U(B_U)\,\mu_V(B_V) = (\mu_U \otimes \mu_V)(B_U \times B_V).$$

These sets form a π-system generating the Borel sets, so the uniqueness part of Corollary P.98 gives $\mu_{U\cup V} = \mu_U \otimes \mu_V$.

Step 2: independence. For Borel sets A, B, Theorem P.89 with $T = (\varepsilon_U, \varepsilon_V)$ and Step 1, then the Fubini–Tonelli theorem (Theorem P.97(i)), give

$$\begin{aligned}\mathsf{P}\{f \in A,\ g \in B\} &= \iint \mathbb{1}\{\tilde f(x) \in A\}\ \mathbb{1}\{\tilde g(y) \in B\}\,\mathrm{d}\mu_U(x)\,\mathrm{d}\mu_V(y)\\ &= \mu_U\{\tilde f \in A\}\,\mu_V\{\tilde g \in B\} = \mathsf{P}\{f \in A\}\ \mathsf{P}\{g \in B\},\end{aligned}$$

the inner integral in x being $\mu_U\{\tilde f \in A\}\,\mathbb{1}\{\tilde g(y) \in B\}$ by Lemma P.77(iv), and the last step being Theorem P.89 again.

Step 3: moments. In the same way, by Theorem P.89, Step 1 and Theorem P.97(i) applied to the non-negative $|\tilde{f}(x)\tilde{g}(y)|$,

$$\begin{aligned}\mathsf{E}\,|fg| &= \iint |\tilde{f}(x)|\,|\tilde{g}(y)|\,\mathrm{d}\mu_U(x)\,\mathrm{d}\mu_V(y)\\ &= \int |\tilde{f}|\,\mathrm{d}\mu_U \int |\tilde{g}|\,\mathrm{d}\mu_V = \mathsf{E}\,|f|\ \mathsf{E}\,|g| < \infty,\end{aligned}$$

the inner integral being $|\tilde{g}(y)| \int |\tilde{f}|\,\mathrm{d}\mu_U$ by Lemma P.77(iii). So fg is integrable, and Theorem P.97(ii) applied to $\tilde{f}(x)\tilde{g}(y)$, with the inner integral $\tilde{g}(y) \int \tilde{f}\,\mathrm{d}\mu_U$ by Theorem P.85(i), gives $\mathsf{E}[fg] = \mathsf{E}[f]\ \mathsf{E}[g]$. □

Lemma P.106 (Borel–Cantelli)**.** If $\sum_k \mathsf{P}(A_k) < \infty$ then $\mathsf{P}\{A_k \text{ infinitely often}\} = 0$.

Proof. Let $N = \sum_k \mathbb{1}_{A_k}$, a measurable function with values in $[0, \infty]$ counting the A_k that contain ω; so $\{A_k \text{ infinitely often}\} = \{N = \infty\}$. By Corollary P.80(i) and Lemma P.77(iv), $\mathsf{E}[N] = \sum_k \mathsf{P}(A_k) < \infty$, so $\mathsf{P}\{N = \infty\} = 0$ by Proposition P.82(v). □

21 Conditional Expectation

Definition P.107 (L^2 and its inner product)**.** $L^2(\Omega, \mathcal{F}, \mathsf{P})$ is the space of random variables with $\mathsf{E}[X^2] < \infty$, with $\langle X, Y\rangle = \mathsf{E}[XY]$, variables equal almost surely being identified.

Theorem P.108 (Conditional expectation in L^2)**.** Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-field and $X \in L^2(\mathcal{F})$. There is a $\mathcal{G}$-measurable $\mathsf{E}[X \mid \mathcal{G}] \in L^2(\mathcal{G})$, unique up to null sets, with

$$\mathsf{E}\big[(X - \mathsf{E}[X \mid \mathcal{G}])\, W\big] = 0 \qquad \text{for every } W \in L^2(\mathcal{G}), \tag{P.17}$$

and it minimises $\mathsf{E}[(X - W)^2]$ over $W \in L^2(\mathcal{G})$.

Proof. Write $\|Y\| = (\mathsf{E}\, Y^2)^{1/2}$. By Lemma P.90, $L^2(\mathcal{F})$ is closed under sums and scalar multiples, $\langle Y, Z\rangle = \mathsf{E}[YZ]$ is finite, bilinear and symmetric, and $\|Y + Z\| \leqslant \|Y\| + \|Z\|$; so $L^2(\mathcal{G})$, the $\mathcal{G}$-measurable members, is a subspace, in particular convex and non-empty.

Step 1: $L^2(\mathcal{G})$ is closed. Let $Y_n \in L^2(\mathcal{G})$ and $Y \in L^2(\mathcal{F})$ with $\mathsf{E}\,|Y_n - Y|^2 \to 0$. Choose $n_1 < n_2 < \cdots$ with $\mathsf{E}\,|Y_{n_k} - Y|^2 \leqslant 8^{-k}$; by Theorem P.100, $\mathsf{P}\{|Y_{n_k} - Y| > 2^{-k}\} \leqslant 4^k\, 8^{-k} =$

2^{-k}, so by Lemma P.106 almost surely $|Y_{n_k} - Y| \leqslant 2^{-k}$ for all large k, and $Y_{n_k} \to Y$ almost surely. The function $Y' = \limsup_k Y_{n_k}$ where this is finite, and 0 elsewhere, is $\mathcal{G}$-measurable by Proposition P.68(ii) and Lemma P.67(ii), and $Y' = Y$ almost surely; so Y, as an element of L^2 (variables equal almost surely being identified), lies in $L^2(\mathcal{G})$.

Step 2: a nearest point. Let $\delta = \inf\{\|X - W\| : W \in L^2(\mathcal{G})\}$ and choose $W_n \in L^2(\mathcal{G})$ with $\|X - W_n\| \to \delta$. The parallelogram law $\|A - B\|^2 = 2\|A\|^2 + 2\|B\|^2 - \|A + B\|^2$ holds in L^2 (expand both sides with the bilinearity of $\langle \cdot, \cdot \rangle$), and with $A = X - W_m$, $B = X - W_n$ and $\frac{1}{2}(W_n + W_m) \in L^2(\mathcal{G})$ it gives, exactly as in the proof of Theorem P.20, $\|W_n - W_m\|^2 \leqslant 2\|X - W_n\|^2 + 2\|X - W_m\|^2 - 4\delta^2$, which is small for large m, n. By Theorem P.91 there is $\hat{X} \in L^2(\mathcal{F})$ with $\|W_n - \hat{X}\| \to 0$; by Step 1, $\hat{X} \in L^2(\mathcal{G})$; and $\delta \leqslant \|X - \hat{X}\| \leqslant \|X - W_n\| + \|W_n - \hat{X}\| \to \delta$, so $\hat{X}$ minimises $\|X - W\|$, equivalently $\mathsf{E}[(X - W)^2]$, over $W \in L^2(\mathcal{G})$.

Step 3: the orthogonality (P.17)**.** For $W \in L^2(\mathcal{G})$ and real t, $\hat{X} + tW \in L^2(\mathcal{G})$, so

$$\|X - \hat{X}\|^2 \leqslant \|X - \hat{X} - tW\|^2 = \|X - \hat{X}\|^2 - 2t\langle X - \hat{X}, W \rangle + t^2\|W\|^2.$$

Hence $2t\langle X - \hat{X}, W \rangle \leqslant t^2\|W\|^2$ for every real t. Dividing by $t > 0$ and letting $t \downarrow 0$ gives $\langle X - \hat{X}, W \rangle \leqslant 0$; dividing by $t < 0$ and letting $t \uparrow 0$ gives $\langle X - \hat{X}, W \rangle \geqslant 0$. So $\mathsf{E}[(X - \hat{X})W] = 0$,

which is (P.17) with $\mathsf{E}[X \mid \mathcal{G}] = \hat{X}$.

Step 4: uniqueness. If $Y \in L^2(\mathcal{G})$ also satisfies (P.17), then subtracting the two identities gives $\mathsf{E}[(\hat{X} - Y)W] = 0$ for every $W \in L^2(\mathcal{G})$, and $W = \hat{X} - Y$ gives $\mathsf{E}[(\hat{X} - Y)^2] = 0$, so $\hat{X} = Y$ almost surely by Proposition P.82(iv). In particular every minimiser, which satisfies (P.17) by the argument of Step 3, equals $\hat{X}$ almost surely. □

Remark. Taking $W = \mathbb{1}_G$ for $G \in \mathcal{G}$ turns (P.17) into the familiar $\mathsf{E}[X\,\mathbb{1}_G] = \mathsf{E}[\mathsf{E}[X \mid \mathcal{G}]\,\mathbb{1}_G]$.

Lemma P.109 (Conditional expectation in L^1)**.** For integrable X there is an integrable $\mathcal{G}$-measurable $\mathsf{E}[X \mid \mathcal{G}]$, unique up to null sets, with $\mathsf{E}[XW] = \mathsf{E}\big[\mathsf{E}[X \mid \mathcal{G}]\,W\big]$ for every bounded $\mathcal{G}$-measurable W. For $X \in L^2$ it is the variable of Theorem P.108, and $X \geqslant 0$ implies $\mathsf{E}[X \mid \mathcal{G}] \geqslant 0$.

Proof. Positivity in L^2. For $X \in L^2$ with $X \geqslant 0$ put $\hat{X} = \mathsf{E}[X \mid \mathcal{G}]$ and $W = \max(-\hat{X}, 0) \in L^2(\mathcal{G})$. By (P.17), $-\mathsf{E}[W^2] = \mathsf{E}[\hat{X}W] = \mathsf{E}[XW] \geqslant 0$, so $W = 0$ and $\hat{X} \geqslant 0$. Since (P.17) is linear in X, uniqueness in Theorem P.108 makes $X \mapsto \mathsf{E}[X \mid \mathcal{G}]$ linear on L^2: $a\,\mathsf{E}[X \mid \mathcal{G}] + b\,\mathsf{E}[X' \mid \mathcal{G}]$ is $\mathcal{G}$-measurable, in L^2, and satisfies (P.17) for $aX + bX'$. Hence it is monotone: if $X \leqslant X'$ then $\mathsf{E}[X' \mid \mathcal{G}] - \mathsf{E}[X \mid \mathcal{G}] = \mathsf{E}[X' - X \mid \mathcal{G}] \geqslant 0$ almost surely by positivity.

Existence. Let $X \geqslant 0$ be integrable. The bounded variables $X \wedge m$ increase in m, so by monotonicity $Y_m = \mathsf{E}[X \wedge m \mid \mathcal{G}]$ increase almost surely to a $\mathcal{G}$-measurable limit Y with values in $[0, \infty]$. For bounded $\mathcal{G}$-measurable $W \geqslant 0$, after redefining every Y_m (and Y) as 0 on the $\mathcal{G}$-measurable null set $\bigcup_m \{Y_m < 0\} \cup \{Y_m > Y_{m+1}\}$, which changes no $\mathsf{E}[Y_m W]$ (Theorem P.85(iv)), monotone convergence (Theorem P.78) on both sides of $\mathsf{E}[(X \wedge m)W] = \mathsf{E}[Y_m W]$ (an identity from Theorem P.108, as $X \wedge m$ is bounded, hence in L^2, and $W \in L^2(\mathcal{G})$), applied to $(X \wedge m)W \uparrow XW$ and $Y_m W \uparrow YW$, gives $\mathsf{E}[XW] = \mathsf{E}[YW]$; with $W \equiv 1$, $\mathsf{E}\, Y = \mathsf{E}\, X < \infty$, so Y is finite almost surely (Proposition P.82(v)) and integrable; redefine it as 0 where it is infinite. A general bounded W is $W^+ - W^-$ with $W^\pm$ bounded, non-negative and $\mathcal{G}$-measurable, and subtracting the two finite identities for W^+ and W^- gives $\mathsf{E}[XW] = \mathsf{E}[YW]$. For general integrable X apply this to X^+ and X^-, obtaining $Y^{(+)}$ and $Y^{(-)}$, and take $\mathsf{E}[X \mid \mathcal{G}] = Y^{(+)} - Y^{(-)}$; subtracting the identities and using Theorem P.85(i) gives the defining identity.

Uniqueness. If Y and Y' both qualify, $W = \mathbb{1}\{Y > Y'\}$ gives $\mathsf{E}[(Y - Y')W] = 0$ with $(Y - Y')W \geqslant 0$, so $Y \leqslant Y'$ almost surely, and symmetrically. For $X \in L^2$ the variable of Theorem P.108 is integrable, as $|x| \leqslant 1 + x^2$, and satisfies the identity, bounded W being in $L^2(\mathcal{G})$; so the two agree. Positivity for $X \geqslant 0$ is built into the construction. □

Proposition P.110 (Tower property)**.** For integrable X, $\mathsf{E}\big[\mathsf{E}[X \mid \mathcal{G}]\big] = \mathsf{E}[X]$.

Proof. Take $W \equiv 1$ in Lemma P.109. □

Proposition P.111 (Conditional variance decomposition)**.** For $X \in L^2$, writing $\operatorname{var}(X \mid \mathcal{G}) = \mathsf{E}[X^2 \mid \mathcal{G}] - (\mathsf{E}[X \mid \mathcal{G}])^2$, with $\mathsf{E}[X^2 \mid \mathcal{G}]$ that of Lemma P.109,

$$\operatorname{var}(X) = \mathsf{E}\big[\operatorname{var}(X \mid \mathcal{G})\big] + \operatorname{var}\big(\mathsf{E}[X \mid \mathcal{G}]\big). \tag{P.18}$$

Proof. By Lemma P.109 with $W \equiv 1$ applied to X^2, and Proposition P.110 applied to X,

$$\operatorname{var}(X) = \mathsf{E}[X^2] - (\mathsf{E}\,X)^2 = \mathsf{E}\big[\mathsf{E}[X^2 \mid \mathcal{G}]\big] - \big(\mathsf{E}\big[\mathsf{E}[X \mid \mathcal{G}]\big]\big)^2.$$

Substituting $\mathsf{E}[X^2 \mid \mathcal{G}] = \operatorname{var}(X \mid \mathcal{G}) + (\mathsf{E}[X \mid \mathcal{G}])^2$, where both terms on the right are integrable ($\mathsf{E}[X \mid \mathcal{G}] \in L^2$ by Theorem P.108, and $\operatorname{var}(X \mid \mathcal{G})$ is a difference of integrable variables), splitting the expectation by Theorem P.85(i), and recognising $\mathsf{E}[(\mathsf{E}[X \mid \mathcal{G}])^2] - (\mathsf{E}[\mathsf{E}[X \mid \mathcal{G}]])^2$ as $\operatorname{var}(\mathsf{E}[X \mid \mathcal{G}])$ gives (P.18). □

Corollary P.112 (Random sums)**.** Let N be a non-negative integer random variable with $\mathsf{E}[N^2] < \infty$ and $Y_1, Y_2, \dots$ independent copies of a square-integrable Y, independent of N, and $S = \sum_{j \leqslant N} Y_j$. Then

$$\mathsf{E}[S] = \mathsf{E}[N]\ \mathsf{E}[Y], \qquad \operatorname{var}(S) = \mathsf{E}[N] \operatorname{var}(Y) + \operatorname{var}(N)(\mathsf{E}\, Y)^2.$$

Proof. Write $\mathsf{E}[\cdot \mid N] = \mathsf{E}[\cdot \mid \sigma(N)]$, and $S_m = \sum_{j \leqslant m} Y_j$, so that $S = S_m$ on $\{N = m\}$.

Step 1: the test functions. A bounded $\sigma(N)$-measurable W is constant on each event $\{N = m\}$: if $\omega, \omega' \in \{N = m\}$, the set $\{W = W(\omega)\}$ lies in $\sigma(N)$, whose members are the sets $\{N \in B\}$ with B Borel (these form a σ-field containing the generators of $\sigma(N)$), so it equals some $\{N \in B\}$; it contains ω, so $m \in B$, so it contains ω'. Hence $W = w(N)$ with $w(m)$ the value on $\{N = m\}$, and $|w| \leqslant c := \sup |W|$.

Step 2: moments of S_m on $\{N = m\}$. The variables $N, Y_1, \dots, Y_m$ are independent, so by Lemma P.105 $\mathbb{1}\{N = m\}$ is independent of S_m and of S_m^2, with $\mathsf{E}[S_m\, \mathbb{1}\{N = m\}] = \mathsf{E}[S_m]\, \mathsf{P}\{N = m\}$ and likewise for S_m^2 and $|S_m|$. By linearity $\mathsf{E}\, S_m = m\, \mathsf{E}\, Y$ and $\mathsf{E}\, |S_m| \leqslant m\, \mathsf{E}\, |Y|$, and, as

$\mathsf{E}[Y_iY_j] = (\mathsf{E}\,Y)^2$ for $i \neq j$ by Lemma P.105,

$$\mathsf{E}\,S_m^2 = \sum_i \mathsf{E}\,Y_i^2 + \sum_{i\neq j} \mathsf{E}[Y_iY_j] = m\,\mathsf{E}[Y^2] + m(m-1)(\mathsf{E}\,Y)^2 = m\,\mathrm{var}(Y) + m^2(\mathsf{E}\,Y)^2.$$

Step 3: summing over m. Pointwise $S\,w(N) = \sum_m w(m)S_m\,\mathbb{1}\{N=m\}$ and $S^2w(N) = \sum_m w(m)S_m^2\,\mathbb{1}\{N=m\}$, only one term being nonzero at each ω. The partial sums over $m \leqslant M$ are dominated by $G_1 = \sum_m c\,|S_m|\,\mathbb{1}\{N=m\}$, respectively $G_2 = \sum_m c\,S_m^2\,\mathbb{1}\{N=m\}$, which are integrable by Corollary P.80(i) and Step 2: $\mathsf{E}\,G_1 \leqslant c\,\mathsf{E}\,|Y| \sum_m m\,\mathsf{P}\{N=m\} = c\,\mathsf{E}\,|Y|\,\mathsf{E}\,N$ and $\mathsf{E}\,G_2 \leqslant c\{\mathsf{E}[Y^2]\,\mathsf{E}\,N + (\mathsf{E}\,Y)^2\,\mathsf{E}[N^2]\}$, finite as $\mathsf{E}\,N \leqslant 1+\mathsf{E}[N^2] < \infty$. Dominated convergence (Theorem P.86) lets the expectation pass through the sums, and Step 2 gives

$$\mathsf{E}[S\,w(N)] = \sum_m w(m)\,m\,\mathsf{E}[Y]\,\mathsf{P}\{N=m\} = \mathsf{E}\big[N\,\mathsf{E}[Y]\,w(N)\big],$$
$$\mathsf{E}[S^2w(N)] = \mathsf{E}\big[\{N\,\mathrm{var}(Y) + N^2(\mathsf{E}\,Y)^2\}w(N)\big],$$

the last equalities by dominated convergence (Theorem P.86) applied to the partial sums over $m \leqslant M$ of $w(m)\,m\,\mathsf{E}[Y]\,\mathbb{1}\{N=m\}$ and of $w(m)\{m\,\mathrm{var}(Y) + m^2(\mathsf{E}\,Y)^2\}\,\mathbb{1}\{N=m\}$, dominated

by the integrable $c|\mathsf{E}\,Y|N$ and $c\{N\,\mathrm{var}(Y)+N^2(\mathsf{E}\,Y)^2\}$.

Step 4: conclusion. With $w\equiv 1$ the second identity gives $\mathsf{E}[S^2]=\mathsf{E}\,N\,\mathrm{var}(Y)+\mathsf{E}[N^2](\mathsf{E}\,Y)^2<\infty$, so $S\in L^2$. The variables $N\,\mathsf{E}[Y]$ and $N\,\mathrm{var}(Y)+N^2(\mathsf{E}\,Y)^2$ are $\sigma(N)$-measurable and integrable, and by Step 1 the two identities hold for every bounded $\sigma(N)$-measurable W; so by the uniqueness in Lemma P.109, $\mathsf{E}[S\mid N]=N\,\mathsf{E}[Y]$ and $\mathsf{E}[S^2\mid N]=N\,\mathrm{var}(Y)+N^2(\mathsf{E}\,Y)^2$, whence $\mathrm{var}(S\mid N)=\mathsf{E}[S^2\mid N]-(\mathsf{E}[S\mid N])^2=N\,\mathrm{var}(Y)$. Proposition P.110 gives $\mathsf{E}\,S=\mathsf{E}[N\,\mathsf{E}\,Y]=\mathsf{E}\,N\;\mathsf{E}\,Y$, and Proposition P.111 gives $\mathrm{var}(S)=\mathsf{E}[N\,\mathrm{var}(Y)]+\mathrm{var}(N\,\mathsf{E}[Y])=\mathsf{E}[N]\,\mathrm{var}(Y)+\mathrm{var}(N)(\mathsf{E}\,Y)^2$, using $\mathrm{var}(aN)=a^2\,\mathrm{var}(N)$ for the constant $a=\mathsf{E}\,Y$. □

22 The Multivariate Normal Distribution

Definition P.113 (Multivariate normal)**.** X in $\mathbb{R}^d$ is $N(\boldsymbol{\mu},\boldsymbol{\Sigma})$ if $X=\boldsymbol{\mu}+\mathbf{A}Z$ where Z has independent standard normal coordinates and $\mathbf{A}\mathbf{A}'=\boldsymbol{\Sigma}$.

The parameters are the mean and the covariance: $\mathsf{E}\,X=\boldsymbol{\mu}$ and $\mathrm{var}(X)=\boldsymbol{\Sigma}$. Indeed, let $Z=(Z_1,\dots,Z_k)$. By Lemma P.99 and Remark 18(d),

$$\mathsf{E}[Z_j^2]=\int x^2\phi(x)\,\mathrm{d}x=1,\qquad \mathsf{E}\,|Z_j|=\int |x|\phi(x)\,\mathrm{d}x<\infty$$

since $|x| \leqslant 1 + x^2$; so $\mathsf{E}\, Z_j = \int x\phi(x)\,\mathrm{d}x$. As $x\phi(x)$ is odd and continuous, $\int x\phi(x)\,\mathbb{1}_{[-n,n]}(x)\,\mathrm{d}x = \int_{-n}^{n} x\phi(x)\,\mathrm{d}x = 0$ (a Riemann integral, Remark 18(c)), and Theorem P.86 with dominating function $|x|\phi(x)$ lets $n \to \infty$: $\mathsf{E}\, Z_j = 0$. For $i \neq j$, Lemma P.105 gives $\mathsf{E}[Z_i Z_j] = \mathsf{E}\, Z_i\ \mathsf{E}\, Z_j = 0$. Hence $\mathsf{E}\, Z = \mathbf{0}$ and $\mathrm{var}(Z) = \mathbf{I}_k$, and linearity (Theorem P.85(i)) gives $\mathsf{E}\, X = \boldsymbol{\mu} + \mathbf{A}\,\mathsf{E}\, Z = \boldsymbol{\mu}$ and Proposition P.102 gives $\mathrm{var}(X) = \mathbf{A}\mathbf{I}_k\mathbf{A}' = \boldsymbol{\Sigma}$.

Theorem P.114 (Closure under affine maps)**.** If $X \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\mathbf{B}$ is a constant matrix and $\mathbf{b}$ a constant vector, then $\mathbf{B}X + \mathbf{b} \sim N(\mathbf{B}\boldsymbol{\mu} + \mathbf{b},\ \mathbf{B}\boldsymbol{\Sigma}\mathbf{B}')$.

Proof. $\mathbf{B}X + \mathbf{b} = (\mathbf{B}\boldsymbol{\mu} + \mathbf{b}) + (\mathbf{B}\mathbf{A})Z$ and $(\mathbf{B}\mathbf{A})(\mathbf{B}\mathbf{A})' = \mathbf{B}\boldsymbol{\Sigma}\mathbf{B}'$, which is Definition P.113. □

23 The χ^2, t and F Distributions

Definition P.115. For independent $Z_1, \ldots, Z_r$ standard normal, $\sum_{j \leqslant r} Z_j^2$ has the **chi-square** law with r degrees of freedom, χ^2_r. If Z is standard normal and $V \sim \chi^2_\nu$ are independent then $Z/\sqrt{V/\nu}$ has **Student's** t law t_ν. If $V_1 \sim \chi^2_{r_1}$ and $V_2 \sim \chi^2_{r_2}$ are independent then $(V_1/r_1)/(V_2/r_2)$ has the F law F_{r_1,r_2}.

Corollary P.116 (The square of a t is an F)**.** If $T \sim t_\nu$ then $T^2 \sim F_{1,\nu}$.

Proof. By Definition P.115, $T = Z/\sqrt{V/\nu}$ with Z standard normal and $V \sim \chi^2_\nu$ independent. Squaring, $T^2 = (Z^2/1)/(V/\nu)$. Now $Z^2 \sim \chi^2_1$ by the same definition with $r = 1$, and it is independent of V by Lemma P.105 (a function of Z and a function of V, with Z, V independent), so the ratio is $F_{1,\nu}$ by the definition with $r_1 = 1$, $r_2 = \nu$. □

Lemma P.117 (Fourth moment of the normal)**.** If Z is standard normal then $\mathsf{E}[Z^4] = 3$, so $\operatorname{var}(Z^2) = 2$; for $\varepsilon \sim N(0, \sigma^2)$ with $\sigma > 0$, $\operatorname{var}(\varepsilon^2/\sigma^2) = 2$.

Proof. With $\phi'(z) = -z\phi(z)$ for the standard normal density, integration by parts on $[-R, R]$ gives

$$\int_{-R}^{R} z^4\phi(z)\,\mathrm{d}z = \int_{-R}^{R} z^3 \cdot z\phi(z)\,\mathrm{d}z = -\int_{-R}^{R} z^3\phi'(z)\,\mathrm{d}z = -2R^3\phi(R) + 3\int_{-R}^{R} z^2\phi(z)\,\mathrm{d}z,$$

as $z^3\phi(z)$ is odd. Let $R \to \infty$. By Theorem P.78 (the integrands $z^4\phi\,\mathbb{1}_{[-R,R]}$ and $z^2\phi\,\mathbb{1}_{[-R,R]}$ increase with R), the left side tends to $\int z^4\phi(z)\,\mathrm{d}z = \mathsf{E}[Z^4]$ (Lemma P.99) and the last integral

to $\int z^2\phi(z)\,\mathrm{d}z = 1$ (Remark 18(d)). The boundary term tends to 0: from the exponential series, whose terms are non-negative, $e^{R^2/2} \geqslant (R^2/2)^2/2 = R^4/8$, so $R^3\phi(R) \leqslant 8(2\pi)^{-1/2}/R \to 0$. Hence $\mathsf{E}[Z^4] = 3$, and $\mathrm{var}(Z^2) = \mathsf{E}[Z^4] - (\mathsf{E}[Z^2])^2 = 3 - 1 = 2$. If $\varepsilon \sim N(0, \sigma^2)$ then by Definition P.113 $\varepsilon = \sum_{j\leqslant k} a_j Z_j$ with $Z_1, \ldots, Z_k$ independent standard normal and $\sum_j a_j^2 = \sigma^2$, and $\mathsf{E}[\varepsilon^2] = \sigma^2$ by the paragraph after Definition P.113. Expand $\mathsf{E}[\varepsilon^4] = \sum_{i,j,l,m} a_i a_j a_l a_m \,\mathsf{E}[Z_i Z_j Z_l Z_m]$. Each product $Z_i Z_j Z_l Z_m$ is a product of powers Z_r^p ($p \leqslant 4$) of distinct coordinates, each integrable as $|Z_r|^p \leqslant 1 + Z_r^4$, so by Lemma P.105 (applied one factor at a time) its expectation is the product of the $\mathsf{E}[Z_r^p]$. If some index occurs exactly once, the factor $\mathsf{E}\, Z_r = 0$ (paragraph after Definition P.113) makes the term 0. What remains are the terms with $i = j = l = m$, each with expectation $\mathsf{E}[Z_i^4] = 3$, and, for each pair $r < s$, the 6 ordered arrangements of r, r, s, s, each with expectation $\mathsf{E}[Z_r^2]\,\mathsf{E}[Z_s^2] = 1$. So $\mathsf{E}[\varepsilon^4] = 3\sum_j a_j^4 + 6\sum_{r<s} a_r^2 a_s^2 = 3\big(\sum_j a_j^2\big)^2 = 3\sigma^4$, and $\mathrm{var}(\varepsilon^2/\sigma^2) = \mathsf{E}[\varepsilon^4]/\sigma^4 - (\mathsf{E}[\varepsilon^2]/\sigma^2)^2 = 3 - 1 = 2$. □

Proposition P.118 (Moments of χ^2)**.** If $V \sim \chi^2_r$ then $\mathsf{E}[V] = r$ and $\mathrm{var}(V) = 2r$.

Proof. $\mathsf{E}[Z^2] = \int z^2\phi(z)\,\mathrm{d}z = 1$ (Lemma P.99 and Remark 18(d)) and, by Lemma P.117, $\mathsf{E}[Z^4] = 3$, so $\operatorname{var}(Z^2) = 3 - 1 = 2$. With $V = \sum_{j\leqslant r} Z_j^2$ as in Definition P.115, linearity gives $\mathsf{E}\,V = \sum_j \mathsf{E}[Z_j^2] = r$. For $i \neq j$, Lemma P.105 gives $\mathsf{E}[Z_i^2 Z_j^2] = \mathsf{E}[Z_i^2]\,\mathsf{E}[Z_j^2]$, that is $\operatorname{cov}(Z_i^2, Z_j^2) = 0$, so expanding $\operatorname{var}(V) = \mathsf{E}[(\sum_j (Z_j^2 - 1))^2]$,

$$\operatorname{var}(V) = \sum_j \operatorname{var}(Z_j^2) + \sum_{i\neq j} \operatorname{cov}(Z_i^2, Z_j^2) = 2r.$$

□

24 Modes of Convergence

$C_b(\mathbb{R}^d)$ denotes the bounded continuous real functions on $\mathbb{R}^d$ and $C_K(\mathbb{R}^d)$ those continuous with compact support.

Definition P.119 (Modes of convergence)**.** Let $X, X_1, X_2, \ldots$ be random vectors in $\mathbb{R}^d$.

(i) $X_n \to X$ **almost surely**, written $X_n \to X$ a.s., if $\mathsf{P}\{\lim_n X_n = X\} = 1$.

(ii) $X_n \xrightarrow{p} X$ **in probability** if $\mathsf{P}\{|X_n - X| > \varepsilon\} \to 0$ for every $\varepsilon > 0$.

(iii) $X_n \xrightarrow{d} X$ **in distribution** if $\mathsf{E}[f(X_n)] \to \mathsf{E}[f(X)]$ for every $f \in C_b(\mathbb{R}^d)$.
Convergence in distribution depends on the X_n only through their laws, so we also write $\mu_n \to \mu$ **weakly** for probability measures.

Lemma P.120 (Almost sure subsequence)**.** If $X_n \xrightarrow{p} X$ then some subsequence converges to X almost surely.

Proof. Choose $n_1 < n_2 < \cdots$ with $\mathsf{P}\{|X_{n_k} - X| > 2^{-k}\} \leqslant 2^{-k}$ recursively: for fixed k, $\mathsf{P}\{|X_n - X| > 2^{-k}\} \to 0$ as $n \to \infty$, so some $n_k > n_{k-1}$ has this probability at most 2^{-k}. These probabilities are summable, so by Lemma P.106 the events $\{|X_{n_k} - X| > 2^{-k}\}$ occur finitely often almost surely; on that event $|X_{n_k} - X| \leqslant 2^{-k}$ for all large k, hence $X_{n_k} \to X$. □

Proposition P.121 (The hierarchy)**.** Almost sure convergence implies convergence in probability, which implies convergence in distribution.

Proof. a.s. $\Rightarrow$ probability. Fix $\varepsilon > 0$ and put $B_n = \{|X_n - X| > \varepsilon\}$. On the event $\{X_n \to X\}$ only finitely many B_n occur, so $\mathbb{1}_{B_n} \to 0$ almost surely; dominated convergence (Theorem P.86) gives $\mathsf{P}(B_n) = \mathsf{E}[\mathbb{1}_{B_n}] \to 0$.

probability $\Rightarrow$ distribution. Let $f \in C_b(\mathbb{R}^d)$ and suppose $\mathsf{E}[f(X_n)] \not\to \mathsf{E}[f(X)]$. Then some subsequence satisfies $|\mathsf{E}[f(X_{n_j})] - \mathsf{E}[f(X)]| \geqslant \delta$ for a $\delta > 0$. That subsequence still converges in probability, so by Lemma P.120 it has a further subsequence converging almost surely; along it $f(X_{n_{j_l}}) \to f(X)$ almost surely by continuity, and dominated convergence (Theorem P.86, with dominating constant $\sup|f|$) gives $\mathsf{E}[f(X_{n_{j_l}})] \to \mathsf{E}[f(X)]$, contradicting the choice of δ. □

Proposition P.122 (Convergence to a constant)**.** If $c \in \mathbb{R}^d$ is constant then $X_n \xrightarrow{d} c$ if and only if $X_n \xrightarrow{p} c$.

Proof. One direction is Proposition P.121. Conversely suppose $X_n \xrightarrow{d} c$ and fix $\varepsilon > 0$. The function $f(x) = \min\{1,\ |x - c|/\varepsilon\}$ is bounded and continuous with $f(c) = 0$, and $\mathbb{1}\{|x - c| >$

$\varepsilon\} \leqslant f(x)$ for every x. Hence

$$\mathsf{P}\{|X_n - c| > \varepsilon\} \leqslant \mathsf{E}[f(X_n)] \to \mathsf{E}[f(c)] = 0. \qquad \square$$

25 Characteristic Functions

Definition P.123 (Characteristic function)**.** The **characteristic function** of a random vector X in $\mathbb{R}^d$, equivalently of its law μ, is

$$\varphi_X(\theta) = \mathsf{E}\big[e^{i\langle\theta,X\rangle}\big] = \int_{\mathbb{R}^d} e^{i\langle\theta,x\rangle}\,\mathrm{d}\mu(x), \qquad \theta \in \mathbb{R}^d. \tag{P.19}$$

The second equality is Theorem P.89.

Proposition P.124 (Elementary properties)**.**

(i) $\varphi_X(0) = 1$ and $|\varphi_X(\theta)| \leqslant 1$ for all θ.

(ii) φ_X is uniformly continuous.

(iii) If X and Y are independent then $\varphi_{X+Y} = \varphi_X \varphi_Y$.

(iv) For a constant matrix $\mathbf{A}$ and vector $\mathbf{b}$, $\varphi_{\mathbf{A}X+\mathbf{b}}(\theta) = e^{i\langle\theta,\mathbf{b}\rangle}\varphi_X(\mathbf{A}'\theta)$.

Proof. (i): $\varphi_X(0) = \mathsf{E}[1] = 1$, and by Theorem P.85(iii), $|\varphi_X(\theta)| \leqslant \mathsf{E}\,|e^{i\langle\theta,X\rangle}| = 1$. For (ii),

$$|\varphi_X(\theta+\eta) - \varphi_X(\theta)| = \big|\mathsf{E}\big[e^{i\langle\theta,X\rangle}(e^{i\langle\eta,X\rangle}-1)\big]\big| \leqslant \mathsf{E}\big|e^{i\langle\eta,X\rangle}-1\big|,$$

a bound free of θ, by Theorem P.85(iii) and $|e^{i\langle\theta,X\rangle}| = 1$. The integrand is at most 2 and tends to 0 pointwise as $\eta \to 0$, so for any sequence $\eta_n \to 0$ dominated convergence (Theorem P.86, dominating constant 2) makes $\mathsf{E}\,|e^{i\langle\eta_n,X\rangle}-1| \to 0$; hence for $\varepsilon > 0$ there is $\delta > 0$ with $\mathsf{E}\,|e^{i\langle\eta,X\rangle}-1| < \varepsilon$ for $|\eta| < \delta$ (otherwise some η_n with $|\eta_n| < 1/n$ would violate it), and then $|\varphi_X(\theta+\eta)-\varphi_X(\theta)| < \varepsilon$ for every θ. For (iii), write $e^{i\langle\theta,X\rangle} = c_X + is_X$ with $c_X = \cos\langle\theta,X\rangle$, $s_X = \sin\langle\theta,X\rangle$, and likewise for Y. Then $e^{i\langle\theta,X+Y\rangle} = (c_Xc_Y - s_Xs_Y) + i(c_Xs_Y + s_Xc_Y)$, and each of the four products is a bounded function of X times a bounded function of Y, whose expectation factorises by Lemma P.105; reassembling, $\mathsf{E}[e^{i\langle\theta,X\rangle}e^{i\langle\theta,Y\rangle}] = \mathsf{E}[e^{i\langle\theta,X\rangle}]\,\mathsf{E}[e^{i\langle\theta,Y\rangle}]$. Finally (iv) follows from $\langle\theta,\mathbf{A}X+\mathbf{b}\rangle = \langle\mathbf{A}'\theta,X\rangle + \langle\theta,\mathbf{b}\rangle$. □

Proposition P.125 (The Gaussian characteristic function)**.** If $X \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ in $\mathbb{R}^d$ then $\varphi_X(\theta) = \exp\bigl(i\langle\theta, \boldsymbol{\mu}\rangle - \tfrac{1}{2}\theta'\boldsymbol{\Sigma}\theta\bigr)$.

Proof. Take first $d = 1$ and Z standard normal, and set $\varphi(t) = \mathsf{E}[e^{itZ}] = \int e^{itz}\phi(z)\,\mathrm{d}z = \int \cos(tz)\phi(z)\,\mathrm{d}z + i\int \sin(tz)\phi(z)\,\mathrm{d}z$ (Lemma P.99, which also gives the second equality in the next display). Differentiating under the integral sign (Theorem P.87, applied to the real and imaginary parts, with $I = \mathbb{R}$), justified because $|\partial_t \cos(tz)|, |\partial_t \sin(tz)| \leqslant |z|$ and $\int |z|\phi(z)\,\mathrm{d}z \leqslant \int (1+z^2)\phi(z)\,\mathrm{d}z = 2$,

$$\varphi'(t) = \mathsf{E}\bigl[iZe^{itZ}\bigr] = i\int ze^{itz}\phi(z)\,\mathrm{d}z = -i\int e^{itz}\phi'(z)\,\mathrm{d}z = i^2 t\int e^{itz}\phi(z)\,\mathrm{d}z = -t\varphi(t),$$

using $\phi'(z) = -z\phi(z)$ and then integration by parts. In detail, on $[-R, R]$, $-\int_{-R}^{R} e^{itz}\phi'(z)\,\mathrm{d}z = -\bigl[e^{itz}\phi(z)\bigr]_{-R}^{R} + it\int_{-R}^{R} e^{itz}\phi(z)\,\mathrm{d}z$; the boundary term has modulus at most $2\phi(R) \to 0$, and as $R \to \infty$ both integrals converge to the integrals over $\mathbb{R}$ by dominated convergence (Theorem P.86, with dominating functions $|z|\phi(z)$ and $\phi(z)$). Hence, by the product rule, $(\varphi(t)e^{t^2/2})' = \varphi'(t)e^{t^2/2} + t\varphi(t)e^{t^2/2} = 0$, and Corollary P.7, applied to the real and imaginary parts on $[0, t]$ or $[t, 0]$, gives

$\varphi(t)e^{t^2/2} = \varphi(0) = 1$, that is $\varphi(t) = e^{-t^2/2}$.
In general $X = \boldsymbol{\mu} + \mathbf{A}Z$ with $Z = (Z_1, \dots, Z_k)$ a vector of independent standard normals and $\mathbf{A}\mathbf{A}' = \boldsymbol{\Sigma}$. For $\eta \in \mathbb{R}^k$, $e^{i\langle \eta, Z\rangle} = \prod_j e^{i\eta_j Z_j}$. By induction on the number of factors, using Proposition P.124(iii) at each step (the sum $\sum_{j<l} \eta_j Z_j$ is a function of $Z_1, \dots, Z_{l-1}$, independent of $\eta_l Z_l$ by Lemma P.105), $\varphi_Z(\eta) = \prod_j \mathsf{E}[e^{i\eta_j Z_j}] = \prod_j e^{-\eta_j^2/2} = e^{-|\eta|^2/2}$, by the case $d = 1$. By (iv), $\varphi_X(\theta) = e^{i\langle \theta, \boldsymbol{\mu}\rangle}\varphi_Z(\mathbf{A}'\theta) = e^{i\langle \theta, \boldsymbol{\mu}\rangle} e^{-|\mathbf{A}'\theta|^2/2}$, with $|\mathbf{A}'\theta|^2 = \theta'\mathbf{A}\mathbf{A}'\theta = \theta'\boldsymbol{\Sigma}\theta$. □

26 Lévy's Continuity Theorem

The Gaussian density on $\mathbb{R}^d$ with variance σ^2 per coordinate is $f_\sigma(x) = (2\pi\sigma^2)^{-d/2} e^{-|x|^2/(2\sigma^2)}$. For $\psi \in C_K(\mathbb{R}^d)$ put

$$\psi_\sigma(x) = \int_{\mathbb{R}^d} f_\sigma(x - y)\, \psi(y)\, \mathrm{d}y. \tag{P.20}$$

Lemma P.126 (Gaussian Fourier inversion)**.** $f_\sigma(z) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{i\langle \theta, z\rangle} e^{-\sigma^2|\theta|^2/2}\, \mathrm{d}\theta$ for every $z \in \mathbb{R}^d$.

Proof. The integrand factors over coordinates, $e^{i\langle\theta,z\rangle}e^{-\sigma^2|\theta|^2/2} = \prod_{j\leqslant d} e^{i\theta_j z_j}e^{-\sigma^2\theta_j^2/2}$, and so does $f_\sigma(z) = \prod_j (2\pi\sigma^2)^{-1/2}e^{-z_j^2/(2\sigma^2)}$. The modulus of the integrand is $\prod_j e^{-\sigma^2\theta_j^2/2}$, integrable over $\mathbb{R}^d$ by Theorem P.97(i), so Theorem P.97(ii), applied $d-1$ times, writes the integral over $\mathbb{R}^d$ as the product of the d one-dimensional integrals. So it suffices to take $d = 1$. By Proposition P.125 applied to $N(0,\sigma^{-2})$, whose density is $(\sigma/\sqrt{2\pi})e^{-\sigma^2t^2/2}$,

$$\int_{\mathbb{R}} e^{itz}\,\frac{\sigma}{\sqrt{2\pi}}e^{-\sigma^2t^2/2}\,\mathrm{d}t = e^{-z^2/(2\sigma^2)},$$

the left side being the characteristic function of $N(0,\sigma^{-2})$ at z (Lemma P.99), and the density of $N(0,\sigma^{-2}) = \sigma^{-1}Z$ following from Remark 18(e) with $a = \sigma^{-1}$. Multiplying by $\sqrt{2\pi}/\sigma$ gives $\int_{\mathbb{R}} e^{itz}e^{-\sigma^2t^2/2}\,\mathrm{d}t = \sqrt{2\pi}\,\sigma^{-1}e^{-z^2/(2\sigma^2)} = 2\pi f_\sigma(z)$. □

Lemma P.127 (Mollification)**.** For $\psi \in C_K(\mathbb{R}^d)$, $\psi_\sigma \to \psi$ uniformly as $\sigma \to 0^+$.

Proof. Let $W = \sigma Z$, where $Z = (Z_1, \dots, Z_d)'$ has independent standard normal components Z_j. Its law is the product of the laws of the σZ_j (Definition P.104), whose densities are $(2\pi\sigma^2)^{-1/2} e^{-u^2/(2\sigma^2)}$ by Remark 18(e); by Theorem P.97(i) on boxes and Corollary P.98, W has density f_σ, so $\int f_\sigma = 1$ and $\int_B f_\sigma = \mathsf{P}\{W \in B\}$. Lebesgue measure is invariant under $u = x - y$: the image of Lebesgue measure under $y \mapsto x - y$ gives a half-open box the product of its side lengths (a point has Lebesgue measure $\lim_n 1/n = 0$, so open, closed and half-open intervals have the same length), hence equals Lebesgue measure by Corollary P.73, and Theorem P.89 gives $\int g(x-y)\,\mathrm{d}y = \int g(u)\,\mathrm{d}u$ for Borel $g \geqslant 0$. In particular $\int f_\sigma(x-y)\,\mathrm{d}y = 1$, so

$$|\psi(x) - \psi_\sigma(x)| \leqslant \int f_\sigma(x-y)\,|\psi(x) - \psi(y)|\,\mathrm{d}y = I_1 + I_2,$$

splitting the integral at $|y-x| \leqslant \delta$ and $|y-x| > \delta$. Given $\varepsilon > 0$ choose δ with $|\psi(x)-\psi(y)| \leqslant \varepsilon$ whenever $|x-y| \leqslant \delta$, possible because ψ is uniformly continuous: if the support of ψ lies in $\{|x| \leqslant R\}$, Theorem P.13 on the compact ball $K = \{|x| \leqslant R+1\}$ gives $\delta \leqslant 1$ that works for $x, y \in K$, and if $|x-y| \leqslant \delta$ with one of them outside K then both lie outside the support and $\psi(x) = \psi(y) = 0$. Then $I_1 \leqslant \varepsilon \int f_\sigma(x-y)\,\mathrm{d}y = \varepsilon$. For the second, $|\psi(x) - \psi(y)| \leqslant 2\|\psi\|_\infty$, the

substitution $u = x - y$, and Markov's inequality (Theorem P.100, applied to $|W|^2$ at the level δ^2), with $\mathsf{E}\,|W|^2 = \sum_j \sigma^2\,\mathsf{E}\,Z_j^2 = d\sigma^2$ (as $\mathsf{E}\,Z_j^2 = \int z^2\phi(z)\,\mathrm{d}z = 1$ by Lemma P.99 and Remark 18(d)), give

$$I_2 \leqslant 2\|\psi\|_\infty \int_{\{|u|>\delta\}} f_\sigma(u)\,\mathrm{d}u = 2\|\psi\|_\infty\,\mathsf{P}\{|W| > \delta\} \leqslant 2\|\psi\|_\infty \frac{\mathsf{E}\,|W|^2}{\delta^2} = 2\|\psi\|_\infty \frac{d\sigma^2}{\delta^2},$$

which tends to 0 as $\sigma \to 0$. Both bounds are free of x. □

Lemma P.128 (Smoothing identity)**.** Let μ be a probability on $\mathbb{R}^d$ with characteristic function φ_μ, and let $\psi \in C_K(\mathbb{R}^d)$. Then

$$\int_{\mathbb{R}^d} \psi_\sigma\,\mathrm{d}\mu = \int_{\mathbb{R}^d} \varphi_\mu(\theta)\,H(\theta)\,\mathrm{d}\theta, \qquad H(\theta) = (2\pi)^{-d} e^{-\sigma^2|\theta|^2/2} \int_{\mathbb{R}^d} \psi(y) e^{-i\langle\theta,y\rangle}\,\mathrm{d}y. \tag{P.21}$$

Proof. Substitute Lemma P.126 into (P.20):

$$\int \psi_\sigma\,\mathrm{d}\mu = \iiint (2\pi)^{-d} e^{-\sigma^2|\theta|^2/2} e^{i\langle\theta,x-y\rangle} \psi(y)\,\mathrm{d}\theta\,\mathrm{d}y\,\mathrm{d}\mu(x).$$

The modulus of the integrand is $(2\pi)^{-d}e^{-\sigma^2|\theta|^2/2}|\psi(y)|$, whose integral over θ, y and x is, by Theorem P.97(i), $(2\pi)^{-d}(2\pi/\sigma^2)^{d/2}\int|\psi(y)|\,\mathrm{d}y\cdot\mu(\mathbb{R}^d)$, finite because ψ is bounded with compact support and μ is a probability (the θ-integral is $\int e^{-\sigma^2|\theta|^2/2}\mathrm{d}\theta=(2\pi)^{d/2}\sigma^{-d}$, as $\int f_{1/\sigma}=1$). The integrand is jointly measurable, being continuous in (θ,y,x) times the Borel $\psi(y)$. Fubini's theorem (Theorem P.97) therefore permits integrating in x first, which produces $\varphi_\mu(\theta)$, and collecting the y-integral into H gives (P.21). □

Lemma P.129 (Compactly supported test functions suffice)**.** Let $\mu,\mu_1,\mu_2,\dots$ be probability measures on $\mathbb{R}^d$ with $\int\psi\,\mathrm{d}\mu_n\to\int\psi\,\mathrm{d}\mu$ for every $\psi\in C_K(\mathbb{R}^d)$. Then $\mu_n\to\mu$ weakly.

Proof. Fix $g\in C_b(\mathbb{R}^d)$ and $\varepsilon>0$. The sets $\{|x|>R\}$, $R=1,2,\dots$, decrease to $\emptyset$, and μ is finite, so continuity from above (Proposition P.65(v)) gives R with $\mu\{|x|>R\}<\varepsilon$. The function $\chi(x)=\min\{1,\max(0,R+1-|x|)\}$ is continuous, vanishes for $|x|\geqslant R+1$, and satisfies $\mathbb{1}\{|x|\leqslant R\}\leqslant\chi\leqslant 1$; so $\chi\in C_K(\mathbb{R}^d)$. Then $\int\chi\,\mathrm{d}\mu\geqslant\mu\{|x|\leqslant R\}\geqslant 1-\varepsilon$, and by hypothesis $\int\chi\,\mathrm{d}\mu_n\to\int\chi\,\mathrm{d}\mu$, so $\int(1-\chi)\,\mathrm{d}\mu_n=1-\int\chi\,\mathrm{d}\mu_n\to 1-\int\chi\,\mathrm{d}\mu\leqslant\varepsilon$.

Now $g\chi \in C_K(\mathbb{R}^d)$, so $\int g\chi \,\mathrm{d}\mu_n \to \int g\chi \,\mathrm{d}\mu$, while

$$\left|\int g(1-\chi)\,\mathrm{d}\mu_n\right| \leqslant \|g\|_\infty \int (1-\chi)\,\mathrm{d}\mu_n, \qquad \left|\int g(1-\chi)\,\mathrm{d}\mu\right| \leqslant \|g\|_\infty\, \varepsilon.$$

Writing $g = g\chi + g(1-\chi)$ and using the triangle inequality,

$$\left|\int g\,\mathrm{d}\mu_n - \int g\,\mathrm{d}\mu\right| \leqslant \left|\int g\chi\,\mathrm{d}\mu_n - \int g\chi\,\mathrm{d}\mu\right| + \|g\|_\infty \int (1-\chi)\,\mathrm{d}\mu_n + \|g\|_\infty\,\varepsilon,$$

and the lim sup of the right side is at most $0 + \|g\|_\infty\varepsilon + \|g\|_\infty\varepsilon$. So $\limsup_n \left|\int g\,\mathrm{d}\mu_n - \int g\,\mathrm{d}\mu\right| \leqslant 2\|g\|_\infty\varepsilon$, and ε was arbitrary. □

Theorem P.130 (Lévy's continuity theorem)**.** Let $\mu, \mu_1, \mu_2, \ldots$ be probabilities on $\mathbb{R}^d$. Then $\mu_n \to \mu$ weakly if and only if $\varphi_{\mu_n}(\theta) \to \varphi_\mu(\theta)$ for every $\theta \in \mathbb{R}^d$.

Proof. Necessity. $x \mapsto e^{i\langle\theta,x\rangle}$ has bounded continuous real and imaginary parts $\cos\langle\theta,x\rangle$ and $\sin\langle\theta,x\rangle$, so weak convergence applies to each, and $\varphi_{\mu_n}(\theta) = \int \cos\langle\theta,x\rangle\,\mathrm{d}\mu_n + i\int \sin\langle\theta,x\rangle\,\mathrm{d}\mu_n \to$

$\varphi_\mu(\theta)$.

Sufficiency. Assume pointwise convergence of the characteristic functions and fix $\psi \in C_K(\mathbb{R}^d)$; by Lemma P.129 it suffices to show $\int \psi \, \mathrm{d}\mu_n \to \int \psi \, \mathrm{d}\mu$. Let $\varepsilon > 0$. By Lemma P.127 choose σ with $\|\psi_\sigma - \psi\|_\infty \leqslant \varepsilon$; since every μ_n and μ is a probability,

$$\left|\int \psi \, \mathrm{d}\mu_n - \int \psi \, \mathrm{d}\mu\right| \leqslant 2\varepsilon + \left|\int \psi_\sigma \, \mathrm{d}\mu_n - \int \psi_\sigma \, \mathrm{d}\mu\right|.$$

For the remaining term use the smoothing identity (P.21) with the same σ for every n. The integrands $\varphi_{\mu_n} H$ converge pointwise to $\varphi_\mu H$ and are dominated in modulus by $(2\pi)^{-d} e^{-\sigma^2|\theta|^2/2} \int |\psi|$ (as $|\varphi_{\mu_n}| \leqslant 1$ by Proposition P.124(i)), which is integrable in θ. Dominated convergence (Theorem P.86) and (P.21) give $\int \psi_\sigma \, \mathrm{d}\mu_n \to \int \psi_\sigma \, \mathrm{d}\mu$, so the lim sup of the left side is at most 2ε. As $\varepsilon > 0$ was arbitrary, $\int \psi \, \mathrm{d}\mu_n \to \int \psi \, \mathrm{d}\mu$. □

Corollary P.131 (Uniqueness)**.** Two probabilities on $\mathbb{R}^d$ with the same characteristic function are equal.

Proof. Apply Theorem P.130 to the constant sequence $\mu_n = \nu$: if $\varphi_\nu = \varphi_\mu$ then $\nu \to \mu$ weakly, so $\int g\,\mathrm{d}\nu = \int g\,\mathrm{d}\mu$ for every $g \in C_b(\mathbb{R}^d)$. Let F be a non-empty closed set and $\rho(x) = \inf_{y\in F}|x-y|$; the triangle inequality gives $|\rho(x) - \rho(x')| \leqslant |x - x'|$, so ρ is continuous, and $\rho(x) = 0$ exactly when $x \in F$ (if $\rho(x) = 0$ there are $y_n \in F$ with $y_n \to x$, and F is closed). The functions $g_k(x) = \max\{0, 1 - k\rho(x)\}$ lie in $C_b(\mathbb{R}^d)$, take values in $[0,1]$, equal 1 on F, and tend to 0 off F; so $g_k \to \mathbb{1}_F$ pointwise, and dominated convergence (Theorem P.86, dominating constant 1) gives $\nu(F) = \lim_k \int g_k\,\mathrm{d}\nu = \lim_k \int g_k\,\mathrm{d}\mu = \mu(F)$; for $F = \emptyset$ both are 0. The closed sets form a π-system (a finite intersection of closed sets is closed), and they generate $\mathcal{B}(\mathbb{R}^d)$, since a σ-field containing them contains their complements, the open sets. Corollary P.73 gives $\nu = \mu$. □

Corollary P.132 (Standardising a normal variable)**.** If $X \sim N(m, \sigma^2)$ in $\mathbb{R}$ with $\sigma > 0$, then $(X - m)/\sigma$ is standard normal, and X has density $\sigma^{-1}\phi\big((x-m)/\sigma\big)$. In particular a $N(0,1)$ variable is standard normal.

Proof. By Theorem P.114 with $\mathbf{B} = 1/\sigma$ and $\mathbf{b} = -m/\sigma$, $W = (X - m)/\sigma \sim N(0,1)$, so by Proposition P.125 its characteristic function is $e^{-t^2/2}$. A standard normal Z is $N(0,1)$ by Definition P.113 with $\mathbf{A} = 1$, so the same proposition gives it the same characteristic function. By

Corollary P.131, W and Z have the same law: $\mathsf{P}\{W \in B\} = \mathsf{P}\{Z \in B\} = \int_B \phi(x)\,dx$ for every Borel B. So W has density ϕ, that is, W is standard normal (Remark 18(d)). Then $X = \sigma W + m$ has density $\sigma^{-1}\phi((x-m)/\sigma)$ by Remark 18(e). The last statement is the case $m = 0$, $\sigma = 1$. □

Corollary P.133 (Cramér–Wold)**.** $X_n \xrightarrow{d} X$ in $\mathbb{R}^d$ if and only if $\langle \theta, X_n \rangle \xrightarrow{d} \langle \theta, X \rangle$ in $\mathbb{R}$ for every $\theta \in \mathbb{R}^d$.

Proof. By Theorem P.130 the left side is equivalent to $\mathsf{E}[e^{i\langle \theta, X_n\rangle}] \to \mathsf{E}[e^{i\langle \theta, X\rangle}]$ for all θ. By the same theorem in $\mathbb{R}$, the right side is equivalent to $\mathsf{E}[e^{it\langle \theta, X_n\rangle}] \to \mathsf{E}[e^{it\langle \theta, X\rangle}]$ for all $t \in \mathbb{R}$ and all θ. Since $t\langle \theta, x\rangle = \langle t\theta, x\rangle$, this is the left condition at the point $t\theta$; conversely, the left condition at θ is the right condition with $t = 1$. So the two conditions are equivalent. □

Theorem P.134 (Uncorrelated blocks are independent)**.** Let $(\mathbf{U}', \mathbf{V}')'$ be jointly normal with cross-covariance $\mathbf{\Sigma}_{UV}$. Then $\mathbf{\Sigma}_{UV} = \mathbf{0}$ if and only if $\mathbf{U}$ and $\mathbf{V}$ are independent.

Proof. Write $\mathbf{U} \in \mathbb{R}^p$, $\mathbf{V} \in \mathbb{R}^q$. If $\mathbf{U}$ and $\mathbf{V}$ are independent, then for each coordinate pair Lemma P.105 gives $\mathsf{E}[U_i V_j] = \mathsf{E}[U_i]\,\mathsf{E}[V_j]$, that is $\operatorname{cov}(U_i, V_j) = 0$: independence always gives zero

covariance. Conversely, let $\boldsymbol{\Sigma}_{UV} = \mathbf{0}$. The vector $(\mathbf{U}', \mathbf{V}')'$ is $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = (\boldsymbol{\mu}_U', \boldsymbol{\mu}_V')'$ and $\boldsymbol{\Sigma}$ block diagonal with blocks $\boldsymbol{\Sigma}_{UU}$, $\boldsymbol{\Sigma}_{VV}$, so for $\theta = (\mathbf{s}', \mathbf{t}')'$, $\langle \theta, \boldsymbol{\mu} \rangle = \langle \mathbf{s}, \boldsymbol{\mu}_U \rangle + \langle \mathbf{t}, \boldsymbol{\mu}_V \rangle$ and $\theta' \boldsymbol{\Sigma} \theta = \mathbf{s}' \boldsymbol{\Sigma}_{UU} \mathbf{s} + \mathbf{t}' \boldsymbol{\Sigma}_{VV} \mathbf{t}$. By Proposition P.125 the joint characteristic function factors,

$$\exp\bigl(i\langle \mathbf{s}, \boldsymbol{\mu}_U \rangle + i\langle \mathbf{t}, \boldsymbol{\mu}_V \rangle - \tfrac{1}{2}\mathbf{s}' \boldsymbol{\Sigma}_{UU} \mathbf{s} - \tfrac{1}{2}\mathbf{t}' \boldsymbol{\Sigma}_{VV} \mathbf{t} \bigr) = \varphi_{\mathbf{U}}(\mathbf{s})\, \varphi_{\mathbf{V}}(\mathbf{t}),$$

where $\varphi_{\mathbf{U}}(\mathbf{s})$ is the joint characteristic function at $(\mathbf{s}', \mathbf{0}')'$, which is the first factor, and likewise $\varphi_{\mathbf{V}}(\mathbf{t})$. The product measure $\mu_{\mathbf{U}} \otimes \mu_{\mathbf{V}}$ (Corollary P.98) has characteristic function

$$\iint e^{i\langle \mathbf{s}, \mathbf{u} \rangle} e^{i\langle \mathbf{t}, \mathbf{v} \rangle}\, \mathrm{d}\mu_{\mathbf{U}}(\mathbf{u})\, \mathrm{d}\mu_{\mathbf{V}}(\mathbf{v}) = \varphi_{\mathbf{U}}(\mathbf{s})\, \varphi_{\mathbf{V}}(\mathbf{t})$$

by Theorem P.97(ii), the integrand having modulus 1. So the joint law of $(\mathbf{U}, \mathbf{V})$ and $\mu_{\mathbf{U}} \otimes \mu_{\mathbf{V}}$ have the same characteristic function, hence are equal by the uniqueness Corollary P.131; that is, $\mathbf{U}$ and $\mathbf{V}$ are independent (Definition P.104). □

Theorem P.135 (Chi-square quadratic forms)**.** If $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{A}$ is symmetric idempotent, then $\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi^2_r$ with $r = \operatorname{rank}(\mathbf{A}) = \operatorname{tr}(\mathbf{A})$.

Proof. By Theorem P.39 write $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ with $\mathbf{P}$ orthogonal; by Lemma P.46 the diagonal of $\mathbf{\Lambda}$ carries r ones and $n - r$ zeros, where $r = \operatorname{rank}(\mathbf{A}) = \operatorname{tr}(\mathbf{A})$; let J be the set of the r indices j with $\lambda_j = 1$. Put $\mathbf{w} = \mathbf{P}'\mathbf{z}$, which is $N(\mathbf{0}, \mathbf{P}'\mathbf{I}\mathbf{P}) = N(\mathbf{0}, \mathbf{I}_n)$ by Theorem P.114.

Each w_j is standard normal. By Proposition P.125, $\varphi_{w_j}(t) = \varphi_{\mathbf{w}}(t\mathbf{e}_j) = e^{-t^2/2}$, which is the characteristic function of $N(0, 1)$ (the case $d = 1$ of the same proposition), so w_j is standard normal by Corollary P.131.

The w_j are independent. For $k = 1, \ldots, n-1$ the vector $(w_k, w_{k+1}, \ldots, w_n)$ is a linear image of $\mathbf{w}$, hence jointly normal by Theorem P.114, with zero covariance between w_k and $(w_{k+1}, \ldots, w_n)$; Theorem P.134 makes w_k independent of $(w_{k+1}, \ldots, w_n)$. Claim: for $k = n, n-1, \ldots, 1$ and Borel $B_k, \ldots, B_n \subseteq \mathbb{R}$,

$$\mathsf{P}\{w_k \in B_k, \ldots, w_n \in B_n\} = \prod_{j=k}^{n} \mathsf{P}\{w_j \in B_j\}.$$

For $k = n$ this is trivial. If it holds for $k+1$, independence of w_k and $(w_{k+1}, \dots, w_n)$, applied to B_k and the Borel set $B_{k+1} \times \cdots \times B_n$, gives $\mathsf{P}\{w_k \in B_k, \dots, w_n \in B_n\} = \mathsf{P}\{w_k \in B_k\}\,\mathsf{P}\{w_{k+1} \in B_{k+1}, \dots, w_n \in B_n\}$, and the hypothesis for $k+1$ completes the step. The case $k = 1$ is Definition P.104.

Conclusion. Then $\mathbf{z}'\mathbf{A}\mathbf{z} = \mathbf{z}'\mathbf{P}\mathbf{\Lambda}\mathbf{P}'\mathbf{z} = \mathbf{w}'\mathbf{\Lambda}\mathbf{w} = \sum_j \lambda_j w_j^2 = \sum_{j \in J} w_j^2$, a sum of squares of r independent standard normals (a subfamily of an independent family is independent), which is χ^2_r by Definition P.115. □

Theorem P.136 (Cochran)**.** If $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{A}_1, \mathbf{A}_2$ are symmetric idempotent with $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$, then $\mathbf{z}'\mathbf{A}_1\mathbf{z}$ and $\mathbf{z}'\mathbf{A}_2\mathbf{z}$ are independent.

Proof. Stack $\mathbf{U} = \mathbf{A}_1\mathbf{z}$ and $\mathbf{V} = \mathbf{A}_2\mathbf{z}$ as $(\mathbf{U}', \mathbf{V}')' = (\mathbf{A}_1', \mathbf{A}_2')'\mathbf{z}$, one linear map of one normal vector, hence jointly normal by Theorem P.114; individual normality would not suffice. By Proposition P.102 the covariance of the stacked vector is $(\mathbf{A}_1', \mathbf{A}_2')'\mathbf{I}(\mathbf{A}_1', \mathbf{A}_2')$, whose off-diagonal block, the cross-covariance, is $\mathbf{A}_1\mathbf{I}\mathbf{A}_2' = \mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$ ($\mathbf{A}_2$ being symmetric), so Theorem P.134 makes $\mathbf{U}$ and $\mathbf{V}$ independent. Finally $\mathbf{z}'\mathbf{A}_i\mathbf{z} = \mathbf{z}'\mathbf{A}_i'\mathbf{A}_i\mathbf{z} = |\mathbf{A}_i\mathbf{z}|^2$, as $\mathbf{A}_i$ is symmetric and idem-

potent, so $\mathbf{z}'\mathbf{A}_1\mathbf{z} = |\mathbf{U}|^2$ and $\mathbf{z}'\mathbf{A}_2\mathbf{z} = |\mathbf{V}|^2$ are functions of $\mathbf{U}$ and of $\mathbf{V}$ alone, independent by Lemma P.105. □

Remark. The statement of Theorem P.136 is often called Craig's theorem; Cochran's theorem proper concerns a decomposition $\sum_i \mathbf{A}_i = \mathbf{I}_n$ whose ranks add up to n.

27 Portmanteau and the Continuous Mapping Theorem

Theorem P.137 (Portmanteau)**.** For probabilities μ, μ_n on $\mathbb{R}^d$ the following are equivalent.

(a) $\mu_n \to \mu$ weakly.

(b) $\liminf_n \int f \, d\mu_n \geqslant \int f \, d\mu$ for every lower semi-continuous f bounded below.

(c) $\int f \, d\mu_n \to \int f \, d\mu$ for every bounded measurable f whose set of discontinuity points is μ-null.

Proof. (c)⇒(a) is trivial, a continuous function having no discontinuities.

(a)⇒(b). A lower semi-continuous f bounded below is the increasing limit of the bounded contin-

uous functions $f_k(x) = \inf_y\{f(y) \wedge k + k|x-y|\}$ (Lemma P.12). For each k,

$$\int f_k \,\mathrm{d}\mu = \lim_n \int f_k \,\mathrm{d}\mu_n \leqslant \liminf_n \int f \,\mathrm{d}\mu_n$$

since $f_k \leqslant f$; letting $k \to \infty$ and using monotone convergence (Theorem P.78) on the left gives (b). (Monotone convergence is applied to $f_k + c \geqslant 0$ with $c = \max(0, -\inf f)$, as $f_k \geqslant \inf f \wedge k$ by Lemma P.12; then c is subtracted, using $\int (f+c)\,\mathrm{d}\mu = \int f\,\mathrm{d}\mu + c$, which follows from Theorem P.79 applied to $f + c = f^+ + (c - f^-)$ with $0 \leqslant c - f^- \leqslant c$.)

(b)⇒(c). If f is upper semi-continuous and bounded above, $-f$ is lower semi-continuous and bounded below, and (b) applied to $-f$ gives $\liminf_n \int (-f)\,\mathrm{d}\mu_n \geqslant \int (-f)\,\mathrm{d}\mu$, that is, the upper form $\limsup_n \int f\,\mathrm{d}\mu_n \leqslant \int f\,\mathrm{d}\mu$. Let f be bounded with μ-null discontinuity set, and take f_*, f^* from Lemma P.11: f_* is lower semi-continuous, f^* upper semi-continuous, both bounded (by the bounds of f), and $f_* \leqslant f \leqslant f^*$. They are Borel: if $f_*(x) > a$, lower semi-continuity gives $r > 0$ with $\inf_{|y-x|<r} f_*(y) > a$, so $\{f_* > a\}$ is open, and Lemma P.67(ii) applies; likewise $\{f^* < a\}$ is open. By Lemma P.11 they agree with f at every continuity point of f, hence off a μ-null set, so all three have the same μ-integral (Theorem P.85(iv)). Then, using $f_* \leqslant f \leqslant f^*$ and

Theorem P.85(ii) for the outer inequalities,

$$\int f\,\mathrm{d}\mu = \int f_*\,\mathrm{d}\mu \leqslant \liminf_n \int f_*\,\mathrm{d}\mu_n \leqslant \liminf_n \int f\,\mathrm{d}\mu_n,$$

$$\int f\,\mathrm{d}\mu = \int f^*\,\mathrm{d}\mu \geqslant \limsup_n \int f^*\,\mathrm{d}\mu_n \geqslant \limsup_n \int f\,\mathrm{d}\mu_n,$$

and the two together give $\int f\,\mathrm{d}\mu \leqslant \liminf_n \int f\,\mathrm{d}\mu_n \leqslant \limsup_n \int f\,\mathrm{d}\mu_n \leqslant \int f\,\mathrm{d}\mu$, so all are equal and $\int f\,\mathrm{d}\mu_n \to \int f\,\mathrm{d}\mu$ by Proposition P.9. □

Theorem P.138 (Continuous mapping)**.** Let $X_n \xrightarrow{d} X$ in $\mathbb{R}^d$ and let $g : \mathbb{R}^d \to \mathbb{R}^m$ be measurable with $\mathsf{P}\{X \in D_g\} = 0$, where D_g is the set of discontinuity points of g. Then $g(X_n) \xrightarrow{d} g(X)$. The same implication holds with $\xrightarrow{d}$ replaced throughout by $\xrightarrow{p}$.

Proof. Continuous g**.** If g is continuous everywhere the proof is one line: for $f \in C_b(\mathbb{R}^m)$ the

composition $f \circ g$ lies in $C_b(\mathbb{R}^d)$, so

$$\mathsf{E}[f(g(X_n))] = \mathsf{E}[(f \circ g)(X_n)] \to \mathsf{E}[(f \circ g)(X)] = \mathsf{E}[f(g(X))].$$

General g. Let $f \in C_b(\mathbb{R}^m)$. If g is continuous at x then so is $f \circ g$, so the discontinuity set of $f \circ g$ is contained in D_g (a Borel set by Remark 18(f)), which is μ-null for μ the law of X; and $f \circ g$ is bounded and measurable (Lemma P.67(iii)). Theorem P.137(c), applied to the laws μ_n of X_n, which converge weakly to μ, gives $\int f \circ g \, \mathrm{d}\mu_n \to \int f \circ g \, \mathrm{d}\mu$, that is $\mathsf{E}[(f \circ g)(X_n)] \to \mathsf{E}[(f \circ g)(X)]$ by Theorem P.89.

In probability. Suppose $X_n \xrightarrow{p} X$. By Lemma P.120 every subsequence has a further subsequence with $X_{n_k} \to X$ almost surely; on the event $\{X \notin D_g\}$, of probability one, $g(X_{n_k}) \to g(X)$ by continuity of g at X. Hence $g(X_{n_k}) \xrightarrow{p} g(X)$ along that sub-subsequence by Proposition P.121. Now fix $\varepsilon > 0$ and put $a_n = \mathsf{P}\{|g(X_n) - g(X)| > \varepsilon\}$. If $a_n \not\to 0$, some subsequence has $a_{n_j} \geqslant \delta > 0$ for all j; but that subsequence has a further subsequence along which $a \to 0$, by what was just shown, a contradiction. So $a_n \to 0$ for every ε: the whole sequence converges in probability. □

Remark. The measurable version is stated because the natural statement of Slutsky's theorem needs a map that is continuous only at the relevant limit.

28 Slutsky's Theorem

Theorem P.139 (Slutsky)**.** Let $X_n \xrightarrow{d} X$ in $\mathbb{R}^d$ and $Y_n \xrightarrow{p} c$ in $\mathbb{R}^m$ with c constant. Then

(i) $(X_n, Y_n) \xrightarrow{d} (X, c)$ in $\mathbb{R}^{d+m}$;

(ii) $h(X_n, Y_n) \xrightarrow{d} h(X, c)$ for every continuous $h : \mathbb{R}^d \times \mathbb{R}^m \to \mathbb{R}^l$;

(iii) in particular, with $m = d$, $X_n + Y_n \xrightarrow{d} X + c$; with $m = 1$, $Y_n X_n \xrightarrow{d} cX$; and if $c \neq 0$, $X_n / Y_n \xrightarrow{d} X/c$, with the convention $x/0 := 0$ on the event $\{Y_n = 0\}$.

Proof. **(i)** By Theorem P.130 it suffices to show the characteristic functions converge. For $\xi \in \mathbb{R}^d$ and $\eta \in \mathbb{R}^m$, split

$$\mathsf{E}\big[e^{i\langle \xi, X_n\rangle} e^{i\langle \eta, Y_n\rangle}\big] = \mathsf{E}\big[e^{i\langle \xi, X_n\rangle}\big] e^{i\langle \eta, c\rangle} + \mathsf{E}\big[e^{i\langle \xi, X_n\rangle}\big(e^{i\langle \eta, Y_n\rangle} - e^{i\langle \eta, c\rangle}\big)\big].$$

The first term tends to $\varphi_X(\xi) e^{i\langle \eta, c\rangle}$, which is the characteristic function of (X, c) at (ξ, η), because

$X_n \xrightarrow{d} X$. The second is bounded in modulus by $\mathsf{E}\, D_n$, $D_n = \left|e^{i\langle \eta, Y_n\rangle} - e^{i\langle \eta, c\rangle}\right|$ (Theorem P.85(iii), with $\left|e^{i\langle \xi, X_n\rangle}\right| = 1$). The function $y \mapsto \left|e^{i\langle \eta, y\rangle} - e^{i\langle \eta, c\rangle}\right|$ is bounded by 2 and continuous, vanishing at $y = c$; since $Y_n \xrightarrow{p} c$, Theorem P.138 gives $D_n \xrightarrow{p} 0$. Then $\mathsf{E}\, D_n \to 0$: otherwise some subsequence has $\mathsf{E}\, D_{n_j} \geqslant \delta > 0$; by Lemma P.120 a further subsequence converges to 0 almost surely, and dominated convergence (Theorem P.86, dominating constant 2) makes its expectations tend to 0, a contradiction. So the joint characteristic functions converge to that of (X, c).

(ii) is Theorem P.138 applied to h and the pair from (i), and **(iii)** consists of three instances: $h(x, y) = x + y$, $h(x, y) = yx$, and $h(x, y) = x/y$ for $y \neq 0$, $h(x, 0) = 0$, the last measurable and continuous at every (x, y) with $y \neq 0$ and so covered by the measurable version of Theorem P.138, the limit (X, c) having $c \neq 0$ with probability one. □

Remark. No independence between X_n and Y_n is assumed: in applications Y_n is typically an estimator computed from the same data as X_n, such as s^2 or $\hat{\tau}^2$.

Proposition P.140 (Quantiles of the t law)**.** For $0 < \alpha < 1$ let z and c_ν be the upper $\alpha/2$ quantiles of $N(0, 1)$ and of t_ν: $\mathsf{P}\{|Z| > z\} = \alpha = \mathsf{P}\{|T_\nu| > c_\nu\}$. Then $c_\nu \geqslant z$ for every $\nu \geqslant 1$,

and $c_\nu \to z$ as $\nu \to \infty$.

Proof. Let $T_\nu = Z/\sqrt{V/\nu}$ with Z standard normal and $V \sim \chi^2_\nu$ independent (Definition P.115), and let Φ and ϕ be the standard normal distribution function and density. For $x \geqslant 0$ and $w > 0$ put $g_x(w) = \mathsf{P}\{|Z| > x\sqrt{w}\} = 2\{1 - \Phi(x\sqrt{w})\}$. By independence and Fubini's theorem (Theorem P.97), with Theorem P.89, $\mathsf{P}\{|T_\nu| > x\} = \mathsf{P}\{|Z| > x\sqrt{V/\nu}\} = \mathsf{E}[g_x(V/\nu)]$, where $V > 0$ with probability one (as $\{V = 0\} = \{Z_1 = \cdots = Z_\nu = 0\}$ and $\mathsf{P}\{Z_1 = 0\} = 0$). As x increases, $g_x(w)$ decreases strictly and continuously from 1 to 0 for each $w > 0$, Φ being continuous and strictly increasing. So for $x < x'$, $g_x(V/\nu) - g_{x'}(V/\nu) > 0$ almost surely, and its expectation is positive by Proposition P.82(iv): the map $x \mapsto \mathsf{P}\{|T_\nu| > x\}$ is strictly decreasing. For $x_k \to x$ (or $x_k \to \infty$), $g_{x_k}(V/\nu) \to g_x(V/\nu)$ (or $\to 0$) almost surely, with dominating constant 1, so by dominated convergence (Theorem P.86) the map is continuous, with value 1 at $x = 0$ and limit 0 at ∞. Write $h(x) = \mathsf{P}\{|T_\nu| > x\}$ and let c be the supremum of $S = \{x \geqslant 0 : h(x) > \alpha\}$, which contains 0 and is bounded as $h \to 0 < \alpha$. If $h(c) > \alpha$, continuity gives $h > \alpha$ on some $[c, c + \eta)$, contradicting $c = \sup S$; if $h(c) < \alpha$, continuity gives $h < \alpha$ on some $(c - \eta, c]$, and as h decreases every point of S is at most $c - \eta$, again a contradiction. So $h(c) = \alpha$: $c_\nu = c$ exists, and it is

unique by strict monotonicity; likewise for z.

Step 1: $c_\nu \geqslant z$**.** Fix $x > 0$. With $\phi'(u) = -u\phi(u)$,

$$g_x'(w) = -\frac{x\,\phi(x\sqrt{w})}{\sqrt{w}}, \qquad g_x''(w) = x\,\phi(x\sqrt{w})\Big(\frac{x^2}{2\sqrt{w}} + \frac{1}{2w^{3/2}}\Big) > 0,$$

so g_x' is increasing by Corollary P.7. The same corollary applied to g_x gives, for $w, y > 0$, some ξ between w and y with $g_x(y) - g_x(w) = g_x'(\xi)(y - w)$; if $y > w$ then $g_x'(\xi) \geqslant g_x'(w)$, and if $y < w$ then $g_x'(\xi) \leqslant g_x'(w)$ while $y - w < 0$; either way $g_x(y) \geqslant g_x(w) + g_x'(w)(y - w)$, so g_x is convex on the open interval $(0, \infty)$ by Proposition P.22. As V/ν takes values in $(0, \infty)$ almost surely (redefine it as 1 on the null set $\{V = 0\}$), Jensen's inequality (Theorem P.101) and $\mathsf{E}[V/\nu] = 1$ (Proposition P.118), $\mathsf{P}\{|T_\nu| > x\} = \mathsf{E}[g_x(V/\nu)] \geqslant g_x(1) = \mathsf{P}\{|Z| > x\}$. At $x = z$ this gives $\mathsf{P}\{|T_\nu| > z\} \geqslant \alpha = \mathsf{P}\{|T_\nu| > c_\nu\}$, and since the left side is strictly decreasing in its argument, $z \leqslant c_\nu$.

Step 2: the limit. $\operatorname{var}(V/\nu) = 2/\nu$ by Proposition P.118, so $V/\nu \xrightarrow{p} 1$ by Chebyshev's inequality (Theorem P.100), hence $\sqrt{V/\nu} \xrightarrow{p} 1$ by Theorem P.138, and Theorem P.139(iii) gives $T_\nu \xrightarrow{d} Z$. The indicator of $(-\infty, x]$ is bounded and discontinuous only at x, a null set for $N(0,1)$,

so Theorem P.137(c) gives $\mathsf{P}\{T_\nu \leqslant x\} \to \Phi(x)$ for every x. Given $\varepsilon > 0$, $\Phi(z+\varepsilon) > 1-\alpha/2$, so $\mathsf{P}\{T_\nu \leqslant z+\varepsilon\} > 1-\alpha/2 = \mathsf{P}\{T_\nu \leqslant c_\nu\}$ for all large ν, the last equality by the symmetry of T_ν, since $\mathsf{P}\{T_\nu > x\} = \mathsf{E}[1-\Phi(x\sqrt{V/\nu})] = \mathsf{P}\{T_\nu < -x\}$, so that $\mathsf{P}\{T_\nu > c_\nu\} = \alpha/2$ and (as $\mathsf{P}\{T_\nu = c_\nu\} = 0$ by continuity of the tail function) $\mathsf{P}\{T_\nu \leqslant c_\nu\} = 1-\alpha/2$. Here $\Phi(z+\varepsilon) > \Phi(z) = 1-\alpha/2$ because $\mathsf{P}\{Z > z\} = \alpha/2$ by the same symmetry and Φ is strictly increasing. As $x \mapsto \mathsf{P}\{T_\nu \leqslant x\}$ is non-decreasing, $c_\nu \geqslant z+\varepsilon$ would give $\mathsf{P}\{T_\nu \leqslant c_\nu\} \geqslant \mathsf{P}\{T_\nu \leqslant z+\varepsilon\} > 1-\alpha/2$; hence $c_\nu < z+\varepsilon$. With Step 1, $z \leqslant c_\nu < z+\varepsilon$ for all large ν, and $\varepsilon > 0$ is arbitrary. □

29 The Lindeberg–Feller Central Limit Theorem

A **triangular array** is a doubly indexed collection $\{\xi_{nj} : 1 \leqslant j \leqslant k_n\}$ of random variables, independent within each row n. Nothing is assumed across rows, and the row entries need not be identically distributed.

Theorem P.141 (Lindeberg–Feller)**.** Let $\{\xi_{nj}\}$ be a triangular array with $\mathsf{E}[\xi_{nj}] = 0$ and

$\sum_j \operatorname{var}(\xi_{nj}) \to 1$. If the **Lindeberg condition**

$$\sum_j \mathsf{E}\big[\xi_{nj}^2\,;\ |\xi_{nj}| > \varepsilon\big] \ \longrightarrow\ 0 \qquad \text{for every } \varepsilon > 0 \tag{P.22}$$

holds, then $\sum_j \xi_{nj} \xrightarrow{d} N(0,1)$ and $\sup_j \operatorname{var}(\xi_{nj}) \to 0$.

Proof. Write $c_{nj} = \mathsf{E}[\xi_{nj}^2]$ and $c_n = \sum_j c_{nj} \to 1$.

Step 1: the variances are uniformly negligible. For any $\varepsilon > 0$, splitting at $|\xi_{nj}| \leqslant \varepsilon$, where $\xi_{nj}^2 \leqslant \varepsilon^2$, so that $c_{nj} = \mathsf{E}[\xi_{nj}^2; |\xi_{nj}| \leqslant \varepsilon] + \mathsf{E}[\xi_{nj}^2; |\xi_{nj}| > \varepsilon]$,

$$\sup_j c_{nj} \leqslant \varepsilon^2 + \sup_j \mathsf{E}\big[\xi_{nj}^2; |\xi_{nj}| > \varepsilon\big] \leqslant \varepsilon^2 + \sum_j \mathsf{E}\big[\xi_{nj}^2; |\xi_{nj}| > \varepsilon\big].$$

Letting $n \to \infty$, (P.22) gives $\limsup_n \sup_j c_{nj} \leqslant \varepsilon^2$, and as $\varepsilon > 0$ is arbitrary, $\sup_j c_{nj} \to 0$. This is the second conclusion.

Step 2: a Gaussian comparison array. Let Z_{nj} be independent standard normal variables

(on a product space, Remark 18(c)) and $\zeta_{nj} = c_{nj}^{1/2} Z_{nj}$, which are independent and $N(0, c_{nj})$ by Definition P.113; their characteristic functions are $\psi_{nj}(t) = e^{-c_{nj}t^2/2}$ by Proposition P.125. By Proposition P.124(iii), applied $k_n - 1$ times (each partial sum is independent of the next summand by Lemma P.105), $\zeta_n = \sum_j \zeta_{nj}$ has characteristic function $\prod_j \psi_{nj}(t) = e^{-c_n t^2/2}$, so ζ_n is $N(0, c_n)$ by Corollary P.131. Since $c_n \to 1$, $e^{-c_n t^2/2} \to e^{-t^2/2}$. In the same way $\sum_j \xi_{nj}$ has characteristic function $\prod_j \varphi_{nj}(t)$. So if, for each fixed t,

$$\left| \prod_j \varphi_{nj}(t) - \prod_j \psi_{nj}(t) \right| \longrightarrow 0, \tag{P.23}$$

φ_{nj} and ψ_{nj} denoting the characteristic functions of ξ_{nj} and ζ_{nj}, then $\prod_j \varphi_{nj}(t) \to e^{-t^2/2}$, the characteristic function of $N(0,1)$, and Theorem P.130 gives $\sum_j \xi_{nj} \xrightarrow{d} N(0,1)$. It remains to prove (P.23).

Step 3: reduce to individual terms. Characteristic functions have modulus at most 1, so Lemma P.16 bounds the left side of (P.23) by $\sum_j |\varphi_{nj}(t) - \psi_{nj}(t)|$, and inserting $1 - \frac{1}{2}t^2 c_{nj}$ and

using the triangle inequality,

$$\Big|\prod_j \varphi_{nj} - \prod_j \psi_{nj}\Big| \leqslant \sum_j \big|\varphi_{nj}(t) - 1 + \tfrac{1}{2}t^2 c_{nj}\big| + \sum_j \big|\psi_{nj}(t) - 1 + \tfrac{1}{2}t^2 c_{nj}\big|. \tag{P.24}$$

Step 4: bound each term. Both ξ_{nj} and ζ_{nj} have mean zero and variance c_{nj}, so consider a generic such variable ξ. Lemma P.15 with $m = 2$ at the point $t\xi$ gives $|e^{it\xi} - 1 - it\xi + \frac{1}{2}t^2\xi^2| \leqslant \min\{t^2\xi^2, |t|^3|\xi|^3/6\}$. Using $\mathsf{E}[\xi] = 0$, linearity (Theorem P.85(i), all terms being integrable as $\mathsf{E}\,\xi^2 < \infty$) and Theorem P.85(iii),

$$\big|\mathsf{E}[e^{it\xi}] - 1 + \tfrac{1}{2}t^2\,\mathsf{E}[\xi^2]\big| = \big|\mathsf{E}\big[e^{it\xi} - 1 - it\xi + \tfrac{1}{2}t^2\xi^2\big]\big| \leqslant \mathsf{E}\Big[\min\Big\{t^2\xi^2,\ \tfrac{|t|^3|\xi|^3}{6}\Big\}\Big] \lesssim_t \mathsf{E}\big[\xi^2(1 \wedge |\xi|)\big],$$

where $\lesssim_t$ means $\leqslant$ up to the factor $C_t = t^2 + |t|^3$, depending on t alone: if $|\xi| \leqslant 1$ the minimum is at most $|t|^3\xi^2|\xi| \leqslant C_t\xi^2(1 \wedge |\xi|)$, and if $|\xi| > 1$ it is at most $t^2\xi^2 \leqslant C_t\xi^2(1 \wedge |\xi|)$. Applying this to each ξ_{nj} and each ζ_{nj} turns (P.24) into

$$\Big|\prod_j \varphi_{nj} - \prod_j \psi_{nj}\Big| \lesssim_t \sum_j \mathsf{E}\big[\xi_{nj}^2(1 \wedge |\xi_{nj}|)\big] + \sum_j \mathsf{E}\big[\zeta_{nj}^2(1 \wedge |\zeta_{nj}|)\big].$$

Step 5: both sums vanish. For the first, split at ε:

$$\sum_j \mathsf{E}\big[\xi_{nj}^2(1 \wedge |\xi_{nj}|)\big] \leqslant \varepsilon \sum_j c_{nj} + \sum_j \mathsf{E}\big[\xi_{nj}^2; |\xi_{nj}| > \varepsilon\big],$$

because $\xi^2(1 \wedge |\xi|) \leqslant \varepsilon\xi^2$ on $\{|\xi| \leqslant \varepsilon\}$ and $\leqslant \xi^2$ on $\{|\xi| > \varepsilon\}$. Its lim sup is at most $\varepsilon \cdot 1$ by $c_n \to 1$ and (P.22), and $\varepsilon > 0$ is arbitrary, so it tends to 0. For the second, bound $1 \wedge |\zeta| \leqslant |\zeta|$ and use the third absolute moment of a centred Gaussian, $\mathsf{E}\,|\zeta_{nj}|^3 = c_{nj}^{3/2}\,\mathsf{E}\,|Z|^3$ with Z standard normal (by Step 2, $\zeta_{nj} = c_{nj}^{1/2} Z_{nj}$, and $\mathsf{E}\,|Z_{nj}|^3 = \int |z|^3 \phi(z)\,\mathrm{d}z = \mathsf{E}\,|Z|^3$ by Lemma P.99), where $\mathsf{E}\,|Z|^3 \leqslant \mathsf{E}[1 + Z^4] = 4$ by Lemma P.117:

$$\sum_j \mathsf{E}\big[\zeta_{nj}^2(1 \wedge |\zeta_{nj}|)\big] \leqslant \mathsf{E}\,|Z|^3 \sum_j c_{nj}^{3/2} \leqslant \mathsf{E}\,|Z|^3\; c_n \sup_j c_{nj}^{1/2} \;\longrightarrow\; 0$$

by Step 1 and $c_n \to 1$, using $c_{nj}^{3/2} = c_{nj}\, c_{nj}^{1/2} \leqslant c_{nj} \sup_i c_{ni}^{1/2}$. Hence (P.23) holds, and the theorem follows. □

Corollary P.142 (Lindeberg–Lévy)**.** If $X_1, X_2, \dots$ are independent and identically distributed with mean μ and finite variance $\sigma^2 > 0$, then $n^{-1/2}\sum_{i\leqslant n}(X_i-\mu)/\sigma \xrightarrow{d} N(0,1)$.

Proof. Put $\xi_{nj} = (X_j-\mu)/(\sigma\sqrt{n})$ for $j \leqslant n$. These are independent within a row with mean 0 and $\sum_j \operatorname{var}(\xi_{nj}) = 1$ exactly. Writing $Y = (X_1-\mu)/\sigma$, the Lindeberg sum is

$$\sum_{j=1}^{n} \mathsf{E}\big[\xi_{nj}^2;\, |\xi_{nj}| > \varepsilon\big] = n \cdot \frac{1}{n}\,\mathsf{E}\big[Y^2;\, |Y| > \varepsilon\sqrt{n}\big] = \mathsf{E}\big[Y^2;\, |Y| > \varepsilon\sqrt{n}\big],$$

the $Y_j = (X_j-\mu)/\sigma$ having the law of Y. This tends to 0 by dominated convergence (Theorem P.86): $Y^2\,\mathbb{1}\{|Y| > \varepsilon\sqrt{n}\} \to 0$ pointwise, Y being finite, and it is dominated by Y^2, with $\mathsf{E}[Y^2] = 1$ finite. Theorem P.141 gives $\sum_j \xi_{nj} = n^{-1/2}\sum_{i\leqslant n}(X_i-\mu)/\sigma \xrightarrow{d} N(0,1)$. □

Corollary P.143 (Weighted sums)**.** Let $\varepsilon_1, \varepsilon_2, \dots$ be independent with mean 0, common variance

$\tau^2 > 0$ and $\sup_i \mathsf{E}[\varepsilon_i^2; |\varepsilon_i| > a] \to 0$ as $a \to \infty$; let $\mathbf{z}_i \in \mathbb{R}^p$ be non-random with

$$n^{-1} \sum_{i \leqslant n} \mathbf{z}_i \mathbf{z}_i' \to \mathbf{Q} \text{ positive definite}, \qquad n^{-1} \max_{i \leqslant n} |\mathbf{z}_i|^2 \to 0.$$

Then $n^{-1/2} \sum_{i \leqslant n} \mathbf{z}_i \varepsilon_i \xrightarrow{d} N(\mathbf{0},\ \tau^2 \mathbf{Q})$.

Proof. By Cramér–Wold (Corollary P.133) it suffices to treat $S_n(\theta) = n^{-1/2} \sum_i \langle \theta, \mathbf{z}_i \rangle \varepsilon_i$ for fixed $\theta \neq \mathbf{0}$. Put $a_{ni} = \langle \theta, \mathbf{z}_i \rangle / \sqrt{n}$ and $\xi_{ni} = a_{ni} \varepsilon_i / v_n$ with $v_n^2 = \tau^2 n^{-1} \sum_i \langle \theta, \mathbf{z}_i \rangle^2 \to \tau^2 \theta' \mathbf{Q} \theta > 0$. Then $\mathsf{E}[\xi_{ni}] = 0$ and $\sum_i \operatorname{var}(\xi_{ni}) = \sum_i a_{ni}^2 \tau^2 / v_n^2 = 1$ (the limit of v_n^2 is $\tau^2 \theta' (n^{-1} \sum_i \mathbf{z}_i \mathbf{z}_i') \theta \to \tau^2 \theta' \mathbf{Q} \theta$, positive as $\mathbf{Q}$ is positive definite, so $v_n > 0$ for large n). By Theorem P.36, $|a_{ni}| \leqslant |\theta|\, n^{-1/2} \max_i |\mathbf{z}_i| =: |\theta|\, \delta_n$ with $\delta_n \to 0$, so $\{|\xi_{ni}| > \varepsilon\} = \{|\varepsilon_i| > \varepsilon v_n / |a_{ni}|\} \subseteq \{|\varepsilon_i| > \varepsilon v_n / (|\theta| \delta_n)\}$ and

$$\sum_i \mathsf{E}\big[\xi_{ni}^2; |\xi_{ni}| > \varepsilon\big] \leqslant \frac{1}{v_n^2} \sum_i a_{ni}^2\, \mathsf{E}\big[\varepsilon_i^2;\ |\varepsilon_i| > \varepsilon v_n / (|\theta| \delta_n)\big] \leqslant \frac{\sum_i a_{ni}^2}{v_n^2} \sup_i \mathsf{E}\big[\varepsilon_i^2; |\varepsilon_i| > b_n\big]$$

with $b_n = \varepsilon v_n/(|\theta|\delta_n) \to \infty$. The prefactor tends to $1/\tau^2$ and the supremum to 0, so the Lindeberg condition holds and Theorem P.141 gives $S_n(\theta)/v_n \xrightarrow{d} N(0,1)$. As $v_n \to v = \tau(\theta'\mathbf{Q}\theta)^{1/2}$, a constant, Theorem P.139(iii) gives $S_n(\theta) = v_n \cdot S_n(\theta)/v_n \xrightarrow{d} vZ \sim N(0, \tau^2\theta'\mathbf{Q}\theta)$, which by Theorem P.114 is the law of $\langle \theta, V\rangle$ for $V \sim N(\mathbf{0}, \tau^2\mathbf{Q})$. For $\theta = \mathbf{0}$ both sides are 0. Corollary P.133 completes the proof. □

Remark. The condition $n^{-1}\max_i |\mathbf{z}_i|^2 \to 0$ says no single observation dominates the design. In regression terms it is implied by $\max_i h_{ii} \to 0$, where $\mathbf{Z}$ has rows $\mathbf{z}_i'$ and $h_{ii} = \mathbf{z}_i'(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{z}_i$: by Corollary P.40 applied to $(\mathbf{Z}'\mathbf{Z})^{-1}$, $h_{ii} \geqslant |\mathbf{z}_i|^2/\lambda_{\max}(\mathbf{Z}'\mathbf{Z})$, and $\lambda_{\max}(\mathbf{Z}'\mathbf{Z})/n \to \lambda_{\max}(\mathbf{Q})$ because, as in the proof of Lemma P.43, $|\mathbf{v}'(n^{-1}\mathbf{Z}'\mathbf{Z} - \mathbf{Q})\mathbf{v}| \to 0$ uniformly over $|\mathbf{v}| = 1$; so $n^{-1}\max_i |\mathbf{z}_i|^2 \leqslant \max_i h_{ii}\, \lambda_{\max}(\mathbf{Z}'\mathbf{Z})/n \to 0$. Without it one observation can carry a non-vanishing share of the sum, the Lindeberg condition fails, and the limit, if there is one, need not be normal.

Corollary P.144 (Bounded summands with unequal variances)**.** Let $\mathbf{W}_1, \mathbf{W}_2, \ldots$ be independent random vectors in $\mathbb{R}^p$ with $\mathsf{E}[\mathbf{W}_i] = \mathbf{0}$ and $|\mathbf{W}_i| \leqslant M$ almost surely for a constant M free of i,

and suppose

$$\mathbf{F}_n = \frac{1}{n}\sum_{i\leqslant n} \operatorname{var}(\mathbf{W}_i) \;\longrightarrow\; \mathbf{F} \quad \text{positive definite.}$$

Then $n^{-1/2}\sum_{i\leqslant n}\mathbf{W}_i \xrightarrow{d} N(\mathbf{0},\mathbf{F})$.

Proof. By Cramér–Wold (Corollary P.133) fix $\theta \neq \mathbf{0}$ and put $\xi_{ni} = \langle \theta, \mathbf{W}_i\rangle/(\sqrt{n}\,v_n)$ with $v_n^2 = \theta'\mathbf{F}_n\theta \to \theta'\mathbf{F}\theta > 0$. Then $\mathsf{E}[\xi_{ni}] = 0$ and $\sum_i \operatorname{var}(\xi_{ni}) = \theta'\mathbf{F}_n\theta/v_n^2 = 1$.
By Cauchy–Schwarz (Theorem P.36), $|\xi_{ni}| \leqslant |\theta| M/(\sqrt{n}\,v_n)$, which tends to 0 uniformly in i. So for each $\varepsilon > 0$ the event $\{|\xi_{ni}| > \varepsilon\}$ has probability 0 for all large n and every i, and the Lindeberg sum (P.22) is exactly 0 (each expectation is over an event of probability 0). Theorem P.141 gives $\sum_i \xi_{ni} \xrightarrow{d} N(0,1)$, and with $v_n \to (\theta'\mathbf{F}\theta)^{1/2}$, Theorem P.139(iii) gives $n^{-1/2}\sum_i \langle \theta, \mathbf{W}_i\rangle = v_n \sum_i \xi_{ni} \xrightarrow{d} N(0, \theta'\mathbf{F}\theta)$, the law of $\langle \theta, V\rangle$ for $V \sim N(\mathbf{0},\mathbf{F})$ by Theorem P.114. Corollary P.133 completes the proof. □

Remark. Corollary P.143 keeps the variance common and lets the weights vary; Corollary P.144 lets the variance vary and pays for it with a uniform bound on the summands. The score of a binary regression, $\sum_i \mathbf{x}_i(y_i - \pi_i)$, has both features and is bounded whenever the design is, so it falls under the second.

Corollary P.145 (Lyapunov)**.** Let $\mathbf{W}_1, \mathbf{W}_2, \dots$ be independent random vectors in $\mathbb{R}^p$ with $\mathsf{E}[\mathbf{W}_i] = \mathbf{0}$ and $\sup_i \mathsf{E}\,|\mathbf{W}_i|^{2+\delta} \leqslant C < \infty$ for some $\delta > 0$, and suppose $\mathbf{F}_n = n^{-1}\sum_{i\leqslant n} \operatorname{var}(\mathbf{W}_i) \to \mathbf{F}$ positive definite. Then $n^{-1/2}\displaystyle\sum_{i\leqslant n} \mathbf{W}_i \xrightarrow{d} N(\mathbf{0}, \mathbf{F})$.

Proof. As before, fix $\theta \neq \mathbf{0}$ and put $\xi_{ni} = \langle \theta, \mathbf{W}_i\rangle/(\sqrt{n}\, v_n)$ with $v_n^2 = \theta'\mathbf{F}_n\theta \to \theta'\mathbf{F}\theta > 0$, so that $\mathsf{E}[\xi_{ni}] = 0$ and $\sum_i \operatorname{var}(\xi_{ni}) = 1$. On $\{|\xi_{ni}| > \varepsilon\}$ one has $1 \leqslant (|\xi_{ni}|/\varepsilon)^{\delta}$, so

$$\sum_{i\leqslant n} \mathsf{E}\big[\xi_{ni}^2; |\xi_{ni}| > \varepsilon\big] \leqslant \frac{1}{\varepsilon^{\delta}} \sum_{i\leqslant n} \mathsf{E}\,|\xi_{ni}|^{2+\delta} \leqslant \frac{|\theta|^{2+\delta} C}{\varepsilon^{\delta} v_n^{2+\delta}} \, \frac{n}{n^{1+\delta/2}},$$

using Cauchy–Schwarz (Theorem P.36) on $|\langle \theta, \mathbf{W}_i\rangle|$, so that $\mathsf{E}\,|\xi_{ni}|^{2+\delta} \leqslant |\theta|^{2+\delta} C/(\sqrt{n}\, v_n)^{2+\delta}$. The

right side is $O(n^{-\delta/2}) \to 0$, as v_n is bounded away from 0, so the Lindeberg condition holds and Theorem P.141 gives $\sum_i \xi_{ni} \xrightarrow{d} N(0,1)$; as in the proof of Corollary P.144, Theorem P.139(iii) turns this into $n^{-1/2}\sum_i \langle \theta, \mathbf{W}_i \rangle \xrightarrow{d} N(0, \theta'\mathbf{F}\theta)$, and Cramér–Wold (Corollary P.133) completes the proof. □

Remark. A Poisson score $\sum_i \mathbf{x}_i(y_i - \mu_i)$ is not bounded, so Corollary P.145 applies to it where Corollary P.144 does not, provided the design and the means μ_i are bounded and $\mathbf{F}_n$ converges to a positive definite limit; any $\delta > 0$ will do, and $\delta = 2$ needs only bounded fourth moments.

30 Dependent Summands

Definition P.146 (Weak stationarity)**.** A sequence $\{X_t\}_{t\in\mathbb{Z}}$ with $\mathsf{E}[X_t^2] < \infty$ is **weakly stationary** if $\mathsf{E}[X_t] = \mu$ and $\operatorname{cov}(X_t, X_{t+k}) = \gamma(k)$ are free of t; γ is its **autocovariance function**. It is **strictly stationary** if the law of $(X_{t_1}, \dots, X_{t_r})$ is unchanged when every index is shifted by the same h.

Lemma P.147 (Variance of a stationary average)**.** For a weakly stationary sequence,

$$\operatorname{var}\Big(\frac{1}{n}\sum_{t=1}^{n} X_t\Big) = \frac{1}{n}\sum_{|k|<n}\Big(1-\frac{|k|}{n}\Big)\gamma(k). \tag{P.25}$$

If $\gamma(k)\to 0$ as $k\to\infty$, the right side tends to 0 and $n^{-1}\sum_{t\leqslant n} X_t \xrightarrow{p} \mu$.

Proof. By bilinearity of the covariance, $\operatorname{var}\big(\sum_{t\leqslant n} X_t\big) = \sum_{s,t\leqslant n}\operatorname{cov}(X_s,X_t) = \sum_{s,t\leqslant n}\gamma(t-s)$, as $\operatorname{cov}(X_s,X_{s+k})=\gamma(k)$ with $k=t-s$ (for $k<0$ too, the definition being for every integer k). For each k with $|k|<n$ there are exactly $n-|k|$ pairs (s,t) in $\{1,\dots,n\}^2$ with $t-s=k$ (for $k\geqslant 0$, $s=1,\dots,n-k$; for $k<0$, $t=1,\dots,n+k$); grouping the terms by k and dividing by n^2 gives (P.25).

Assume $\gamma(k)\to 0$ and fix $\varepsilon>0$. Choose K with $|\gamma(k)|\leqslant\varepsilon$ for $|k|>K$. Since $|\gamma(k)|\leqslant\gamma(0)$ by Cauchy–Schwarz (Lemma P.90 with $\mu=\mathsf{P}$, applied to the centred X_t and X_{t+k}, each of variance $\gamma(0)$) and $0\leqslant 1-|k|/n\leqslant 1$, the $2K+1$ terms with $|k|\leqslant K$ contribute at most $n^{-1}(2K+1)\gamma(0)$ to the modulus of the right side of (P.25), and the fewer than $2n$ terms with $K<|k|<n$ at

most $n^{-1} \cdot 2n\varepsilon = 2\varepsilon$. Letting $n \to \infty$, the lim sup is at most 2ε, and $\varepsilon > 0$ is arbitrary, so the variance tends to 0. The average has mean μ, so Chebyshev's inequality (Theorem P.100) gives $\mathsf{P}\{|n^{-1}\sum_t X_t - \mu| \geqslant \eta\} \leqslant \operatorname{var}(n^{-1}\sum_t X_t)/\eta^2 \to 0$ for every $\eta > 0$. □

Remark. Independence is not needed, only that the correlation between distant terms dies out. Without that the conclusion fails: the constant sequence $X_t = X_0$ with $\operatorname{var} X_0 > 0$ is weakly stationary with $\gamma(k) = \operatorname{var} X_0$ for every k, and its average is X_0, which does not converge to μ.

Lemma P.148 (Approximation)**.** Let S_n and $S_n^{(p)}$, $n, p \geqslant 1$, be real random variables such that

(i) for each fixed p, $S_n^{(p)} \xrightarrow{d} Z_p$ as $n \to \infty$;

(ii) $Z_p \xrightarrow{d} Z$ as $p \to \infty$;

(iii) $\lim_{p\to\infty} \limsup_{n\to\infty} \mathsf{P}\{|S_n - S_n^{(p)}| > \varepsilon\} = 0$ for every $\varepsilon > 0$.

Then $S_n \xrightarrow{d} Z$.

Proof. By Theorem P.130 it is enough that $\mathsf{E}[e^{itS_n}] \to \mathsf{E}[e^{itZ}]$ for each fixed t. Write, for any p,

$$\bigl|\mathsf{E}[e^{itS_n}] - \mathsf{E}[e^{itZ}]\bigr| \leqslant \mathsf{E}\bigl|e^{itS_n} - e^{itS_n^{(p)}}\bigr| + \bigl|\mathsf{E}[e^{itS_n^{(p)}}] - \mathsf{E}[e^{itZ_p}]\bigr| + \bigl|\mathsf{E}[e^{itZ_p}] - \mathsf{E}[e^{itZ}]\bigr|.$$

For the first term, $|e^{ia} - e^{ib}| = |e^{i(a-b)} - 1| \leqslant \min\{2, |a-b|\}$ by Lemma P.15 with $m = 0$, so for every $\varepsilon > 0$

$$\mathsf{E}\bigl|e^{itS_n} - e^{itS_n^{(p)}}\bigr| \leqslant \mathsf{E}\bigl[\min\{2,\ |t|\,|S_n - S_n^{(p)}|\}\bigr] \leqslant |t|\varepsilon + 2\,\mathsf{P}\bigl\{|S_n - S_n^{(p)}| > \varepsilon\bigr\}.$$

The second term vanishes as $n \to \infty$ by (i) and Theorem P.130. Taking $\limsup_n$ therefore leaves

$$\limsup_n \bigl|\mathsf{E}[e^{itS_n}] - \mathsf{E}[e^{itZ}]\bigr| \leqslant |t|\varepsilon + 2\limsup_n \mathsf{P}\bigl\{|S_n - S_n^{(p)}| > \varepsilon\bigr\} + \bigl|\mathsf{E}[e^{itZ_p}] - \mathsf{E}[e^{itZ}]\bigr|.$$

The left side does not depend on p or ε. Now let $p \to \infty$: the second term vanishes by (iii) and the third by (ii) with Theorem P.130, leaving the bound $|t|\varepsilon$. Finally let $\varepsilon \to 0$: the lim sup is 0. □

Definition P.149 (m-dependence)**.** $\{X_t\}$ is **m-dependent** if for every t and all finite index sets $I \subseteq \{s : s \leqslant t\}$ and $J \subseteq \{s : s \geqslant t+m+1\}$ the random vectors $(X_s)_{s\in I}$ and $(X_s)_{s\in J}$ are independent.

Lemma P.150 (Shift functions of an independent sequence)**.** Let $\{\varepsilon_t\}_{t\in\mathbb{Z}}$ be independent and identically distributed and put $X_t = g(\varepsilon_{t-m}, \dots, \varepsilon_t)$ for one measurable $g : \mathbb{R}^{m+1} \to \mathbb{R}$. Then $\{X_t\}$ is strictly stationary and m-dependent.

Proof. Fix indices $t_1, \dots, t_r$ and h, and let $A = \bigcup_i \{t_i - m, \dots, t_i\}$. The families $(\varepsilon_u)_{u\in A}$ and $(\varepsilon_{u+h})_{u\in A}$ have the same law, being independent with a common marginal and indexed in the same order: both laws are the product of $|A|$ copies of the law of ε_0 (Definition P.104), unique by Corollary P.98. And $(X_{t_1}, \dots, X_{t_r})$ and $(X_{t_1+h}, \dots, X_{t_r+h})$ are the images of the two families under the same measurable map G (built from g coordinate by coordinate), so for Borel B, $\mathsf{P}\{(X_{t_i})_i \in B\} = \mathsf{P}\{(\varepsilon_u)_{u\in A} \in G^{-1}(B)\} = \mathsf{P}\{(\varepsilon_{u+h})_{u\in A} \in G^{-1}(B)\} = \mathsf{P}\{(X_{t_i+h})_i \in B\}$. That is strict stationarity. For m-dependence, $(X_s)_{s\in I}$ with $I \subseteq \{s \leqslant t\}$ finite is a function of $(\varepsilon_u)_{u\in U}$ with $U \subseteq \{u \leqslant t\}$ finite (each X_s, $s \leqslant t$, uses $\varepsilon_{s-m}, \dots, \varepsilon_s$, all with index $\leqslant t$), while

$(X_s)_{s\in J}$ with $J \subseteq \{s \geqslant t+m+1\}$ finite is a function of $(\varepsilon_u)_{u\in V}$ with $V \subseteq \{u \geqslant t+1\}$ finite (indices $\geqslant s-m \geqslant t+1$); the two index sets are disjoint, so Lemma P.105, applied to the finitely many ε_u with $u \in U \cup V$, gives the independence. □

Theorem P.151 (Central limit theorem for m-dependent sequences)**.** Let $\{X_t\}$ be strictly stationary and m-dependent with $\mathsf{E}[X_t] = 0$ and $\mathsf{E}[X_t^2] < \infty$, and write $\gamma(k) = \mathsf{E}[X_t X_{t+k}]$. Then $\gamma(k) = 0$ for $|k| > m$,

$$\sigma^2 \;=\; \gamma(0) + 2\sum_{k=1}^{m} \gamma(k) \;\geqslant\; 0, \tag{P.26}$$

and $n^{-1/2}\sum_{t=1}^{n} X_t \xrightarrow{d} N(0,\sigma^2)$, where $N(0,0)$ is read as the point mass at 0.

Proof. Strict stationarity with finite second moments makes $\mathsf{E}[X_t X_{t+k}]$ free of t, so $\{X_t\}$ is weakly stationary and $\gamma(-k) = \mathsf{E}[X_{t-k}X_t] = \gamma(k)$. For $k > m$, Definition P.149 with $I = \{t\}$ and $J = \{t+k\}$ makes X_t and X_{t+k} independent, so $\gamma(k) = \mathsf{E}[X_t]\,\mathsf{E}[X_{t+k}] = 0$ by Lemma P.105

(with the two independent variables X_t, X_{t+k} in the role of the ε); and $\gamma(-k) = \gamma(k) = 0$. Hence, by Lemma P.147 applied to a block of l consecutive terms (its first display multiplied by l^2),

$$\operatorname{var}\Big(\sum_{t=1}^{l} X_t\Big) = \sum_{|k|<l}(l-|k|)\gamma(k) = \sum_{|k|\leqslant m\wedge(l-1)}(l-|k|)\gamma(k). \tag{P.27}$$

Dividing by l, $l^{-1}\operatorname{var}(\sum_{t\leqslant l} X_t) = \sum_{|k|\leqslant m}(1-|k|/l)\gamma(k)$ for $l > m$, which tends to $\sum_{|k|\leqslant m}\gamma(k) = \gamma(0) + 2\sum_{k=1}^{m}\gamma(k) = \sigma^2$ as $l \to \infty$; a limit of non-negative numbers is non-negative, so $\sigma^2 \geqslant 0$. Put $C = \sum_{|k|\leqslant m}|\gamma(k)|$, so $\operatorname{var}(\sum_{t\leqslant l} X_t) \leqslant \sum_{|k|\leqslant m} l|\gamma(k)| = lC$ for every l. Write $S_n = n^{-1/2}\sum_{t\leqslant n} X_t$.

Blocks. Fix an integer $p > m$ and let $r = \lfloor n/(p+m)\rfloor$. Cut $\{1,\dots,n\}$ into r consecutive stretches of length $p+m$ followed by a remainder, and inside the ith stretch let U_i be the sum of the first p terms and V_i the sum of the last m; let R_n be the sum over the remainder, which has fewer than $p+m$ terms. Then

$$\sum_{t=1}^{n} X_t = \sum_{i=1}^{r} U_i + \sum_{i=1}^{r} V_i + R_n.$$

The big blocks are independent and identically distributed. The index set of U_i ends at $(i-1)(p+m)+p$ and that of U_{i+1} begins $m+1$ places later, so Definition P.149, with $t=(i-1)(p+m)+p$, makes $(U_1,\dots,U_i)$, a function of $(X_s)_{s\leqslant t}$, independent of $(U_{i+1},\dots,U_r)$, a function of $(X_s)_{s\geqslant t+m+1}$, for each i. Iterating, by downward induction on i: if $\mathsf{P}\{U_{i+1}\in B_{i+1},\dots,U_r\in B_r\}=\prod_{j>i}\mathsf{P}\{U_j\in B_j\}$, then the independence of U_i and $(U_{i+1},\dots,U_r)$ (a function of the independent pair just described) gives the same product formula from i on; at $i=1$ this is full independence. Strict stationarity makes them identically distributed, each U_i being the same function of p consecutive terms shifted by $(i-1)(p+m)$. The same argument, with the gap now the big block of length $p>m$, applies to $V_1,\dots,V_r$.

Limit of the big-block sum. By (P.27) with $l=p>m$,

$$\operatorname{var}(U_1)=\sum_{|k|\leqslant m}(p-|k|)\gamma(k)=p\,\sigma^2-c,\qquad c=\sum_{|k|\leqslant m}|k|\gamma(k),$$

which is free of n. The blocks $U_1,U_2,\dots$ are fixed (they do not depend on n), and only their number $r=r(n)\to\infty$ grows with n. If $\operatorname{var}(U_1)=0$ then each U_i, having mean 0, vanishes almost surely (Proposition P.82(iv) applied to U_i^2), and $S_n^{(p)}:=n^{-1/2}\sum_i U_i$ is 0 almost surely; otherwise

Corollary P.142 gives $k^{-1/2}\sum_{i\leqslant k} U_i/\operatorname{sd}(U_1) \xrightarrow{d} N(0,1)$ as $k\to\infty$, hence along $k=r(n)$, and multiplying by $\operatorname{sd}(U_1)$ (Proposition P.124(iv) and Theorem P.130) $r^{-1/2}\sum_i U_i \xrightarrow{d} N(0,\operatorname{var}(U_1))$. Since $n/(p+m)-1<r\leqslant n/(p+m)$, $r/n\to(p+m)^{-1}$, and Theorem P.139(iii) yields in both cases

$$S_n^{(p)} = \sqrt{r/n}\left(r^{-1/2}\sum_{i\leqslant r} U_i\right) \xrightarrow{d} Z_p \sim N(0,\sigma_p^2), \qquad \sigma_p^2 = \frac{p\,\sigma^2 - c}{p+m}.$$

As $p\to\infty$, $\sigma_p^2\to\sigma^2$, so $\mathsf{E}[e^{itZ_p}] = e^{-\sigma_p^2 t^2/2}\to e^{-\sigma^2 t^2/2}$ and $Z_p \xrightarrow{d} N(0,\sigma^2)$ by Theorem P.130.

The small blocks and the remainder are negligible. The V_i are independent with mean 0 and, by (P.27) with $l=m$, $\operatorname{var}(V_1)\leqslant mC$; their cross moments vanish by Lemma P.105, so $\mathsf{E}[(\sum_i V_i)^2] = r\operatorname{var}(V_1)$. The remainder R_n is a sum of fewer than $p+m$ consecutive terms, so $\operatorname{var}(R_n)\leqslant (p+m)C$. Hence

$$\mathsf{E}\big[(S_n - S_n^{(p)})^2\big] = \frac{1}{n}\mathsf{E}\Big[\Big(\sum_i V_i + R_n\Big)^2\Big] \leqslant \frac{2}{n}\big\{r\,\operatorname{var}(V_1) + \operatorname{var}(R_n)\big\} \leqslant \frac{2mC}{p+m} + \frac{2(p+m)C}{n},$$

using $(a+b)^2\leqslant 2a^2+2b^2$ and $r\leqslant n/(p+m)$. By Chebyshev's inequality (Theorem P.100),

$\limsup_n \mathsf{P}\{|S_n - S_n^{(p)}| > \varepsilon\} \leqslant 2mC/\{(p+m)\varepsilon^2\}$, which tends to 0 as $p \to \infty$. Lemma P.148 completes the proof. □

Definition P.152 (The past space)**.** Let $\{\varepsilon_t\}_{t\in\mathbb{Z}}$ be independent and identically distributed with mean 0 and variance $\tau^2 \in (0,\infty)$. Let $\mathcal{L}_t$ be the closure in L^2 (Definition P.107) of the set of square-integrable measurable functions of finitely many ε_u with $u \leqslant t$.

Each $\mathcal{L}_t$ is a closed subspace of L^2 and $\mathcal{L}_s \subseteq \mathcal{L}_t$ for $s \leqslant t$. (It is a subspace because L^2-limits of sums and multiples are the sums and multiples of the limits, by the triangle inequality of Lemma P.90; it is closed by the diagonal argument used for $\overline{E}$ in the proof of Theorem P.26, with the L^2 distance of the same lemma in place of $|\cdot|$.) If $(c_j)_{j\geqslant 0}$ is square summable then the partial sums of $\sum_{j\geqslant 0} c_j\varepsilon_{t-j}$ are Cauchy in L^2. Indeed, the increments are orthogonal and

$$\mathsf{E}\Big[\Big(\sum_{M<j\leqslant N} c_j\varepsilon_{t-j}\Big)^2\Big] = \tau^2 \sum_{M<j\leqslant N} c_j^2.$$

Thus the series has an L^2 limit and that limit lies in $\mathcal{L}_t$.

Lemma P.153 (Innovation orthogonality)**.** Let $v \in \mathcal{L}_{t-1}$ and put $X = v\,\varepsilon_t$. Then $X \in \mathcal{L}_t$ and

$$\mathsf{E}[X] = 0, \qquad \mathsf{E}[X^2] = \tau^2\ \mathsf{E}[v^2], \qquad \mathsf{E}[ZX] = 0 \ \text{ for every } Z \in \mathcal{L}_{t-1}.$$

Proof. Choose $v^{(M)} \to v$ in L^2, each a square-integrable measurable function of finitely many ε_u with $u \leqslant t-1$. For fixed M, Lemma P.105 makes $v^{(M)}$ and ε_t independent, so $v^{(M)}\varepsilon_t$ is square integrable with

$$\mathsf{E}\big[v^{(M)}\varepsilon_t\big] = 0, \qquad \mathsf{E}\big[(v^{(M)}\varepsilon_t)^2\big] = \tau^2\ \mathsf{E}\big[(v^{(M)})^2\big].$$

Applying the second identity to $v^{(M)} - v^{(M')}$ shows $\{v^{(M)}\varepsilon_t\}_M$ is Cauchy in L^2, hence convergent there; its limit is $v\varepsilon_t$, because L^2 convergence implies convergence in probability by Chebyshev's inequality (Theorem P.100) and therefore almost sure convergence along a subsequence (Lemma P.120), along which $v^{(M)}\varepsilon_t \to v\varepsilon_t$, while (by the same two results) a further subsequence converges almost surely to the L^2 limit, so the two limits agree almost surely. Each $v^{(M)}\varepsilon_t$ is a square-integrable function of finitely many ε_u with $u \leqslant t$, so $X \in \mathcal{L}_t$, the closure. The two displayed identities pass to the limit: $|\,\mathsf{E}[Y_M] - \mathsf{E}[Y]| \leqslant \mathsf{E}\,|Y_M - Y| \leqslant (\mathsf{E}\,|Y_M - Y|^2)^{1/2} \to 0$ by Lemma P.90, and $\mathsf{E}[Y_M^2] \to \mathsf{E}[Y^2]$ because $\|Y_M\| \to \|Y\|$ by the triangle inequality of the same

lemma; apply these to $Y_M = v^{(M)}\varepsilon_t$ and to $Y_M = v^{(M)}$.
For the third claim take $Z^{(M')} \to Z$ in L^2 of the same kind. Then $Z^{(M')}v^{(M)}$ is a measurable function of finitely many ε_u with $u \leqslant t-1$, and it is integrable by Cauchy–Schwarz (Lemma P.90), so Lemma P.105 gives $\mathsf{E}[Z^{(M')}v^{(M)}\varepsilon_t] = \mathsf{E}[Z^{(M')}v^{(M)}]\,\mathsf{E}[\varepsilon_t] = 0$. Letting $M, M' \to \infty$ gives $\mathsf{E}[ZX] = 0$, since by Lemma P.90

$$\mathsf{E}\,|Z^{(M')}v^{(M)}\varepsilon_t - ZX| \leqslant \|Z^{(M')} - Z\|\,\|v^{(M)}\varepsilon_t\| + \|Z\|\,\|v^{(M)}\varepsilon_t - X\| \to 0,$$

with $\|Y\| = (\mathsf{E}\,Y^2)^{1/2}$, as $\|v^{(M)}\varepsilon_t\|$ stays bounded. □

Corollary P.154 (Orthogonality across time)**.** If $v_{t-1} \in \mathcal{L}_{t-1}$ for every t and $X_t = v_{t-1}\varepsilon_t$, then $\mathsf{E}[X_t] = 0$, $\mathsf{E}[X_sX_t] = 0$ for $s \neq t$, and

$$\mathsf{E}\Big[\Big(\sum_{t=1}^{n} X_t\Big)^2\Big] = \tau^2 \sum_{t=1}^{n} \mathsf{E}[v_{t-1}^2]. \tag{P.28}$$

Proof. For $s < t$, $X_s \in \mathcal{L}_s \subseteq \mathcal{L}_{t-1}$ by Lemma P.153, so the third statement of that lemma (with $Z = X_s$) gives $\mathsf{E}[X_s X_t] = 0$; $\mathsf{E}[X_t] = 0$ is its first statement. Expanding the square, $\mathsf{E}[(\sum_t X_t)^2] = \sum_t \mathsf{E}[X_t^2] + \sum_{s \neq t} \mathsf{E}[X_s X_t] = \sum_t \mathsf{E}[X_t^2]$, and $\mathsf{E}[X_t^2] = \tau^2 \, \mathsf{E}[v_{t-1}^2]$ by the second statement gives (P.28). □

Theorem P.155 (Central limit theorem for autoregressive scores)**.** Let $\{\varepsilon_t\}_{t \in \mathbb{Z}}$ be independent and identically distributed with mean 0 and variance $\tau^2 \in (0, \infty)$, let $\mathbf{a}_0, \mathbf{a}_1, \ldots$ lie in $\mathbb{R}^d$ with $\sum_{j \geqslant 0} |\mathbf{a}_j|^2 < \infty$, and let

$$\mathbf{u}_t = \sum_{j \geqslant 0} \mathbf{a}_j \, \varepsilon_{t-j}$$

be the L^2 limit of its partial sums. Then $\mathbf{\Gamma} := \mathsf{E}[\mathbf{u}_t \mathbf{u}_t'] = \tau^2 \sum_{j \geqslant 0} \mathbf{a}_j \mathbf{a}_j'$ is free of t and

$$\frac{1}{\sqrt{n}} \sum_{t=1}^{n} \mathbf{u}_{t-1} \, \varepsilon_t \xrightarrow{d} N(\mathbf{0}, \, \tau^2 \mathbf{\Gamma}). \tag{P.29}$$

Proof. The limit variance. Write $\mathbf{u}_t^{(m)} = \sum_{j<m} \mathbf{a}_j \varepsilon_{t-j}$. The ε are uncorrelated with mean 0 ($\mathsf{E}[\varepsilon_{t-j}\varepsilon_{t-k}] = \tau^2$ for $j = k$ and 0 otherwise, by Lemma P.105), so expanding $\mathsf{E}[\mathbf{u}_t^{(m)}\mathbf{u}_t^{(m)\prime}] = \sum_{j,k<m} \mathbf{a}_j\mathbf{a}_k' \,\mathsf{E}[\varepsilon_{t-j}\varepsilon_{t-k}] = \tau^2 \sum_{j<m} \mathbf{a}_j\mathbf{a}_j'$. As $m \to \infty$ the right side converges entrywise, the (r,s) entries being bounded by $|\mathbf{a}_j|^2$, which is summable; and $\mathbf{u}_t^{(m)} \to \mathbf{u}_t$ in L^2 coordinatewise, so for each pair of coordinates $|\,\mathsf{E}[AB] - \mathsf{E}[A'B']| \leqslant \|A - A'\|\,\|B\| + \|A'\|\,\|B - B'\|$ (Lemma P.90) passes the identity to the limit. The limit $\tau^2 \sum_j \mathbf{a}_j\mathbf{a}_j'$ is free of t.

Reduction to one dimension. By Cramér–Wold (Corollary P.133) it suffices to show, for each fixed $\theta \in \mathbb{R}^d$, that $n^{-1/2}\sum_t \langle \theta, \mathbf{u}_{t-1}\rangle \varepsilon_t \xrightarrow{d} N(0, \tau^2\theta'\mathbf{\Gamma}\theta)$. Put $a_j = \langle \theta, \mathbf{a}_j\rangle$ and $u_t = \langle \theta, \mathbf{u}_t\rangle = \sum_j a_j \varepsilon_{t-j}$, a square-summable combination with $\mathsf{E}[u_t^2] = \tau^2 \sum_j a_j^2 = \theta'\mathbf{\Gamma}\theta$. So it is enough to prove the scalar statement

$$S_n = \frac{1}{\sqrt{n}} \sum_{t=1}^{n} u_{t-1}\varepsilon_t \xrightarrow{d} N\Big(0,\ \tau^4 \sum_{j\geqslant 0} a_j^2\Big).$$

Truncation. Put $u_{t-1}^{(m)} = \sum_{j<m} a_j \varepsilon_{t-1-j}$, $X_t^{(m)} = u_{t-1}^{(m)}\varepsilon_t$ and $S_n^{(m)} = n^{-1/2}\sum_{t\leqslant n} X_t^{(m)}$. Since

$X_t^{(m)} = g(\varepsilon_{t-m},\dots,\varepsilon_t)$ with $g(x_0,\dots,x_m) = x_m \sum_{j<m} a_j x_{m-1-j}$, Lemma P.150 makes $\{X_t^{(m)}\}$ strictly stationary and m-dependent. Corollary P.154 gives $\mathsf{E}[X_t^{(m)}] = 0$, $\gamma^{(m)}(k) = 0$ for $k \neq 0$ and $\gamma^{(m)}(0) = \tau^2\,\mathsf{E}[(u^{(m)})^2] = \tau^4 \sum_{j<m} a_j^2$. So Theorem P.151 applies with $\sigma^2 = \gamma^{(m)}(0)$:

$$S_n^{(m)} \xrightarrow{d} Z_m \sim N\Big(0,\ \tau^4 \sum_{j<m} a_j^2\Big) \quad (n\to\infty), \qquad Z_m \xrightarrow{d} N\Big(0,\ \tau^4 \sum_{j\geqslant 0} a_j^2\Big) \quad (m\to\infty),$$

the second by Theorem P.130 as in the previous proof.

Negligibility. The difference is $X_t - X_t^{(m)} = w_{t-1}^{(m)} \varepsilon_t$ with $w_{t-1}^{(m)} = \sum_{j\geqslant m} a_j \varepsilon_{t-1-j} \in \mathcal{L}_{t-1}$ and $\mathsf{E}[(w_{t-1}^{(m)})^2] = \tau^2 \sum_{j\geqslant m} a_j^2$, free of t. By Corollary P.154,

$$\mathsf{E}\big[(S_n - S_n^{(m)})^2\big] = \frac{1}{n}\cdot n\,\tau^2\ \mathsf{E}\big[(w^{(m)})^2\big] = \tau^4 \sum_{j\geqslant m} a_j^2,$$

free of n and tending to 0 as $m\to\infty$. Chebyshev's inequality (Theorem P.100) gives $\mathsf{P}\{|S_n - S_n^{(m)}| > \varepsilon\} \leqslant \tau^4 \sum_{j\geqslant m} a_j^2/\varepsilon^2$ for every n, which is hypothesis (iii) of Lemma P.148; the truncation

step supplies (i) and (ii). The lemma gives the scalar statement, whose limit $N(0, \tau^2 \theta' \mathbf{\Gamma} \theta)$ is, by Theorem P.114, the law of $\langle \theta, V \rangle$ for $V \sim N(\mathbf{0}, \tau^2 \mathbf{\Gamma})$; Corollary P.133 gives (P.29). □

Remark. The summands $u_{t-1}\varepsilon_t$ are uncorrelated by Corollary P.154 but far from independent, and $\operatorname{var}(S_n) = \tau^2 \, \mathsf{E}[u^2]$ exactly, for every n. What the theorem adds is that the limit law is the normal one with that variance. Two of the course's statistics are of this form: with $a_j = \phi^j$, $|\phi| < 1$, it is the score of an AR(1) least squares fit, and with $a_j = \mathbb{1}\{j = k - 1\}$ it is the numerator of the lag k sample autocorrelation of white noise.

31 The Delta Method

Theorem P.156 (Delta method)**.** Let $r_n \to \infty$ and $r_n(T_n - \vartheta) \xrightarrow{d} T$ in $\mathbb{R}^d$, and let $g : \mathbb{R}^d \to \mathbb{R}^m$ be measurable and differentiable at ϑ with Jacobian $\mathbf{D}_g(\vartheta)$. Then

$$r_n\big(g(T_n) - g(\vartheta)\big) \xrightarrow{d} \mathbf{D}_g(\vartheta)\, T.$$

In particular, if $\sqrt{n}(T_n - \vartheta) \xrightarrow{d} N(\mathbf{0}, \mathbf{\Sigma})$ then $\sqrt{n}\{g(T_n) - g(\vartheta)\} \xrightarrow{d} N\big(\mathbf{0},\ \mathbf{D}_g \mathbf{\Sigma} \mathbf{D}_g'\big)$.

Proof. Differentiability at ϑ means $g(\vartheta+h)-g(\vartheta)=\mathbf{D}_g(\vartheta)h+R(h)$ with $|R(h)|/|h|\to 0$ as $h\to 0$. Since $r_n(T_n-\vartheta)$ converges in distribution and $r_n\to\infty$, Proposition P.122 and Theorem P.139(iii) give $T_n-\vartheta=r_n^{-1}\cdot r_n(T_n-\vartheta)\xrightarrow{p}\mathbf{0}$. Writing $H_n=T_n-\vartheta$,

$$r_n\big(g(T_n)-g(\vartheta)\big)=\mathbf{D}_g(\vartheta)\,r_nH_n+r_nR(H_n).$$

The first term converges in distribution to $\mathbf{D}_g(\vartheta)T$ by Theorem P.138, the linear map $x\mapsto\mathbf{D}_g(\vartheta)x$ being continuous. For the second, define $\rho(h)=|R(h)|/|h|$ for $h\neq\mathbf{0}$ and $\rho(\mathbf{0})=0$; ρ is Borel, as $R(h)=g(\vartheta+h)-g(\vartheta)-\mathbf{D}_g(\vartheta)h$ is, it is continuous at $\mathbf{0}$ by differentiability, and $|r_nR(H_n)|=\rho(H_n)\,|r_nH_n|$. Here $\rho(H_n)\xrightarrow{p}\rho(\mathbf{0})=0$ by Theorem P.138 (in probability, the limit $\mathbf{0}$ of H_n not being a discontinuity point of ρ), and $|r_nH_n|\xrightarrow{d}|T|$ by the same theorem; so the product converges in distribution to $0\cdot|T|=0$ by Theorem P.139(iii), hence in probability by Proposition P.122. One more application of Theorem P.139(iii), to the sum of a sequence converging in distribution and one converging in probability to $\mathbf{0}$, gives $r_n\{g(T_n)-g(\vartheta)\}\xrightarrow{d}\mathbf{D}_g(\vartheta)T$. In the particular case, $r_n=\sqrt{n}$, $T\sim N(\mathbf{0},\boldsymbol{\Sigma})$, and $\mathbf{D}_g(\vartheta)T\sim N(\mathbf{0},\mathbf{D}_g\boldsymbol{\Sigma}\mathbf{D}_g')$ by Theorem P.114. □

Theorem P.157 (Quadratic forms of asymptotically normal vectors)**.** Let $V_n \xrightarrow{d} N(\mathbf{0}, \mathbf{\Sigma})$ in $\mathbb{R}^p$ with $\mathbf{\Sigma}$ positive definite, and let $\hat{\mathbf{\Sigma}}_n \xrightarrow{p} \mathbf{\Sigma}$ with $\hat{\mathbf{\Sigma}}_n$ symmetric. Then, with the convention $\mathbf{A}^{-1} := \mathbf{0}$ for singular $\mathbf{A}$ (so that the statistic is defined for every n),

$$V_n' \hat{\mathbf{\Sigma}}_n^{-1} V_n \xrightarrow{d} \chi^2_p.$$

Proof. On the symmetric matrices, identified with $\mathbb{R}^{p(p+1)/2}$, inversion so extended is measurable: by Corollary P.40 the smallest eigenvalue of $\mathbf{A}^2$ is $\min_{|\mathbf{u}|=1} |\mathbf{A}\mathbf{u}|^2$, so the singular symmetric matrices are those whose $\mathbf{A}^2$ has smallest eigenvalue 0; this set is closed, because the minimum is continuous in $\mathbf{A}$ ($|\mathbf{u}'(\mathbf{B}-\mathbf{C})\mathbf{u}| \leqslant \sum_{r,s} |(\mathbf{B}-\mathbf{C})_{rs}|$ for $|\mathbf{u}| = 1$, as in the proof of Lemma P.43); and off it $\mathbf{A}^2$ is positive definite (Proposition P.42) and $\mathbf{A}^{-1} = (\mathbf{A}^2)^{-1}\mathbf{A}$ is continuous by Lemma P.43. So for an open set U of matrices, the preimage of U under the extended inversion is the union of the Borel set $\{\mathbf{A}$ singular$\}$ (if $\mathbf{0} \in U$; otherwise nothing) and the preimage of U under the continuous inversion on the open set of non-singular matrices, which is open; so the extended inversion is measurable. It is also continuous at $\mathbf{\Sigma}$: Lemma P.43 makes matrices converging to $\mathbf{\Sigma}$ invertible eventually,

with inverses converging to $\boldsymbol{\Sigma}^{-1}$. So $\hat{\boldsymbol{\Sigma}}_n^{-1} \xrightarrow{p} \boldsymbol{\Sigma}^{-1}$ by Theorem P.138, applied in probability with the constant limit $\boldsymbol{\Sigma}$ (Proposition P.122). By Theorem P.139(i), $(V_n, \hat{\boldsymbol{\Sigma}}_n^{-1}) \xrightarrow{d} (V, \boldsymbol{\Sigma}^{-1})$ with $V \sim N(\mathbf{0}, \boldsymbol{\Sigma})$, and $(\mathbf{v}, \mathbf{A}) \mapsto \mathbf{v}'\mathbf{A}\mathbf{v}$ is continuous, so Theorem P.138 gives $V_n'\hat{\boldsymbol{\Sigma}}_n^{-1}V_n \xrightarrow{d} V'\boldsymbol{\Sigma}^{-1}V$. It remains to identify the limit law. Let $\boldsymbol{\Sigma}^{1/2}$ be the symmetric positive definite square root and $W = \boldsymbol{\Sigma}^{-1/2}V$. By Proposition P.125 and Proposition P.124(iv), $W \sim N(\mathbf{0}, \mathbf{I}_p)$, with characteristic function $e^{-|\mathbf{t}|^2/2} = \prod_j e^{-t_j^2/2}$, which is that of a vector of independent standard normals; by Corollary P.131 its coordinates are independent standard normals, and since $\boldsymbol{\Sigma}^{-1} = (\boldsymbol{\Sigma}^{1/2}\boldsymbol{\Sigma}^{1/2})^{-1} = \boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}^{-1/2}$ with $\boldsymbol{\Sigma}^{-1/2}$ symmetric, $V'\boldsymbol{\Sigma}^{-1}V = |W|^2 = \sum_{j \leqslant p} W_j^2$, which is χ_p^2 by Definition P.115. □

Part IV
Results Used in the Course

32 Maximum Likelihood

32.1 Score and information

Definition P.158 (Log-likelihood, score, information)**.** Let $\{f(\cdot;\boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta\}$ be a family of densities with respect to a measure μ on $\mathcal{X}$, with $\Theta \subseteq \mathbb{R}^p$ open. For an observation Y the **log-likelihood** is $\ell(\boldsymbol{\theta}) = \ln f(Y;\boldsymbol{\theta})$, the **score** is $\mathbf{U}(\boldsymbol{\theta}) = \nabla\ell(\boldsymbol{\theta})$, and the **information** at $\boldsymbol{\theta}$ is $\mathcal{I}(\boldsymbol{\theta}) = \operatorname{var}_{\boldsymbol{\theta}}\big(\mathbf{U}(\boldsymbol{\theta})\big)$.

Lemma P.159 (Score identities)**.** Suppose $f(y;\boldsymbol{\theta}) > 0$ for all $y, \boldsymbol{\theta}$, that $\boldsymbol{\theta} \mapsto f(y;\boldsymbol{\theta})$ is twice

continuously differentiable on Θ, and that there are μ-integrable g_1, g_2 with

$$\left|\frac{\partial f}{\partial \theta_j}(y;\boldsymbol{\theta})\right| \leqslant g_1(y), \qquad \left|\frac{\partial^2 f}{\partial \theta_j \partial \theta_k}(y;\boldsymbol{\theta})\right| \leqslant g_2(y)$$

for every j, k, every y and every $\boldsymbol{\theta} \in \Theta$. Then, with Y distributed according to $f(\cdot;\boldsymbol{\theta})$ and provided $\mathsf{E}_{\boldsymbol{\theta}}\,|\mathbf{U}(\boldsymbol{\theta})|^2 < \infty$,

$$\mathsf{E}_{\boldsymbol{\theta}}\big[\mathbf{U}(\boldsymbol{\theta})\big] = \mathbf{0}, \qquad \mathcal{I}(\boldsymbol{\theta}) = \mathsf{E}_{\boldsymbol{\theta}}\big[\mathbf{U}(\boldsymbol{\theta})\mathbf{U}(\boldsymbol{\theta})'\big] = -\,\mathsf{E}_{\boldsymbol{\theta}}\big[\nabla^2 \ell(\boldsymbol{\theta})\big]. \tag{P.30}$$

Proof. Apply Theorem P.87 to $\int f(y;\boldsymbol{\theta})\,\mathrm{d}\mu(y) = 1$, one coordinate of $\boldsymbol{\theta}$ at a time with the others fixed, the hypothesis on g_1 supplying the domination:

$$\mathbf{0} = \nabla \int f(y;\boldsymbol{\theta})\,\mathrm{d}\mu(y) = \int \nabla f(y;\boldsymbol{\theta})\,\mathrm{d}\mu(y) = \int \frac{\nabla f(y;\boldsymbol{\theta})}{f(y;\boldsymbol{\theta})} f(y;\boldsymbol{\theta})\,\mathrm{d}\mu(y) = \mathsf{E}_{\boldsymbol{\theta}}[\mathbf{U}(\boldsymbol{\theta})],$$

since $\mathbf{U} = \nabla \ln f = \nabla f / f$ (the last equality being the expectation of a function of Y written as an integral against its density, Lemma P.99, whose proof uses nothing about $\mathbb{R}^d$ and so applies on $\mathcal{X}$). Here Theorem P.87 applies to $t \mapsto f(y;\boldsymbol{\theta} + t\mathbf{e}_j)$ on an open interval of t keeping $\boldsymbol{\theta} + t\mathbf{e}_j \in \Theta$:

$f(\cdot;\boldsymbol{\theta})$ is integrable, being a density, and the t-derivative is dominated by g_1. Differentiating the identity $\int \partial_j f(y;\boldsymbol{\theta})\,\mathrm{d}\mu(y) = 0$ in θ_k in the same way, now with $\partial_j f$ integrable as $|\partial_j f| \leqslant g_1$ and its θ_k-derivative dominated by g_2,

$$\mathbf{0} = \int \nabla^2 f(y;\boldsymbol{\theta})\,\mathrm{d}\mu(y).$$

The quotient rule gives $\partial_k(\partial_j f/f) = \partial_k\partial_j f/f - (\partial_j f)(\partial_k f)/f^2$, that is $\nabla^2 \ln f = \nabla^2 f/f - (\nabla f/f)(\nabla f/f)' = \nabla^2 f/f - \mathbf{U}\mathbf{U}'$. Both terms on the right are integrable under $f(\cdot;\boldsymbol{\theta})\,\mathrm{d}\mu$: the first because $\int |\partial_k\partial_j f/f|\, f\,\mathrm{d}\mu \leqslant \int g_2\,\mathrm{d}\mu$, the second because $\mathsf{E}_{\boldsymbol{\theta}}\,|\mathbf{U}|^2 < \infty$. So, taking expectations (Lemma P.99 again),

$$\mathsf{E}_{\boldsymbol{\theta}}\big[\nabla^2\ell(\boldsymbol{\theta})\big] = \int \nabla^2 f(y;\boldsymbol{\theta})\,\mathrm{d}\mu(y) - \mathsf{E}_{\boldsymbol{\theta}}[\mathbf{U}\mathbf{U}'] = -\,\mathsf{E}_{\boldsymbol{\theta}}[\mathbf{U}\mathbf{U}'].$$

Finally $\mathrm{var}(\mathbf{U}) = \mathsf{E}[(\mathbf{U} - \mathsf{E}\,\mathbf{U})(\mathbf{U} - \mathsf{E}\,\mathbf{U})'] = \mathsf{E}[\mathbf{U}\mathbf{U}']$ because $\mathsf{E}[\mathbf{U}] = \mathbf{0}$. □

Remark. When μ is counting measure on a finite set, as for the Bernoulli and multinomial families, the integrals are finite sums, so $\mathsf{E}_{\boldsymbol{\theta}} |\mathbf{U}(\boldsymbol{\theta})|^2 < \infty$, and the domination hypothesis holds with Θ replaced by any open ball whose closure lies in Θ, by continuity of the derivatives on that compact closure; this suffices, because the conclusion at $\boldsymbol{\theta}$ involves the family only near $\boldsymbol{\theta}$. The equality of the two expressions for $\mathcal{I}$ in (P.30) is the **information equality**; it is a property of the model at the true parameter, not an identity in $\boldsymbol{\theta}$ for a misspecified one.

32.2 The maximum likelihood estimator

Theorem P.160 (Concave maximum likelihood)**.** Let $Y_1, Y_2, \ldots$ be independent, let $\Theta \subseteq \mathbb{R}^p$ be open and convex, let $\boldsymbol{\theta}_0 \in \Theta$, and let

$$\ell_n(\boldsymbol{\theta}) = \sum_{i=1}^{n} \ell_i(\boldsymbol{\theta}), \qquad \mathbf{U}_n = \nabla \ell_n(\boldsymbol{\theta}_0), \qquad \mathbf{H}_n(\boldsymbol{\theta}) = -\nabla^2 \ell_n(\boldsymbol{\theta}).$$

Assume:

(L1) each ℓ_i is concave and twice continuously differentiable on Θ, whatever the value of Y_i, where $\ell_i(\boldsymbol{\theta}) = \ell_i(\boldsymbol{\theta}; Y_i)$ depends on the data only through Y_i, and the closed ball $\{\boldsymbol{\theta} : |\boldsymbol{\theta} - \boldsymbol{\theta}_0| \leqslant$

$\delta_0\} \subseteq \Theta$ for some $\delta_0 > 0$;

(L2) $\mathsf{E}[\nabla \ell_i(\boldsymbol{\theta}_0)] = \mathbf{0}$, $\mathbf{F}_n = n^{-1}\sum_{i\leqslant n} \operatorname{var}(\nabla \ell_i(\boldsymbol{\theta}_0)) \to \mathbf{F}$ positive definite, and $n^{-1/2}\mathbf{U}_n \xrightarrow{d} N(\mathbf{0}, \mathbf{F})$;

(L3) $n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0) \xrightarrow{p} \mathbf{G}$ positive definite, and there are random variables L_n with $\|\mathbf{H}_n(\boldsymbol{\theta}) - \mathbf{H}_n(\boldsymbol{\theta}_0)\| \leqslant nL_n|\boldsymbol{\theta} - \boldsymbol{\theta}_0|$ for every $|\boldsymbol{\theta} - \boldsymbol{\theta}_0| \leqslant \delta_0$, **bounded in probability**: for every $\eta > 0$ there is a constant L with $\mathsf{P}(L_n > L) < \eta$ for every n. A constant $L_n = L$ is the case met most often.

Then, writing $\lambda = \lambda_{\min}(\mathbf{G}) > 0$:

(i) with probability tending to one ℓ_n has a maximiser $\hat{\boldsymbol{\theta}}_n$ over Θ, and it is unique whenever some ℓ_i is strictly concave;

(ii) $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$;

(iii) $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0},\ \mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1})$;

(iv) if moreover $\mathbf{F} = \mathbf{G}$, then $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{F}^{-1})$ and $2\{\ell_n(\hat{\boldsymbol{\theta}}_n) - \ell_n(\boldsymbol{\theta}_0)\} \xrightarrow{d} \chi^2_p$;

(v) $n^{-1}\mathbf{H}_n(\hat{\boldsymbol{\theta}}_n) \xrightarrow{p} \mathbf{G}$;

(vi) with $\mathbf{Z}_n = n^{-1/2}\mathbf{U}_n$, $2\{\ell_n(\hat{\boldsymbol{\theta}}_n) - \ell_n(\boldsymbol{\theta}_0)\} - \mathbf{Z}_n'\mathbf{G}^{-1}\mathbf{Z}_n \xrightarrow{p} 0$.

Remark. G is the limiting curvature of the criterion and **F** the limiting variance of its gradient. When ℓ_n is the log-likelihood of a correctly specified model satisfying the domination hypotheses of Lemma P.159, and $n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0) - \mathsf{E}[n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0)] \xrightarrow{p} \mathbf{0}$, the two agree: the information equality of Lemma P.159 gives $\mathbf{F}_n = \mathsf{E}[n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0)]$, so $\mathbf{G} = \lim_n \mathbf{F}_n = \mathbf{F}$, and (iv) applies. When only the mean is correctly specified they need not, and (iii) leaves the **sandwich** $\mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1}$.

Proof. Step 1: a curvature floor near $\boldsymbol{\theta}_0$. Fix $\eta > 0$ and choose L with $\mathsf{P}(L_n > L) < \eta$ for every n, by (L3). Put $\delta_1 = \min\{\delta_0,\ \lambda/(4L)\}$ and $A_n = \{\lambda_{\min}(n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0)) > \tfrac{3}{4}\lambda\} \cap \{L_n \leqslant L\}$. The smallest eigenvalue of a symmetric matrix is $\min_{|\mathbf{u}|=1} \mathbf{u}'\mathbf{A}\mathbf{u}$ by Corollary P.40, hence 1-Lipschitz in the operator norm: for symmetric $\mathbf{A}, \mathbf{B}$ and $|\mathbf{u}| = 1$, $|\mathbf{u}'(\mathbf{A}-\mathbf{B})\mathbf{u}| \leqslant |\mathbf{u}|\,|(\mathbf{A}-\mathbf{B})\mathbf{u}| \leqslant \|\mathbf{A}-\mathbf{B}\|$ by Theorem P.36, so $\mathbf{u}'\mathbf{A}\mathbf{u} \geqslant \mathbf{u}'\mathbf{B}\mathbf{u} - \|\mathbf{A}-\mathbf{B}\|$; minimising over $\mathbf{u}$ gives $\lambda_{\min}(\mathbf{A}) \geqslant \lambda_{\min}(\mathbf{B}) - \|\mathbf{A}-\mathbf{B}\|$, and exchanging $\mathbf{A}$ and $\mathbf{B}$ gives $|\lambda_{\min}(\mathbf{A}) - \lambda_{\min}(\mathbf{B})| \leqslant \|\mathbf{A}-\mathbf{B}\|$. For every $\mathbf{A} \in \mathbb{R}^{p\times p}$, $\|\mathbf{A}\|^2 \leqslant \operatorname{tr}(\mathbf{A}'\mathbf{A})$, the sum of the squared entries of $\mathbf{A}$: this is trivial for $\mathbf{A} = \mathbf{0}$, and otherwise, by Proposition P.53, Theorem P.51 and Proposition P.35(ii),(iv), $\|\mathbf{A}\|^2 = d_1^2 \leqslant \sum_j d_j^2 = \operatorname{tr}(\mathbf{V}\mathbf{D}^2\mathbf{V}') = \operatorname{tr}(\mathbf{A}'\mathbf{A})$. So convergence in probability of the entries implies it in the operator norm: each squared entry of $n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0) - \mathbf{G}$ tends to 0 in probability by Theorem P.138, and so does their finite sum

by Theorem P.139(iii) and Proposition P.122. Hence $\lambda_{\min}(n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0)) \geqslant \lambda - \|n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0) - \mathbf{G}\|$ exceeds $\frac{3}{4}\lambda$ with probability tending to 1, and $\mathsf{P}(A_n^c) \leqslant \mathsf{P}\{\lambda_{\min}(n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0)) \leqslant \frac{3}{4}\lambda\} + \mathsf{P}(L_n > L)$ gives $\liminf \mathsf{P}(A_n) \geqslant 1-\eta$. On A_n, for $|\boldsymbol{\theta} - \boldsymbol{\theta}_0| \leqslant \delta_1$ and $|\mathbf{u}| = 1$, using (L3) in the form $\|n^{-1}\mathbf{H}_n(\boldsymbol{\theta}) - n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0)\| \leqslant L_n|\boldsymbol{\theta} - \boldsymbol{\theta}_0| \leqslant L\delta_1 \leqslant \lambda/4$,

$$\mathbf{u}'\tfrac{1}{n}\mathbf{H}_n(\boldsymbol{\theta})\mathbf{u} \geqslant \mathbf{u}'\tfrac{1}{n}\mathbf{H}_n(\boldsymbol{\theta}_0)\mathbf{u} - \left\|\tfrac{1}{n}\mathbf{H}_n(\boldsymbol{\theta}) - \tfrac{1}{n}\mathbf{H}_n(\boldsymbol{\theta}_0)\right\| \geqslant \tfrac{3}{4}\lambda - L\delta_1 \geqslant \tfrac{1}{2}\lambda. \tag{P.31}$$

Step 2: the score is of order $\sqrt{n}$. The summands of $\mathbf{U}_n$ are independent with mean zero by (L2) and finite variances, so the cross terms $\mathsf{E}[\nabla\ell_i(\boldsymbol{\theta}_0)'\nabla\ell_k(\boldsymbol{\theta}_0)]$, $i \neq k$, vanish by Lemma P.105 (each ℓ_i depends on Y_i alone) and $\mathsf{E}\,|\mathbf{U}_n|^2 = \sum_i \mathsf{E}\,|\nabla\ell_i(\boldsymbol{\theta}_0)|^2 = \sum_i \operatorname{tr}\operatorname{var}(\nabla\ell_i(\boldsymbol{\theta}_0)) = n\operatorname{tr}\mathbf{F}_n$. Markov's inequality (Theorem P.100) applied to $|\mathbf{U}_n|^2$ at the level $K^2 n$ gives

$$\mathsf{P}\{|\mathbf{U}_n| > K\sqrt{n}\} \leqslant \operatorname{tr}\mathbf{F}_n/K^2. \tag{P.32}$$

Step 3: consistency. With η and δ_1 of Step 1, fix $0 < \delta \leqslant \delta_1$ and choose K with $\sup_n \operatorname{tr}\mathbf{F}_n/K^2 < \eta$, possible since $\operatorname{tr}\mathbf{F}_n \to \operatorname{tr}\mathbf{F}$ makes $(\operatorname{tr}\mathbf{F}_n)$ bounded. Let $C_n = A_n \cap \{|\mathbf{U}_n| \leqslant K\sqrt{n}\}$, so $\liminf \mathsf{P}(C_n) \geqslant 1-2\eta$ by Step 1 and (P.32). For a unit vector $\mathbf{u}$ apply Taylor's theorem

(Theorem P.14, $m = 1$) to $g(t) = \ell_n(\boldsymbol{\theta}_0 + t\delta\mathbf{u})$, whose derivatives are $g'(t) = \delta\mathbf{u}'\nabla\ell_n(\boldsymbol{\theta}_0 + t\delta\mathbf{u})$ and $g''(t) = -\delta^2\mathbf{u}'\mathbf{H}_n(\boldsymbol{\theta}_0 + t\delta\mathbf{u})\mathbf{u}$ by Proposition P.59: for some $s \in (0,1)$,

$$\ell_n(\boldsymbol{\theta}_0 + \delta\mathbf{u}) - \ell_n(\boldsymbol{\theta}_0) = \delta\,\mathbf{u}'\mathbf{U}_n - \tfrac{1}{2}\delta^2\,\mathbf{u}'\mathbf{H}_n(\boldsymbol{\theta}_0 + s\delta\mathbf{u})\mathbf{u} \leqslant \delta K\sqrt{n} - \tfrac{1}{4}n\lambda\delta^2$$

on C_n, using $\mathbf{u}'\mathbf{U}_n \leqslant |\mathbf{U}_n| \leqslant K\sqrt{n}$ (Cauchy–Schwarz) and, as $|s\delta\mathbf{u}| \leqslant \delta \leqslant \delta_1$, $\mathbf{u}'\mathbf{H}_n(\boldsymbol{\theta}_0 + s\delta\mathbf{u})\mathbf{u} \geqslant \frac{1}{2}n\lambda$ by (P.31). The right side is $\delta\sqrt{n}(K - \frac{1}{4}\sqrt{n}\lambda\delta)$, negative once $n > (4K/(\lambda\delta))^2$, and the bound does not depend on $\mathbf{u}$.

Concavity extends this beyond the sphere: if $\boldsymbol{\theta} \in \Theta$ has $|\boldsymbol{\theta} - \boldsymbol{\theta}_0| = r > \delta$ then $\boldsymbol{\theta}_\delta = (1 - \delta/r)\boldsymbol{\theta}_0 + (\delta/r)\boldsymbol{\theta}$ lies on the sphere ($|\boldsymbol{\theta}_\delta - \boldsymbol{\theta}_0| = (\delta/r)r = \delta$) and in Θ, and concavity of ℓ_n, a sum of concave functions, gives $\ell_n(\boldsymbol{\theta}_\delta) \geqslant (1 - \delta/r)\ell_n(\boldsymbol{\theta}_0) + (\delta/r)\ell_n(\boldsymbol{\theta})$; since $\ell_n(\boldsymbol{\theta}_\delta) < \ell_n(\boldsymbol{\theta}_0)$ this forces $\ell_n(\boldsymbol{\theta}) < \ell_n(\boldsymbol{\theta}_0)$. So on C_n and for n large, ℓ_n is strictly below $\ell_n(\boldsymbol{\theta}_0)$ outside the open ball of radius δ.

The closed ball of radius δ is compact (Theorem P.5) and ℓ_n continuous, so a maximum M_n over it is attained (Theorem P.13), at a point $\hat{\boldsymbol{\theta}}_n$ that can be chosen measurably by Remark 18(g). As $M_n \geqslant \ell_n(\boldsymbol{\theta}_0)$ and, by the previous paragraph, $\ell_n < \ell_n(\boldsymbol{\theta}_0)$ on the sphere and outside the ball, $\hat{\boldsymbol{\theta}}_n$ lies in the open ball and $\ell_n(\hat{\boldsymbol{\theta}}_n) = M_n \geqslant \ell_n(\boldsymbol{\theta})$ for every $\boldsymbol{\theta} \in \Theta$: it is a maximum over all of

Θ. This holds on C_n for all large n, an event of probability at least $1-3\eta$ eventually, and η is arbitrary; this proves (i). Also $\liminf \mathsf{P}\{|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0| < \delta\} \geqslant 1-2\eta$ with η arbitrary and δ arbitrarily small proves (ii). If some ℓ_i is strictly concave then so is ℓ_n, a strictly concave function plus concave ones, and a strictly concave function has at most one maximiser: two distinct maximisers $\boldsymbol{\theta}, \boldsymbol{\theta}'$ with common value M would give $\ell_n(\frac{1}{2}\boldsymbol{\theta} + \frac{1}{2}\boldsymbol{\theta}') > M$.

Step 4: the linearisation. On C_n and for large n the maximiser is interior, so $\nabla \ell_n(\hat{\boldsymbol{\theta}}_n) = \mathbf{0}$ (as in the proof of Proposition P.22). Write $\mathbf{h} = \hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0$. Applying the mean value theorem (Corollary P.7) to $t \mapsto \partial_j \ell_n(\boldsymbol{\theta}_0 + t\mathbf{h})$, whose derivative is $\sum_k \partial_k \partial_j \ell_n(\boldsymbol{\theta}_0 + t\mathbf{h}) h_k = -(\mathbf{H}_n(\boldsymbol{\theta}_0 + t\mathbf{h})\mathbf{h})_j$ by Proposition P.59, gives, for each j, an $s_j \in (0,1)$ with

$$0 - \big(\mathbf{U}_n\big)_j = \big(\nabla \ell_n(\hat{\boldsymbol{\theta}}_n) - \nabla \ell_n(\boldsymbol{\theta}_0)\big)_j = -\big(\mathbf{H}_n(\boldsymbol{\theta}_0 + s_j\mathbf{h})\,\mathbf{h}\big)_j,$$

that is $\mathbf{U}_n = \tilde{\mathbf{H}}_n \mathbf{h}$, where row j of $\tilde{\mathbf{H}}_n$ is row j of $\mathbf{H}_n(\boldsymbol{\theta}_0 + s_j\mathbf{h})$. By Remark 18(g), applied to the closed set of $s \in [0,1]$ at which the display holds, the s_j can be chosen measurably, possibly at an endpoint, which changes nothing below. When $|\mathbf{h}| \leqslant \delta_0$, an event of probability tending to one by (ii), row j of $\tilde{\mathbf{H}}_n - \mathbf{H}_n(\boldsymbol{\theta}_0)$ is row j of $\mathbf{B}_j = \mathbf{H}_n(\boldsymbol{\theta}_0 + s_j\mathbf{h}) - \mathbf{H}_n(\boldsymbol{\theta}_0)$, of operator norm at most $nL_n s_j |\mathbf{h}| \leqslant nL_n|\mathbf{h}|$ by (L3). A row of a matrix $\mathbf{B}$ has length at most $\|\mathbf{B}\|$: for the

symmetric $\mathbf{B}_j$, $|\mathbf{e}_j'\mathbf{B}_j| = |\mathbf{B}_j\mathbf{e}_j| \leqslant \|\mathbf{B}_j\|$. And a matrix $\mathbf{M}$ with rows $\mathbf{m}_j'$ has $|\mathbf{M}\mathbf{x}|^2 = \sum_j (\mathbf{m}_j'\mathbf{x})^2 \leqslant \sum_j |\mathbf{m}_j|^2|\mathbf{x}|^2 \leqslant p \max_j |\mathbf{m}_j|^2|\mathbf{x}|^2$ by Theorem P.36. So $\|n^{-1}\tilde{\mathbf{H}}_n - n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0)\| \leqslant \sqrt{p}\, L_n|\mathbf{h}|$. This tends to 0 in probability: given η and ε, choose L with $\mathsf{P}(L_n > L) < \eta$, and then $\mathsf{P}(\sqrt{p}L_n|\mathbf{h}| > \varepsilon) \leqslant \eta + \mathsf{P}(|\mathbf{h}| > \varepsilon/(\sqrt{p}L))$, whose second term tends to 0 by (ii). With (L3) again, the triangle inequality $\|n^{-1}\tilde{\mathbf{H}}_n - \mathbf{G}\| \leqslant \|n^{-1}\tilde{\mathbf{H}}_n - n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0)\| + \|n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0) - \mathbf{G}\|$ and the bound $|a_{rs}| \leqslant \|\mathbf{A}\|$ on the entries give

$$\mathbf{G}_n := n^{-1}\tilde{\mathbf{H}}_n \xrightarrow{p} \mathbf{G}. \tag{P.33}$$

The same argument with $\mathbf{H}_n(\hat{\boldsymbol{\theta}}_n)$ in place of $\tilde{\mathbf{H}}_n$, using $\|\mathbf{H}_n(\hat{\boldsymbol{\theta}}_n) - \mathbf{H}_n(\boldsymbol{\theta}_0)\| \leqslant nL_n|\mathbf{h}|$ directly, gives (v).

Step 5: asymptotic normality. Write $\mathbf{Z}_n = n^{-1/2}\mathbf{U}_n$, which converges in distribution to $N(\mathbf{0}, \mathbf{F})$ by (L2). By the bound $\|\mathbf{A}\|^2 \leqslant \operatorname{tr}(\mathbf{A}'\mathbf{A})$ of Step 1, (P.33) gives $\|\mathbf{G}_n - \mathbf{G}\| \xrightarrow{p} 0$, and the event $D_n = \{\|\mathbf{G}_n - \mathbf{G}\| \leqslant \lambda/2\}$ has probability tending to one. Since $\|\mathbf{G}^{-1}\| = 1/\lambda$ by Proposition P.53 ($\mathbf{G}$ being symmetric positive definite), $|\mathbf{v}| = |\mathbf{G}^{-1}\mathbf{G}\mathbf{v}| \leqslant |\mathbf{G}\mathbf{v}|/\lambda$, so on D_n, $|\mathbf{G}_n\mathbf{v}| \geqslant |\mathbf{G}\mathbf{v}| - \|\mathbf{G}_n - \mathbf{G}\|\,|\mathbf{v}| \geqslant \lambda|\mathbf{v}| - \frac{1}{2}\lambda|\mathbf{v}| = \lambda|\mathbf{v}|/2$ for every $\mathbf{v}$. So $\mathbf{G}_n\mathbf{v} = \mathbf{0}$ forces $\mathbf{v} = \mathbf{0}$, the square matrix $\mathbf{G}_n$ is invertible (Theorem P.32), and with $\mathbf{w} = \mathbf{G}_n\mathbf{v}$, $|\mathbf{G}_n^{-1}\mathbf{w}| = |\mathbf{v}| \leqslant 2|\mathbf{w}|/\lambda$, that is

$\|\mathbf{G}_n^{-1}\| \leqslant 2/\lambda$. On $C_n \cap D_n$, $\mathbf{U}_n = \tilde{\mathbf{H}}_n\mathbf{h} = n\mathbf{G}_n\mathbf{h}$, so $\sqrt{n}\,\mathbf{h} = \mathbf{G}_n^{-1}\mathbf{Z}_n$. There $\mathbf{G}_n^{-1} - \mathbf{G}^{-1} = \mathbf{G}_n^{-1}(\mathbf{G} - \mathbf{G}_n)\mathbf{G}^{-1}$ has norm at most $\|\mathbf{G}_n^{-1}\|\,\|\mathbf{G} - \mathbf{G}_n\|\,\|\mathbf{G}^{-1}\| \leqslant 2\|\mathbf{G}_n - \mathbf{G}\|/\lambda^2$, so $\mathbf{G}_n^{-1} \xrightarrow{p} \mathbf{G}^{-1}$ (with the convention $\mathbf{A}^{-1} := \mathbf{0}$ off D_n). By Theorem P.139(i) $(\mathbf{Z}_n, \mathbf{G}_n^{-1}) \xrightarrow{d} (Z, \mathbf{G}^{-1})$ with $Z \sim N(\mathbf{0}, \mathbf{F})$, and the continuous map $(\mathbf{z}, \mathbf{A}) \mapsto \mathbf{A}\mathbf{z}$ with Theorem P.138 gives $\mathbf{G}_n^{-1}\mathbf{Z}_n \xrightarrow{d} \mathbf{G}^{-1}Z$. By Theorem P.114 that law is $N(\mathbf{0}, \mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1})$. Finally $\sqrt{n}\,\mathbf{h}$ and $\mathbf{G}_n^{-1}\mathbf{Z}_n$ agree on $C_n \cap D_n$, so for $f \in C_b(\mathbb{R}^p)$, $|\,\mathsf{E}\, f(\sqrt{n}\,\mathbf{h}) - \mathsf{E}\, f(\mathbf{G}_n^{-1}\mathbf{Z}_n)| \leqslant 2\sup|f|\;\mathsf{P}\{(C_n \cap D_n)^c\}$, whose lim sup is at most $2\sup|f| \cdot 2\eta$; as η is arbitrary, $\mathsf{E}\, f(\sqrt{n}\,\mathbf{h}) \to \mathsf{E}\, f(\mathbf{G}^{-1}Z)$, which is (iii). If $\mathbf{F} = \mathbf{G}$ the variance collapses to $\mathbf{G}^{-1}\mathbf{G}\mathbf{G}^{-1} = \mathbf{F}^{-1}$.

Step 6: the likelihood ratio at $\boldsymbol{\theta}_0$. Taylor's theorem applied to $t \mapsto \ell_n(\boldsymbol{\theta}_0 + t\mathbf{h})$ gives an $s \in (0,1)$, chosen measurably as in Step 4, with

$$2\{\ell_n(\hat{\boldsymbol{\theta}}_n) - \ell_n(\boldsymbol{\theta}_0)\} = 2\mathbf{U}_n'\mathbf{h} - \mathbf{h}'\mathbf{H}_n(\boldsymbol{\theta}_0 + s\mathbf{h})\mathbf{h} = 2\mathbf{Z}_n'\mathbf{G}_n^{-1}\mathbf{Z}_n - (\mathbf{G}_n^{-1}\mathbf{Z}_n)'\mathbf{K}_n(\mathbf{G}_n^{-1}\mathbf{Z}_n),$$

the second equality holding on $C_n \cap D_n$, where $\mathbf{U}_n = \sqrt{n}\mathbf{Z}_n$ and $\mathbf{h} = n^{-1/2}\mathbf{G}_n^{-1}\mathbf{Z}_n$, and where $\mathbf{K}_n = n^{-1}\mathbf{H}_n(\boldsymbol{\theta}_0 + s\mathbf{h}) \xrightarrow{p} \mathbf{G}$ by the argument of Step 4 ($|s\mathbf{h}| \leqslant |\mathbf{h}|$). By Theorem P.139(i), $(\mathbf{Z}_n, \mathbf{G}_n^{-1}, \mathbf{K}_n) \xrightarrow{d} (Z, \mathbf{G}^{-1}, \mathbf{G})$, and the continuous map $(\mathbf{z}, \mathbf{A}, \mathbf{K}) \mapsto 2\mathbf{z}'\mathbf{A}\mathbf{z} - (\mathbf{A}\mathbf{z})'\mathbf{K}(\mathbf{A}\mathbf{z})$ with

Theorem P.138 gives the limit $2Z'\mathbf{G}^{-1}Z - Z'\mathbf{G}^{-1}\mathbf{G}\mathbf{G}^{-1}Z = Z'\mathbf{G}^{-1}Z$, the complement of $C_n \cap D_n$ being handled as at the end of Step 5. Under $\mathbf{F} = \mathbf{G}$ this is $Z'\mathbf{F}^{-1}Z$ with $Z \sim N(\mathbf{0}, \mathbf{F})$, which is χ^2_p by Theorem P.157, completing (iv). For (vi), the display minus $\mathbf{Z}_n'\mathbf{G}^{-1}\mathbf{Z}_n$ is $\mathbf{Z}_n'\mathbf{M}_n\mathbf{Z}_n$ with $\mathbf{M}_n = 2\mathbf{G}_n^{-1} - (\mathbf{G}_n^{-1})'\mathbf{K}_n\mathbf{G}_n^{-1} - \mathbf{G}^{-1}$, and $\mathbf{M}_n \xrightarrow{p} \mathbf{0}$ since $\mathbf{G}_n^{-1} \xrightarrow{p} \mathbf{G}^{-1}$, $(\mathbf{G}_n^{-1})' \xrightarrow{p} \mathbf{G}^{-1}$ ($\mathbf{G}$ is symmetric) and $\mathbf{K}_n \xrightarrow{p} \mathbf{G}$, by Theorem P.138; Slutsky and the continuous mapping theorem give $\mathbf{Z}_n'\mathbf{M}_n\mathbf{Z}_n \xrightarrow{d} Z'\mathbf{0}Z = 0$, which is convergence in probability by Proposition P.122. □

Corollary P.161 (Wald)**.** Under Theorem P.160 with $\mathbf{F} = \mathbf{G}$, let $\mathbf{R}$ be a fixed $q \times p$ matrix of rank q and $\hat{\mathbf{G}}_n = n^{-1}\mathbf{H}_n(\hat{\boldsymbol{\theta}}_n)$. If $\mathbf{R}\boldsymbol{\theta}_0 = \mathbf{r}$ then

$$W_n = (\mathbf{R}\hat{\boldsymbol{\theta}}_n - \mathbf{r})'\left\{\tfrac{1}{n}\mathbf{R}\hat{\mathbf{G}}_n^{-1}\mathbf{R}'\right\}^{-1}(\mathbf{R}\hat{\boldsymbol{\theta}}_n - \mathbf{r}) \xrightarrow{d} \chi^2_q,$$

both inverses taken with the convention $\mathbf{A}^{-1} := \mathbf{0}$ for singular $\mathbf{A}$ of Theorem P.157. The same holds without $\mathbf{F} = \mathbf{G}$ if $\hat{\mathbf{G}}_n^{-1}$ is replaced by $\hat{\mathbf{G}}_n^{-1}\hat{\mathbf{F}}_n\hat{\mathbf{G}}_n^{-1}$ for any symmetric $\hat{\mathbf{F}}_n \xrightarrow{p} \mathbf{F}$.

Proof. By (iv), Theorem P.138 (the linear map $\mathbf{R}$ is continuous) and Theorem P.114, $V_n := \sqrt{n}(\mathbf{R}\hat{\boldsymbol{\theta}}_n - \mathbf{r}) = \mathbf{R}\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{R}\mathbf{F}^{-1}\mathbf{R}')$, using $\mathbf{R}\boldsymbol{\theta}_0 = \mathbf{r}$. The matrix $\mathbf{R}\mathbf{F}^{-1}\mathbf{R}'$ is positive definite: $\mathbf{F}^{-1}$ is positive definite (its eigenvalues are the reciprocals of those of $\mathbf{F}$, Proposition P.42), so $\mathbf{v}'\mathbf{R}\mathbf{F}^{-1}\mathbf{R}'\mathbf{v} = (\mathbf{R}'\mathbf{v})'\mathbf{F}^{-1}(\mathbf{R}'\mathbf{v}) > 0$ unless $\mathbf{R}'\mathbf{v} = \mathbf{0}$, and $\mathbf{R}'$ has rank q, so $\mathcal{N}(\mathbf{R}') = \{\mathbf{0}\}$ by Theorem P.32. By (v), $\hat{\mathbf{G}}_n \xrightarrow{p} \mathbf{G} = \mathbf{F}$, and $\hat{\mathbf{G}}_n$ is symmetric, a Hessian of a twice continuously differentiable function; by Lemma P.43 (inversion, extended as in Theorem P.157, is measurable and continuous at the positive definite $\mathbf{F}$ on the symmetric matrices) and Theorem P.138, $\hat{\boldsymbol{\Sigma}}_n := \mathbf{R}\hat{\mathbf{G}}_n^{-1}\mathbf{R}' \xrightarrow{p} \mathbf{R}\mathbf{F}^{-1}\mathbf{R}'$, a symmetric matrix. Since $\{\frac{1}{n}\mathbf{A}\}^{-1} = n\mathbf{A}^{-1}$, also under the convention for singular $\mathbf{A}$, $W_n = V_n'\hat{\boldsymbol{\Sigma}}_n^{-1}V_n$, and Theorem P.157 gives $W_n \xrightarrow{d} \chi^2_q$. In the general case the same argument runs with the limiting variance $\mathbf{R}\mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1}\mathbf{R}'$ of V_n, by (iii), positive definite as before since $\mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1}$ is ($\mathbf{v}'\mathbf{G}^{-1}\mathbf{F}\mathbf{G}^{-1}\mathbf{v} = (\mathbf{G}^{-1}\mathbf{v})'\mathbf{F}(\mathbf{G}^{-1}\mathbf{v}) > 0$ for $\mathbf{v} \neq \mathbf{0}$), and its consistent estimate $\mathbf{R}\hat{\mathbf{G}}_n^{-1}\hat{\mathbf{F}}_n\hat{\mathbf{G}}_n^{-1}\mathbf{R}'$ (Theorem P.138), symmetric as $\hat{\mathbf{F}}_n$ and $\hat{\mathbf{G}}_n$ are. □

Theorem P.162 (Likelihood ratio)**.** Under Theorem P.160 with $\mathbf{F} = \mathbf{G}$, partition $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)$ with $\boldsymbol{\theta}_2$ of dimension q, and let $\hat{\boldsymbol{\theta}}_n^R$ maximise ℓ_n over $\Theta \cap \{\boldsymbol{\theta}_2 = \boldsymbol{\theta}_{2,0}\}$, where $\boldsymbol{\theta}_{2,0}$ is the true value.

Suppose the hypotheses of Theorem P.160 also hold for the restricted problem, again with $\mathbf{F} = \mathbf{G}$ and with limiting information $\mathbf{F}_{11}$, the leading $(p-q)$ block of $\mathbf{F}$. Then

$$\mathrm{LR}_n = 2\{\ell_n(\hat{\boldsymbol{\theta}}_n) - \ell_n(\hat{\boldsymbol{\theta}}_n^R)\} \xrightarrow{d} \chi^2_q.$$

Proof. Applying Theorem P.160(vi), with $\mathbf{G} = \mathbf{F}$, to the full problem and, in the subspace $\{\boldsymbol{\theta}_2 = \boldsymbol{\theta}_{2,0}\}$, to the restricted one,

$$2\{\ell_n(\hat{\boldsymbol{\theta}}_n) - \ell_n(\boldsymbol{\theta}_0)\} = \mathbf{Z}_n'\mathbf{F}^{-1}\mathbf{Z}_n + R_n, \qquad 2\{\ell_n(\hat{\boldsymbol{\theta}}_n^R) - \ell_n(\boldsymbol{\theta}_0)\} = \mathbf{Z}_{n,1}'\mathbf{F}_{11}^{-1}\mathbf{Z}_{n,1} + R_n^R,$$

with $R_n \xrightarrow{p} 0$ and $R_n^R \xrightarrow{p} 0$, where $\mathbf{Z}_n = n^{-1/2}\mathbf{U}_n \xrightarrow{d} Z \sim N(\mathbf{0}, \mathbf{F})$ and $\mathbf{Z}_{n,1}$ is its first $p-q$ coordinates, the score of the restricted model (the gradient in $\boldsymbol{\theta}_1$ at $\boldsymbol{\theta}_0$); the restricted curvature $\mathbf{G}$ is the leading block of the full one, which equals $\mathbf{F}_{11}$. Subtracting,

$$\mathrm{LR}_n = \mathbf{Z}_n'\mathbf{F}^{-1}\mathbf{Z}_n - \mathbf{Z}_{n,1}'\mathbf{F}_{11}^{-1}\mathbf{Z}_{n,1} + R_n - R_n^R \xrightarrow{d} Z'\mathbf{F}^{-1}Z - Z_1'\mathbf{F}_{11}^{-1}Z_1$$

by the continuous mapping theorem (Theorem P.138, applied to the continuous map $\mathbf{z} \mapsto \mathbf{z}'\mathbf{F}^{-1}\mathbf{z} - \mathbf{z}_1'\mathbf{F}_{11}^{-1}\mathbf{z}_1$) and Slutsky (Theorem P.139(iii), as $R_n - R_n^R \xrightarrow{p} 0$).

Write $\mathbf{A} = [\mathbf{I}_{p-q} \quad \mathbf{0}]$, so $Z_1 = \mathbf{A}Z$ and $\mathbf{F}_{11} = \mathbf{AFA}'$, and put $W = \mathbf{F}^{-1/2}Z$, which is $N(\mathbf{0}, \mathbf{I}_p)$ by Theorem P.114 and the symmetric square root of Theorem P.44 (its covariance is $\mathbf{F}^{-1/2}\mathbf{F}\mathbf{F}^{-1/2} = \mathbf{I}_p$). Then $Z = \mathbf{F}^{1/2}W$, $\mathbf{F}^{-1} = \mathbf{F}^{-1/2}\mathbf{F}^{-1/2}$, $Z'\mathbf{F}^{-1}Z = |W|^2$, and $Z_1'\mathbf{F}_{11}^{-1}Z_1 = (\mathbf{AF}^{1/2}W)'(\mathbf{AFA}')^{-1}(\mathbf{AF}^{1/2}W) = W'\mathbf{P}W$ with

$$\mathbf{P} = \mathbf{F}^{1/2}\mathbf{A}'(\mathbf{AFA}')^{-1}\mathbf{AF}^{1/2},$$

where $\mathbf{AFA}' = \mathbf{F}_{11}$ is invertible, being positive definite ($\mathbf{v}'\mathbf{AFA}'\mathbf{v} = (\mathbf{A}'\mathbf{v})'\mathbf{F}(\mathbf{A}'\mathbf{v}) > 0$ for $\mathbf{v} \neq \mathbf{0}$, as $\mathbf{A}'\mathbf{v} \neq \mathbf{0}$). It is symmetric, as $(\mathbf{AFA}')^{-1}$ is, and, using $\mathbf{F}^{1/2}\mathbf{F}^{1/2} = \mathbf{F}$ and the cyclic property of the trace (Proposition P.35(ii)),

$$\begin{aligned}\mathbf{P}^2 &= \mathbf{F}^{1/2}\mathbf{A}'(\mathbf{AFA}')^{-1}(\mathbf{AFA}')(\mathbf{AFA}')^{-1}\mathbf{AF}^{1/2} = \mathbf{P},\\ \operatorname{tr}\mathbf{P} &= \operatorname{tr}\{(\mathbf{AFA}')^{-1}\mathbf{AFA}'\} = \operatorname{tr}\mathbf{I}_{p-q} = p-q.\end{aligned}$$

Hence $\mathbf{I}-\mathbf{P}$ is symmetric, idempotent ($(\mathbf{I}-\mathbf{P})^2 = \mathbf{I}-2\mathbf{P}+\mathbf{P}^2 = \mathbf{I}-\mathbf{P}$), with trace $p-(p-q) = q$, hence rank q by Lemma P.46; and the limit is $|W|^2 - W'\mathbf{P}W = W'(\mathbf{I}-\mathbf{P})W \sim \chi^2_q$ by Theorem P.135. □

Remark. The two statistics use different parts of the same expansion: Wald uses $\hat{\boldsymbol{\theta}}_n$ and the curvature at it, the likelihood ratio uses the heights of the two fitted log-likelihoods, and both are χ^2_q in the limit. They are not equal in finite samples.

33 The Statistics of This Course

Remark (The Breusch–Pagan statistic)**.** In Unit 4 the errors ε_i are independent and identically distributed with mean 0, variance σ^2 and $\mathsf{E}\,\varepsilon_i^4 < \infty$, with ε_i^2 not almost surely constant (so that $\kappa > 0$ below), $\mathbf{g}$ is the vector with entries $g_i = \varepsilon_i^2/\sigma^2 - 1$, and $\mathbf{Z}$ is the $n \times p$ matrix whose rows are the variance regressors $\mathbf{z}_i'$, centred so that $\mathbf{1}'\mathbf{Z} = \mathbf{0}'$ and hence $\mathbf{M}_0\mathbf{Z} = \mathbf{Z}$. There $V_n = n^{-1/2}\mathbf{Z}'\mathbf{g}$ satisfies $V_n \xrightarrow{d} N(\mathbf{0},\ \kappa\mathbf{Q})$ by Corollary P.143, whose proof uses the $\mathbf{z}_i$ only through the weights a_{ni} and so applies verbatim to the centred $\mathbf{z}_i - \bar{\mathbf{z}}_n$, assumed to satisfy its two design conditions: the summands $\mathbf{z}_i(\varepsilon_i^2/\sigma^2 - 1)$ are independent, mean zero, and carry non-identical weights, so the Lindeberg–Feller form is required and the identically-distributed Corollary P.142 does not apply. The constant is $\kappa = \operatorname{var}(g_i) = \operatorname{var}(\varepsilon_i^2)/\sigma^4$, the common variance that Corollary P.143 calls τ^2, with $\kappa = 2$ under normality by Lemma P.117. The empirical $n^{-1}\mathbf{Z}'\mathbf{M}_0\mathbf{Z} = n^{-1}\mathbf{Z}'\mathbf{Z}$ converges to

the $\mathbf{Q}$ of Corollary P.143, so Theorem P.157 delivers the χ^2_p limit, and Slutsky's theorem lets the unknown σ^2 and κ be replaced by consistent estimates without changing the limit: κ enters only the weight matrix, and for σ^2, which sits inside g_i, $\mathbf{1}'\mathbf{Z} = \mathbf{0}'$ makes V_n computed with $\hat{\sigma}^2$ equal to $(\sigma^2/\hat{\sigma}^2)V_n$.

34 Numerical Illustration

The array below has $\sum_j \operatorname{var}(\xi_{nj}) = 1$ for every n and mean zero, but one summand carries a fixed share of the variance, violating the conclusion $\sup_j c_{nj} \to 0$ of Step 1 of the proof of Theorem P.141.

```
set.seed(1)
one_row <- function(n, share) {
  # summand 1 carries `share` of the total variance; the rest split the remainder
  v1 <- share; vr <- (1 - share) / (n - 1)
  c(rnorm(1, sd = sqrt(v1)), rnorm(n - 1, sd = sqrt(vr)))
}
lindeberg_sum <- function(n, share, eps = 0.3, B = 4000) {
  v1 <- share; vr <- (1 - share)/(n - 1)
  sds <- c(sqrt(v1), rep(sqrt(vr), n - 1))
  # E[xi^2 ; |xi|>eps] for a centred normal with sd s
```

```r
  contrib <- sapply(sds, function(s)
    2 * integrate(function(x) x^2 * dnorm(x, sd = s), eps, Inf)$value)
  sum(contrib)
}
c(share_equal = lindeberg_sum(400, 1/400),  # equal weights: condition holds
  share_half = lindeberg_sum(400, 0.5))     # one dominant term: condition fails
```

```
##   share_equal    share_half
## 0.00000007488 0.49037575671
```

With equal weights the Lindeberg sum is numerically zero and the row sums are close to normal; with one dominant summand it is bounded away from zero and the dominant summand never becomes negligible.

The two design conditions of Corollary P.143 are limits and so cannot be verified on a single sample; what a sample does show is the finite-n quantity behind the second of them, the largest share of $\sum_i |\mathbf{z}_i|^2$ carried by one observation. If that share is small the design is far from the degenerate case, and if it is close to 1 no central limit theorem applies however large n is.

```r
suppressMessages(library(ISLR2))
z  <- Boston$lstat; nn <- length(z)
```

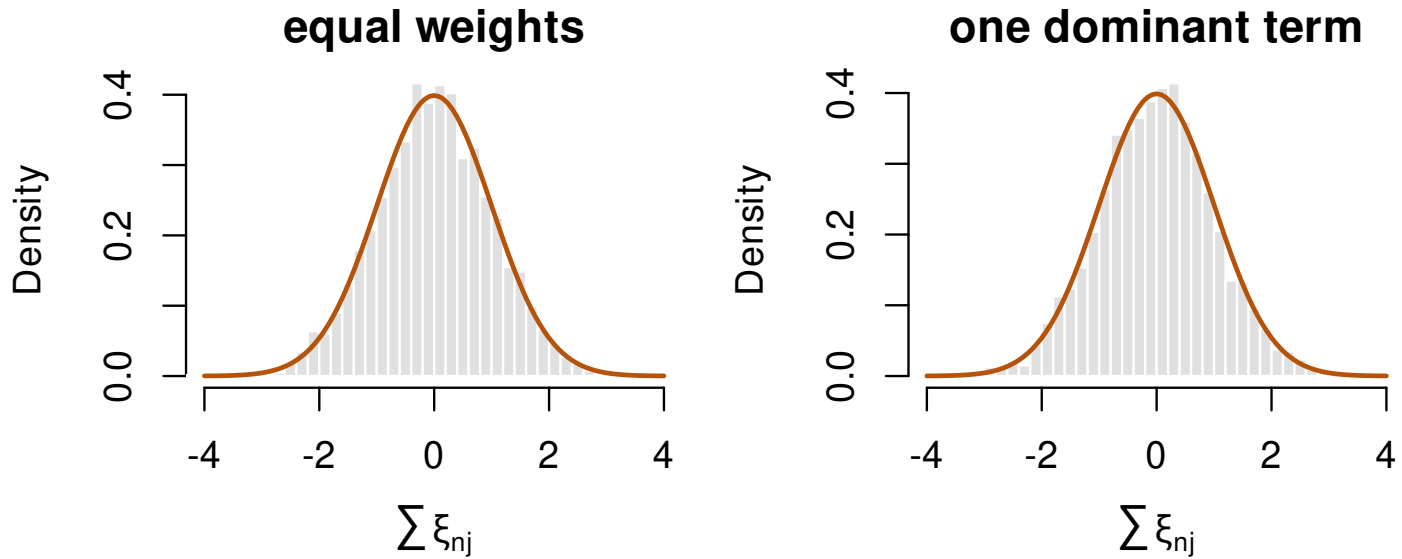


Figure P.1: Row sums of a triangular array against the $N(0,1)$ density. Both rows have total variance 1; only the left satisfies (P.22). In the right panel the sum is a sum of two normals and so is itself exactly normal. The illustration is that the limit is not reached **by accumulation**, the dominant term never becoming negligible.

```
Zc <- z - mean(z)                                   # centred, as Z'MOZ requires
```

```
c(Q_n            = sum(Zc^2)/nn,                  # estimates Q; must be positive
  max_share      = max(Zc^2)/sum(Zc^2),           # largest single share
  share_if_equal = 1/nn,                          # the value under equal weights
  ratio          = max(Zc^2)/sum(Zc^2)*nn)        # how many times the equal share
```

```
##            Q_n      max_share share_if_equal          ratio
##      50.893979       0.024889       0.001976      12.593774
```

The largest observation carries about 2.5% of the total, roughly twelve times an equal share: unequal, but nowhere near domination by one point.

```
set.seed(3)
B <- 4000
Tn <- replicate(B, {
  eps <- rnorm(nn)                                # H0: homoscedastic normal
  g   <- eps^2 - 1
  Vn  <- sum(Zc * g)/sqrt(nn)
  Vn^2 / (2 * sum(Zc^2)/nn)                       # Theorem thm:quadform with p = 1
})
c(mean_stat = mean(Tn), chisq1_mean = 1,
  var_stat  = var(Tn),  chisq1_var  = 2,
```

```
  rejection_rate_5pct = mean(Tn > qchisq(0.95, 1)))
```

```
##          mean_stat         chisq1_mean            var_stat
##            1.03319             1.00000             2.09932
##         chisq1_var rejection_rate_5pct
##            2.00000             0.05375
```

The simulated statistic has the mean, variance and upper tail of a χ^2_1, which is Theorem P.157 with the covariance $\kappa\mathbf{Q}$ and $\kappa = 2$ supplied by Corollary P.143 under normality.

The variance of $n^{-1/2}\sum_t u_{t-1}\varepsilon_t$ is $\tau^4\sum_j a_j^2$ **exactly**, for every n, by Corollary P.154; what the theorem adds is that the law is asymptotically normal. Both are checked below for $a_j = \phi^j$ with $\phi = 0.8$, the score of a first-order autoregression, together with the truncation bound $\mathsf{E}[(S_n - S_n^{(m)})^2] = \tau^4\sum_{j\geqslant m} a_j^2$ that drives the proof.

```
set.seed(5); phi <- 0.8; B <- 4000; ms <- c(1, 4, 16)
sim <- function(n) t(replicate(B, {
  e <- rnorm(n + 80); k <- 81:(n + 80)
  u <- as.numeric(stats::filter(e, phi, "recursive"))
  S <- sum(u[k - 1]*e[k])/sqrt(n)
  c(S, sapply(ms, function(m) {
```

```
      um <- as.numeric(stats::filter(e, phi^(0:(m - 1)), sides = 1))
      S - sum(um[k - 1]*e[k])/sqrt(n) })) }))
R <- sim(400)
t(sapply(seq_along(ms), function(i)
  c(m = ms[i], mean_squared_difference = mean(R[, i + 1]^2),
    theory_tail_sum = phi^(2*ms[i])/(1 - phi^2))))
```

```
##       m mean_squared_difference theory_tail_sum
## [1,]  1                1.802536        1.777778
## [2,]  4                0.436599        0.466034
## [3,] 16                0.002291        0.002201
```

The truncation error is what the approximation lemma needs to be small uniformly in n, and it is: at $m = 16$ the m-dependent array of the proof reproduces the score to within 0.0023 in mean square, against a variance of 2.78.

```
t(sapply(c(50, 200, 800), function(n) {
  S <- sim(n)[, 1]
  c(n = n, variance = var(S), theory = 1/(1 - phi^2),
    skewness = mean((S - mean(S))^3)/sd(S)^3,
    kurtosis = mean((S - mean(S))^4)/var(S)^2) }))
```

```
##        n variance theory skewness kurtosis
## [1,]  50    2.723  2.778   0.9377    4.755
## [2,] 200    2.835  2.778   0.5092    3.368
## [3,] 800    2.767  2.778   0.2295    3.136
```

The variance matches at every n, as it must. The skewness and kurtosis fall towards their normal values 0 and 3: the skewness roughly halves each time n is quadrupled, which is the $n^{-1/2}$ rate, and the excess kurtosis falls faster, by a factor of roughly four, which is the n^{-1} rate (the last step, a factor of 2.7, is within simulation error: at $n = 800$ a kurtosis estimate from 4000 draws has standard error about 0.1). The approach is not fast: at $n = 50$ the kurtosis is still 4.76, so normality here is an asymptotic statement and not a small-sample one.

35 Exercises

An exercise with a source in its title is from Knapp, **Basic Real Analysis** (digital second edition, 2016), Abadir and Magnus, **Matrix Algebra** (2005), or Baldi, **Probability: An Introduction Through Theory and Exercises** (2023), under that book's number. Its solution is based on the book's hint or solution, adapted to the notation and results of these notes.

Analysis

Exercise P.1 (Knapp, problem I.2)**.** Let $a > 0$, $x_0 = 1$ and $x_{n+1} = (x_n^2 + a)/(2x_n)$ for $n \geqslant 0$ (Newton's method for $x^2 = a$). Prove that (x_n) converges and that its limit is $\sqrt{a}$.

Solution. By induction every x_n is positive: $x_0 = 1$, and $x_n > 0$ gives $x_{n+1} > 0$. For $n \geqslant 0$,

$$x_{n+1} - \sqrt{a} = \frac{x_n^2 - 2\sqrt{a}\,x_n + a}{2x_n} = \frac{(x_n - \sqrt{a})^2}{2x_n} \geqslant 0,$$

so $x_n \geqslant \sqrt{a}$ for every $n \geqslant 1$. Then for $n \geqslant 1$, $x_n - x_{n+1} = (x_n^2 - a)/(2x_n) \geqslant 0$, so $(x_n)_{n\geqslant 1}$ is decreasing and bounded below by $\sqrt{a}$. Theorem P.2, applied to the increasing bounded sequence $(-x_n)_{n\geqslant 1}$, shows that x_n converges to some $L \geqslant \sqrt{a} > 0$. Letting $n \to \infty$ in $2x_n x_{n+1} = x_n^2 + a$

gives $2L^2 = L^2 + a$, so $L^2 = a$ and, L being positive, $L = \sqrt{a}$. ■

Exercise P.2 (Knapp, problem II.5)**.** Let $U \subseteq \mathbb{R}^d$ be open and let $E_1 \supseteq E_2 \supseteq \cdots$ be closed bounded subsets of $\mathbb{R}^d$ with $\bigcap_n E_n \subseteq U$.

(a) Show that $E_N \subseteq U$ for some N.

(b) Show by examples that the conclusion fails if the E_n are only assumed bounded, or only assumed closed.

Solution. (a) Suppose not. Then for every n there is $x_n \in E_n$ with $x_n \notin U$. All x_n lie in the bounded set E_1, so by Theorem P.3 a subsequence x_{n_k} converges to some x. Fix m. For every k with $n_k \geqslant m$, $x_{n_k} \in E_{n_k} \subseteq E_m$, and E_m is closed, so $x \in E_m$. Hence $x \in \bigcap_m E_m \subseteq U$. As U is open there is $\delta > 0$ with $\{y : |y - x| < \delta\} \subseteq U$, and $|x_{n_k} - x| < \delta$ for all large k, so $x_{n_k} \in U$ for those k, contradicting the choice of the x_n.

(b) In both examples $d = 1$ and $U = \emptyset$, which is open. The bounded sets $E_n = \mathbb{Q} \cap [\sqrt{2}, \sqrt{2} + 1/n]$ decrease and $\bigcap_n E_n = \emptyset$, since the only real number in every $[\sqrt{2}, \sqrt{2} + 1/n]$ is the irrational $\sqrt{2}$; they are not closed, which is where (a) fails: a limit of points of E_n may be $\sqrt{2} \notin E_n$. The closed sets $E_n = [n, \infty)$ decrease and $\bigcap_n E_n = \emptyset$; they are unbounded, and the points $x_n = n$ have

no convergent subsequence. Every E_N is non-empty: $[N,\infty)$ plainly, and $\mathbb{Q}\cap[\sqrt{2},\sqrt{2}+1/N]$ contains the rational $\lceil 10^k\sqrt{2}\rceil/10^k$ once $10^{-k}<1/N$. So in neither case is any E_N contained in $U=\emptyset$. ■

Exercise P.3 (Knapp, problem I.4)**.** For real sequences (a_n) and (b_n) prove that

$$\limsup_n (a_n+b_n)\leqslant \limsup_n a_n+\limsup_n b_n,$$

provided the right side is not of the form $\infty-\infty$. Give an example in which the inequality is strict.

Solution. Let $A=\limsup_n a_n$ and $B=\limsup_n b_n$ (Definition P.8). If $A=\infty$ or $B=\infty$ the right side is ∞ (the other term not being $-\infty$) and there is nothing to prove. Otherwise put $\alpha_m=\sup_{n\geqslant m}a_n$ and $\beta_m=\sup_{n\geqslant m}b_n$; these decrease to A and B, and they are finite for all large m, say $m\geqslant m_0$, because $A,B<\infty$ (if $\alpha_m=\infty$ for every m then $A=\infty$) and $\alpha_m\geqslant a_m>-\infty$.

For $n \geqslant m$, $a_n + b_n \leqslant \alpha_m + \beta_m$, so $\sup_{n \geqslant m}(a_n + b_n) \leqslant \alpha_m + \beta_m$ and therefore

$$\limsup_n (a_n + b_n) = \inf_m \sup_{n \geqslant m} (a_n + b_n) \leqslant \alpha_m + \beta_m \qquad (m \geqslant m_0).$$

If A and B are finite, $\alpha_m + \beta_m \to A + B$. If, say, $A = -\infty$, then $\beta_m \leqslant \beta_{m_0} < \infty$ while $\alpha_m \to -\infty$, so $\alpha_m + \beta_m \to -\infty = A + B$. In either case letting $m \to \infty$ gives the claim.
For strict inequality take $a_n = (-1)^n$ and $b_n = -a_n$: then $a_n + b_n = 0$, so the left side is 0, while $\limsup_n a_n = \limsup_n b_n = 1$. ■

Exercise P.4 (Knapp, problem I.7)**. (Dini's theorem.)** Let $K \subseteq \mathbb{R}^d$ be compact (for instance $K = [a, b]$), let $f_n : K \to \mathbb{R}$ be continuous with $f_1 \leqslant f_2 \leqslant \cdots$, and suppose that $f(x) = \lim_n f_n(x)$ is finite and continuous on K. Prove that $\sup_{x \in K} |f(x) - f_n(x)| \to 0$.

Solution. Put $g_n = f - f_n$. Each g_n is continuous and non-negative, $g_n \geqslant g_{n+1}$, and $g_n(x) \to 0$ for every x. By Theorem P.13 each g_n attains its maximum $M_n = g_n(x_n)$ at some $x_n \in K$, and $M_n \geqslant M_{n+1}$ since $g_n \geqslant g_{n+1}$. Suppose $M_n \not\to 0$. The sequence (M_n) is decreasing, so then $M_n \geqslant \varepsilon$ for every n, for some $\varepsilon > 0$. By compactness (Definition P.4) a subsequence x_{n_k} converges

to some $x \in K$. Fix m; for $n_k \geqslant m$,

$$g_m(x_{n_k}) \geqslant g_{n_k}(x_{n_k}) = M_{n_k} \geqslant \varepsilon,$$

and letting $k \to \infty$, continuity of g_m gives $g_m(x) \geqslant \varepsilon$. This holds for every m, contradicting $g_m(x) \to 0$. Hence $\sup_K |f - f_n| = M_n \to 0$.
Continuity of the limit cannot be dropped: on $K = [0,1]$, $f_n(x) = 1 - x^n$ increases to the function equal to 1 on $[0,1)$ and 0 at 1, and $\sup_K |f - f_n| = \sup_{x<1} x^n = 1$ for every n. ■

Exercise P.5 (Knapp, problem I.9)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be infinitely differentiable with $|f^{(n)}(x)| \leqslant 1$ for all $n \geqslant 0$ and all x. Prove that $f(x) = \sum_{n=0}^{\infty} f^{(n)}(0)x^n/n!$ for every $x \in \mathbb{R}$.

Solution. The case $x = 0$ is trivial; fix $x \neq 0$. By Theorem P.14, for each m there is ξ_m between 0 and x with

$$\left| f(x) - \sum_{n=0}^{m} \frac{f^{(n)}(0)}{n!} x^n \right| = \frac{|f^{(m+1)}(\xi_m)|}{(m+1)!} |x|^{m+1} \leqslant c_{m+1}, \qquad c_m = \frac{|x|^m}{m!}.$$

It remains to show $c_m \to 0$. Let M be an integer with $M \geqslant 2|x|$. For $m \geqslant M$, $c_{m+1}/c_m = |x|/(m+1) \leqslant \frac{1}{2}$, so, multiplying these ratio bounds for $M, M+1, \dots, m-1$, $c_m \leqslant c_M 2^{-(m-M)} \to 0$. Hence the partial sums converge to $f(x)$. The functions sin and cos satisfy the hypothesis. ■

Exercise P.6 (Knapp, problem II.8)**. (Contraction mapping principle.)** Let $C \subseteq \mathbb{R}^d$ be non-empty and closed, let $0 \leqslant r < 1$, and let $f : C \to C$ satisfy $|f(x) - f(y)| \leqslant r|x - y|$ for all $x, y \in C$. Prove that f has exactly one fixed point $x^\star \in C$, and that for every $x_0 \in C$ the iterates $x_{n+1} = f(x_n)$ satisfy $|x_n - x^\star| \leqslant r^n |x_1 - x_0|/(1 - r)$.

Solution. By induction $|x_{k+1} - x_k| \leqslant r^k |x_1 - x_0|$, the step being $|x_{k+1} - x_k| = |f(x_k) - f(x_{k-1})| \leqslant r|x_k - x_{k-1}|$. For $m > n$ the triangle inequality gives

$$|x_m - x_n| \leqslant \sum_{k=n}^{m-1} r^k |x_1 - x_0| \leqslant \frac{r^n}{1 - r} |x_1 - x_0|. \tag{P.34}$$

With $n = 0$ this shows that (x_n) is bounded, so by Theorem P.3 a subsequence x_{n_j} converges to some $x^\star$, which lies in C because C is closed. The whole sequence converges to $x^\star$: given $\varepsilon > 0$,

(P.34) gives N with $|x_m - x_n| < \varepsilon$ for $m, n \geqslant N$; choosing j with $n_j \geqslant N$ and $|x_{n_j} - x^\star| < \varepsilon$ gives $|x_n - x^\star| < 2\varepsilon$ for $n \geqslant N$. Since $|f(x_n) - f(x^\star)| \leqslant r|x_n - x^\star| \to 0$,

$$x^\star = \lim_n x_{n+1} = \lim_n f(x_n) = f(x^\star).$$

If also $f(y) = y$ then $|x^\star - y| = |f(x^\star) - f(y)| \leqslant r|x^\star - y|$, which forces $|x^\star - y| = 0$ as $r < 1$. Letting $m \to \infty$ in (P.34) gives the error bound. ■

Exercise P.7 (Knapp, problem III.20)**. (Arithmetic-geometric mean inequality.)** Prove that $(a_1 a_2 \cdots a_n)^{1/n} \leqslant (a_1 + \cdots + a_n)/n$ for all $a_1, \ldots, a_n \geqslant 0$, with equality if and only if all a_i are equal, by applying Theorem P.31 to

$$\text{minimise } -\sum_{i=1}^n \log x_i \quad \text{subject to} \quad \sum_{i=1}^n x_i - s \leqslant 0, \qquad x \in (0, \infty)^n,$$

where $s = a_1 + \cdots + a_n$.

Solution. If some $a_i = 0$ the left side is 0 and the inequality holds; equality then forces $s = 0$, that is, all $a_i = 0$. Assume all $a_i > 0$, so $s > 0$.

The problem is convex. $\mathcal{D} = (0, \infty)^n$ is open and convex. The function $-\log$ has second derivative $1/t^2 > 0$ on $(0, \infty)$, so it is strictly convex by Lemma P.19, and a sum of convex functions of separate coordinates is convex by Definition P.18; so $f_0(x) = -\sum_i \log x_i$ is convex, and it is strictly convex because for $x \neq y$ some coordinate differs and contributes a strict inequality. The constraint $f_1(x) = \sum_i x_i - s$ is affine, hence convex.

A KKT pair. Take $x^\star = (s/n, \ldots, s/n)$ and $\lambda^\star = n/s$. Then $f_1(x^\star) = 0 \leqslant 0$, $\lambda^\star \geqslant 0$, $\lambda^\star f_1(x^\star) = 0$, and the ith coordinate of $\nabla f_0(x^\star) + \lambda^\star \nabla f_1(x^\star)$ is $-n/s + n/s = 0$. By the converse part of Theorem P.31, $x^\star$ is primal optimal: $-\sum_i \log x_i \geqslant -n \log(s/n)$ for every feasible x.

Conclusion. The point $a = (a_1, \ldots, a_n)$ is feasible, so $\sum_i \log a_i \leqslant n \log(s/n)$; dividing by n and exponentiating gives the inequality. If equality holds, a is a second minimiser. Two distinct minimisers are impossible: their midpoint is feasible, the feasible set being convex, and strict convexity of f_0 gives it a strictly smaller value. Hence $a = x^\star$, all a_i being equal to s/n; the converse is immediate. Slater's condition holds here (take $x = x^\star/2$) but was not needed: the converse part of Theorem P.31 assumes only convexity. ■

Linear algebra

Exercise P.8 (Abadir–Magnus, exercise 4.13)**.** Let $\mathbf{A} \in \mathbb{R}^{n\times p}$. Show that

(a) $\mathcal{N}(\mathbf{A}') = \mathcal{N}(\mathbf{A}\mathbf{A}')$;

(b) $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{A}\mathbf{A}')$;

(c) $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}\mathbf{A}') = \operatorname{rank}(\mathbf{A}'\mathbf{A}) = \operatorname{rank}(\mathbf{A}')$.

In particular the column rank of a matrix equals its row rank.

Solution. (a) is Proposition P.33 applied to $\mathbf{A}'$ in place of $\mathbf{A}$, since $(\mathbf{A}')'\mathbf{A}' = \mathbf{A}\mathbf{A}'$.
(b) $\mathcal{C}(\mathbf{A}\mathbf{A}') \subseteq \mathcal{C}(\mathbf{A})$ because $\mathbf{A}\mathbf{A}'\mathbf{v} = \mathbf{A}(\mathbf{A}'\mathbf{v})$. Conversely let $\mathbf{y} \in \mathcal{C}(\mathbf{A})$ and let $\hat{\mathbf{y}}$ be its orthogonal projection onto the subspace $\mathcal{C}(\mathbf{A}\mathbf{A}')$ (Theorem P.37), so $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to $\mathcal{C}(\mathbf{A}\mathbf{A}')$. Then $(\mathbf{A}\mathbf{A}'\mathbf{v})'\mathbf{r} = \mathbf{v}'\mathbf{A}\mathbf{A}'\mathbf{r} = 0$ for every $\mathbf{v}$, so $\mathbf{A}\mathbf{A}'\mathbf{r} = \mathbf{0}$, and by (a) $\mathbf{A}'\mathbf{r} = \mathbf{0}$; thus $\mathbf{r}'\mathbf{A}\mathbf{u} = (\mathbf{A}'\mathbf{r})'\mathbf{u} = 0$ for every $\mathbf{u}$, that is, $\mathbf{r} \perp \mathcal{C}(\mathbf{A})$. But $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}} \in \mathcal{C}(\mathbf{A})$, both terms lying in $\mathcal{C}(\mathbf{A})$. Hence $\mathbf{r} \perp \mathbf{r}$, $\mathbf{r} = \mathbf{0}$ and $\mathbf{y} = \hat{\mathbf{y}} \in \mathcal{C}(\mathbf{A}\mathbf{A}')$.
(c) By (b), $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}\mathbf{A}')$. Proposition P.33 gives $\operatorname{rank}(\mathbf{A}'\mathbf{A}) = \operatorname{rank}(\mathbf{A})$, and applied to $\mathbf{A}'$ it gives $\operatorname{rank}(\mathbf{A}\mathbf{A}') = \operatorname{rank}(\mathbf{A}')$. ■

Exercise P.9 (Abadir–Magnus, exercise 12.5)**.** Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m\times n}$. Show that

(a) $(\operatorname{tr}\mathbf{A}'\mathbf{B})^2 \leqslant \operatorname{tr}(\mathbf{A}'\mathbf{A})\operatorname{tr}(\mathbf{B}'\mathbf{B})$, with equality if and only if one of $\mathbf{A}$, $\mathbf{B}$ is a multiple of the other;

(b) $\operatorname{tr}\big((\mathbf{A}'\mathbf{B})^2\big) \leqslant \operatorname{tr}(\mathbf{A}'\mathbf{A}\mathbf{B}'\mathbf{B})$, with equality if and only if $\mathbf{A}\mathbf{B}'$ is symmetric.

Solution. (a) The (j,j) entry of $\mathbf{A}'\mathbf{B}$ is $\sum_i a_{ij}b_{ij}$, so $\operatorname{tr}(\mathbf{A}'\mathbf{B}) = \sum_{i,j} a_{ij}b_{ij} = \langle \mathbf{a}, \mathbf{b}\rangle$, where $\mathbf{a}, \mathbf{b} \in \mathbb{R}^{mn}$ list the entries of $\mathbf{A}$ and $\mathbf{B}$ in the same order; likewise $\operatorname{tr}(\mathbf{A}'\mathbf{A}) = |\mathbf{a}|^2$ and $\operatorname{tr}(\mathbf{B}'\mathbf{B}) = |\mathbf{b}|^2$ (Proposition P.35(iv)). Theorem P.36 gives the inequality, with equality exactly when $\mathbf{a}$ and $\mathbf{b}$ are proportional, that is, when one matrix is a multiple of the other.

(b) Put $\mathbf{C} = \mathbf{A}\mathbf{B}' \in \mathbb{R}^{m\times m}$. By Proposition P.35(ii), $\operatorname{tr}(\mathbf{C}'\mathbf{C}) = \operatorname{tr}(\mathbf{B}\mathbf{A}'\mathbf{A}\mathbf{B}') = \operatorname{tr}(\mathbf{A}'\mathbf{A}\mathbf{B}'\mathbf{B})$, and $\operatorname{tr}(\mathbf{C}^2) = \operatorname{tr}(\mathbf{A}\mathbf{B}'\mathbf{A}\mathbf{B}') = \operatorname{tr}(\mathbf{B}'\mathbf{A}\mathbf{B}'\mathbf{A}) = \operatorname{tr}\big(((\mathbf{A}'\mathbf{B})')^2\big) = \operatorname{tr}\big((\mathbf{A}'\mathbf{B})^2\big)$, a trace being unchanged by transposition. Expanding,

$$\operatorname{tr}\big((\mathbf{C}-\mathbf{C}')'(\mathbf{C}-\mathbf{C}')\big) = \operatorname{tr}(\mathbf{C}'\mathbf{C}) - \operatorname{tr}(\mathbf{C}'\mathbf{C}') - \operatorname{tr}(\mathbf{C}\mathbf{C}) + \operatorname{tr}(\mathbf{C}\mathbf{C}') = 2\operatorname{tr}(\mathbf{C}'\mathbf{C}) - 2\operatorname{tr}(\mathbf{C}^2),$$

using $\operatorname{tr}(\mathbf{C}'\mathbf{C}') = \operatorname{tr}(\mathbf{C}^2)$ and $\operatorname{tr}(\mathbf{C}\mathbf{C}') = \operatorname{tr}(\mathbf{C}'\mathbf{C})$. The left side is non-negative and vanishes only if $\mathbf{C} = \mathbf{C}'$, by Proposition P.35(iv). Hence $\operatorname{tr}\big((\mathbf{A}'\mathbf{B})^2\big) = \operatorname{tr}(\mathbf{C}^2) \leqslant \operatorname{tr}(\mathbf{C}'\mathbf{C}) = \operatorname{tr}(\mathbf{A}'\mathbf{A}\mathbf{B}'\mathbf{B})$, with

equality if and only if $\mathbf{AB}'$ is symmetric. ■

Exercise P.10 (Abadir–Magnus, exercise 13.57)**. (Gauss–Markov.)** Let $\mathbf{X} \in \mathbb{R}^{n\times k}$ have rank k and $\mathbf{G}_0 = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. Show that every $\mathbf{G} \in \mathbb{R}^{k\times n}$ with $\mathbf{GX} = \mathbf{I}_k$ satisfies: $\mathbf{GG}' - \mathbf{G}_0\mathbf{G}_0'$ is positive semi-definite, and $\operatorname{tr}(\mathbf{GG}') \geqslant \operatorname{tr}(\mathbf{G}_0\mathbf{G}_0')$ with equality only for $\mathbf{G} = \mathbf{G}_0$. Deduce that if $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\operatorname{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$, then among all estimators $\mathbf{Gy}$ with $\mathsf{E}[\mathbf{Gy}] = \boldsymbol{\beta}$ for every $\boldsymbol{\beta}$, the least squares estimator $\mathbf{G}_0\mathbf{y}$ has the smallest covariance matrix.

Solution. $\mathbf{X}'\mathbf{X}$ is invertible by Proposition P.33, and $\mathbf{G}_0\mathbf{X} = \mathbf{I}_k$. Let $\mathbf{D} = \mathbf{G} - \mathbf{G}_0$, so $\mathbf{DX} = \mathbf{0}$ and $\mathbf{G}_0\mathbf{D}' = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{DX})' = \mathbf{0}$. Hence

$$\mathbf{GG}' = \mathbf{G}_0\mathbf{G}_0' + \mathbf{G}_0\mathbf{D}' + \mathbf{DG}_0' + \mathbf{DD}' = \mathbf{G}_0\mathbf{G}_0' + \mathbf{DD}'.$$

$\mathbf{DD}'$ is positive semi-definite, as $\mathbf{v}'\mathbf{DD}'\mathbf{v} = |\mathbf{D}'\mathbf{v}|^2$, and $\operatorname{tr}(\mathbf{DD}') = \sum_{i,j} d_{ij}^2$ (Proposition P.35(iv)) is positive unless $\mathbf{D} = \mathbf{0}$.

For the estimators: $\mathsf{E}[\mathbf{Gy}] = \mathbf{GX}\boldsymbol{\beta}$, and this equals $\boldsymbol{\beta}$ for every $\boldsymbol{\beta}$ if and only if $\mathbf{GX} = \mathbf{I}_k$ (take $\boldsymbol{\beta} = \mathbf{e}_j$, the jth standard basis vector, for each j). By Proposition P.102, $\operatorname{var}(\mathbf{Gy}) = \sigma^2\mathbf{GG}'$, so $\operatorname{var}(\mathbf{Gy}) - \operatorname{var}(\mathbf{G}_0\mathbf{y}) = \sigma^2\mathbf{DD}'$ is positive semi-definite: every linear combination $\mathbf{c}'\mathbf{Gy}$ has variance

at least that of $\mathbf{c}'\mathbf{G}_0\mathbf{y}$. ■

Exercise P.11 (Abadir–Magnus, exercise 12.49)**. (Ky Fan.)** Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1 \geqslant \cdots \geqslant \lambda_n$, and let $1 \leqslant r \leqslant n$.

(a) Show that over $\mathbf{X} \in \mathbb{R}^{n\times r}$ with $\mathbf{X}'\mathbf{X} = \mathbf{I}_r$,

$$\max \operatorname{tr}(\mathbf{X}'\mathbf{A}\mathbf{X}) = \sum_{k=1}^{r} \lambda_k, \qquad \min \operatorname{tr}(\mathbf{X}'\mathbf{A}\mathbf{X}) = \sum_{k=1}^{r} \lambda_{n-r+k}.$$

(b) Deduce that the leading $r \times r$ block $\mathbf{A}_{(r)}$ of $\mathbf{A}$ satisfies $\sum_{k=1}^{r} \lambda_{n-r+k} \leqslant \operatorname{tr} \mathbf{A}_{(r)} \leqslant \sum_{k=1}^{r} \lambda_k$.

Solution. (a) Write $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ (Theorem P.39), with the diagonal of $\mathbf{\Lambda}$ in the order $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ after the reordering in the proof of Corollary P.40, and $\mathbf{W} = \mathbf{P}'\mathbf{X}$, so $\mathbf{W}'\mathbf{W} = \mathbf{X}'\mathbf{P}\mathbf{P}'\mathbf{X} = \mathbf{I}_r$ and, by Proposition P.35(ii),

$$\operatorname{tr}(\mathbf{X}'\mathbf{A}\mathbf{X}) = \operatorname{tr}(\mathbf{W}'\mathbf{\Lambda}\mathbf{W}) = \sum_{j=1}^{n} \lambda_j h_j, \qquad h_j = \sum_{l=1}^{r} w_{jl}^2 = (\mathbf{W}\mathbf{W}')_{jj}.$$

First, $\sum_j h_j = \operatorname{tr}(\mathbf{W}\mathbf{W}') = \operatorname{tr}(\mathbf{W}'\mathbf{W}) = r$. Second, $0 \leqslant h_j \leqslant 1$: the matrix $\mathbf{H} = \mathbf{W}\mathbf{W}'$ is symmetric and idempotent, as $\mathbf{W}\mathbf{W}'\mathbf{W}\mathbf{W}' = \mathbf{W}\mathbf{W}'$, so it is an orthogonal projection (Proposition P.47), and $h_j = \mathbf{e}_j'\mathbf{H}\mathbf{e}_j = \mathbf{e}_j'\mathbf{H}'\mathbf{H}\mathbf{e}_j = |\mathbf{H}\mathbf{e}_j|^2 \leqslant |\mathbf{e}_j|^2 = 1$ by Theorem P.38 applied to $\mathbf{e}_j = \mathbf{H}\mathbf{e}_j + (\mathbf{e}_j - \mathbf{H}\mathbf{e}_j)$. Therefore

$$\sum_j \lambda_j h_j - \sum_{j\leqslant r} \lambda_j = \sum_{j\leqslant r} \lambda_j (h_j - 1) + \sum_{j>r} \lambda_j h_j \leqslant \lambda_r \sum_{j\leqslant r} (h_j - 1) + \lambda_r \sum_{j>r} h_j = \lambda_r \Big(\sum_j h_j - r \Big) = 0,$$

using $h_j - 1 \leqslant 0$ and $\lambda_j \geqslant \lambda_r$ for $j \leqslant r$, and $h_j \geqslant 0$ and $\lambda_j \leqslant \lambda_r$ for $j > r$. The bound is attained by the first r columns of $\mathbf{P}$, for which $\mathbf{W}$ consists of the first r columns of $\mathbf{I}_n$ and $h_j = \mathbb{1}\{j \leqslant r\}$. The minimum follows by applying the maximum to $-\mathbf{A}$, whose eigenvalues in decreasing order are $-\lambda_n \geqslant \cdots \geqslant -\lambda_1$.

(b) Take $\mathbf{X} = (\mathbf{I}_r, \mathbf{0})'$, for which $\mathbf{X}'\mathbf{X} = \mathbf{I}_r$ and $\mathbf{X}'\mathbf{A}\mathbf{X} = \mathbf{A}_{(r)}$. For $r = 1$, (a) is Corollary P.40; for general r it says that the first r principal components capture the largest possible total variance. ■

Exercise P.12 (Abadir–Magnus, exercise 12.16)**.** Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$ be positive definite. Show that $\mathbf{A}-\mathbf{B}$ is positive definite if and only if $\mathbf{B}^{-1}-\mathbf{A}^{-1}$ is, and that the same holds with "positive semi-definite" in both places.

Solution. **Step 1: congruence.** If $\mathbf{T}$ is invertible and $\mathbf{M}$ symmetric, then $\mathbf{M}$ is positive definite if and only if $\mathbf{T}'\mathbf{M}\mathbf{T}$ is: $\mathbf{v}'\mathbf{T}'\mathbf{M}\mathbf{T}\mathbf{v} = (\mathbf{T}\mathbf{v})'\mathbf{M}(\mathbf{T}\mathbf{v})$, and $\mathbf{T}\mathbf{v} \neq \mathbf{0}$ exactly when $\mathbf{v} \neq \mathbf{0}$, while every $\mathbf{u}$ is $\mathbf{T}\mathbf{v}$ with $\mathbf{v} = \mathbf{T}^{-1}\mathbf{u}$; so $\mathbf{u}'\mathbf{M}\mathbf{u}$ over $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v}'\mathbf{T}'\mathbf{M}\mathbf{T}\mathbf{v}$ over $\mathbf{v} \neq \mathbf{0}$ take the same values. The same argument applies to semi-definiteness.

Step 2. Let $\mathbf{B}^{1/2}$ and $\mathbf{B}^{-1/2}$ be as in Theorem P.44; both are symmetric, and $\mathbf{B}^{-1/2}\mathbf{B}^{-1/2} = (\mathbf{B}^{1/2}\mathbf{B}^{1/2})^{-1} = \mathbf{B}^{-1}$. Put $\mathbf{C} = \mathbf{B}^{-1/2}\mathbf{A}\mathbf{B}^{-1/2}$, positive definite by Step 1, with spectral decomposition $\mathbf{C} = \mathbf{P}\boldsymbol{\Gamma}\mathbf{P}'$, $\boldsymbol{\Gamma} = \operatorname{diag}(\gamma_1, \ldots, \gamma_n)$, all $\gamma_i > 0$. Then

$$\mathbf{A}-\mathbf{B} = \mathbf{B}^{1/2}(\mathbf{C}-\mathbf{I})\mathbf{B}^{1/2}, \qquad \mathbf{B}^{-1}-\mathbf{A}^{-1} = \mathbf{B}^{-1/2}(\mathbf{I}-\mathbf{C}^{-1})\mathbf{B}^{-1/2},$$

the second because $\mathbf{C}^{-1} = \mathbf{B}^{1/2}\mathbf{A}^{-1}\mathbf{B}^{1/2}$. By Step 1 and Proposition P.42, $\mathbf{A}-\mathbf{B}$ is positive definite iff $\mathbf{C}-\mathbf{I} = \mathbf{P}(\boldsymbol{\Gamma}-\mathbf{I})\mathbf{P}'$ is, iff every $\gamma_i > 1$; and $\mathbf{B}^{-1}-\mathbf{A}^{-1}$ is positive definite iff $\mathbf{I}-\mathbf{C}^{-1} = \mathbf{P}(\mathbf{I}-\boldsymbol{\Gamma}^{-1})\mathbf{P}'$ is, iff every $1/\gamma_i < 1$. The two conditions coincide. With $\geqslant$ in place of $>$

the same chain gives the semi-definite statement. Thus a larger information matrix has a smaller inverse, which is how asymptotic efficiency comparisons are read. ■

Exercise P.13 (Abadir–Magnus, exercise 8.64)**.** Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$ be idempotent.
(a) Show that $\mathbf{A}+\mathbf{B}$ is idempotent if and only if $\mathbf{AB} = \mathbf{BA} = \mathbf{0}$.
(b) If $\mathbf{A}+\mathbf{B}$ is idempotent, show that $\operatorname{rank}(\mathbf{A}+\mathbf{B}) = \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B})$.

Solution. (a) $(\mathbf{A}+\mathbf{B})^2 = \mathbf{A}+\mathbf{B}+\mathbf{AB}+\mathbf{BA}$, so $\mathbf{A}+\mathbf{B}$ is idempotent if and only if $\mathbf{AB}+\mathbf{BA} = \mathbf{0}$. This clearly holds if $\mathbf{AB} = \mathbf{BA} = \mathbf{0}$. Conversely, multiplying $\mathbf{AB}+\mathbf{BA} = \mathbf{0}$ by $\mathbf{A}$ on the left and on the right gives $\mathbf{AB}+\mathbf{ABA} = \mathbf{0}$ and $\mathbf{ABA}+\mathbf{BA} = \mathbf{0}$; subtracting, $\mathbf{AB} = \mathbf{BA}$, and then $2\mathbf{AB} = \mathbf{0}$.
(b) By Lemma P.46 applied to $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{A}+\mathbf{B}$, and linearity of the trace, $\operatorname{rank}(\mathbf{A}+\mathbf{B}) = \operatorname{tr}(\mathbf{A}+\mathbf{B}) = \operatorname{tr}\mathbf{A} + \operatorname{tr}\mathbf{B} = \operatorname{rank}\mathbf{A} + \operatorname{rank}\mathbf{B}$. For instance, with $\mathbf{H}$ a projection, $\mathbf{H} + (\mathbf{I}-\mathbf{H}) = \mathbf{I}$ splits n into $\operatorname{rank}\mathbf{H} + \operatorname{rank}(\mathbf{I}-\mathbf{H})$, the degrees-of-freedom count behind Theorem P.136. ■

Exercise P.14 (Abadir–Magnus, exercise 8.74)**.** For $n \geqslant 2$ and real t let $\mathbf{R} = (1-t)\mathbf{I}_n + t\mathbf{1}\mathbf{1}'$, the equicorrelation matrix.

(a) Show that its eigenvalues are $1+(n-1)t$, with eigenvector $\mathbf{1}$, and $1-t$, with multiplicity $n-1$ (for $t \neq 0$) and eigenvectors the vectors orthogonal to $\mathbf{1}$.
(b) For which t is $\mathbf{R}$ positive definite?
(c) Find $\mathbf{R}^{-1}$ when it exists, and check the answer with Lemma P.50.

Solution. (a) Let $\mathbf{J} = n^{-1}\mathbf{1}\mathbf{1}'$. It is symmetric and idempotent ($\mathbf{1}'\mathbf{1} = n$), so is $\mathbf{I}-\mathbf{J}$, and $\mathbf{J}(\mathbf{I}-\mathbf{J}) = \mathbf{0}$. Since $t\mathbf{1}\mathbf{1}' = nt\mathbf{J}$,

$$\mathbf{R} = \alpha_1\mathbf{J} + \alpha_2(\mathbf{I}-\mathbf{J}), \qquad \alpha_1 = 1+(n-1)t, \quad \alpha_2 = 1-t.$$

$\mathbf{J}\mathbf{1} = \mathbf{1}$ gives $\mathbf{R}\mathbf{1} = \alpha_1\mathbf{1}$, and $\mathbf{v} \perp \mathbf{1}$ gives $\mathbf{J}\mathbf{v} = \mathbf{0}$ and $\mathbf{R}\mathbf{v} = \alpha_2\mathbf{v}$. The orthogonal complement of $\mathbf{1}$ is $\mathcal{N}(\mathbf{1}')$, of dimension $n-1$ by Theorem P.32. These are all the eigenvalues: if $\mathbf{R}\mathbf{x} = \mu\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, multiplying by $\mathbf{J}$ and by $\mathbf{I}-\mathbf{J}$ and using $\mathbf{J}^2 = \mathbf{J}$, $\mathbf{J}(\mathbf{I}-\mathbf{J}) = \mathbf{0}$ gives $(\alpha_1-\mu)\mathbf{J}\mathbf{x} = \mathbf{0}$ and $(\alpha_2-\mu)(\mathbf{I}-\mathbf{J})\mathbf{x} = \mathbf{0}$; as $\mathbf{J}\mathbf{x} + (\mathbf{I}-\mathbf{J})\mathbf{x} = \mathbf{x} \neq \mathbf{0}$, one of the two is non-zero, so $\mu \in \{\alpha_1, \alpha_2\}$. For the multiplicities (the multiplicity of an eigenvalue being the number of diagonal entries of $\mathbf{\Lambda}$ in Theorem P.39 equal to it; the count below does not depend on the choice of $\mathbf{P}$), if $t = 0$ then $\mathbf{R} = \mathbf{I}$ and $\alpha_1 = \alpha_2 = 1$, so the two eigenvalues coincide and 1 has multiplicity n; let

$t \neq 0$, so $\alpha_1 \neq \alpha_2$, and let $\mathbf{R} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ as in Theorem P.39, the columns $\mathbf{p}_1, \ldots, \mathbf{p}_n$ of $\mathbf{P}$ being orthonormal eigenvectors. A column with eigenvalue α_1 satisfies $(\mathbf{I}-\mathbf{J})\mathbf{p}_i = \mathbf{0}$ (by the relations just derived, with $\mu = \alpha_1 \neq \alpha_2$), so $\mathbf{p}_i = \mathbf{J}\mathbf{p}_i = n^{-1}(\mathbf{1}'\mathbf{p}_i)\mathbf{1}$; two such columns $a\mathbf{1}$, $b\mathbf{1}$ would have $0 = \mathbf{p}_i'\mathbf{p}_l = abn$, so one of them would be $\mathbf{0}$, which is impossible; hence at most one column has eigenvalue α_1. A column with eigenvalue α_2 satisfies $\mathbf{J}\mathbf{p}_i = \mathbf{0}$, that is $\mathbf{p}_i \in \mathcal{N}(\mathbf{1}')$; orthonormal vectors are linearly independent (if $\sum_i c_i\mathbf{p}_i = \mathbf{0}$, multiply by $\mathbf{p}_l'$ to get $c_l = 0$), so at most $n-1$ columns lie in this $(n-1)$-dimensional subspace. The n columns are thus split as exactly one with eigenvalue α_1 and $n-1$ with eigenvalue α_2: the multiplicities are 1 and $n-1$.

(b) By Proposition P.42, $\mathbf{R}$ is positive definite iff $\alpha_1 > 0$ and $\alpha_2 > 0$, that is, $-1/(n-1) < t < 1$. Equal correlations among n variables can be negative only down to $-1/(n-1)$.

(c) $\mathbf{R}$ is invertible iff $\alpha_1 \neq 0 \neq \alpha_2$. If $\alpha_1 = 0$ then $\mathbf{R}\mathbf{1} = \mathbf{0}$, and if $\alpha_2 = 0$ then $\mathbf{R}\mathbf{v} = \mathbf{0}$ for any non-zero $\mathbf{v} \in \mathcal{N}(\mathbf{1}')$, which exists as this space has dimension $n-1 \geqslant 1$; either way $\mathbf{R}$ is not invertible. Otherwise $\mathbf{R}^{-1} = \alpha_1^{-1}\mathbf{J} + \alpha_2^{-1}(\mathbf{I}-\mathbf{J})$, since by $\mathbf{J}^2 = \mathbf{J}$, $\mathbf{J}(\mathbf{I}-\mathbf{J}) = \mathbf{0}$ and $(\mathbf{I}-\mathbf{J})^2 = \mathbf{I}-\mathbf{J}$ the product with $\mathbf{R}$ is $\mathbf{J} + (\mathbf{I}-\mathbf{J}) = \mathbf{I}$. Writing $\mathbf{R} = \mathbf{A} - \mathbf{u}\mathbf{v}'$ with $\mathbf{A} = (1-t)\mathbf{I}$, $\mathbf{u} = -t\mathbf{1}$, $\mathbf{v} = \mathbf{1}$,

one has $1 - \mathbf{v}'\mathbf{A}^{-1}\mathbf{u} = 1 + nt/(1-t) = \alpha_1/\alpha_2$, and (P.12) gives

$$\mathbf{R}^{-1} = \frac{1}{1-t}\Big(\mathbf{I} - \frac{t}{1+(n-1)t}\mathbf{1}\mathbf{1}'\Big),$$

which agrees with the first expression, as $\alpha_1^{-1} - \alpha_2^{-1} = -nt/(\alpha_1\alpha_2)$. ■

Exercise P.15 (Abadir–Magnus, exercise 8.47)**.** Let $\mathbf{Z} = \left(\begin{smallmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{B}' & \mathbf{D}\end{smallmatrix}\right)$ be symmetric with $\mathbf{A} \in \mathbb{R}^{m\times m}$. Show that $\mathbf{Z}$ is positive definite if and only if $\mathbf{A}$ and $\mathbf{S} = \mathbf{D} - \mathbf{B}'\mathbf{A}^{-1}\mathbf{B}$ are both positive definite.

Solution. If $\mathbf{Z}$ is positive definite, so is $\mathbf{A}$: for $\mathbf{u} \neq \mathbf{0}$, $\mathbf{u}'\mathbf{A}\mathbf{u} = (\mathbf{u}', \mathbf{0}')\mathbf{Z}(\mathbf{u}', \mathbf{0}')' > 0$. So in either direction $\mathbf{A}$ is positive definite, hence invertible (Proposition P.42), and $\mathbf{T} = \left(\begin{smallmatrix}\mathbf{I} & -\mathbf{A}^{-1}\mathbf{B}\\ \mathbf{0} & \mathbf{I}\end{smallmatrix}\right)$ is defined and invertible, with inverse $\left(\begin{smallmatrix}\mathbf{I} & \mathbf{A}^{-1}\mathbf{B}\\ \mathbf{0} & \mathbf{I}\end{smallmatrix}\right)$. As $\mathbf{A}$ is symmetric, $(\mathbf{A}^{-1})'\mathbf{A} = (\mathbf{A}\mathbf{A}^{-1})' = \mathbf{I}$, so $(\mathbf{A}^{-1})' = \mathbf{A}^{-1}$; this gives the form of $\mathbf{T}'$ below and makes $\mathbf{S}$ symmetric ($\mathbf{D}$ is, as $\mathbf{Z}$ is). Blockwise,

$$\mathbf{Z}\mathbf{T} = \begin{pmatrix}\mathbf{A} & \mathbf{0}\\ \mathbf{B}' & \mathbf{S}\end{pmatrix}, \qquad \mathbf{T}'\mathbf{Z}\mathbf{T} = \begin{pmatrix}\mathbf{I} & \mathbf{0}\\ -\mathbf{B}'\mathbf{A}^{-1} & \mathbf{I}\end{pmatrix}\begin{pmatrix}\mathbf{A} & \mathbf{0}\\ \mathbf{B}' & \mathbf{S}\end{pmatrix} = \begin{pmatrix}\mathbf{A} & \mathbf{0}\\ \mathbf{0} & \mathbf{S}\end{pmatrix}.$$

By Step 1 of the solution of Exercise P.12, $\mathbf{Z}$ is positive definite iff the block-diagonal matrix

is, and $(\mathbf{u}', \mathbf{w}')\,\mathrm{diag}(\mathbf{A}, \mathbf{S})(\mathbf{u}', \mathbf{w}')' = \mathbf{u}'\mathbf{A}\mathbf{u} + \mathbf{w}'\mathbf{S}\mathbf{w}$ is positive for all $(\mathbf{u}, \mathbf{w}) \neq \mathbf{0}$ iff $\mathbf{A}$ and $\mathbf{S}$ are positive definite. With Theorem P.48 this shows that the lower-right block $\mathbf{S}^{-1}$ of $\mathbf{Z}^{-1}$ is positive definite: it is symmetric as $\mathbf{S}$ is (by the argument just given for $\mathbf{A}^{-1}$), and for $\mathbf{w} \neq \mathbf{0}$, $\mathbf{w}'\mathbf{S}^{-1}\mathbf{w} = (\mathbf{S}^{-1}\mathbf{w})'\mathbf{S}(\mathbf{S}^{-1}\mathbf{w}) > 0$ since $\mathbf{S}^{-1}\mathbf{w} \neq \mathbf{0}$. ■

Exercise P.16 (Abadir–Magnus, exercise 5.17)**. (Woodbury.)** Let $\mathbf{A} \in \mathbb{R}^{m\times m}$ and $\mathbf{D} \in \mathbb{R}^{n\times n}$ be invertible, $\mathbf{B} \in \mathbb{R}^{m\times n}$, $\mathbf{C} \in \mathbb{R}^{n\times m}$, and suppose $\mathbf{S} = \mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$ is invertible. Show that $\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}$ is invertible with

$$(\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1} = \mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1},$$

and that Lemma P.50 is the case $n = 1$.

Solution. Multiply the proposed inverse by $\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}$ on the left:

$$\mathbf{I} + \mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}\mathbf{A}^{-1} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1}\mathbf{C}\mathbf{A}^{-1} = \mathbf{I} + \mathbf{B}\mathbf{K}\mathbf{C}\mathbf{A}^{-1},$$

where $\mathbf{K} = \mathbf{S}^{-1} - \mathbf{D}^{-1} - \mathbf{D}^{-1}\mathbf{C}\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1} = \mathbf{D}^{-1}(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B})\mathbf{S}^{-1} - \mathbf{D}^{-1} = \mathbf{D}^{-1}\mathbf{S}\mathbf{S}^{-1} - \mathbf{D}^{-1} = \mathbf{0}$. So the proposed matrix is a right inverse of the square matrix $\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}$, hence its inverse

(Remark 1). For $n = 1$ take $\mathbf{B} = \mathbf{u}$, $\mathbf{C} = \mathbf{v}'$ and $\mathbf{D} = 1$: then $\mathbf{S} = 1 - \mathbf{v}'\mathbf{A}^{-1}\mathbf{u}$ and the formula is (P.12). ■

Exercise P.17 (Abadir–Magnus, exercise 8.40)**. (Polar decomposition.)** Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ have full column rank n. Show that $\mathbf{A} = \mathbf{QV}$ with $\mathbf{Q} \in \mathbb{R}^{m\times n}$, $\mathbf{Q}'\mathbf{Q} = \mathbf{I}_n$, and $\mathbf{V}$ positive definite, and that necessarily $\mathbf{V} = (\mathbf{A}'\mathbf{A})^{1/2}$, so the factorisation is unique. State the analogue for full row rank.

Solution. By Theorem P.51 with $r = n$, $\mathbf{A} = \mathbf{UDW}'$ with $\mathbf{U} \in \mathbb{R}^{m\times n}$, $\mathbf{U}'\mathbf{U} = \mathbf{I}_n$, $\mathbf{W} \in \mathbb{R}^{n\times n}$, $\mathbf{W}'\mathbf{W} = \mathbf{I}_n$, and $\mathbf{D}$ diagonal with positive entries. $\mathbf{W}$ is square, so $\mathbf{WW}' = \mathbf{I}_n$ as well (Remark 1). Put $\mathbf{Q} = \mathbf{UW}'$ and $\mathbf{V} = \mathbf{WDW}'$. Then $\mathbf{Q}'\mathbf{Q} = \mathbf{WU}'\mathbf{UW}' = \mathbf{I}_n$, $\mathbf{V}$ is symmetric with eigenvalues $d_1, \dots, d_n > 0$, hence positive definite, and $\mathbf{QV} = \mathbf{UW}'\mathbf{WDW}' = \mathbf{UDW}' = \mathbf{A}$.
If $\mathbf{A} = \mathbf{QV}$ is any such factorisation, $\mathbf{A}'\mathbf{A} = \mathbf{VQ}'\mathbf{QV} = \mathbf{V}^2$, so $\mathbf{V}$ is a positive semi-definite square root of $\mathbf{A}'\mathbf{A}$ and equals $(\mathbf{A}'\mathbf{A})^{1/2}$ by the uniqueness in Theorem P.44; then $\mathbf{Q} = \mathbf{AV}^{-1}$. If $\mathbf{A}$ has full row rank m, applying this to $\mathbf{A}'$ gives $\mathbf{A} = \mathbf{VQ}'$ with $\mathbf{V} = (\mathbf{AA}')^{1/2}$ and $\mathbf{Q}'\mathbf{Q} = \mathbf{I}_m$. ■

Exercise P.18 (Knapp, problem III.1)**.** For $\mathbf{A} \in \mathbb{R}^{m\times n}$ let $|\mathbf{A}|_F = (\operatorname{tr} \mathbf{A}'\mathbf{A})^{1/2} = (\sum_{i,j} a_{ij}^2)^{1/2}$ be the Hilbert–Schmidt (Frobenius) norm. Show that

(a) $|\mathbf{AB}|_F \leqslant |\mathbf{A}|_F |\mathbf{B}|_F$ for $\mathbf{B} \in \mathbb{R}^{n\times k}$;

(b) $|\mathbf{I}_n|_F = \sqrt{n}$;

(c) $\|\mathbf{A}\| \leqslant |\mathbf{A}|_F \leqslant \sqrt{\operatorname{rank}(\mathbf{A})}\, \|\mathbf{A}\|$, with $\|\cdot\|$ the operator norm of Definition P.52.

Solution. (a) The (i,j) entry of $\mathbf{AB}$ is $\sum_l a_{il}b_{lj}$, the inner product of the ith row of $\mathbf{A}$ with the jth column of $\mathbf{B}$. By Theorem P.36 its square is at most $(\sum_l a_{il}^2)(\sum_l b_{lj}^2)$; summing over i and j, $|\mathbf{AB}|_F^2 \leqslant (\sum_{i,l} a_{il}^2)(\sum_{l,j} b_{lj}^2) = |\mathbf{A}|_F^2 |\mathbf{B}|_F^2$.

(b) $\operatorname{tr}(\mathbf{I}_n'\mathbf{I}_n) = n$.

(c) For $\mathbf{A} = \mathbf{0}$ all three are 0. Otherwise, with $\mathbf{A} = \mathbf{UDV}'$ from Theorem P.51, $\mathbf{A}'\mathbf{A} = \mathbf{VD}^2\mathbf{V}'$ and, by Proposition P.35(ii), $|\mathbf{A}|_F^2 = \operatorname{tr}(\mathbf{D}^2\mathbf{V}'\mathbf{V}) = \sum_{j\leqslant r} d_j^2$, where $r = \operatorname{rank}(\mathbf{A})$. This lies between d_1^2 and rd_1^2, and $d_1 = \|\mathbf{A}\|$ by Proposition P.53. Part (b) shows that the right inequality can be an equality, and that $|\cdot|_F$ differs from the operator norm, which gives the identity the norm $\sup_{|\mathbf{x}|=1} |\mathbf{x}| = 1$. ■

Exercise P.19 (Abadir–Magnus, exercise 13.56)**. (Constrained least squares.)** Let $\mathbf{X} \in \mathbb{R}^{n\times k}$ have rank k, $\mathbf{\Omega} \in \mathbb{R}^{n\times n}$ be positive definite, $\mathbf{R} \in \mathbb{R}^{k\times r}$ have rank r, $\mathbf{c} \in \mathbb{R}^r$, $\mathbf{y} \in \mathbb{R}^n$, and put $\mathbf{M} = \mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X}$ and $\hat{\boldsymbol{\beta}} = \mathbf{M}^{-1}\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{y}$. Using Theorem P.31, with the constraint $\mathbf{R}'\boldsymbol{\beta} = \mathbf{c}$ written as the two inequalities $\mathbf{R}'\boldsymbol{\beta} - \mathbf{c} \leqslant \mathbf{0}$ and $\mathbf{c} - \mathbf{R}'\boldsymbol{\beta} \leqslant \mathbf{0}$, show that the unique minimiser of $Q(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\mathbf{\Omega}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ subject to $\mathbf{R}'\boldsymbol{\beta} = \mathbf{c}$ is

$$\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} - \mathbf{M}^{-1}\mathbf{R}(\mathbf{R}'\mathbf{M}^{-1}\mathbf{R})^{-1}(\mathbf{R}'\hat{\boldsymbol{\beta}} - \mathbf{c}).$$

Solution. **Step 1: the inverses exist.** $\mathbf{\Omega}^{-1}$ is positive definite, its eigenvalues being the reciprocals of those of $\mathbf{\Omega}$. For $\mathbf{v} \neq \mathbf{0}$, $\mathbf{Xv} \neq \mathbf{0}$ by full column rank (Theorem P.32 gives $\dim \mathcal{N}(\mathbf{X}) = k - k = 0$), so $\mathbf{v}'\mathbf{M}\mathbf{v} = (\mathbf{Xv})'\mathbf{\Omega}^{-1}(\mathbf{Xv}) > 0$: $\mathbf{M}$ and hence $\mathbf{M}^{-1}$ are positive definite. For $\mathbf{w} \neq \mathbf{0}$, $\mathbf{Rw} \neq \mathbf{0}$, so $\mathbf{w}'\mathbf{R}'\mathbf{M}^{-1}\mathbf{R}\mathbf{w} > 0$ and $\mathbf{R}'\mathbf{M}^{-1}\mathbf{R}$ is invertible.
Step 2: convexity and the gradient. Expanding, $Q(\boldsymbol{\beta}) = \mathbf{y}'\mathbf{\Omega}^{-1}\mathbf{y} - 2\mathbf{y}'\mathbf{\Omega}^{-1}\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{M}\boldsymbol{\beta}$, so by Lemma P.58 $\nabla Q(\boldsymbol{\beta}) = 2(\mathbf{M}\boldsymbol{\beta} - \mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{y})$, and for every $\mathbf{h}$

$$Q(\boldsymbol{\beta} + \mathbf{h}) = Q(\boldsymbol{\beta}) + \nabla Q(\boldsymbol{\beta})'\mathbf{h} + \mathbf{h}'\mathbf{M}\mathbf{h}. \tag{P.35}$$

As $\mathbf{h}'\mathbf{M}\mathbf{h} \geqslant 0$, the gradient inequality of Proposition P.22 holds and Q is convex on the open set

$\mathbb{R}^k$. The constraint functions are affine, hence convex.

Step 3: a KKT pair. Let $\boldsymbol{\nu} = 2(\mathbf{R}'\mathbf{M}^{-1}\mathbf{R})^{-1}(\mathbf{R}'\hat{\boldsymbol{\beta}} - \mathbf{c})$, and let $\boldsymbol{\lambda}_+$, $\boldsymbol{\lambda}_-$ be its coordinatewise positive and negative parts, so $\boldsymbol{\lambda}_\pm \geqslant \mathbf{0}$ and $\boldsymbol{\lambda}_+ - \boldsymbol{\lambda}_- = \boldsymbol{\nu}$. Since $\mathbf{M}\hat{\boldsymbol{\beta}} = \mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{y}$, $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} - \frac{1}{2}\mathbf{M}^{-1}\mathbf{R}\boldsymbol{\nu}$ satisfies

$$\nabla Q(\tilde{\boldsymbol{\beta}}) + \mathbf{R}\boldsymbol{\lambda}_+ - \mathbf{R}\boldsymbol{\lambda}_- = 2\mathbf{M}(\tilde{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}) + \mathbf{R}\boldsymbol{\nu} = \mathbf{0},$$

which is the stationarity condition (P.10), and $\mathbf{R}'\tilde{\boldsymbol{\beta}} = \mathbf{R}'\hat{\boldsymbol{\beta}} - (\mathbf{R}'\hat{\boldsymbol{\beta}} - \mathbf{c}) = \mathbf{c}$, so both constraint vectors vanish at $\tilde{\boldsymbol{\beta}}$: feasibility and complementary slackness hold. By the converse part of Theorem P.31, $\tilde{\boldsymbol{\beta}}$ minimises Q on the feasible set.

Step 4: uniqueness. If $\boldsymbol{\beta}$ is feasible, $\mathbf{h} = \boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}$ has $\mathbf{R}'\mathbf{h} = \mathbf{0}$, so $\nabla Q(\tilde{\boldsymbol{\beta}})'\mathbf{h} = -\boldsymbol{\nu}'\mathbf{R}'\mathbf{h} = 0$ and (P.35) gives $Q(\boldsymbol{\beta}) = Q(\tilde{\boldsymbol{\beta}}) + \mathbf{h}'\mathbf{M}\mathbf{h} > Q(\tilde{\boldsymbol{\beta}})$ unless $\mathbf{h} = \mathbf{0}$. No point satisfies both inequalities strictly, so Slater's condition fails and the necessity half of Theorem P.31 is unavailable; the sufficiency half needs only convexity. ■

Measure-theoretic probability

Exercise P.20 (Knapp, problems V.3–V.4)**.** Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, $E_1, E_2, \ldots \in \mathcal{F}$, $A = \bigcap_m \bigcup_{k \geqslant m} E_k$ and $B = \bigcup_m \bigcap_{k \geqslant m} E_k$.

(a) Show that $\mathbb{1}_A = \limsup_n \mathbb{1}_{E_n}$ and $\mathbb{1}_B = \liminf_n \mathbb{1}_{E_n}$, so $B \subseteq A$.

(b) If μ is finite and $A = B =: E$, show that $\mu(E_n) \to \mu(E)$.

(c) Show that finiteness of μ cannot be dropped in (b).

Solution. (a) $\omega \in A$ iff for every m some $k \geqslant m$ has $\omega \in E_k$, iff $\sup_{k \geqslant m} \mathbb{1}_{E_k}(\omega) = 1$ for every m, iff $\limsup_n \mathbb{1}_{E_n}(\omega) = 1$; otherwise that lim sup is 0. Likewise $\omega \in B$ iff ω lies in all E_k from some index on, iff $\liminf_n \mathbb{1}_{E_n}(\omega) = 1$. Proposition P.9 gives $\mathbb{1}_B \leqslant \mathbb{1}_A$.

(b) By (a), $\limsup_n \mathbb{1}_{E_n} = \mathbb{1}_E = \liminf_n \mathbb{1}_{E_n}$, so $\mathbb{1}_{E_n} \to \mathbb{1}_E$ at every point (Proposition P.9). The constant 1 dominates every $\mathbb{1}_{E_n}$ and is integrable because $\mu(\Omega) < \infty$, so Theorem P.86 and Lemma P.77(iv) give $\mu(E_n) = \int \mathbb{1}_{E_n}\, d\mu \to \int \mathbb{1}_E\, d\mu = \mu(E)$.

(c) For the counting measure on the positive integers and $E_n = \{n, n+1, \ldots\}$, $A = B = \emptyset$ while $\mu(E_n) = \infty$ for every n. In probability, A is the event $\{E_n \text{ infinitely often}\}$ of Lemma P.106. ■

Exercise P.21 (Baldi, exercise 1.11)**.** Let $(w_n)_{n\geqslant 1}$ be real and $\varphi(t)=\sum_{n=1}^{\infty} w_n e^{-tn}$ for $t>0$.
(a) If (w_n) is bounded, show that φ is differentiable on $(0,\infty)$ with $\varphi'(t)=-\sum_{n=1}^{\infty} n w_n e^{-tn}$.
(b) Show that the same holds for $w_n=\sqrt{n}$ and for $w_n=e^{\sqrt{n}}$.

Solution. Sums as integrals. Let κ be the counting measure on the positive integers with all subsets measurable. For $g\geqslant 0$, $g=\sum_n g(n)\,\mathbb{1}_{\{n\}}$, so Corollary P.80(i) and Lemma P.77(iv) give $\int g\,\mathrm{d}\kappa=\sum_n g(n)$. Hence g is κ-integrable iff $\sum_n |g(n)|<\infty$, and then $\int g\,\mathrm{d}\kappa=\sum_n g(n)$, applying the above to $g^{\pm}$.
(a) Let $|w_n|\leqslant W$, fix $a>0$, and apply Theorem P.87 on $I=(a,\infty)$ to $f(t,n)=w_n e^{-tn}$. (i) $\sum_n |w_n|e^{-tn}\leqslant W\sum_n e^{-an}<\infty$. (ii) $\partial f/\partial t(t,n)=-n w_n e^{-tn}$. (iii) $|\partial f/\partial t(t,n)|\leqslant g(n):=Wne^{-an}$ for $t\in I$, and $\sum_n ne^{-an}<\infty$: $ne^{-an/2}\leqslant 8/(a^2 n)\to 0$ (from the exponential series, whose terms are non-negative, $e^{an/2}\geqslant (an/2)^2/2$), so $ne^{-an}\leqslant Ce^{-an/2}$ for a constant C, a geometric bound. The theorem gives the formula on (a,∞), and $a>0$ was arbitrary.
(b) The same three checks go through once $\sum_n |w_n|ne^{-an}<\infty$ for every $a>0$ (this sum also dominates the one in (i)). For $w_n=\sqrt{n}$ the terms are $n^{3/2}e^{-an}\leqslant Ce^{-an/2}$, as $n^{3/2}e^{-an/2}\leqslant 48/(a^3 n^{3/2})\to 0$ by $e^{an/2}\geqslant (an/2)^3/6$. For $w_n=e^{\sqrt{n}}$, once $n\geqslant 16/a^2$ one has $\sqrt{n}\leqslant an/4$, so

$ne^{\sqrt{n}-an} \leqslant ne^{-3an/4}$, again summable. ■

Exercise P.22 (Baldi, exercise 3.8)**.** Let $X_1, X_2, \ldots$ be random variables with $\sum_k \mathsf{E}\,|X_k| < \infty$ and $S_n = \sum_{k\leqslant n} X_k$. Show that $\sum_k X_k$ converges absolutely almost surely to an integrable S, that $\mathsf{E}\,|S_n - S| \to 0$, and that $\mathsf{E}[S] = \sum_k \mathsf{E}[X_k]$.

Solution. Let $G = \sum_k |X_k|$, a measurable function with values in $[0,\infty]$. By Corollary P.80(i), $\mathsf{E}[G] = \sum_k \mathsf{E}\,|X_k| < \infty$, so $N = \{G = \infty\}$ is null by Proposition P.82(v). Off N the partial sums of X_k^+ and of X_k^- increase and are bounded by G, so both converge (Theorem P.2) and so does S_n. Define $S = \lim_n S_n$ off N and $S = 0$ on N; it is measurable as the pointwise limit of $S_n \mathbb{1}_{N^c}$ (Proposition P.68(ii)). Now $|S_n| \leqslant G$ and $|S| \leqslant G$ almost surely, with G integrable, so Theorem P.86 gives $\mathsf{E}\,|S_n - S| \to 0$ and $\sum_{k\leqslant n} \mathsf{E}[X_k] = \mathsf{E}[S_n] \to \mathsf{E}[S]$. ■

Exercise P.23 (Knapp, problem V.20)**. (Scheffé.)** Let $f, f_1, f_2, \ldots \geqslant 0$ be integrable with $f_n \to f$ almost everywhere and $\int f_n \,\mathrm{d}\mu \to \int f \,\mathrm{d}\mu$. Show that $\int |f_n - f| \,\mathrm{d}\mu \to 0$. Deduce that if probability densities p_n converge almost everywhere to a probability density p, then the corresponding laws satisfy $\sup_B |P_n(B) - P(B)| \to 0$, and in particular $P_n \to P$ weakly.

Solution. With $x = f_n - f$, the identity $|x| = x + 2x^-$ reads $|f_n - f| = (f_n - f) + 2(f - f_n)^+$. The functions $(f - f_n)^+$ are measurable, satisfy $0 \leqslant (f - f_n)^+ \leqslant f$, and tend to 0 almost everywhere, so $\int (f - f_n)^+ \,\mathrm{d}\mu \to 0$ by Theorem P.86. By Theorem P.85(i), $\int (f_n - f)\,\mathrm{d}\mu = \int f_n \,\mathrm{d}\mu - \int f \,\mathrm{d}\mu \to 0$. Adding, $\int |f_n - f| \,\mathrm{d}\mu \to 0$.
For densities, $\int p_n \,\mathrm{d}\mu = 1 = \int p \,\mathrm{d}\mu$, so the first part applies, and for every Borel B, by Theorem P.85(i),(iii), $|P_n(B) - P(B)| = |\int_B (p_n - p)\,\mathrm{d}\mu| \leqslant \int |p_n - p| \,\mathrm{d}\mu$, a bound free of B. Likewise $|\int g \,\mathrm{d}P_n - \int g \,\mathrm{d}P| \leqslant \sup|g| \int |p_n - p| \,\mathrm{d}\mu \to 0$ for $g \in C_b$, where $\int g \,\mathrm{d}P_n = \int g p_n \,\mathrm{d}\mu$ by Lemma P.99, applied to a random vector with law P_n (and likewise for P). ■

Exercise P.24 (Knapp, problem V.14)**.** Let $(\Omega, \mathcal{F}, \mu)$ be σ-finite, λ Lebesgue measure on $\mathbb{R}$, and f a non-negative measurable function with values in $[0, \infty]$. Show that the region under the graph, $R = \{(x, y) : 0 \leqslant y < f(x)\}$, lies in $\mathcal{F} \otimes \mathcal{B}(\mathbb{R})$ and that $(\mu \otimes \lambda)(R) = \int f \,\mathrm{d}\mu$. Deduce that every random variable $X \geqslant 0$ satisfies

$$\mathsf{E}[X] = \int_{[0,\infty)} \mathsf{P}\{X > y\} \,\mathrm{d}y.$$

Solution. **Measurability.** $y < f(x)$ holds iff some rational q has $y < q < f(x)$, also when $f(x) = \infty$, so

$$R = \big(\Omega \times [0,\infty)\big) \cap \bigcup_{q\in\mathbb{Q}} \big(\{f > q\} \times (-\infty, q)\big),$$

a countable union and intersection of measurable rectangles (Lemma P.67(ii)).

Measure. The section $R_x = [0, f(x))$ has $\lambda(R_x) = f(x)$: by Proposition P.65(iv),(v), $\lambda(\{0\}) = \lim_k \lambda((-1/k, 0]) = 0$, $\lambda((0,a)) = \lim_k \lambda((0, a - 1/k]) = a$ for $0 < a < \infty$, so $\lambda([0,a)) = a$ by Proposition P.65(i), and $\lambda([0,\infty)) = \lim_k \lambda([0,k)) = \infty$. As λ is σ-finite (Remark 18), Theorem P.96 gives $(\mu \otimes \lambda)(R) = \int \lambda(R_x)\, \mu(\mathrm{d}x) = \int f \,\mathrm{d}\mu$.

Expectation. Take $\mu = \mathsf{P}$ and $f = X$. The other formula in (P.14) uses the sections $R^y = \{\omega : X(\omega) > y\}$ for $y \geqslant 0$ and $R^y = \emptyset$ for $y < 0$, so $\mathsf{E}[X] = (\mathsf{P}\otimes\lambda)(R) = \int_{[0,\infty)} \mathsf{P}\{X > y\}\,\mathrm{d}y$. ■

Exercise P.25 (Knapp, problem III.7)**.** Let $f(x,y) = e^{-xy} - 2e^{-2xy}$ on $[1,\infty) \times (0,1]$, with

Lebesgue measure on each factor. Prove or disprove:

$$\int_{(0,1]}\Big[\int_{[1,\infty)} f(x,y)\,\mathrm{d}x\Big]\mathrm{d}y = \int_{[1,\infty)}\Big[\int_{(0,1]} f(x,y)\,\mathrm{d}y\Big]\mathrm{d}x.$$

Reconcile the answer with Theorem P.97.

Solution. For $c>0$ and $y>0$, $\int_{[1,N]} e^{-cxy}\,\mathrm{d}x = (e^{-cy}-e^{-cNy})/(cy)$: the integrand is continuous, so by Remark 18(c) this is a Riemann integral, computed by calculus (Remark 1). Letting $N\to\infty$ with Theorem P.78, $\int_{[1,\infty)} e^{-cxy}\,\mathrm{d}x = e^{-cy}/(cy)$. Both terms of $f(\cdot,y)$ being integrable,

$$\int_{[1,\infty)} f(x,y)\,\mathrm{d}x = \frac{e^{-y}-e^{-2y}}{y} > 0 \qquad (0<y\leqslant 1).$$

This is at most 1, since $e^{-y}-e^{-2y} = ye^{-\xi}$ for some $\xi\in(y,2y)$ by Corollary P.7; so the left side is the integral of a bounded positive function over $(0,1]$, which is positive by Proposition P.82(iv).

In the other order, for $x \geqslant 1$, $\int_{(0,1]} e^{-cxy}\,\mathrm{d}y = (1-e^{-cx})/(cx)$, so

$$\int_{(0,1]} f(x,y)\,\mathrm{d}y = \frac{(1-e^{-x})-(1-e^{-2x})}{x} = \frac{e^{-2x}-e^{-x}}{x} < 0,$$

bounded in modulus by e^{-x}, which is integrable on $[1,\infty)$; the right side is negative. The two iterated integrals differ (numerically they are 0.523 and -0.170), so the equality is false.
Let π be the product of the two Lebesgue measures. The function f is product-measurable: it is the restriction of the continuous $(x,y) \mapsto e^{-xy} - 2e^{-2xy}$ on $\mathbb{R}^2$, which is Borel by Lemma P.67(iii), that is $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$-measurable by Lemma P.93. So Theorem P.97(ii) would force both iterated integrals to equal $\int f\,\mathrm{d}\pi$ if $\int |f|\,\mathrm{d}\pi < \infty$. Hence $\int |f|\,\mathrm{d}\pi = \infty$: the integrability hypothesis of Fubini's theorem cannot be dropped, while Tonelli's theorem covers only non-negative integrands. ■

Exercise P.26 (Baldi, exercise 2.4)**.** Let X and Y be independent real random variables with $\mathsf{E}\,|X+Y| < \infty$. Prove that $\mathsf{E}\,|X| < \infty$ and $\mathsf{E}\,|Y| < \infty$. Show that independence cannot be dropped.

Solution. By Definition P.104 the law of (X, Y) is $\mu_X \otimes \mu_Y$. By Theorem P.89 and Theorem P.97(i),

$$\infty > \mathsf{E}\,|X+Y| = \int |x+y|\,\mathrm{d}(\mu_X \otimes \mu_Y)(x,y) = \int h(y)\,\mu_Y(\mathrm{d}y), \qquad h(y) = \int |x+y|\,\mu_X(\mathrm{d}x).$$

By Proposition P.82(v), $h < \infty$ outside a μ_Y-null set, which therefore is not all of $\mathbb{R}$; pick y_0 with $h(y_0) < \infty$. By Theorem P.89, $\mathsf{E}\,|X + y_0| = h(y_0) < \infty$, and $|X| \leqslant |X + y_0| + |y_0|$, so X is integrable. By symmetry so is Y. Without independence take any X with $\mathsf{E}\,|X| = \infty$, for instance $\mathsf{P}\{X = 2^k\} = 2^{-k}$ for $k \geqslant 1$, so that $\mathsf{E}\,|X| = \sum_k 2^k 2^{-k} = \infty$ (Corollary P.80(i)), and $Y = -X$: then $X + Y = 0$. ■

Exercise P.27 (Baldi, exercise 3.3)**.** Call events $A_1, A_2, \ldots$ independent if their indicators are independent (Definition P.104). A random variable is **uniform on** $[0,1]$ if its law is $B \mapsto \lambda(B \cap [0,1])$.

(a) Let X be uniform on $[0,1]$ and $A_n = \{X \leqslant 1/n\}$. Show that $\sum_n \mathsf{P}(A_n) = \infty$ but $\mathsf{P}\{A_n \text{ infinitely often}\} = 0$.

(b) **(Second Borel–Cantelli lemma.)** Show that if $A_1, A_2, \ldots$ are independent and $\sum_n \mathsf{P}(A_n) = \infty$, then $\mathsf{P}\{A_n \text{ infinitely often}\} = 1$.

(c) Let $X_1, X_2, \ldots$ be independent and uniform on $[0,1]$. Show that $\mathsf{P}\{X_n \leqslant 1/n \text{ i.o.}\} = 1$ and $\mathsf{P}\{X_n \leqslant 1/n^2 \text{ i.o.}\} = 0$.

Solution. (a) $\mathsf{P}(A_n) = \lambda([0,1/n]) = 1/n$, and $\sum_n 1/n = \infty$, each block $2^j < n \leqslant 2^{j+1}$ contributing at least $\frac{1}{2}$. If $X(\omega) > 0$ then $X(\omega) > 1/n$ for all $n > 1/X(\omega)$, so $\{A_n \text{ i.o.}\} = \{X \leqslant 0\}$, which has probability $\lambda(\{0\}) = 0$. Lemma P.106 has no converse without independence.
(b) The complement of $\{A_n \text{ i.o.}\}$ is $\bigcup_m \bigcap_{n \geqslant m} A_n^c$. For $m \leqslant N$, independence (with the Borel sets $\{0\}$ for the indicators) gives

$$\mathsf{P}\Big(\bigcap_{n=m}^{N} A_n^c\Big) = \prod_{n=m}^{N} \big(1 - \mathsf{P}(A_n)\big) \leqslant \exp\Big(-\sum_{n=m}^{N} \mathsf{P}(A_n)\Big) \xrightarrow[N\to\infty]{} 0,$$

using $1 - x \leqslant e^{-x}$, the tangent line inequality of Proposition P.22 for the function e^{-x} at 0, convex by Lemma P.19 as its second derivative e^{-x} is positive. By Proposition P.65(v), $\mathsf{P}(\bigcap_{n \geqslant m} A_n^c) = 0$ for each m, and a countable union of null sets is null.

(c) The indicators of $\{X_n \leqslant c_n\}$ are functions of the independent X_n, hence independent: with $h_n = \mathbb{1}_{(-\infty, c_n]}$, Borel, and any Borel B_n, $\{h_n(X_n) \in B_n\} = \{X_n \in h_n^{-1}(B_n)\}$, so the product rule of Definition P.104 for any finitely many of the X_n gives it for their indicators. With $c_n = 1/n$ the probabilities sum to ∞ and (b) applies; with $c_n = 1/n^2$ they sum to $\pi^2/6$ (Lemma P.17) and Lemma P.106 applies. ■

Exercise P.28 (Baldi, exercise 3.5)**.** Let $Z, Z_1, Z_2, \ldots$ be independent, identically distributed and non-negative.

(a) Show that $\sum_{n\geqslant 1} \mathsf{P}\{Z \geqslant n\} \leqslant \mathsf{E}[Z] \leqslant \sum_{n\geqslant 0} \mathsf{P}\{Z \geqslant n\}$.

(b) Show that $\mathsf{P}\{Z_n \geqslant n \text{ i.o.}\}$ is 0 if $\mathsf{E}[Z] < \infty$ and 1 if $\mathsf{E}[Z] = \infty$.

(c) Let $X_1, X_2, \ldots$ be independent standard normal. Show that $\limsup_n |X_n|/\sqrt{\log n} = \sqrt{2}$ almost surely (the ratio taken for $n \geqslant 2$).

Solution. (a) By Exercise P.24, $\mathsf{E}[Z] = \int_{[0,\infty)} \mathsf{P}\{Z > y\}\, dy$. Splitting $[0, \infty)$ into the intervals

$[n, n+1)$ (Corollary P.80(i)) and using $\mathsf{P}\{Z \geqslant n+1\} \leqslant \mathsf{P}\{Z > y\} \leqslant \mathsf{P}\{Z \geqslant n\}$ for $y \in [n, n+1)$,

$$\sum_{n\geqslant 0} \mathsf{P}\{Z \geqslant n+1\} \leqslant \mathsf{E}[Z] \leqslant \sum_{n\geqslant 0} \mathsf{P}\{Z \geqslant n\}.$$

(b) $\mathsf{P}\{Z_n \geqslant n\} = \mathsf{P}\{Z \geqslant n\}$. If $\mathsf{E}[Z] < \infty$ these sum to at most $\mathsf{E}[Z]$ by (a), and Lemma P.106 applies. If $\mathsf{E}[Z] = \infty$ then by (a) $\sum_{n\geqslant 0} \mathsf{P}\{Z \geqslant n\} = \infty$, hence also the sum over $n \geqslant 1$; the events $\{Z_n \geqslant n\}$ are independent (their indicators are Borel functions of the independent Z_n, as in the solution of Exercise P.27(c)), and Exercise P.27(b) gives probability 1.
(c) For $0 < \theta < \frac{1}{2}$, $\mathsf{E}[e^{\theta X_1^2}] = \int (2\pi)^{-1/2} e^{-(1-2\theta)z^2/2}\,\mathrm{d}z = (1-2\theta)^{-1/2}$, by Lemma P.99 and Remark 18(e) with $a = (1-2\theta)^{-1/2}$; for $\theta \geqslant \frac{1}{2}$ the integrand is at least $(2\pi)^{-1/2}$ and $\mathsf{E}[e^{\theta X_1^2}] = \infty$. Apply (b) to the independent, identically distributed $Z_n = e^{\theta X_n^2}$ (independent as in the solution of Exercise P.27(c), and identically distributed as images of one law under one Borel map). If $\theta < \frac{1}{2}$, almost surely $e^{\theta X_n^2} < n$, that is $X_n^2/\log n < 1/\theta$ $(n \geqslant 2)$, for all large n, so $\limsup_n X_n^2/\log n \leqslant 1/\theta$. If $\theta > \frac{1}{2}$, almost surely $X_n^2/\log n \geqslant 1/\theta$ infinitely often, so $\limsup_n X_n^2/\log n \geqslant 1/\theta$. Letting θ run through rationals tending to $\frac{1}{2}$ from either side and intersecting the countably many events of probability one (Proposition P.82(vi)), $\limsup_n X_n^2/\log n = 2$, and taking square roots gives the

claim. ■

Exercise P.29 (Baldi, exercise 4.1)**.** Let X, Y be independent with $\mathsf{P}\{X = 1\} = \mathsf{P}\{Y = 1\} = p = 1 - \mathsf{P}\{X = 0\} = 1 - \mathsf{P}\{Y = 0\}$, $0 < p < 1$, and let $\mathcal{G} = \{\emptyset, A_0, A_1, \Omega\}$ with $A_0 = \{X + Y = 0\}$ and $A_1 = A_0^c$, the σ-field generated by $\mathbb{1}_{A_0}$. Compute $\mathsf{E}[X \mid \mathcal{G}]$ and $\mathsf{E}[Y \mid \mathcal{G}]$ and their laws. Are they independent?

Solution. **Conditioning on a two-set partition.** Let $A_0, A_1 \in \mathcal{F}$ partition Ω with positive probabilities and let X be integrable. A bounded $\mathcal{G}$-measurable W is constant on each A_i, since $\{W = w\} \in \mathcal{G}$ for every w; write $W = w_0 \mathbb{1}_{A_0} + w_1 \mathbb{1}_{A_1}$. Put $V = \sum_i \alpha_i \mathbb{1}_{A_i}$ with $\alpha_i = \mathsf{E}[X \mathbb{1}_{A_i}] / \mathsf{P}(A_i)$. Then V is $\mathcal{G}$-measurable and $\mathsf{E}[VW] = \sum_i w_i \alpha_i \mathsf{P}(A_i) = \sum_i w_i \mathsf{E}[X \mathbb{1}_{A_i}] = \mathsf{E}[XW]$, so $V = \mathsf{E}[X \mid \mathcal{G}]$ by Lemma P.109.

Here. $\mathsf{P}(A_0) = (1-p)^2$ by independence and $\mathsf{P}(A_1) = 1 - (1-p)^2 = p(2-p)$. On A_0, $X = 0$, so $\alpha_0 = 0$; and $\{X = 1\} \subseteq A_1$, so $\mathsf{E}[X \mathbb{1}_{A_1}] = p$ and $\alpha_1 = 1/(2-p)$. Hence

$$\mathsf{E}[X \mid \mathcal{G}] = \frac{1}{2-p} \mathbb{1}_{A_1},$$

which takes the value $1/(2-p)$ with probability $p(2-p)$ and 0 with probability $(1-p)^2$; its mean is $p = \mathsf{E}[X]$, as Proposition P.110 requires. The formula is symmetric in X and Y, so $\mathsf{E}[Y \mid \mathcal{G}]$ is the same variable U. It is not independent of itself: $\mathsf{P}\{U = 0, U = 0\} = (1-p)^2 \neq (1-p)^4 = \mathsf{P}\{U = 0\}^2$. ■

Exercise P.30 (Baldi, exercise 4.5)**.** For a random vector Z in $\mathbb{R}^k$ let $\sigma(Z) = \{Z^{-1}(B) : B \in \mathcal{B}(\mathbb{R}^k)\}$ and $\mathsf{E}[\cdot \mid Z] = \mathsf{E}[\cdot \mid \sigma(Z)]$.

(a) Show that to identify $\mathsf{E}[X \mid \mathcal{G}]$ it suffices to check the identity of Lemma P.109 for $W = \mathbb{1}_G$, $G \in \mathcal{G}$.

(b) Let X, Y be real and suppose (X, Z) and (Y, Z) have the same law. Show that $\mathsf{E}[h(X) \mid Z] = \mathsf{E}[h(Y) \mid Z]$ almost surely for every Borel h with $h(X)$ integrable.

(c) Let $T_1, \ldots, T_n$ be independent, identically distributed and integrable, and $T = T_1 + \cdots + T_n$. Show that $\mathsf{E}[T_1 \mid T] = T/n$.

Solution. (a) $\sigma(Z)$ is a σ-field, by the argument in the proof of Lemma P.67(i). Suppose V is $\mathcal{G}$-measurable and integrable and $\mathsf{E}[X\,\mathbb{1}_G] = \mathsf{E}[V\,\mathbb{1}_G]$ for all $G \in \mathcal{G}$. By linearity the identity extends

to simple $\mathcal{G}$-measurable W. For bounded $\mathcal{G}$-measurable $W \geqslant 0$, Theorem P.70 gives simple $\mathcal{G}$-measurable $s_m \uparrow W$, with $|Xs_m| \leqslant |X| \sup W$ and $|Vs_m| \leqslant |V| \sup W$, so Theorem P.86 passes the identity to W; a general bounded W is $W^+ - W^-$.
(b) Let $V = \mathsf{E}[h(X) \mid Z]$. For $G = Z^{-1}(B) \in \sigma(Z)$, Theorem P.89 applied to the function $(x, z) \mapsto h(x)\,\mathbb{1}_B(z)$ and the equal laws of (X, Z) and (Y, Z) give $\mathsf{E}[h(Y)\,\mathbb{1}_G] = \mathsf{E}[h(X)\,\mathbb{1}_G] = \mathsf{E}[V\,\mathbb{1}_G]$; also $h(Y)$ is integrable, by the same theorem applied to $|h(x)|$. By (a), $V = \mathsf{E}[h(Y) \mid Z]$.
(c) Fix j. By Lemma P.105, T_j is independent of $T_{(j)} = \sum_{i \neq j} T_i$, so the law of $(T_j, T_{(j)})$ is the product of their laws. Both factors are the same for every j: the T_i share one law, and the law of $(T_i)_{i \neq j}$, a product of identical factors (Corollary P.98), does not depend on j. So (T_j, T), the image of $(T_j, T_{(j)})$ under $(s, t) \mapsto (s, s+t)$, has the same law for every j, and by (b) $\mathsf{E}[T_j \mid T] = \mathsf{E}[T_1 \mid T]$. The conditional expectation is linear (by the uniqueness in Lemma P.109) and $\mathsf{E}[T \mid T] = T$, T being $\sigma(T)$-measurable. Hence $n\,\mathsf{E}[T_1 \mid T] = \sum_j \mathsf{E}[T_j \mid T] = T$. For Bernoulli trials this says that, given the number of successes, each trial succeeds with probability T/n. ■

Exercise P.31 (Baldi, exercise 4.23)**.** Let $(X, Y)' \sim N\big(\mathbf{0}, \left(\begin{smallmatrix} 1 & r \\ r & 1 \end{smallmatrix}\right)\big)$ with $|r| < 1$. Show that X and Y are $N(0, 1)$, that $\mathsf{E}[X \mid Y] = rY$ and $\mathsf{E}[X \mid X + Y] = (X + Y)/2$, and that $\mathsf{E}[(X - \mathsf{E}[X \mid$

$Y])^2] = 1 - r^2$.

Solution. By Theorem P.114 with $\mathbf{B} = (1,0)$ or $(0,1)$, X and Y are $N(0,1)$. The pair $(X - rY, Y)$ is a linear image of (X,Y), hence jointly normal by Theorem P.114, with $\operatorname{cov}(X - rY, Y) = r - r = 0$; by Theorem P.134, $X - rY$ and Y are independent. For $G = Y^{-1}(B) \in \sigma(Y)$, Lemma P.105 gives $\mathsf{E}[(X - rY)\,\mathbb{1}_B(Y)] = \mathsf{E}[X - rY]\ \mathsf{P}\{Y \in B\} = 0$, that is, $\mathsf{E}[X\,\mathbb{1}_G] = \mathsf{E}[rY\,\mathbb{1}_G]$. Since rY is $\sigma(Y)$-measurable and integrable, Exercise P.30(a) gives $\mathsf{E}[X \mid Y] = rY$.
In the same way $X - \frac{1}{2}(X+Y) = \frac{1}{2}(X-Y)$ and $X+Y$ are jointly normal with covariance $\frac{1}{2}(\operatorname{var} X - \operatorname{var} Y) = 0$, so $\mathsf{E}[X \mid X+Y] = (X+Y)/2$, whatever r. Finally $\mathsf{E}[(X - rY)^2] = 1 - 2r^2 + r^2 = 1 - r^2$: the regression of X on Y removes the fraction r^2 of the variance. ■

Exercise P.32 (Baldi, exercise 2.56)**.** Let $X \sim N(0,1)$ and Y with $\mathsf{P}\{Y = 1\} = \mathsf{P}\{Y = -1\} = \frac{1}{2}$ be independent, and $Z = XY$. Show that $Z \sim N(0,1)$ and $\operatorname{cov}(X,Z) = 0$, yet X and Z are not independent; compute the characteristic function of $X + Z$ and deduce that (X,Z) is not jointly normal. Which hypothesis of Theorem P.134 fails?

Solution. For Borel B, by independence, $\mathsf{P}\{Z \in B\} = \frac{1}{2}\mathsf{P}\{X \in B\} + \frac{1}{2}\mathsf{P}\{-X \in B\} = \mathsf{P}\{X \in B\}$, because $-X$ has the density $\phi(-x) = \phi(x)$ (Remark 18(e); X has the law of a standard normal, hence the density ϕ, by Proposition P.125 and Corollary P.131); so $Z \sim N(0,1)$. By Lemma P.105, $\operatorname{cov}(X,Z) = \mathsf{E}[X^2Y] = \mathsf{E}[X^2]\,\mathsf{E}[Y] = 0$. They are not independent: $|Z| = |X|$, so $\mathsf{P}\{|X| \leqslant 1, |Z| > 1\} = 0$, while $\mathsf{P}\{|X| \leqslant 1\}\,\mathsf{P}\{|Z| > 1\} > 0$.
Splitting on the value of Y ($X + Z = 0$ on $\{Y = -1\}$, and $\mathsf{E}[e^{2i\theta X}\,\mathbb{1}\{Y = 1\}] = \frac{1}{2}\mathsf{E}[e^{2i\theta X}]$ by Lemma P.105 applied to the real and imaginary parts) and using Proposition P.125,

$$\mathsf{E}\big[e^{i\theta(X+Z)}\big] = \tfrac{1}{2}\,\mathsf{E}\big[e^{2i\theta X}\big] + \tfrac{1}{2} = \tfrac{1}{2}e^{-2\theta^2} + \tfrac{1}{2}.$$

If (X,Z) were jointly normal, $X+Z$ would be $N(m,s^2)$ for some m, s (Theorem P.114), with characteristic function of modulus $e^{-s^2\theta^2/2}$. The modulus above stays above $\frac{1}{2}$ as $\theta \to \infty$, forcing $s = 0$, but then the modulus would be identically 1, while at $\theta = 1$ it is $\frac{1}{2}(1 + e^{-2}) < 1$. So the joint normality required by Theorem P.134 fails, although each marginal is normal. ■

Exercise P.33 (Baldi, exercise 2.53)**.** Let $X \sim N(\mathbf{0}, \mathbf{C})$ in $\mathbb{R}^m$, where $\mathbf{C}$ has eigenvalues $\lambda_1, \dots, \lambda_m \geqslant 0$. Show that $|X|^2$ has the law of $\sum_k \lambda_k Z_k^2$ with $Z_1, \dots, Z_m$ independent standard normal, that $\mathsf{E}\,|X|^2 = \operatorname{tr} \mathbf{C}$ and that $\operatorname{var} |X|^2 = 2 \operatorname{tr}(\mathbf{C}^2)$. What does this say when $\mathbf{C} = \mathbf{I}_m$?

Solution. Write $\mathbf{C} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ (Theorem P.39) and let $Z = (Z_1, \dots, Z_m)'$ have independent standard normal coordinates. Then $\tilde{X} = \mathbf{P}\mathbf{\Lambda}^{1/2} Z$ is $N(\mathbf{0}, \mathbf{C})$ by Definition P.113, as $(\mathbf{P}\mathbf{\Lambda}^{1/2})(\mathbf{P}\mathbf{\Lambda}^{1/2})' = \mathbf{C}$. By Proposition P.125, X and $\tilde{X}$ have the same characteristic function, hence the same law (Corollary P.131), so $|X|^2$ and $|\tilde{X}|^2$ have the same law. Since $\mathbf{P}'\mathbf{P} = \mathbf{I}$,

$$|\tilde{X}|^2 = Z' \mathbf{\Lambda}^{1/2} \mathbf{P}' \mathbf{P} \mathbf{\Lambda}^{1/2} Z = \sum_{k=1}^{m} \lambda_k Z_k^2.$$

So $\mathsf{E}\,|X|^2 = \sum_k \lambda_k = \operatorname{tr} \mathbf{\Lambda} = \operatorname{tr} \mathbf{C}$ (Proposition P.35(iii)); the same follows from Lemma P.103 with $\mathbf{A} = \mathbf{I}$. The Z_k^2 are independent with variance 2 (Lemma P.117), so their covariances vanish (Lemma P.105) and $\operatorname{var} |X|^2 = 2 \sum_k \lambda_k^2 = 2 \operatorname{tr}(\mathbf{\Lambda}^2) = 2 \operatorname{tr}(\mathbf{C}^2)$, as $\mathbf{C}^2 = \mathbf{P}\mathbf{\Lambda}^2\mathbf{P}'$. For $\mathbf{C} = \mathbf{I}_m$ the law is χ^2_m with mean m and variance $2m$ (Proposition P.118); for a symmetric idempotent $\mathbf{C}$ of rank r it is χ^2_r, as in Theorem P.135, since its eigenvalues lie in $\{0, 1\}$ and sum to $\operatorname{tr} \mathbf{C} = \operatorname{rank} \mathbf{C} = r$

(Lemma P.46 and Proposition P.35(iii)), so that r of them equal 1 and the rest 0. ■

Exercise P.34 (Baldi, exercise 3.9)**. (Dominated convergence in probability.)** Let $X_n \xrightarrow{p} X$ and $|X_n| \leqslant Y$ with $\mathsf{E}[Y] < \infty$. Show that $\mathsf{E}\,|X_n - X| \to 0$, so $\mathsf{E}[X_n] \to \mathsf{E}[X]$.

Solution. By Lemma P.120 a subsequence converges to X almost surely, so $|X| \leqslant Y$ almost surely and $|X_n - X| \leqslant 2Y$. Suppose $\mathsf{E}\,|X_n - X| \not\to 0$; then $\mathsf{E}\,|X_{n_j} - X| \geqslant \delta$ along a subsequence, for some $\delta > 0$. That subsequence still converges in probability, so a further subsequence converges almost surely, and along it Theorem P.86, with dominating function $2Y$, gives $\mathsf{E}\,|X_{n_{j_l}} - X| \to 0$, a contradiction. Finally $|\mathsf{E}\,X_n - \mathsf{E}\,X| \leqslant \mathsf{E}\,|X_n - X|$ (Theorem P.85(iii)). ■

Exercise P.35. Give a sequence converging in distribution but not in probability, and one converging in probability but not almost surely.

Solution. For the first, let $X \sim N(0,1)$ and $X_n = (-1)^n X$. Each X_n has the $N(0,1)$ law, so $X_n \xrightarrow{d} X$; but $|X_n - X| = 2|X|$ for odd n, which does not tend to 0 in probability. This also shows convergence in distribution says nothing about the joint behaviour of X_n and X, only about laws.

For the second, take the sliding indicators on $[0,1]$ with Lebesgue measure: for $n = 2^k + r$ with $0 \leqslant r < 2^k$, let $X_n = \mathbb{1}_{[r2^{-k},(r+1)2^{-k}]}$. Then $\mathsf{P}\{X_n \neq 0\} = 2^{-k} \to 0$, so $X_n \xrightarrow{p} 0$; but every ω lies in infinitely many of the intervals (at least one of each level k), and, for $k \geqslant 2$, in at most two of the 2^k intervals of level k, so $X_n(\omega) = 1$ and $X_n(\omega) = 0$ each for infinitely many n, and $X_n(\omega)$ does not converge for any ω. ■

Exercise P.36 (Baldi, exercise 2.31)**.** Let φ be the characteristic function of a real random variable X. Show that $\overline{\varphi}$, φ^2 and $|\varphi|^2$ are characteristic functions, and that $|\varphi|^2$ is real and even.

Solution. Taking complex conjugates commutes with the integral, which is defined through real and imaginary parts (Definition P.84), so $\overline{\varphi(t)} = \mathsf{E}[e^{-itX}] = \varphi_{-X}(t)$. Let μ be the law of X; on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2), \mu \otimes \mu)$ (Corollary P.98) the coordinate maps X_1, X_2 are independent copies of X. By Proposition P.124(iii), $\varphi_{X_1+X_2} = \varphi^2$, and since X_1 and $-X_2$ are independent, $\varphi_{X_1-X_2} = \varphi\,\overline{\varphi} = |\varphi|^2$. This is real, and even because $\varphi(-t) = \overline{\varphi(t)}$; the symmetrised variable $X_1 - X_2$ thus has a real characteristic function. ■

Exercise P.37 (Baldi, exercise 2.38)**.** Show that $\theta \mapsto e^{-\theta^4}$ is not the characteristic function of any real random variable.

Solution. Suppose $\mathsf{E}[e^{i\theta X}] = e^{-\theta^4}$. Taking real parts, $\mathsf{E}[1-\cos(\theta X)] = 1 - e^{-\theta^4}$. By Theorem P.14 with $m = 1$ applied to cos, $1 - \cos u = \frac{u^2}{2}\cos\xi$ for some ξ between 0 and u, so $k^2\{1-\cos(x/k)\} \to x^2/2$ for every x, through non-negative values. By Theorem P.83 and $1 - e^{-s} \leqslant s$ (the tangent line inequality of Proposition P.22 for e^{-s} at 0, as in the solution of Exercise P.27(b)),

$$\mathsf{E}\Big[\frac{X^2}{2}\Big] \leqslant \liminf_k k^2\ \mathsf{E}\big[1-\cos(X/k)\big] = \liminf_k k^2\big(1-e^{-k^{-4}}\big) \leqslant \liminf_k k^{-2} = 0.$$

So $X = 0$ almost surely (Proposition P.82(iv)) and its characteristic function is identically 1, whereas $e^{-1} \neq 1$. The argument shows more: if $1 - \operatorname{Re}\varphi(\theta) = o(\theta^2)$ as $\theta \to 0$, then $X = 0$ almost surely. ∎

Exercise P.38 (Baldi, exercise 3.14)**.** Let $\mu_n \to \mu$ weakly on $\mathbb{R}^d$ and $\nu_n \to \nu$ weakly on $\mathbb{R}^m$.

(a) Show that $\mu_n \otimes \nu_n \to \mu \otimes \nu$ weakly on $\mathbb{R}^{d+m}$.

(b) For $d = m$, let $X_n \sim \mu_n$ be independent of $Y_n \sim \nu_n$ and $X \sim \mu$ of $Y \sim \nu$. Show that $X_n + Y_n \xrightarrow{d} X + Y$.

(c) Let $Y_n \sim N(\mathbf{0}, n^{-1}\mathbf{I}_d)$ be independent of X. Show that $X + Y_n \xrightarrow{d} X$.

Solution. (a) For $\theta_1 \in \mathbb{R}^d$ and $\theta_2 \in \mathbb{R}^m$ the integrand $e^{i\langle\theta_1,x\rangle}e^{i\langle\theta_2,y\rangle}$ is bounded, hence integrable against the probability $\mu_n \otimes \nu_n$, and Theorem P.97(ii) factorises the characteristic function at (θ_1, θ_2) as $\varphi_{\mu_n}(\theta_1)\varphi_{\nu_n}(\theta_2)$. By Theorem P.130 (necessity) this tends to $\varphi_\mu(\theta_1)\varphi_\nu(\theta_2)$, the characteristic function of $\mu \otimes \nu$, and Theorem P.130 (sufficiency) gives the claim.

(b) By Definition P.104 the laws of (X_n, Y_n) and (X, Y) are $\mu_n \otimes \nu_n$ and $\mu \otimes \nu$, so $(X_n, Y_n) \xrightarrow{d} (X, Y)$ by (a), and Theorem P.138 with $(x, y) \mapsto x + y$ gives $X_n + Y_n \xrightarrow{d} X + Y$.

(c) The characteristic function of Y_n is $e^{-|\theta|^2/(2n)} \to 1$ (Proposition P.125), that of the point mass at $\mathbf{0}$, so $Y_n \xrightarrow{d} \mathbf{0}$ by Theorem P.130, and (b) with $X_n = X$ applies. Alternatively, $Y_n \xrightarrow{p} \mathbf{0}$ by Proposition P.122, and Theorem P.139(iii) gives this without independence. Adding a small

independent Gaussian perturbation is the smoothing behind Lemma P.127. ■

Exercise P.39 (Baldi, exercise 3.21)**.** Let $X_1, X_2, \ldots$ be independent and uniform on $[0,1]$ (Exercise P.27) and $Z_n = \min(X_1, \ldots, X_n)$.

(a) Show that $Z_n \to 0$ almost surely.

(b) Show that $nZ_n \xrightarrow{d} E$, where E has density e^{-t} on $[0, \infty)$.

(c) Deduce that $\mathsf{P}\{Z_n \leqslant 2/n\} \to 1 - e^{-2}$.

Solution. By independence, $\mathsf{P}\{Z_n > z\} = \prod_{i \leqslant n} \mathsf{P}\{X_i > z\} = (1-z)^n$ for $z \in [0,1]$.

(a) Z_n decreases in n and is non-negative, so it converges pointwise to some $Z_\infty \geqslant 0$. For $\varepsilon \in (0,1)$, $\mathsf{P}\{Z_\infty > \varepsilon\} \leqslant \mathsf{P}\{Z_n > \varepsilon\} = (1-\varepsilon)^n \to 0$, so $Z_\infty = 0$ almost surely.

(b) Let $g_n(t) = (1 - t/n)^{n-1}$ for $0 \leqslant t \leqslant n$ and $g_n(t) = 0$ otherwise. For $0 \leqslant a \leqslant n$, $\mathsf{P}\{nZ_n \leqslant a\} = 1 - (1 - a/n)^n = \int_{[0,a]} g_n(t)\,\mathrm{d}t$ by calculus, and both sides equal 0 for $a < 0$ and 1 for $a > n$. The half-lines $(-\infty, a]$ form a π-system generating $\mathcal{B}(\mathbb{R})$ (Lemma P.63(iii)), so by Corollary P.73 nZ_n has density g_n. For $t \geqslant 0$, $g_n(t) \to e^{-t}$, as $(n-1)\log(1 - t/n) \to -t$; and $1 - x \leqslant e^{-x}$ (as in the solution of Exercise P.27(b)) gives $g_n(t) \leqslant e^{-t(n-1)/n} \leqslant e^{-t/2}$ for $n \geqslant 2$. For

$f \in C_b(\mathbb{R})$, Theorem P.89 and Theorem P.86, with dominating function $\sup|f|\, e^{-t/2}$ on $[0,\infty)$, give

$$\mathsf{E}[f(nZ_n)] = \int_{[0,\infty)} f(t) g_n(t)\,\mathrm{d}t \longrightarrow \int_{[0,\infty)} f(t) e^{-t}\,\mathrm{d}t = \mathsf{E}[f(E)].$$

(The integral against the density is Lemma P.99.)

(c) The indicator of $(-\infty, 2]$ is bounded and discontinuous only at 2, and $\mathsf{P}\{E = 2\} = 0$, so Theorem P.137(c) gives $\mathsf{P}\{nZ_n \leqslant 2\} \to \int_0^2 e^{-t}\,\mathrm{d}t = 1 - e^{-2} \approx 0.865$. ■

Exercise P.40 (Baldi, exercise 3.23)**.** (a) Let $\mu_n = (1-a_n)\delta_0 + a_n\delta_n$ with $a_n \in [0,1]$, $a_n \to 0$, where δ_x is the point mass at x. Show that $\mu_n \to \delta_0$ weakly.

(b) Choose a_n so that the means of μ_n tend to ∞; and so that the means tend to 0 but the variances tend to ∞.

(c) Show that $X_n \xrightarrow{d} X$ implies $\liminf_n \mathsf{E}\,|X_n| \geqslant \mathsf{E}\,|X|$ and $\liminf_n \mathsf{E}[X_n^2] \geqslant \mathsf{E}[X^2]$.

Solution. (a) For $f \in C_b(\mathbb{R})$, $\int f\,\mathrm{d}\mu_n = (1-a_n)f(0) + a_n f(n) \to f(0)$, as $|a_n f(n)| \leqslant a_n \sup|f|$.

(b) μ_n has mean na_n, second moment $n^2 a_n$ and variance $n^2 a_n(1-a_n)$. With $a_n = n^{-1/2}$ the mean

is $n^{1/2} \to \infty$. With $a_n = n^{-3/2}$ the mean is $n^{-1/2} \to 0$ while the variance is $n^{1/2}(1-n^{-3/2}) \to \infty$. The limit δ_0 has mean and variance 0.
(c) $x \mapsto |x|$ and $x \mapsto x^2$ are continuous, hence lower semi-continuous, and bounded below by 0; Theorem P.137(b) applied to the laws gives both inequalities, with Theorem P.89. Part (b) shows that they can be strict: an asymptotic variance, the variance of the limit law, need not be the limit of the variances. ■

Exercise P.41. Show by example that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ do not imply $X_n + Y_n \xrightarrow{d} X + Y$, and identify which hypothesis of Theorem P.139 rules the example out.

Solution. Let $X \sim N(0,1)$, $X_n = X$ and $Y_n = (-1)^n X$. Then $X_n \xrightarrow{d} N(0,1)$ and $Y_n \xrightarrow{d} N(0,1)$, yet $X_n + Y_n$ equals $2X$ for even n and 0 for odd n, so its law alternates between $N(0,4)$ and δ_0 and does not converge: for $f(x) = \min(|x|, 1) \in C_b(\mathbb{R})$, $\mathsf{E}\, f(X_n + Y_n)$ alternates between $\mathsf{E} \min(2|X|, 1) > 0$ (Proposition P.82(iv); $\mathsf{P}\{X = 0\} = 0$ because X has the law of a standard normal, hence the density ϕ, by Proposition P.125 and Corollary P.131) and 0.
Slutsky requires the second sequence to converge to a **constant**. That is exactly what fails:

convergence in distribution constrains only marginal laws, and the sum depends on the joint law, which is unconstrained. Convergence to a constant pins the joint law down, which is the content of part (i) of the theorem. ■

Exercise P.42 (Baldi, exercise 3.27)**.** Let $X_1, \dots, X_{12}$ be independent and uniform on $[0,1]$ and $W = X_1 + \cdots + X_{12} - 6$, once used to generate approximately standard normal numbers.

(a) Show that $\mathsf{E}[W] = 0$ and $\operatorname{var}(W) = 1$, and explain the approximation with Corollary P.142.

(b) Show that $\mathsf{E}[W^4] = 2.9$, to be compared with $\mathsf{E}[Z^4] = 3$ for $Z \sim N(0,1)$.

Solution. (a) $\mathsf{E}[X_i] = \int_0^1 x\,\mathrm{d}x = \frac{1}{2}$ and $\mathsf{E}[X_i^2] = \frac{1}{3}$ (Lemma P.99, with density $\mathbb{1}_{[0,1]}$ against Lebesgue measure), so $\operatorname{var}(X_i) = \frac{1}{12}$. Then $\mathsf{E}[W] = 0$ and, the covariances vanishing by Lemma P.105, $\operatorname{var}(W) = 12 \cdot \frac{1}{12} = 1$. Thus $W = n^{-1/2} \sum_{i \leqslant n} (X_i - \frac{1}{2})/\sigma$ with $n = 12$ and $\sigma^2 = \frac{1}{12}$, exactly the standardised sum of Corollary P.142, whose law approaches $N(0,1)$ as $n \to \infty$.

(b) $\mathsf{E}[Z^4] = 3$ is Lemma P.117, as Z has the law of a standard normal by Proposition P.125 and Corollary P.131. Let $U_i = X_i - \frac{1}{2}$, independent and uniform on $[-\frac{1}{2}, \frac{1}{2}]$, with $\mathsf{E}[U_i] = \mathsf{E}[U_i^3] = 0$, $\mathsf{E}[U_i^2] = \frac{1}{12}$ and $\mathsf{E}[U_i^4] = \int_{-1/2}^{1/2} u^4\,\mathrm{d}u = \frac{1}{80}$ (Lemma P.99 again). Expand $W^4 = \sum_{i,j,k,l} U_i U_j U_k U_l$.

By Lemma P.105 the expectation of a term factorises over its distinct indices, so it vanishes when some index occurs exactly once or exactly three times. The surviving terms have all four indices equal (twelve terms, each $\frac{1}{80}$) or two distinct indices each occurring twice: for each of the 66 pairs $\{i, j\}$ there are $\binom{4}{2} = 6$ placements, each with expectation $\frac{1}{144}$. Hence

$$\mathsf{E}[W^4] = \frac{12}{80} + \frac{6 \cdot 66}{144} = 0.15 + 2.75 = 2.9.$$

The kurtosis $2.9 < 3$ and the bound $|W| \leqslant 6$ show the tails of W are lighter than normal, which matters when many extreme draws are needed. ■

Exercise P.43. Verify the Lindeberg condition for $\xi_{ni} = a_{ni}\varepsilon_i$, $i = 1, \dots, k_n$, with ε_i i.i.d. mean zero, variance 1, where $\sum_i a_{ni}^2 = 1$ and $\max_i |a_{ni}| \to 0$.

Solution. $\mathsf{E}[\xi_{ni}] = 0$ and $\sum_i \operatorname{var}(\xi_{ni}) = \sum_i a_{ni}^2 = 1$. Writing $\delta_n = \max_i |a_{ni}|$, the event

$\{|\xi_{ni}| > \varepsilon\}$ is contained in $\{|\varepsilon_i| > \varepsilon/\delta_n\}$, so

$$\sum_i \mathsf{E}\big[\xi_{ni}^2; |\xi_{ni}| > \varepsilon\big] \leqslant \sum_i a_{ni}^2 \, \mathsf{E}\big[\varepsilon_1^2; |\varepsilon_1| > \varepsilon/\delta_n\big] = \mathsf{E}\big[\varepsilon_1^2; |\varepsilon_1| > \varepsilon/\delta_n\big],$$

using that the ε_i are identically distributed. Since $\varepsilon/\delta_n \to \infty$ and $\mathsf{E}[\varepsilon_1^2] = 1 < \infty$, dominated convergence (Theorem P.86) sends this to 0. The condition $\max_i |a_{ni}| \to 0$ is essential: with $a_{n1} = 1$ and the rest zero the sum is $\mathsf{E}[\varepsilon_1^2; |\varepsilon_1| > \varepsilon]$, which for $\varepsilon < 1$ is $1 - \mathsf{E}[\varepsilon_1^2; |\varepsilon_1| \leqslant \varepsilon] \geqslant 1 - \varepsilon^2 > 0$ and so does not tend to 0. ■

Exercise P.44 (Baldi, exercise 3.11)**.** Let $\{\varepsilon_t\}_{t\in\mathbb{Z}}$ be independent and identically distributed with mean 0 and variance $\sigma^2 > 0$, and $Y_t = \varepsilon_t \varepsilon_{t+1}$.

(a) Show that $\mathsf{E}[Y_t] = 0$, $\operatorname{var}(Y_t) = \sigma^4$ and $\operatorname{cov}(Y_s, Y_t) = 0$ for $s \neq t$.

(b) Show that $n^{-1} \sum_{t\leqslant n} Y_t \xrightarrow{p} 0$.

(c) Show that $n^{-1/2} \sum_{t\leqslant n} Y_t \xrightarrow{d} N(0, \sigma^4)$.

Solution. (a) By Lemma P.105, $\mathsf{E}[Y_t] = \mathsf{E}[\varepsilon_t]\,\mathsf{E}[\varepsilon_{t+1}] = 0$ and $\mathsf{E}[Y_t^2] = \mathsf{E}[\varepsilon_t^2]\,\mathsf{E}[\varepsilon_{t+1}^2] = \sigma^4$. Let $s < t$. If $t = s+1$, $Y_s Y_t = \varepsilon_s \cdot \varepsilon_{s+1}^2 \varepsilon_{s+2}$, a product of two independent integrable factors (the second by the same lemma), so $\mathsf{E}[Y_s Y_t] = \mathsf{E}[\varepsilon_s]\,\mathsf{E}[\varepsilon_{s+1}^2 \varepsilon_{s+2}] = 0$. If $t > s+1$, Y_s and Y_t are functions of disjoint sets of indices and $\mathsf{E}[Y_s Y_t] = \mathsf{E}[Y_s]\,\mathsf{E}[Y_t] = 0$.
(b) By (a), $\{Y_t\}$ is weakly stationary (Definition P.146) with $\gamma(0) = \sigma^4$ and $\gamma(k) = 0$ for $k \neq 0$, so Lemma P.147 applies.
(c) $Y_t = g(\eta_{t-1}, \eta_t)$ with $\eta_t = \varepsilon_{t+1}$ and $g(u, v) = uv$, so by Lemma P.150 $\{Y_t\}$ is strictly stationary and 1-dependent, with mean 0 and finite variance. Theorem P.151 gives the limit $N(0, \gamma(0) + 2\gamma(1)) = N(0, \sigma^4)$. The Y_t are uncorrelated but not independent (Y_t and Y_{t+1} share ε_{t+1}), so Corollary P.142 does not apply directly. ■

Exercise P.45. Let $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2)$ with $\theta > 0$. Find the limit law of $\sqrt{n}(\log \hat{\theta} - \log \theta)$ and of $\sqrt{n}(\hat{\theta}^2 - \theta^2)$.

Solution. By Theorem P.156 with $g(x) = \log x$ for $x > 0$ and $g(x) = 0$ otherwise, which is measurable and differentiable at $\theta > 0$, $g'(\theta) = 1/\theta$, so $\sqrt{n}(\log \hat{\theta} - \log \theta) \xrightarrow{d} N(0, \sigma^2/\theta^2)$,

reading $\log\hat\theta$ as $g(\hat\theta)$; the two differ only on $\{\hat\theta \leqslant 0\}$, whose probability tends to 0 since $\hat\theta - \theta = n^{-1/2}\sqrt{n}(\hat\theta - \theta) \xrightarrow{d} 0$ by Theorem P.139(iii), hence $\hat\theta \xrightarrow{p} \theta > 0$ by Proposition P.122. The asymptotic variance of a log-transformed estimator is $(\sigma/\theta)^2$, the squared coefficient of variation. With $g(x) = x^2$, $g'(\theta) = 2\theta$, giving $N(0, 4\theta^2\sigma^2)$. Note the method fails where $g'(\theta) = 0$: at $\theta = 0$ the limit $N(0, 4\theta^2\sigma^2)$ would be degenerate, and the correct normalisation is then n rather than $\sqrt{n}$: $n\hat\theta^2 = (\sqrt{n}\,\hat\theta)^2 \xrightarrow{d} \sigma^2\chi^2_1$ by the continuous mapping theorem. ■

Exercise P.46 (Baldi, exercise 3.31)**.** Let $X_n \sim \chi^2_n$.

(a) Show that $(X_n - n)/\sqrt{2n} \xrightarrow{d} N(0,1)$.

(b) **(Fisher's approximation.)** Show that $\sqrt{2X_n} - \sqrt{2n-1} \xrightarrow{d} N(0,1)$.

(c) Use (a) and (b) to approximate the 0.95 quantile of χ^2_{100}, which is 124.34.

Solution. (a) Convergence in distribution depends only on the laws, and by Definition P.115 X_n has the law of $S_n = \sum_{i\leqslant n} V_i$, where $V_i = Z_i^2$ for independent standard normal $Z_1, Z_2, \ldots$. The V_i are independent, identically distributed, with mean 1 and variance 2 (Proposition P.118), so Corollary P.142 gives $(S_n - n)/\sqrt{2n} \xrightarrow{d} N(0,1)$.

(b) With $T_n = S_n/n$, (a) and Theorem P.138 give $\sqrt{n}(T_n - 1) \xrightarrow{d} N(0,2)$. Let $g(t) = \sqrt{2t}$ for $t \geqslant 0$ and $g(t) = 0$ for $t < 0$, measurable and differentiable at 1 with $g'(1) = 1/\sqrt{2}$. Theorem P.156 gives $\sqrt{n}\{g(T_n) - g(1)\} = \sqrt{2S_n} - \sqrt{2n} \xrightarrow{d} N(0, 2 \cdot \frac{1}{2}) = N(0,1)$. Since $\sqrt{2n} - \sqrt{2n-1} = 1/(\sqrt{2n} + \sqrt{2n-1}) \to 0$, Theorem P.139 gives the claim, and it transfers to X_n because the law of $\sqrt{2X_n} - \sqrt{2n-1}$ is that of $\sqrt{2S_n} - \sqrt{2n-1}$.
(c) Let $z = 1.6449$ be the 0.95 quantile of $N(0,1)$. Reading (a) as $\mathsf{P}\{X_n \leqslant x\} \approx \Phi((x-n)/\sqrt{2n})$ gives $x \approx n + z\sqrt{2n} = 123.26$ for $n = 100$. Reading (b) as $\mathsf{P}\{X_n \leqslant x\} \approx \Phi(\sqrt{2x} - \sqrt{2n-1})$ gives $x \approx \frac{1}{2}(z + \sqrt{2n-1})^2 = 124.06$, much closer: the square root removes most of the skewness of χ^2_n. ■

Exercise P.47 (Baldi, exercise 3.33)**.** Let $X_1, X_2, \ldots$ be independent with $\mathsf{P}\{X_i = 1\} = \mathsf{P}\{X_i = -1\} = \frac{1}{2}$ and $\bar{X}_n = n^{-1} \sum_{i \leqslant n} X_i$. Find the limits in distribution of (a) $\sqrt{n} \sin \bar{X}_n$, (b) $\sqrt{n}(1 - \cos \bar{X}_n)$ and (c) $n(1 - \cos \bar{X}_n)$.

Solution. $\mathsf{E}[X_i] = 0$ and $\operatorname{var}(X_i) = 1$, so $\sqrt{n}\,\bar{X}_n \xrightarrow{d} N(0,1)$ by Corollary P.142, and $\bar{X}_n = n^{-1/2} \cdot \sqrt{n}\,\bar{X}_n \xrightarrow{d} 0$ by Theorem P.139(iii), hence $\bar{X}_n \xrightarrow{p} 0$ by Proposition P.122.

(a) By Theorem P.156 with $g = \sin$, $g'(0) = 1$: the limit is $N(0,1)$.
(b) With $g(x) = 1 - \cos x$, $g'(0) = 0$, and Theorem P.156 (first form, $T \sim N(0,1)$) gives the limit $0 \cdot T = 0$: the point mass at 0.
(c) Let $q(x) = (1 - \cos x)/x^2$ for $x \neq 0$ and $q(0) = \frac{1}{2}$; q is measurable and continuous at 0 (Theorem P.14 as in Exercise P.37), so $q(\bar{X}_n) \xrightarrow{p} \frac{1}{2}$ by Theorem P.138. Also $(\sqrt{n}\bar{X}_n)^2 \xrightarrow{d} Z^2$ with $Z \sim N(0,1)$, by Theorem P.138. Since $n(1 - \cos \bar{X}_n) = (\sqrt{n}\bar{X}_n)^2 q(\bar{X}_n)$, Theorem P.139 gives the limit $\frac{1}{2}Z^2$, that is, $\frac{1}{2}\chi^2_1$. As in Exercise P.45, a vanishing derivative calls for the normalisation n and a second-order limit. ■

Results used in the course

Exercise P.48 (Baldi, exercise 2.24)**. (Kullback–Leibler divergence.)** Let f and g be probability densities with respect to a measure μ on $\mathcal{B}(\mathbb{R}^d)$, and define

$$H(f;g) = \int_{\{f>0\}} f \log \frac{f}{g} \, \mathrm{d}\mu,$$

the integrand being $+\infty$ where $g = 0 < f$.

(a) Show that the integral exists in $[0,\infty]$ and that $H(f;g)=0$ if and only if $f=g$ μ-almost everywhere.

(b) Compute H for f,g the binomial probabilities $B(n,q)\{j\} = \binom{n}{j}q^j(1-q)^{n-j}$ and $B(n,p)\{j\} = \binom{n}{j}p^j(1-p)^{n-j}$, $0<p,q<1$ (μ the counting measure on $\{0,\dots,n\}$), and for f,g the exponential densities $\rho e^{-\rho t}$ and $\beta e^{-\beta t}$ on $(0,\infty)$.

(c) Let Y have density $f(\cdot;\boldsymbol{\theta}_0)$ in a family as in Definition P.158 with $\mathcal{X}=\mathbb{R}^d$, so that H of (a) applies, and suppose $\ell(\boldsymbol{\theta})=\log f(Y;\boldsymbol{\theta})$ and $\ell(\boldsymbol{\theta}_0)$ are integrable. Deduce $\mathsf{E}[\ell(\boldsymbol{\theta})]\leqslant\mathsf{E}[\ell(\boldsymbol{\theta}_0)]$, with equality only if $f(\cdot;\boldsymbol{\theta})=f(\cdot;\boldsymbol{\theta}_0)$ μ-almost everywhere.

Solution. (a) **An inequality.** Let $h(u)=u-1-\log u$ for $u>0$. Then $h(1)=0$ and $h'(u)=1-1/u$, and by Corollary P.7 $h(u)=h'(\xi)(u-1)$ for some ξ between 1 and u, where $h'(\xi)$ has the sign of $u-1$; so $h(u)>0$ for $u\neq 1$. That is, $\log u\leqslant u-1$, with equality only at $u=1$.

Existence and sign. On $\{f>0\}$ let $k=f\log(f/g)$. Where $g>0$, the inequality with $u=g/f$ gives $k\geqslant f-g$, and where $g=0$, $k=\infty\geqslant f-g$. Thus $k^-\leqslant(f-g)^-\leqslant g$, which is

integrable, so the integral of k exists in $(-\infty,\infty]$ (Definition P.84), and

$$H(f;g) \geqslant \int_{\{f>0\}} (f-g)\,\mathrm{d}\mu = 1 - \int_{\{f>0\}} g\,\mathrm{d}\mu \geqslant 0.$$

Equality. If $H(f;g)=0$ then k is integrable and $\int_{\{f>0\}}\{k-(f-g)\}\,\mathrm{d}\mu \leqslant 0$ with a non-negative integrand, so by Proposition P.82(iv) $k = f-g$ almost everywhere on $\{f>0\}$, which forces $g/f=1$ there; and $\int_{\{f>0\}} g\,\mathrm{d}\mu = 1$ forces $g=0$ almost everywhere on $\{f=0\}$. Hence $f=g$ almost everywhere. The converse is clear.

(b) For the binomials, $\log(f/g)(j) = j\log(q/p) + (n-j)\log\{(1-q)/(1-p)\}$, and the $B(n,q)$ mean is $\sum_j j\binom{n}{j}q^j(1-q)^{n-j} = nq\sum_{j\geqslant 1}\binom{n-1}{j-1}q^{j-1}(1-q)^{n-j} = nq$, by $j\binom{n}{j} = n\binom{n-1}{j-1}$ and the binomial theorem, so

$$H = n\Big\{q\log\frac{q}{p} + (1-q)\log\frac{1-q}{1-p}\Big\}.$$

For the exponentials, $\log(f/g)(t) = \log(\rho/\beta) - (\rho-\beta)t$ and $\int t\rho e^{-\rho t}\,\mathrm{d}t = 1/\rho$, so $H = \log(\rho/\beta) - (\rho-\beta)/\rho = h(\beta/\rho)$, positive unless $\beta=\rho$, as (a) requires.

(c) Almost surely $f(Y;\boldsymbol{\theta}_0) > 0$, as $\mathsf{P}\{f(Y;\boldsymbol{\theta}_0)=0\} = \int_{\{f(\cdot;\boldsymbol{\theta}_0)=0\}} f(\cdot;\boldsymbol{\theta}_0)\,\mathrm{d}\mu = 0$ (Lemma P.99),

and $\ell(\boldsymbol{\theta}_0)-\ell(\boldsymbol{\theta}) = \log\{f(Y;\boldsymbol{\theta}_0)/f(Y;\boldsymbol{\theta})\}$ is integrable, so by Lemma P.99, integrating against the density $f(\cdot;\boldsymbol{\theta}_0)$, $\mathsf{E}[\ell(\boldsymbol{\theta}_0)]-\mathsf{E}[\ell(\boldsymbol{\theta})] = H(f(\cdot;\boldsymbol{\theta}_0); f(\cdot;\boldsymbol{\theta})) \geqslant 0$, with equality only in the stated case by (a). The expected log-likelihood is maximised at the true parameter; this is why the maximiser of ℓ_n targets $\boldsymbol{\theta}_0$, and why the model must be identifiable for that maximiser to be unique. ■

Exercise P.49 (Baldi, exercise 2.46)**. (Exponential families.)** Let μ be a probability on $\mathbb{R}$ with $L(t) = \int e^{tx}\,\mu(\mathrm{d}x) < \infty$ for $t \in (a,b)$, where $a < 0 < b$, and let $\psi = \log L$.

(a) Show that L is twice differentiable on (a,b) with $L'(t) = \int x e^{tx}\,\mu(\mathrm{d}x)$ and $L''(t) = \int x^2 e^{tx}\,\mu(\mathrm{d}x)$, and that μ has mean $\psi'(0)$ and variance $\psi''(0)$.

(b) For $\gamma \in (a,b)$ let $\mu_\gamma(\mathrm{d}x) = e^{\gamma x}\mu(\mathrm{d}x)/L(\gamma)$. Show that μ_γ is a probability with $L_\gamma(t) = L(t+\gamma)/L(\gamma)$ for $t+\gamma \in (a,b)$, mean $\psi'(\gamma)$ and variance $\psi''(\gamma)$; deduce that ψ is convex and the mean of μ_γ is non-decreasing in γ, strictly increasing unless μ is a point mass.

(c) Identify μ_γ when $\mu = N(0,\sigma^2)$ and when $\mu = B(n,p)$.

Solution. (a) Fix $t \in (a,b)$ and choose $a < c' < c < t < d < d' < b$. For $\delta > 0$ and $y \geqslant 0$, $ye^{-\delta y} \leqslant 1/(e\delta)$ and $y^2e^{-\delta y} \leqslant 4/(e\delta)^2$ (maximise in y). For $s \in (c,d)$: if $x \geqslant 0$,

$|x|e^{sx} \leqslant xe^{-(d'-d)x}e^{d'x}$; if $x<0$, $|x|e^{sx} \leqslant |x|e^{-(c-c')|x|}e^{c'x}$. Hence, with constants K_1, K_2,

$$|x|e^{sx} \leqslant K_1(e^{c'x}+e^{d'x}), \qquad x^2e^{sx} \leqslant K_2(e^{c'x}+e^{d'x}),$$

and the right sides are μ-integrable because $L(c'), L(d') < \infty$. Theorem P.87 on $I=(c,d)$, applied to e^{sx} (integrable, derivative xe^{sx} dominated by the first bound) and then to xe^{sx} (integrable by the first bound, derivative x^2e^{sx} dominated by the second), gives both formulas. Then $\psi' = L'/L$ and $\psi'' = L''/L - (L'/L)^2$, and since $L(0)=1$, $\psi'(0) = \int x\,d\mu$ and $\psi''(0) = \int x^2\,d\mu - (\int x\,d\mu)^2$.
(b) μ_γ is a measure by Corollary P.80(ii) with total mass $L(\gamma)/L(\gamma) = 1$. For Borel $h \geqslant 0$, $\int h\,d\mu_\gamma = \int he^{\gamma x}\,d\mu/L(\gamma)$, by Lemma P.99 applied to the identity map on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_\gamma)$, whose law μ_γ has density $e^{\gamma x}/L(\gamma)$ with respect to μ. With $h = e^{tx}$ this gives $L_\gamma(t) = L(t+\gamma)/L(\gamma)$, finite for t in the interval $(a-\gamma, b-\gamma)$, which contains 0, and $\psi_\gamma(t) = \psi(t+\gamma) - \psi(\gamma)$. Part (a) applied to μ_γ gives mean $\psi_\gamma'(0) = \psi'(\gamma)$ and variance $\psi_\gamma''(0) = \psi''(\gamma) \geqslant 0$. So $\psi'' \geqslant 0$ on (a,b), ψ is convex and ψ' is non-decreasing (Lemma P.19 and Corollary P.7). If $\psi''(\gamma) = 0$ then μ_γ is a point mass δ_m (zero variance and Proposition P.82(iv)), and since $e^{\gamma x} > 0$, μ and μ_γ have the same null sets, so $\mu = \delta_m$. Otherwise $\psi'' > 0$ and ψ' is strictly increasing.
(c) For $\mu = N(0,\sigma^2)$ with $\sigma = 0$, $\mu = \delta_0$, so $L \equiv 1$ and $\mu_\gamma = \delta_0 = N(\sigma^2\gamma, \sigma^2)$. For $\sigma > 0$, complet-

ing the square, $e^{\gamma x}e^{-x^2/(2\sigma^2)} = e^{\sigma^2\gamma^2/2}e^{-(x-\sigma^2\gamma)^2/(2\sigma^2)}$; integrating against the density (Lemma P.99 and Remark 18(e)) gives $L(\gamma) = e^{\sigma^2\gamma^2/2}$ for all γ, and μ_γ has the density of $N(\sigma^2\gamma, \sigma^2)$, so $\psi'(\gamma) = \sigma^2\gamma$. For $\mu = B(n,p)$, the binomial theorem gives $L(\gamma) = (1-p+pe^\gamma)^n$ and

$$\mu_\gamma\{j\} = \binom{n}{j}\frac{(pe^\gamma)^j(1-p)^{n-j}}{(1-p+pe^\gamma)^n} = \binom{n}{j}p_\gamma^j(1-p_\gamma)^{n-j}, \qquad p_\gamma = \frac{pe^\gamma}{1-p+pe^\gamma},$$

so $\mu_\gamma = B(n,p_\gamma)$, with $\operatorname{logit} p_\gamma = \operatorname{logit} p + \gamma$ and mean $\psi'(\gamma) = np_\gamma$: tilting the binomial shifts its log-odds, the canonical link of logistic regression. ■

Exercise P.50. Let $\sqrt{n}(\hat{\boldsymbol\beta} - \boldsymbol\beta) \xrightarrow{d} N(\mathbf{0}, \boldsymbol\Sigma)$ with $\boldsymbol\Sigma$ positive definite and let $\hat{\boldsymbol\Sigma}$ be symmetric with $\hat{\boldsymbol\Sigma} \xrightarrow{p} \boldsymbol\Sigma$. Show that under $H_0\colon \mathbf{R}\boldsymbol\beta = \mathbf{r}$ with $\mathbf{R}$ of full row rank q,

$$n(\mathbf{R}\hat{\boldsymbol\beta} - \mathbf{r})'\left(\mathbf{R}\hat{\boldsymbol\Sigma}\mathbf{R}'\right)^{-1}(\mathbf{R}\hat{\boldsymbol\beta} - \mathbf{r}) \xrightarrow{d} \chi^2_q,$$

with the convention $\mathbf{A}^{-1} := \mathbf{0}$ for singular $\mathbf{A}$ of Theorem P.157.

Solution. Under H_0, $\sqrt{n}(\mathbf{R}\hat{\boldsymbol{\beta}}-\mathbf{r}) = \mathbf{R}\sqrt{n}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}) \xrightarrow{d} N(\mathbf{0}, \mathbf{R}\boldsymbol{\Sigma}\mathbf{R}')$ by Theorem P.138 applied to the linear map $\mathbf{v} \mapsto \mathbf{R}\mathbf{v}$, the image of $V \sim N(\mathbf{0}, \boldsymbol{\Sigma})$ being $\mathbf{R}V \sim N(\mathbf{0}, \mathbf{R}\boldsymbol{\Sigma}\mathbf{R}')$ by Theorem P.114. The limiting covariance is positive definite: for $\mathbf{v} \neq \mathbf{0}$, $\mathbf{R}'\mathbf{v} \neq \mathbf{0}$ because $\mathbf{R}$ has full row rank, so $\mathbf{v}'\mathbf{R}\boldsymbol{\Sigma}\mathbf{R}'\mathbf{v} > 0$. Also $\mathbf{R}\hat{\boldsymbol{\Sigma}}\mathbf{R}' \xrightarrow{p} \mathbf{R}\boldsymbol{\Sigma}\mathbf{R}'$ by Theorem P.138. Theorem P.157 with $p = q$ then gives the χ^2_q limit. This is the Wald statistic of Units 7–8. ■

Exercise P.51. In Theorem P.141 normality is nowhere assumed of the ξ_{nj}, yet the proof introduces Gaussian variables ζ_{nj}. Explain their role, and say where in the Breusch–Pagan derivation of Unit 4 normality of the **errors** does real work. (There the errors ε_i are independent and identically distributed with mean 0, variance σ^2, $\mathsf{E}\,\varepsilon_i^4 < \infty$ and $\kappa = \operatorname{var}(\varepsilon_i^2/\sigma^2)$, as in the Breusch–Pagan remark; SSR_z and R_z^2 are the regression sum of squares and the R^2 of the regression of ε_i^2/σ^2, in practice of squared residuals over an estimate of σ^2, on an intercept and variance regressors $\mathbf{z}_i$; the Breusch–Pagan statistic is $LM = \mathrm{SSR}_z/2$, and Koenker's statistic is nR_z^2.)

Solution. The ζ_{nj} are a comparison array, not an assumption: they are constructed to have the same variances c_{nj} as the ξ_{nj}, so that their sum is exactly $N(0, c_n)$ and the problem reduces to

showing the two products of characteristic functions are close. Normality enters only as a device for producing a reference sum with a known law.

In Unit 4 normality of the errors is used in exactly one place: to evaluate $\operatorname{var}(\varepsilon_i^2/\sigma^2) = 2$, which fixes the divisor 2 in $LM = \mathrm{SSR}_z/2$. Without it that variance is κ and the correctly scaled statistic is Koenker's nR_z^2. The central limit theorem behind both is the same, and needs no normality at all. ■